
Abelian Varieties

— *EYNTKA* Series —

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Part I

Complex Abelian Varieties

An abelian variety wears three faces at once: analytically a complex torus $\mathbb{C}^g/\Lambda$, cohomologically a polarized integral Hodge structure of weight one, and algebraically a projective group variety. This Part develops the first two faces in full, working entirely over the complex numbers. The torus description turns every question into linear algebra over a lattice refined by a complex structure: polarizations become positive Hermitian forms, line bundles become theta functions, and the dual variety and the endomorphism algebra become explicit constructions on the pair (V, Λ) .

The one fact the analytic viewpoint cannot supply from within is that a polarizable torus is genuinely projective; that is the work of the theta functions in Chapter 4. Everything else is a computation with (V, Λ) , and Part II rebuilds the same class of objects over an arbitrary field, where no lattice is available and the theorem of the cube takes its place.

Complex Tori

An elliptic curve over $\mathbb{C}$ is two things at once: a projective cubic carved out by a Weierstrass equation, and a quotient $\mathbb{C}/\Lambda$ of the complex plane by a lattice. The second description is the transparent one. The group law is ordinary addition, the torsion points are the obvious fractions of periods, and the whole object is a compact complex Lie group of the simplest imaginable kind. This chapter takes that second description as the starting point for everything that follows and asks what survives in higher dimension: replace $\mathbb{C}$ by $\mathbb{C}^g$ and Λ by a lattice of rank $2g$, and one obtains a *complex torus* $X = \mathbb{C}^g/\Lambda$, a compact complex Lie group of dimension g .

Complex tori are easy to write down and easy to map between: a homomorphism is nothing more than a $\mathbb{C}$ -linear map of the universal covers carrying one lattice into the other, so the entire category is governed by linear algebra over $\mathbb{Z}$ and $\mathbb{C}$. What is not easy, and what occupies the rest of Part I, is the question of which complex tori are *algebraic*, that is, embed into projective space. For $g = 1$ the answer is every one of them; for $g \geq 2$ it is a delicate condition on the lattice, and most tori fail it. The present chapter develops the linear-algebraic anatomy of a complex torus, its endomorphisms, and the Hodge structure on its first cohomology, and closes by making the algebraicity question precise, so that Chapters 2 and 3 can answer it.

1.1 Complex Tori and Their Morphisms

The starting object of the entire analytic theory is the simplest quotient of a complex vector space by a discrete subgroup: a complex torus. Its transparency is the whole point. A complex torus is built from a g -dimensional vector space and a lattice of rank $2g$ inside it, and every question about the torus that the first chapters will ask reduces, sooner or later, to a question about that lattice together with the complex structure on the ambient space. This section fixes the object, establishes that it is a compact connected complex Lie group, and then settles the first structural question one can ask: what are the maps between complex tori? The answer, that every holomorphic homomorphism is

induced by a $\mathbb{C}$ -linear map of the ambient vector spaces carrying one lattice into the other, is the linearization principle that makes the category of complex tori governable by linear algebra over $\mathbb{Z}$.

The reader who has met the $g = 1$ case in [3] will recognize everything here as the higher-dimensional shadow of the picture $E = \mathbb{C}/\Lambda$: a compact Riemann surface of genus one, a complex Lie group, whose endomorphisms are exactly the complex numbers scaling the lattice into itself. The passage from $\mathbb{C}$ to $\mathbb{C}^g$ costs nothing conceptually and buys the whole subject.

A lattice in a real vector space is a discrete subgroup of maximal possible rank. Recording this precisely is the first order of business, because the rank condition, that the lattice has rank $2g$ in a space of real dimension $2g$, is exactly what makes the quotient compact.

Definition 1.1.1: Full Lattice

Let $V = \mathbb{C}^g$, regarded as a real vector space of dimension $2g$. A *full lattice* in V is a discrete subgroup $\Lambda \subseteq V$ of rank $2g$ over $\mathbb{Z}$; equivalently, $\Lambda = \mathbb{Z}\lambda_1 \oplus \cdots \oplus \mathbb{Z}\lambda_{2g}$ for real-linearly independent vectors $\lambda_1, \dots, \lambda_{2g} \in V$, so that $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$. The vectors $\lambda_1, \dots, \lambda_{2g}$ form a $\mathbb{Z}$ -basis of Λ .

The two conditions in the definition, discreteness and rank $2g$, are the ones that matter. Discreteness alone permits a subgroup of smaller rank, such as $\mathbb{Z} \subseteq \mathbb{C}$; the rank condition forces the lattice to span V over $\mathbb{R}$, so that the quotient V/Λ is compact rather than a cylinder or a line. Every full lattice in V is the image of the standard lattice $\mathbb{Z}^{2g} \subseteq \mathbb{R}^{2g}$ under a real-linear isomorphism $\mathbb{R}^{2g} \xrightarrow{\sim} V$, and two $\mathbb{Z}$ -bases of the same Λ differ by an element of $\mathrm{GL}_{2g}(\mathbb{Z})$, exactly as in the $g = 1$ case where the change-of-basis group is $\mathrm{GL}_2(\mathbb{Z})$.

The quotient of V by a full lattice is the object of study.

Definition 1.1.2: Complex Torus

Let $V = \mathbb{C}^g$ and let $\Lambda \subseteq V$ be a full lattice. The *complex torus* of dimension g associated to (V, Λ) is the quotient group

$$X = V/\Lambda$$

equipped with the quotient topology and the quotient complex structure. Its points are the cosets $v + \Lambda$; the projection $\pi: V \rightarrow X$, $\pi(v) = v + \Lambda$, is the quotient map.

A complex torus is at once a group, a topological space, and a complex manifold, and the content of the definition is that these three structures are compatible. The group structure is the additive one inherited from V : the sum of $v + \Lambda$ and $w + \Lambda$ is $(v + w) + \Lambda$, well defined because Λ is a subgroup. The complex structure comes from V through the covering π , which is a local biholomorphism, so that a function on an open set of X is holomorphic precisely when its pullback along π is holomorphic on the corresponding open set of V . That these two structures cohere, that the group operations are holomorphic, is the assertion that X is a complex Lie group, which is the next proposition.

The name “torus” is topological. As a real manifold, X is diffeomorphic to the product of $2g$ circles: choosing a $\mathbb{Z}$ -basis $\lambda_1, \dots, \lambda_{2g}$ of Λ identifies V with $\mathbb{R}^{2g}$ carrying Λ to $\mathbb{Z}^{2g}$, and $\mathbb{R}^{2g}/\mathbb{Z}^{2g} = (\mathbb{R}/\mathbb{Z})^{2g} \cong (S^1)^{2g}$. The complex structure is the additional datum: the same real torus $(S^1)^{2g}$ carries many inequivalent complex structures once $g \geq 1$, coming from the many ways a rank- $2g$ lattice can sit inside $\mathbb{C}^g$, and distinguishing them is one of the threads of this chapter.

Proposition 1.1.3: A Complex Torus Is a Compact Connected Complex Lie Group

Let $X = V/\Lambda$ be a complex torus of dimension g . Then X is a compact, connected, abelian complex Lie group of complex dimension g : the multiplication $m: X \times X \rightarrow X$, $m(x, y) = x + y$, and the inversion $\iota: X \rightarrow X$, $\iota(x) = -x$, are holomorphic, and X is compact and connected.

Proof:

Step 1: The complex structure and dimension. The projection $\pi: V \rightarrow X$ is a covering map of topological spaces, since Λ acts on $V = \mathbb{C}^g$ by translations, properly discontinuously and freely (discreteness of Λ gives, for each v , a neighborhood U with $(U + \lambda) \cap U = \emptyset$ for all nonzero $\lambda \in \Lambda$, and $\pi|_U$ a homeomorphism onto its image). Declaring $\pi|_U$ to be a biholomorphism transports the complex structure of V to X ; the transition maps between two such charts are translations $v \mapsto v + \lambda$ of V , which are biholomorphic, so the charts are compatible and X is a complex manifold of complex dimension g . This is the standard construction of the quotient complex structure ([10]).

Step 2: Compactness and connectedness. Fix a $\mathbb{Z}$ -basis $\lambda_1, \dots, \lambda_{2g}$ of Λ and consider the closed fundamental parallelootope

$$P = \left\{ \sum_{j=1}^{2g} t_j \lambda_j \mid 0 \leq t_j \leq 1 \right\}$$

Since every $v \in V$ is congruent modulo Λ to a point of P , the restriction $\pi|_P: P \rightarrow X$ is surjective. As P is compact (a continuous image of the cube $[0, 1]^{2g}$) and π is continuous, $X = \pi(P)$ is compact. Connectedness is immediate: V is connected and π is a continuous surjection, so its image X is connected.

Step 3: The group operations are holomorphic. The addition $\tilde{m}: V \times V \rightarrow V$, $(v, w) \mapsto v + w$, is holomorphic, and it descends to $m: X \times X \rightarrow X$ because $\tilde{m}$ carries $\Lambda \times \Lambda$ into Λ . To see m is holomorphic near a point (x, y) , lift to points $v, w \in V$ with $\pi(v) = x$, $\pi(w) = y$; on a suitable product of chart neighborhoods, m reads in local coordinates as $\tilde{m}$ followed by a chart of X , hence is holomorphic. The same argument applied to $\tilde{\iota}: V \rightarrow V$, $v \mapsto -v$, which carries Λ into Λ , shows ι is holomorphic. Commutativity of m is inherited from the commutativity of addition in V . Thus X is an abelian complex Lie group, completing the proof.

The proposition is the reason complex tori are the natural first examples in the analytic theory of abelian varieties: they are group objects in the category of compact complex manifolds, produced by nothing more than a quotient. The group is written additively throughout, with identity element $0 = 0 + \Lambda$ and inverse $-x$; this matches the $g = 1$ convention of [3], where the group law on $E = \mathbb{C}/\Lambda$ is the descended addition of $\mathbb{C}$. The Lie algebra of X is V itself, with the trivial (abelian) bracket, and the exponential map $\text{Lie}(X) = V \rightarrow X$ is simply the projection π ; this is the analytic incarnation of the fact that a complex torus is its own “linearization”, the observation that governs the study of endomorphisms in the next section.

The first concrete instance is the $g = 1$ torus, which is an elliptic curve; it is the running example that every later construction specializes to.

Example 1.1.4: The Elliptic Curve as a Complex Torus

Take $g = 1$, so $V = \mathbb{C}$, and let $\tau \in \mathbb{H}$ lie in the upper half-plane. The lattice

$$\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau = \{m + n\tau : m, n \in \mathbb{Z}\}$$

is a full lattice in $\mathbb{C}$: the generators 1 and τ are $\mathbb{R}$ -linearly independent precisely because $\text{Im}\tau > 0$. The quotient

$$E = \mathbb{C}/\Lambda$$

is a complex torus of dimension 1. By proposition 1.1.3 it is a compact connected abelian complex Lie group, and as a Riemann surface it has genus 1: the holomorphic 1-form dz on $\mathbb{C}$ is Λ -invariant, so it descends to a nowhere-vanishing holomorphic 1-form on E , forcing the genus to be 1.

The content that goes beyond the group and manifold structure is that E is algebraic. The Weierstrass uniformization theorem provides an isomorphism of complex Lie groups

$$\mathbb{C}/\Lambda \xrightarrow{\sim} E(\mathbb{C}), \quad z + \Lambda \mapsto [\wp(z) : \wp'(z) : 1]$$

between the torus and the complex points of the projective plane cubic $y^2 = 4x^3 - g_2x - g_3$, where $\wp$ is the Weierstrass function of Λ and g_2, g_3 are the associated invariants. The construction of $\wp$, the proof that this map is a biholomorphism, and the verification that it is a group homomorphism are carried out in [10] and [3]; they are not reproduced here. What matters for the present chapter is that in dimension one the complex torus and the algebraic curve are the same object, and that this coincidence, automatic when $g = 1$, becomes a nontrivial condition, the Riemann relations, once $g \geq 2$.

Products of complex tori are again complex tori, and the smallest higher-dimensional example is the square of an elliptic curve. It will recur throughout the book as the concrete non-simple abelian surface against which general constructions are tested.

Example 1.1.5: The Product $E \times E$

Let $E = \mathbb{C}/\Lambda$ be the elliptic curve of example 1.1.4, with $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$. Set

$$V = \mathbb{C}^2, \quad \Lambda' = \Lambda \times \Lambda = \{(m_1 + n_1\tau, m_2 + n_2\tau) : m_i, n_i \in \mathbb{Z}\}$$

Then Λ' is a full lattice in $\mathbb{C}^2$: a $\mathbb{Z}$ -basis is

$$(1, 0), \quad (\tau, 0), \quad (0, 1), \quad (0, \tau)$$

four vectors, real-linearly independent because $\{1, \tau\}$ is $\mathbb{R}$ -linearly independent in $\mathbb{C}$. The associated complex torus is

$$X = V/\Lambda' = (\mathbb{C}/\Lambda) \times (\mathbb{C}/\Lambda) = E \times E$$

a complex torus of dimension 2. Its period matrix, the 2×4 matrix whose columns are the chosen $\mathbb{Z}$ -basis expressed in the standard $\mathbb{C}$ -basis of $\mathbb{C}^2$, is

$$\Pi = \begin{pmatrix} 1 & \tau & 0 & 0 \\ 0 & 0 & 1 & \tau \end{pmatrix}$$

As a group, X is $E \times E$ with coordinatewise addition, and as a complex manifold it is the product of E with itself. This torus is the prototype of a non-simple abelian surface: it is isogenous to a product, it carries the diagonal and antidiagonal copies of E as subtori, and its endomorphism ring is the ring $M_2(\text{End}(E))$ of 2×2 matrices over $\text{End}(E)$, a point developed in section 1.2.

With the object in hand, the natural question is what its morphisms are. A morphism of complex tori should respect both structures they carry: the complex manifold structure (holomorphy) and the group structure (homomorphism). The following definition names the maps that respect both.

Definition 1.1.6: Homomorphism of Complex Tori

Let $X = V/\Lambda$ and $X' = V'/\Lambda'$ be complex tori. A *homomorphism* $f: X \rightarrow X'$ is a map that is simultaneously holomorphic and a homomorphism of abelian groups, so that $f(x + y) = f(x) + f(y)$ for all $x, y \in X$. An *isomorphism* is a homomorphism admitting a holomorphic inverse homomorphism. When $X = X'$, a homomorphism $X \rightarrow X$ is an *endomorphism*, and the endomorphisms form a ring $\text{End}(X)$ under pointwise addition and composition.

A homomorphism automatically preserves the identity, $f(0) = 0$. This is the constraint that distinguishes a homomorphism from an arbitrary holomorphic map of the underlying manifolds, and the next result shows exactly how much it constrains: a holomorphic homomorphism is a $\mathbb{C}$ -linear map of the universal covers that carries one lattice into the other, nothing more. The endomorphism ring $\text{End}(X)$ is the central invariant of an abelian variety; the linearization theorem below is what makes it computable, by identifying $\text{End}(X)$ with a ring of matrices preserving a lattice.

The mechanism behind the theorem is the interplay of two facts. First, V is the universal cover of X (it is simply connected, being a vector space, and $\pi: V \rightarrow X$ is a covering), so a based map out of X lifts uniquely to a map out of V . Second, holomorphy plus the homomorphism property force the lift to be not just holomorphic but $\mathbb{C}$ -linear, because its complex derivative is a Λ -periodic holomorphic function on the compact torus, hence constant. The clean statement separates the general holomorphic map (a translate of a homomorphism) from the homomorphism itself.

Proposition 1.1.7: Holomorphic Maps of Tori Are Linear

Let $X = V/\Lambda$ and $X' = V'/\Lambda'$ be complex tori, with projections $\pi: V \rightarrow X$ and $\pi': V' \rightarrow X'$.

1. Every holomorphic homomorphism $f: X \rightarrow X'$ lifts to a unique $\mathbb{C}$ -linear map $F: V \rightarrow V'$ satisfying $F(\Lambda) \subseteq \Lambda'$ and $\pi' \circ F = f \circ \pi$. Conversely, every $\mathbb{C}$ -linear $F: V \rightarrow V'$ with $F(\Lambda) \subseteq \Lambda'$ descends to a holomorphic homomorphism $f: X \rightarrow X'$. These two operations are mutually inverse, so

$$\text{Hom}(X, X') \cong \{F \in \text{Hom}_{\mathbb{C}}(V, V') : F(\Lambda) \subseteq \Lambda'\}$$

an isomorphism of abelian groups; for $X = X'$ it is an isomorphism of rings $\text{End}(X) \cong \{F \in \text{End}_{\mathbb{C}}(V) : F(\Lambda) \subseteq \Lambda\}$.

2. Every holomorphic map $h: X \rightarrow X'$ (not assumed a homomorphism) is a translate of a homomorphism: there is a unique homomorphism $f: X \rightarrow X'$ and a unique point $c = h(0) \in X'$ with $h = t_c \circ f$, where t_c is translation by c . In particular h is a homomorphism if and only if $h(0) = 0$.

Proof:

The two parts are proved in turn; part (2) is deduced from part (1) at the end.

1. *Existence and uniqueness of the lift.* Since V' is simply connected and $\pi': V' \rightarrow X'$ is a covering map, the composite $f \circ \pi: V \rightarrow X'$ lifts through π' : because V is simply connected, the lifting criterion for covering spaces ([10]) produces a continuous $F: V \rightarrow V'$ with $\pi' \circ F = f \circ \pi$, and this lift is unique once we normalize $F(0) = 0$ (using $f(0) = 0$, which holds as f is a homomorphism). The map F is holomorphic: locally π and π' are biholomorphisms and f is holomorphic, so $F = (\pi')^{-1} \circ f \circ \pi$ is holomorphic on each chart.

Next, F is a homomorphism of the additive group V . For fixed $w \in V$, consider

$$g_w(v) = F(v + w) - F(v) - F(w)$$

Applying π' and using that f is a homomorphism together with $\pi' \circ F = f \circ \pi$, one computes $\pi'(g_w(v)) = f(\pi(v + w)) - f(\pi(v)) - f(\pi(w)) = 0$ in X' , so $g_w(v) \in \Lambda'$ for all v . Thus $g_w: V \rightarrow \Lambda'$ is a continuous map from the connected space V into the discrete set Λ' , hence constant; its value at $v = 0$ is $g_w(0) = F(w) - F(0) - F(w) = 0$. Therefore $g_w \equiv 0$, that is, $F(v + w) = F(v) + F(w)$ for all v, w , so F is additive.

The lift is $\mathbb{C}$ -linear. F is holomorphic and additive; additivity gives $\mathbb{R}$ -linearity (a continuous additive map of real vector spaces is $\mathbb{R}$ -linear). It remains to promote $\mathbb{R}$ -linearity to $\mathbb{C}$ -linearity, and this is where holomorphy enters. Because F is $\mathbb{R}$ -linear its differential dF is constant, equal to F itself under the identification of a linear map with its derivative. Concretely, each component of F is a holomorphic function on $V = \mathbb{C}^g$ whose first-order Taylor expansion is exact, so the Jacobian matrix $\partial F_i / \partial z_j$ is a constant matrix and the antiholomorphic derivatives $\partial F_i / \partial \bar{z}_j$ vanish identically by the Cauchy-Riemann equations. A real-linear map whose antiholomorphic part vanishes is $\mathbb{C}$ -linear. Hence $F \in \text{Hom}_{\mathbb{C}}(V, V')$, and $F(\Lambda) \subseteq \Lambda'$ because for $\lambda \in \Lambda$ we have $\pi'(F(\lambda)) = f(\pi(\lambda)) = f(0) = 0$, so $F(\lambda) \in \Lambda'$.

The converse and the bijection. Conversely, let $F \in \text{Hom}_{\mathbb{C}}(V, V')$ satisfy $F(\Lambda) \subseteq \Lambda'$. Then F carries cosets to cosets: if $v - v' \in \Lambda$ then $F(v) - F(v') = F(v - v') \in \Lambda'$, so the rule $f(v + \Lambda) = F(v) + \Lambda'$ is well defined. It is a group homomorphism because F is additive, and holomorphic because F is holomorphic and π, π' are local biholomorphisms; it satisfies $\pi' \circ F = f \circ \pi$ by construction. The two assignments $f \mapsto F$ (lift) and $F \mapsto f$ (descent) are mutually inverse: the lift of the descent of F is F by uniqueness of lifts, and the descent of the lift of f is f since both agree after composing with the surjection π . Both operations are additive in their argument, so the bijection $\text{Hom}(X, X') \cong \{F : F(\Lambda) \subseteq \Lambda'\}$ is an isomorphism of abelian groups. When $X = X'$, composition of homomorphisms corresponds to composition of the linear lifts, and the identity endomorphism corresponds to id_V , so the bijection is a ring isomorphism.

2. *A general holomorphic map is a translate of a homomorphism.* Let $h: X \rightarrow X'$ be holomorphic and set $c = h(0) \in X'$. Let $t_{-c}: X' \rightarrow X'$ be translation by $-c$, a biholomorphism of X' (its lift is the translation $v' \mapsto v' - \gamma$ of V' for any $\gamma \in V'$ with $\pi'(\gamma) = c$). Then $f := t_{-c} \circ h: X \rightarrow X'$ is holomorphic and satisfies $f(0) = h(0) - c = 0$. It remains to see that a holomorphic map f of tori with $f(0) = 0$ is a homomorphism; granting

this, $h = t_c \circ f$ is the asserted decomposition, and c is forced by evaluating at 0, giving uniqueness of both c and $f = t_{-c} \circ h$.

To prove f is a homomorphism, lift f to a holomorphic map $F: V \rightarrow V'$ with $F(0) = 0$ and $\pi' \circ F = f \circ \pi$, exactly as in the first paragraph of part (1); this used only that V, V' are simply connected covers and $f(0) = 0$, not that f is a homomorphism. Fix $\lambda \in \Lambda$ and consider $\delta_\lambda(v) = F(v + \lambda) - F(v)$. Applying π' gives $\pi'(\delta_\lambda(v)) = f(\pi(v + \lambda)) - f(\pi(v)) = 0$, since $\pi(v + \lambda) = \pi(v)$; thus $\delta_\lambda(v) \in \Lambda'$ for all v . As δ_λ is continuous on the connected space V with values in the discrete set Λ' , it is constant. Differentiating the identity $F(v + \lambda) = F(v) + \delta_\lambda$ shows the Jacobian satisfies $dF(v + \lambda) = dF(v)$ for all $\lambda \in \Lambda$, so each entry of dF is a Λ -periodic holomorphic function on V . Such a function descends to a holomorphic function on the compact connected torus X , hence is constant by the maximum principle ([10]). A holomorphic map F with constant Jacobian and $F(0) = 0$ is $\mathbb{C}$ -linear, given by $v \mapsto dF \cdot v$. In particular F is additive, so f is a homomorphism, completing the proof.

The proposition is the organizing principle of the whole analytic theory. It says that the seemingly transcendental category of complex tori and holomorphic homomorphisms is equivalent to a purely linear-algebraic category: objects are pairs (V, Λ) of a complex vector space with a full lattice, and morphisms are $\mathbb{C}$ -linear maps carrying lattice into lattice. Two structures are being remembered at once, the complex-linear structure on V and the integral structure Λ , and a homomorphism must respect both. This is why the subject sits at the meeting point of complex analysis and the arithmetic of lattices: the complex structure is analytic, the lattice is arithmetic, and a morphism is exactly a map compatible with the two.

The mechanism of the proof deserves a second look, because it recurs. The key step, that the lift F is $\mathbb{C}$ -linear, splits into two independent facts. That F is additive is topology: a continuous map into a discrete group from a connected space is constant, applied to the “defect” $F(v + w) - F(v) - F(w)$. That F is then $\mathbb{C}$ -linear, not just $\mathbb{R}$ -linear, is complex analysis: the antiholomorphic derivative of a holomorphic map vanishes. Part (2) isolates the same periodic-derivative argument in its cleanest form, where the derivative of the lift is a holomorphic function on the compact torus and so must be constant, since the only holomorphic functions on a compact connected complex manifold are the constants. This last fact, that X carries no nonconstant global holomorphic functions, is a recurring obstruction: it is the reason abelian varieties are studied through their line bundles and meromorphic functions rather than through holomorphic functions, a theme taken up in Chapter 2.

For the elliptic curve of example 1.1.4, the proposition recovers a computation the reader has already seen in [3]: the endomorphism ring of $E = \mathbb{C}/\Lambda$ is

$$\text{End}(E) \cong \{\alpha \in \mathbb{C} : \alpha\Lambda \subseteq \Lambda\}$$

the set of complex numbers scaling the lattice into itself, since a $\mathbb{C}$ -linear endomorphism of $V = \mathbb{C}$ is multiplication by a scalar $\alpha \in \mathbb{C}$ and the lattice condition reads $\alpha\Lambda \subseteq \Lambda$. For a generic τ this set is just $\mathbb{Z}$, acting by $\alpha = n$, and $\text{End}(E) = \mathbb{Z}$; for special τ satisfying a quadratic equation over $\mathbb{Q}$ it is an order in an imaginary quadratic field, the case of complex multiplication. The higher-dimensional generalization of this scalar description, the analytic and rational representations of an endomorphism, is the subject of section 1.2.

One consequence of proposition 1.1.7 organizes the philosophy of the chapter. The category of complex tori is governed by linear algebra over $\mathbb{Z}$, refined by a complex structure. An isomorphism $X \cong X'$ is a $\mathbb{C}$ -linear isomorphism $V \xrightarrow{\sim} V'$ carrying Λ onto Λ' ; fixing $V = V' = \mathbb{C}^g$, this means the

two lattices are related by an element of $\mathrm{GL}_g(\mathbb{C})$, and the moduli of complex tori of dimension g is the space of full lattices in $\mathbb{C}^g$ modulo this action. All of the finer structure the book develops, polarizations, the Néron-Severi group, theta functions, isogenies, sits on top of this linear-algebraic skeleton, adding a positivity condition (the Riemann relations) that most complex tori fail. The reader should hold this contrast in mind against the algebraic picture that Part II builds: there a complex torus is replaced by a projective group scheme over an arbitrary field, with no lattice and no complex structure available, and the linear-algebraic control of the present section is replaced by the geometric rigidity of the theorem of the cube. Over $\mathbb{C}$ the two descriptions agree, and moving between them is where the subject's leverage lies; but the agreement is a theorem, not a definition, and it holds only for the tori that satisfy the algebraicity condition of Chapter 3.

1.2 The Analytic and Rational Representations

An endomorphism of a complex torus $X = V/\Lambda$ is, by proposition 1.1.7, nothing more exotic than a $\mathbb{C}$ -linear map of the covering space V that carries the lattice Λ into itself. This double life is the source of everything that makes the endomorphism theory of complex tori tractable: a single endomorphism can be read either on V , where it is a $\mathbb{C}$ -linear operator on a g -dimensional complex vector space, or on Λ , where it is a $\mathbb{Z}$ -linear operator on a free abelian group of rank $2g$. These two readings are the analytic and the rational representation. Neither sees the whole endomorphism on its own, but together they constrain it tightly, and the comparison between them is the mechanism that forces the endomorphism ring of a complex torus to be small and arithmetically rigid. This section sets up the two representations, proves the comparison relating them, and extracts from it the characteristic-polynomial identity that will govern degrees of isogenies and, in Part II, the whole arithmetic of endomorphisms.

Throughout, $X = V/\Lambda$ is a complex torus of dimension g , with $V = \mathbb{C}^g$ and $\Lambda \subset V$ a full lattice (definition 1.1.2).

The endomorphisms of X form a ring under composition and pointwise addition, and it is this ring, rather than any single endomorphism, that carries the arithmetic content. Composition is composition of maps; the sum of two endomorphisms f, h is the map $x \mapsto f(x) + h(x)$, which is again a homomorphism because X is abelian. The zero endomorphism is the constant map to the origin, and the identity is the identity map. The ring and its rationalization are recorded below.

Definition 1.2.1: Endomorphism Ring and Endomorphism Algebra

Let $X = V/\Lambda$ be a complex torus. The *endomorphism ring* of X is

$$\mathrm{End}(X) = \{ f: X \rightarrow X \mid f \text{ holomorphic, } f(0) = 0 \}$$

with addition induced pointwise from the group law of X and multiplication given by composition. The *endomorphism algebra* of X is the $\mathbb{Q}$ -algebra

$$\mathrm{End}^0(X) = \mathrm{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$$

Requiring $f(0) = 0$ is what makes $\mathrm{End}(X)$ a ring: by proposition 1.1.7 the holomorphic maps $X \rightarrow X$ fixing the origin are exactly the group homomorphisms, and a general holomorphic map is such a homomorphism followed by a translation. Discarding the translations leaves a set closed under both

composition and addition. Passing to $\text{End}^0(X)$ discards the finer integral structure and keeps only the $\mathbb{Q}$ -algebra, which is the object Albert's classification will eventually pin down in Chapter 6; the integral ring $\text{End}(X)$ sits inside it as an order. The rank of $\text{End}(X)$ as an abelian group, equivalently $\dim_{\mathbb{Q}} \text{End}^0(X)$, will turn out to be at most $2g^2$, and the two representations below are what bound it.

The first representation lifts an endomorphism to the universal cover. By proposition 1.1.7 every $f \in \text{End}(X)$ is induced by a unique $\mathbb{C}$ -linear map $F: V \rightarrow V$ satisfying $F(\Lambda) \subseteq \Lambda$; conversely such an F descends to an endomorphism of X . Reading f through F is the analytic representation.

Definition 1.2.2: Analytic Representation

The *analytic representation* of $\text{End}(X)$ is the ring homomorphism

$$\rho_a: \text{End}(X) \longrightarrow \text{End}_{\mathbb{C}}(V)$$

sending $f \in \text{End}(X)$ to the unique $\mathbb{C}$ -linear map $\rho_a(f) = F: V \rightarrow V$ with $F(\Lambda) \subseteq \Lambda$ inducing f . Extending scalars, ρ_a denotes also its $\mathbb{Q}$ -linear extension $\text{End}^0(X) \rightarrow \text{End}_{\mathbb{C}}(V)$ and its $\mathbb{R}$ -linear extension to $\text{End}(X) \otimes_{\mathbb{Z}} \mathbb{R}$.

The analytic representation records how an endomorphism moves the tangent directions at the origin: $\rho_a(f)$ is exactly the derivative of f at 0, viewed as a $\mathbb{C}$ -linear operator on the tangent space $V = T_0X$. It is injective, since an endomorphism is determined by its lift, so $\text{End}(X)$ embeds into the g^2 -dimensional complex algebra $\text{End}_{\mathbb{C}}(V) \cong M_g(\mathbb{C})$. What this embedding does *not* do is bound the size of $\text{End}^0(X)$: an injection of $\text{End}(X)$ into a complex vector space forces neither finite generation of $\text{End}(X)$ as an abelian group nor finite-dimensionality of $\text{End}^0(X)$ as a $\mathbb{Q}$ -algebra, because a $\mathbb{Q}$ -subspace of a finite-dimensional real vector space can be infinite-dimensional over $\mathbb{Q}$ (already $\mathbb{Q}(\pi) \subset \mathbb{C} = M_1(\mathbb{C})$ in the case $g = 1$), and a subgroup of a vector space need not be discrete. Both bounds instead come from the rational representation defined below, which lands in $M_{2g}(\mathbb{Z})$: its additive group $\mathbb{Z}^{4g^2}$ is finitely generated, so $\text{End}(X)$ has finite rank, whence $\text{End}^0(X) = \text{End}(X) \otimes \mathbb{Q}$ is finite-dimensional over $\mathbb{Q}$. What the analytic representation cannot see is exactly this lattice: it forgets that F preserves Λ , and so it forgets the integrality that will do the real work. Choosing a $\mathbb{C}$ -basis of V identifies $\rho_a(f)$ with a matrix in $M_g(\mathbb{C})$, whose entries are the complex numbers by which f scales the coordinates $z_1, \dots, z_g$.

The second representation forgets the complex structure entirely and remembers only the action on the lattice. Since f is a group homomorphism carrying Λ to itself, restricting the lift F to Λ gives a $\mathbb{Z}$ -linear endomorphism of a free abelian group of rank $2g$.

Definition 1.2.3: Rational Representation

The *rational representation* of $\text{End}(X)$ is the ring homomorphism

$$\rho_r: \text{End}(X) \longrightarrow \text{End}_{\mathbb{Z}}(\Lambda) \cong M_{2g}(\mathbb{Z})$$

sending $f \in \text{End}(X)$ to the restriction $\rho_r(f) = F|_{\Lambda}: \Lambda \rightarrow \Lambda$ of its analytic lift. Extending scalars gives the $\mathbb{Q}$ -linear extension $\rho_r: \text{End}^0(X) \rightarrow \text{End}_{\mathbb{Q}}(\Lambda \otimes \mathbb{Q})$ and the $\mathbb{R}$ -linear extension $\rho_r \otimes \mathbb{R}: \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{R} \rightarrow \text{End}_{\mathbb{R}}(\Lambda \otimes \mathbb{R})$, where $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$ as real vector spaces.

The name “rational” reflects the object it naturally lives on: after tensoring with $\mathbb{Q}$, the representation

acts on the $2g$ -dimensional $\mathbb{Q}$ -vector space $\Lambda \otimes \mathbb{Q}$, and ρ_r itself is the integral representation of $\text{End}(X)$ on the rank- $2g$ lattice Λ that this action packages. Fixing a $\mathbb{Z}$ -basis $\lambda_1, \dots, \lambda_{2g}$ of Λ presents $\rho_r(f)$ as an integer matrix in $M_{2g}(\mathbb{Z})$. This is the source of every integrality constraint on $\text{End}(X)$: whatever f does, it does it as an integer matrix of size $2g$, so its trace and determinant are integers and its characteristic polynomial has integer coefficients. The analytic representation, by contrast, produces a complex matrix of half the size and carries no such constraint on its own. The comparison between the two is what transfers the integrality from ρ_r to the eigenvalues seen by ρ_a .

Before comparing them, it helps to ground both representations on the endomorphism every complex torus possesses.

Example 1.2.4: Multiplication by n

For $n \in \mathbb{Z}$, the multiplication-by- n map $[n]: X \rightarrow X$, $x \mapsto nx$, is an endomorphism: it fixes the origin and is a homomorphism because X is abelian. Its analytic lift is scalar multiplication by n on V , so

$$\rho_a([n]) = n \cdot \text{id}_V = n I_g \in \text{End}_{\mathbb{C}}(V), \quad \rho_r([n]) = n \cdot \text{id}_{\Lambda} = n I_{2g} \in \text{End}_{\mathbb{Z}}(\Lambda)$$

The analytic representation is a $g \times g$ scalar matrix and the rational representation a $2g \times 2g$ scalar matrix, both with the single eigenvalue n . Consequently the characteristic polynomial of $\rho_a([n])$ is $(t - n)^g$ and that of $\rho_r([n])$ is

$$\det(t I_{2g} - n I_{2g}) = (t - n)^{2g}$$

The kernel of $[n]$ on X is the n -torsion $X[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$, a group of order n^{2g} . This integer n^{2g} is the *analytic degree* of $[n]$, and it is exactly $|\det \rho_r([n])| = |n^{2g}|$, the absolute value of the determinant of the rational representation. In Part II this coincidence becomes a theorem: for any isogeny the degree equals the determinant of the rational representation, and the case $f = [n]$ yields the degree formula $\deg[n] = n^{2g}$ obtained there by the theorem of the cube.

The computation for $[n]$ exhibits in miniature the pattern the comparison proposition will establish in general: the characteristic polynomial of the rational representation is a product of a degree- g polynomial and its complex conjugate, here a literal square $((t - n)^g)^2$ because n is real, and its roots come in the pairs one expects when a complex g -dimensional picture is doubled into a real $2g$ -dimensional one. For $[n]$ the two “halves” coincide because n is real; the content of the next result is that in general the second half is the complex conjugate of the first.

The two representations are related through a single linear-algebra fact: the real vector space underlying $\Lambda \otimes \mathbb{R}$ is V itself, and V carries a complex structure. An $\mathbb{R}$ -linear operator on V that happens to be $\mathbb{C}$ -linear is determined, after complexifying, by how it acts on the holomorphic and antiholomorphic directions, and these two actions are conjugate. This is what ties ρ_r to ρ_a and its conjugate.

Proposition 1.2.5: Comparison of the Analytic and Rational Representations

Let $X = V/\Lambda$ be a complex torus of dimension g . Complexifying the rational representation gives an isomorphism

$$\rho_r \otimes_{\mathbb{Z}} \mathbb{C} \cong \rho_a \oplus \bar{\rho}_a$$

of complex representations of $\text{End}(X)$ on the $2g$ -dimensional complex vector space $V \oplus \bar{V}$, where $\bar{\rho}_a$ denotes the complex-conjugate representation $f \mapsto \overline{\rho_a(f)}$. Before complexifying, the underlying real statement is that $\rho_r \otimes_{\mathbb{Z}} \mathbb{R}$ is the realification $(\rho_a)_{\mathbb{R}}$ of the analytic representation, that is, the representation $f \mapsto \rho_a(f)_{\mathbb{R}}$ on the $2g$ -dimensional real vector space $\Lambda \otimes \mathbb{R} = V$; the splitting into ρ_a and its conjugate emerges only after tensoring this real representation with $\mathbb{C}$.

Proof:

Step 1: Identify the real space the rational representation acts on. Since Λ is a full lattice, $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$ as real vector spaces (definition 1.1.2); this is the defining property of a full lattice, that its real span is all of V . For $f \in \text{End}(X)$ the operator $\rho_r(f)$ is by definition the restriction of the $\mathbb{C}$ -linear lift $F = \rho_a(f)$ to Λ , so under $\Lambda \otimes \mathbb{R} = V$ the $\mathbb{R}$ -linear extension of $\rho_r(f)$ is precisely F regarded as an $\mathbb{R}$ -linear operator on the $2g$ -dimensional real vector space V . Thus $\rho_r \otimes_{\mathbb{Z}} \mathbb{R}$ is the representation $f \mapsto \rho_a(f)_{\mathbb{R}}$, where $(\cdot)_{\mathbb{R}}$ denotes an operator viewed as $\mathbb{R}$ -linear by forgetting the complex structure. It therefore suffices to compare, for a single $\mathbb{C}$ -linear operator F on V , its realification $F_{\mathbb{R}}$ with the pair $(F, \bar{F})$.

Step 2: Complexify and split into holomorphic and antiholomorphic parts. Complexify the real vector space V to $V \otimes_{\mathbb{R}} \mathbb{C}$. The complex structure J on V , that is, multiplication by i , is an $\mathbb{R}$ -linear operator with $J^2 = -\text{id}$, so on $V \otimes_{\mathbb{R}} \mathbb{C}$ it has the two eigenvalues $\pm i$, and

$$V \otimes_{\mathbb{R}} \mathbb{C} = V^{1,0} \oplus V^{0,1}$$

is the decomposition into the $(+i)$ -eigenspace $V^{1,0}$ and the $(-i)$ -eigenspace $V^{0,1}$ of J . Each summand has complex dimension g . The two natural $\mathbb{C}$ -linear maps

$$V^{1,0} \xrightarrow{\sim} V, \quad V^{0,1} \xrightarrow{\sim} \bar{V}$$

are isomorphisms: $V^{1,0}$ is $\mathbb{C}$ -linearly isomorphic to (V, J) via $v \otimes 1 - Jv \otimes i \mapsto v$, and $V^{0,1}$ is $\mathbb{C}$ -linearly isomorphic to the conjugate space $\bar{V} = (V, -J)$ via $v \otimes 1 + Jv \otimes i \mapsto v$. These are the standard holomorphic and antiholomorphic identifications for the underlying real space of a complex vector space.

Step 3: A $\mathbb{C}$ -linear operator acts on the two parts as F and $\bar{F}$. Let $F: V \rightarrow V$ be $\mathbb{C}$ -linear, and let $F_{\mathbb{R}}$ be F regarded as $\mathbb{R}$ -linear on V . Its complexification $F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ is $\mathbb{C}$ -linear on $V \otimes_{\mathbb{R}} \mathbb{C}$ and commutes with J , because F commutes with J precisely because F is $\mathbb{C}$ -linear ($F(Jv) = F(iv) = iF(v) = JF(v)$). An operator commuting with J preserves each eigenspace of J , so $F_{\mathbb{R}} \otimes \mathbb{C}$ preserves both $V^{1,0}$ and $V^{0,1}$ and splits as

$$F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} = (F_{\mathbb{R}} \otimes \mathbb{C})|_{V^{1,0}} \oplus (F_{\mathbb{R}} \otimes \mathbb{C})|_{V^{0,1}}$$

Under the isomorphism $V^{1,0} \cong V$ of Step 2 the restriction to $V^{1,0}$ is F itself (this is the holomorphic direction, on which the complex structure agrees with that of V). Under $V^{0,1} \cong \bar{V}$

the restriction to $V^{0,1}$ is F acting on the conjugate space, which is by definition $\bar{F}$: if F has matrix A in a $\mathbb{C}$ -basis of V , then F acting on $\bar{V}$ has matrix $\bar{A}$. Therefore

$$F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \cong F \oplus \bar{F}$$

as $\mathbb{C}$ -linear operators on $V \oplus \bar{V}$.

Step 4: Assemble the representations. Steps 1 and 3 hold naturally in f : the identifications in Step 2 are chosen once for the vector space V and do not depend on f , so applying Step 3 to $F = \rho_a(f)$ for every $f \in \text{End}(X)$ simultaneously gives an isomorphism of representations. Combining with Step 1, which identifies $(\rho_r \otimes_{\mathbb{Z}} \mathbb{R})(f)$ with $\rho_a(f)_{\mathbb{R}}$, complexification yields

$$\rho_r \otimes_{\mathbb{Z}} \mathbb{C} = (\rho_r \otimes_{\mathbb{Z}} \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C} \cong \rho_a \oplus \bar{\rho}_a$$

as complex representations on $V \oplus \bar{V}$, a $2g$ -dimensional complex vector space. Before this final complexification, Step 1 identifies $\rho_r \otimes_{\mathbb{Z}} \mathbb{R}$ with the realification $(\rho_a)_{\mathbb{R}}$ on the $2g$ -dimensional real space V ; the conjugate splitting is precisely what the passage to $\mathbb{C}$ produces, completing the proof.

The comparison is the structural heart of the section. The rational representation is real, of dimension $2g$, and carries all the integrality; the analytic representation is complex, of dimension g , and carries the geometry. The proposition says that once the rational representation is complexified, it becomes the analytic representation doubled by its conjugate. Every subsequent statement about how $\text{End}(X)$ is constrained is a shadow of this single fact. In particular the eigenvalues of $\rho_r(f)$ are exactly the eigenvalues of $\rho_a(f)$ together with their complex conjugates, which is the precise sense in which the $2g$ real dimensions split into g holomorphic and g antiholomorphic ones. For a self-map that is already defined over $\mathbb{R}$, such as $[n]$, the conjugate half is identical to the original half, which is why $\rho_r([n])$ and $\rho_a([n])$ shared the single eigenvalue n in example 1.2.4.

The eigenvalue statement is best packaged as an identity of characteristic polynomials, since that is the form in which it feeds the degree theory of isogenies and, in Part II, the study of Frobenius over finite fields. Recall that for a $\mathbb{C}$ -linear operator F with matrix A in some basis, the characteristic polynomial is $P(t) = \det(tI - A)$, and $\bar{P}(t)$ denotes the polynomial obtained by conjugating the coefficients, so that $\bar{P}(t) = \det(tI - \bar{A})$.

Proposition 1.2.6: Characteristic Polynomial of the Rational Representation

Let $X = V/\Lambda$ be a complex torus of dimension g and $f \in \text{End}(X)$. Let

$$P_a(t) = \det(t I_g - \rho_a(f)) \in \mathbb{C}[t]$$

be the characteristic polynomial of the analytic representation of f , a monic polynomial of degree g , and let

$$P_r(t) = \det(t I_{2g} - \rho_r(f)) \in \mathbb{Z}[t]$$

be that of the rational representation. Then

$$P_r(t) = P_a(t) \overline{P_a(t)}$$

Consequently P_r is a monic polynomial of degree $2g$ with integer coefficients whose roots are the eigenvalues of $\rho_a(f)$ together with their complex conjugates, occurring in conjugate pairs.

Proof:

Step 1: The characteristic polynomial is unchanged by complexification. The characteristic polynomial of $\rho_r(f)$ computed over $\mathbb{Z}$ (equivalently over $\mathbb{Q}$, or over $\mathbb{R}$) is unchanged when the operator is regarded as acting on the complexification $\Lambda \otimes \mathbb{C}$, since $\det(t I - M)$ depends only on the matrix M and not on the field over which one computes the determinant. Thus $P_r(t) = \det(t I_{2g} - (\rho_r(f) \otimes \mathbb{C}))$.

Step 2: Apply the comparison. By proposition 1.2.5, $\rho_r(f) \otimes \mathbb{C}$ is conjugate, as a $\mathbb{C}$ -linear operator, to $\rho_a(f) \oplus \overline{\rho_a(f)}$ acting on $V \oplus \bar{V}$. Conjugate operators have equal characteristic polynomials, and the characteristic polynomial of a direct sum is the product of the characteristic polynomials of the summands. Therefore

$$P_r(t) = \det(t I_g - \rho_a(f)) \cdot \det(t I_g - \overline{\rho_a(f)}) = P_a(t) \overline{P_a(t)}$$

where the second factor is $\overline{P_a(t)}$ because conjugating the matrix A of $\rho_a(f)$ conjugates every coefficient of $\det(t I_g - A)$.

Step 3: Integrality and the conjugate-pair structure. Both P_a and $\overline{P_a}$ are monic of degree g , so their product P_r is monic of degree $2g$. Since $\rho_r(f)$ is represented by an integer matrix in $M_{2g}(\mathbb{Z})$ in any $\mathbb{Z}$ -basis of Λ , its characteristic polynomial $P_r(t) = \det(t I_{2g} - \rho_r(f))$ has coefficients in $\mathbb{Z}$, so $P_r \in \mathbb{Z}[t]$. Finally, if $\alpha_1, \dots, \alpha_g$ are the roots of P_a , that is, the eigenvalues of $\rho_a(f)$, then the roots of $\overline{P_a}$ are $\overline{\alpha_1}, \dots, \overline{\alpha_g}$, so the multiset of roots of P_r is $\{\alpha_1, \dots, \alpha_g, \overline{\alpha_1}, \dots, \overline{\alpha_g}\}$, which is stable under complex conjugation and hence pairs each root with its conjugate, as we sought to show.

The identity $P_r = P_a \overline{P_a}$ is the engine that converts the analytic picture into arithmetic. On the one hand P_r has integer coefficients, forced by the rational representation; on the other its roots are the analytic eigenvalues $\alpha_1, \dots, \alpha_g$ and their conjugates. Together these say that the eigenvalues of $\rho_a(f)$ are algebraic integers stable, as a set-with-conjugates, under complex conjugation, and that their symmetric functions satisfy integrality relations coming from $\mathbb{Z}[t]$. Two invariants of f acquire clean expressions: the trace of the rational representation is $\text{tr} \rho_r(f) = \text{tr} \rho_a(f) + \text{tr} \overline{\rho_a(f)} = 2 \text{Re}(\text{tr} \rho_a(f))$, an

integer, and its determinant is $\det \rho_r(f) = \det \rho_a(f) \cdot \overline{\det \rho_a(f)} = |\det \rho_a(f)|^2 \geq 0$. The nonnegativity of $\det \rho_r(f)$ is not a formality: when f is an isogeny the determinant is strictly positive, so the rational representation is orientation-preserving on $\Lambda \otimes \mathbb{R} = V$, which is the analytic reason the degree of an isogeny, computed in Part II as this determinant, is a positive integer rather than a nonzero integer of either sign. When f is not an isogeny the determinant is zero and the operator is degenerate. For $f = [n]$ one recovers $P_r(t) = (t - n)^g \cdot \overline{(t - n)^g} = (t - n)^{2g}$ and $\det \rho_r([n]) = |n^g|^2 = n^{2g}$, matching example 1.2.4.

The two representations together explain why the endomorphism algebras of complex tori, and of abelian varieties, are so constrained. The analytic representation embeds $\text{End}^0(X)$ into $M_g(\mathbb{C})$, exhibiting each endomorphism as a g -dimensional complex operator; the rational representation embeds it into $M_{2g}(\mathbb{Q})$, which forces $\text{End}(X)$ to have finite rank and hence bounds $\dim_{\mathbb{Q}} \text{End}^0(X)$, and, through proposition 1.2.6, forces every element to satisfy a monic integer polynomial of degree $2g$, so every element of $\text{End}(X)$ is an algebraic integer and $\text{End}(X)$ is an order in the finite-dimensional $\mathbb{Q}$ -algebra $\text{End}^0(X)$. An endomorphism cannot scale the holomorphic tangent directions by arbitrary complex numbers: the eigenvalues it produces must assemble, with their conjugates, into the roots of an integer polynomial. This integrality is exactly the rigidity that will let Albert's classification, taken up in Chapter 6, enumerate the possible endomorphism algebras of a simple abelian variety, cutting the list down to four types of division algebra with involution rather than the continuum a g -dimensional complex operator might suggest.

1.3 Homology, Cohomology, and the Hodge Structure on H^1

A complex torus $X = V/\Lambda$ (definition 1.1.2) is built from two pieces of linear data: the lattice Λ , a free abelian group of rank $2g$, and the complex structure on the ambient space $V = \mathbb{C}^g$. This section shows that both pieces are visible in cohomology, and that they interact there in a way that turns out to control the entire theory. The lattice is the topology: the singular homology and cohomology of X are exterior algebras manufactured from Λ alone, with no reference to the complex structure at all. The complex structure reappears one level up, as a splitting of the first cohomology into holomorphic and antiholomorphic parts. That splitting is a Hodge structure of weight one, and the passage from a complex torus to its weight-one Hodge structure is the bridge to Hodge theory that organizes everything downstream: a polarization, the subject of Chapter 3, is exactly a piece of extra structure on this Hodge structure, and the entire complex-analytic theory of Part I can be read as the study of that extra structure.

The topological facts are quickly settled because X is a product of circles. As a smooth manifold, $X = V/\Lambda \cong (S^1)^{2g}$: choosing a $\mathbb{Z}$ -basis $\lambda_1, \dots, \lambda_{2g}$ of Λ identifies V with $\mathbb{R}^{2g}$ carrying $\Lambda = \mathbb{Z}^{2g}$, and $\mathbb{R}^{2g}/\mathbb{Z}^{2g}$ is the $2g$ -fold product of $\mathbb{R}/\mathbb{Z} = S^1$. Everything topological about X is therefore the topology of a torus.

Proposition 1.3.1: First Homology of a Complex Torus

Let $X = V/\Lambda$ be a complex torus of dimension g . The quotient map $V \rightarrow X$ induces a canonical isomorphism of abelian groups

$$\Lambda \xrightarrow{\sim} H_1(X, \mathbb{Z})$$

sending a lattice vector $\lambda \in \Lambda$ to the homology class of the loop $t \mapsto t\lambda$, $t \in [0, 1]$. In particular $H_1(X, \mathbb{Z})$ is free of rank $2g$.

Proof:

The universal cover of X is the projection $\pi: V \rightarrow V/\Lambda = X$, a covering map because Λ acts on V freely, properly discontinuously, by translations. Since $V \cong \mathbb{R}^{2g}$ is contractible, it is the universal cover, and the deck-transformation group is Λ ; hence $\pi_1(X, 0) \cong \Lambda$, the isomorphism sending $\lambda \in \Lambda$ to the homotopy class of the loop obtained by projecting any path in V from 0 to λ . The straight path $t \mapsto t\lambda$ is one such path, so λ corresponds to the loop $t \mapsto \pi(t\lambda)$.

Because Λ is abelian, the fundamental group $\pi_1(X, 0) \cong \Lambda$ equals its own abelianization. By the Hurewicz theorem ([8]), the Hurewicz map $\pi_1(X, 0) \rightarrow H_1(X, \mathbb{Z})$ is the abelianization map, hence here an isomorphism. Tracing the two identifications, $\lambda \in \Lambda$ maps to the homology class of the loop $t \mapsto \pi(t\lambda)$, which is the assertion. As Λ is free abelian of rank $2g$, so is $H_1(X, \mathbb{Z})$, completing the proof.

The content of this proposition is that the lattice is not an auxiliary choice but an intrinsic invariant of the torus: $\Lambda \cong H_1(X, \mathbb{Z})$ is recovered from X as a topological space, before any mention of the complex structure. The identification is the analytic origin of a theme that runs through the whole book. Under the analytic representation of section 1.2, an endomorphism of X acts on Λ ; that action is now revealed to be the action on $H_1(X, \mathbb{Z})$, and its ℓ -adic shadow is the action on the Tate module studied in Part II. The lattice is where the arithmetic lives.

Higher homology and cohomology follow from the product structure. Rather than re-derive the (co)homology of a product of circles, the computation is imported from algebraic topology and applied.

Theorem 1.3.2: Cohomology of a Complex Torus

Let $X = V/\Lambda$ be a complex torus of dimension g , and write $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ for the dual lattice. For every $n \geq 0$ there is a canonical isomorphism

$$H^n(X, \mathbb{Z}) \cong \bigwedge^n \Lambda^\vee$$

and, taken together, these assemble into an isomorphism of graded rings

$$H^\bullet(X, \mathbb{Z}) \cong \bigwedge^\bullet \Lambda^\vee$$

from the singular cohomology ring of X (cup product) to the exterior algebra on $\Lambda^\vee$ (wedge product). In particular $H^n(X, \mathbb{Z})$ is free of rank $\binom{2g}{n}$, and $H^1(X, \mathbb{Z}) \cong \Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$.

Proof:

As a smooth manifold $X \cong (S^1)^{2g}$, the $2g$ -fold product of circles. The integral cohomology ring of a single circle is $H^\bullet(S^1, \mathbb{Z}) = \mathbb{Z}[e]/(e^2)$, the exterior algebra on one degree-one generator e . By the Künneth theorem for a finite product of spaces whose integral cohomology is free of finite rank in each degree ([8]), the cross product induces a ring isomorphism

$$H^\bullet(X, \mathbb{Z}) \cong \bigotimes_{j=1}^{2g} H^\bullet(S^1, \mathbb{Z}) = \bigotimes_{j=1}^{2g} (\mathbb{Z} \oplus \mathbb{Z}e_j)$$

where the tensor product is the graded-commutative tensor product of graded rings. Because each factor is the exterior algebra $\bigwedge(\mathbb{Z}e_j)$ on one generator and the tensor product of exterior algebras is the exterior algebra on the direct sum of the generating modules, the right-hand side is $\bigwedge^\bullet(\bigoplus_{j=1}^{2g} \mathbb{Z}e_j)$. The generators $e_1, \dots, e_{2g}$ are the degree-one classes dual to the $2g$ circle factors, so $\bigoplus_j \mathbb{Z}e_j = H^1(X, \mathbb{Z})$ and the isomorphism reads $H^\bullet(X, \mathbb{Z}) \cong \bigwedge^\bullet H^1(X, \mathbb{Z})$.

It remains to identify $H^1(X, \mathbb{Z})$ with $\Lambda^\vee$. Cohomology in degree one is dual to homology in degree one for a space with free homology: the universal-coefficients theorem ([8]) gives $H^1(X, \mathbb{Z}) \cong \text{Hom}(H_1(X, \mathbb{Z}), \mathbb{Z})$, the Ext term vanishing because $H_0(X, \mathbb{Z}) = \mathbb{Z}$ is free. Combined with the identification $H_1(X, \mathbb{Z}) \cong \Lambda$ of proposition 1.3.1, this yields $H^1(X, \mathbb{Z}) \cong \text{Hom}(\Lambda, \mathbb{Z}) = \Lambda^\vee$. Substituting into the Künneth isomorphism gives $H^n(X, \mathbb{Z}) \cong \bigwedge^n \Lambda^\vee$ in each degree and the graded-ring statement overall. The rank of $\bigwedge^n \Lambda^\vee$ is $\binom{2g}{n}$ since $\Lambda^\vee$ is free of rank $2g$, as we sought to show.

The cohomology ring of a complex torus is thus the exterior algebra on a single rank- $2g$ lattice, with no input beyond the group Λ . Two consequences are worth isolating. First, the total dimension $\sum_n \dim H^n(X, \mathbb{Q}) = 2^{2g}$, and the Poincaré polynomial is $(1+t)^{2g}$, recording that a torus is as topologically simple as a compact manifold of its dimension can be. Second, and more important for what follows, degree one carries all the information: every cohomology group is a wedge power of $H^1(X, \mathbb{Z})$, so any structure placed on H^1 propagates to the whole ring by functoriality of $\bigwedge$. The complex structure will now supply exactly such a structure on H^1 .

To see where the complex structure enters, pass to complex coefficients. By theorem 1.3.2 and the

universal-coefficients theorem,

$$H^1(X, \mathbb{C}) \cong \text{Hom}(\Lambda, \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{C} \cong \text{Hom}_{\mathbb{Z}}(\Lambda, \mathbb{C}) = \text{Hom}_{\mathbb{R}}(V, \mathbb{C})$$

the last equality because $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$, so a $\mathbb{Z}$ -linear map $\Lambda \rightarrow \mathbb{C}$ extends uniquely to an $\mathbb{R}$ -linear map $V \rightarrow \mathbb{C}$. The space $\text{Hom}_{\mathbb{R}}(V, \mathbb{C})$ of $\mathbb{R}$ -linear functionals on V is where the complex structure asserts itself: a functional $f: V \rightarrow \mathbb{C}$ that is $\mathbb{R}$ -linear need be neither $\mathbb{C}$ -linear nor $\mathbb{C}$ -antilinear, and the two extremes span the whole space. This is the splitting that defines the Hodge structure.

The holomorphic side of the splitting has an independent description as the space of global holomorphic 1-forms, which merits a name of its own. The torus X inherits from $V = \mathbb{C}^g$ the standard holomorphic coordinates $z_1, \dots, z_g$: these are not functions on X (they are not Λ -periodic), but their differentials $dz_1, \dots, dz_g$ are Λ -invariant, since translation by a lattice vector is a translation in V and $d(z_k + c) = dz_k$ for any constant c . Hence each dz_k descends to a global holomorphic 1-form on X , and likewise each $d\bar{z}_k$ descends to a global antiholomorphic 1-form.

Definition 1.3.3: The Hodge Structure on H^1

Let $X = V/\Lambda$ be a complex torus. The *Hodge decomposition* of $H^1(X, \mathbb{C}) = \text{Hom}_{\mathbb{R}}(V, \mathbb{C})$ (the identification derived above) is the direct-sum decomposition

$$H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$$

into

- $H^{1,0} = V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$, the $\mathbb{C}$ -linear functionals on V , called the *holomorphic part*; and
- $H^{0,1} = \overline{V^*} = \text{Hom}_{\bar{\mathbb{C}}}(V, \mathbb{C})$, the $\mathbb{C}$ -antilinear functionals on V , called the *antiholomorphic part*.

The subspace $H^{1,0}$ is identified with the space $H^0(X, \Omega_X^1)$ of global holomorphic 1-forms on X , and $H^{0,1} = \overline{H^{1,0}}$ is its complex conjugate. This decomposition makes $H^1(X, \mathbb{Z})$ an integral *Hodge structure of weight one*.

The decomposition is the linear-algebra statement that an $\mathbb{R}$ -linear map $V \rightarrow \mathbb{C}$ splits uniquely into a $\mathbb{C}$ -linear part and a $\mathbb{C}$ -antilinear part. That the two summands are complex conjugates of one another, $H^{0,1} = \overline{H^{1,0}}$, and that they have equal dimension g , is the defining feature of a weight-one Hodge structure: the language of Hodge structures, developed in general in [4], is here entirely explicit and requires no transcendental input. A complex torus is, from this vantage, nothing but a weight-one Hodge structure wearing a geometric disguise, and the correspondence between the two is the reason abelian varieties and Hodge theory are the same subject viewed from two sides. The next proposition proves that the decomposition of definition 1.3.3 is correct as stated, including the identification of $H^{1,0}$ with holomorphic 1-forms.

Proposition 1.3.4: Explicit Hodge Decomposition

Let $X = V/\Lambda$ with $V = \mathbb{C}^g$ and holomorphic coordinates $z_1, \dots, z_g$. Then:

1. Every $\mathbb{R}$ -linear functional $f: V \rightarrow \mathbb{C}$ is uniquely the sum $f = f' + f''$ of a $\mathbb{C}$ -linear functional $f' \in V^*$ and a $\mathbb{C}$ -antilinear functional $f'' \in \overline{V^*}$, giving $H^1(X, \mathbb{C}) = V^* \oplus \overline{V^*} = H^{1,0} \oplus H^{0,1}$ with $\dim_{\mathbb{C}} H^{1,0} = \dim_{\mathbb{C}} H^{0,1} = g$.
2. The classes of $dz_1, \dots, dz_g$ form a basis of $H^{1,0}$ and the classes of $d\bar{z}_1, \dots, d\bar{z}_g$ form a basis of $H^{0,1}$; the identification $H^{1,0} = H^0(X, \Omega_X^1)$ sends a $\mathbb{C}$ -linear functional to the holomorphic 1-form it defines.
3. The two summands are exchanged by complex conjugation: $\overline{H^{1,0}} = H^{0,1}$.

Proof:

1. *The splitting.* Let $J: V \rightarrow V$ denote multiplication by i , the complex structure on V . For an $\mathbb{R}$ -linear $f: V \rightarrow \mathbb{C}$, define

$$f'(v) = \frac{1}{2}(f(v) - i f(iv)), \quad f''(v) = \frac{1}{2}(f(v) + i f(iv))$$

Then $f = f' + f''$ by inspection. The functional f' is $\mathbb{C}$ -linear: $f'(iv) = \frac{1}{2}(f(iv) - i f(-v)) = \frac{1}{2}(f(iv) + i f(v)) = i f'(v)$, using $f(-v) = -f(v)$. The functional f'' is $\mathbb{C}$ -antilinear: the same computation gives $f''(iv) = \frac{1}{2}(f(iv) - i f(v)) = -i f''(v)$. For uniqueness, suppose $f = g' + g''$ with g' $\mathbb{C}$ -linear and g'' $\mathbb{C}$ -antilinear; then $f' - g' = g'' - f''$ is simultaneously $\mathbb{C}$ -linear and $\mathbb{C}$ -antilinear, hence satisfies both $h(iv) = ih(v)$ and $h(iv) = -ih(v)$, forcing $h = 0$. Thus the decomposition $\text{Hom}_{\mathbb{R}}(V, \mathbb{C}) = V^* \oplus \overline{V^*}$ is direct. A $\mathbb{C}$ -linear functional on $V = \mathbb{C}^g$ is determined by its values on a $\mathbb{C}$ -basis, so $\dim_{\mathbb{C}} V^* = g$; likewise $\dim_{\mathbb{C}} \overline{V^*} = g$, consistent with $\dim_{\mathbb{C}} H^1(X, \mathbb{C}) = 2g$.

2. *The holomorphic forms.* A global holomorphic 1-form on X pulls back along $\pi: V \rightarrow X$ to a Λ -invariant holomorphic 1-form ω on V . Writing $\omega = \sum_{k=1}^g h_k(z) dz_k$ with h_k holomorphic, Λ -invariance forces each $h_k(z + \lambda) = h_k(z)$, so each h_k is a Λ -periodic holomorphic function on V , hence bounded (its values are attained on the compact fundamental domain), hence constant by Liouville's theorem ([10]). Thus $\omega = \sum_k c_k dz_k$ with $c_k \in \mathbb{C}$: the global holomorphic 1-forms are exactly the constant-coefficient combinations of $dz_1, \dots, dz_g$, and $\dim_{\mathbb{C}} H^0(X, \Omega_X^1) = g$ with basis $dz_1, \dots, dz_g$. A constant form $\sum_k c_k dz_k$ is precisely a $\mathbb{C}$ -linear functional $V \rightarrow \mathbb{C}$, namely $v \mapsto \sum_k c_k z_k(v)$; this is the identification $H^0(X, \Omega_X^1) = V^* = H^{1,0}$. Under the de Rham comparison $H^1(X, \mathbb{C}) \cong H_{\text{dR}}^1(X)$ ([8]), the class of a translation-invariant closed 1-form is its own de Rham class, and the invariant forms $dz_1, \dots, dz_g, d\bar{z}_1, \dots, d\bar{z}_g$ are closed and represent a basis of $H^1(X, \mathbb{C})$; the holomorphic ones span $H^{1,0}$ and the antiholomorphic ones span $H^{0,1}$.
3. *Conjugation.* Complex conjugation on the coefficient field $\mathbb{C}$ acts on $H^1(X, \mathbb{C}) = H^1(X, \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{C}$ through the second factor, fixing the real structure $H^1(X, \mathbb{R})$. On functionals it sends f to $\bar{f}: v \mapsto \overline{f(v)}$. If f is $\mathbb{C}$ -linear then $\bar{f}(iv) = \overline{f(iv)} = \overline{if(v)} = -i \overline{f(v)} = -i \bar{f}(v)$, so $\bar{f}$ is $\mathbb{C}$ -antilinear; thus conjugation carries $V^* = H^{1,0}$ onto $\overline{V^*} = H^{0,1}$, and $\overline{H^{1,0}} = H^{0,1}$. Concretely $\overline{dz_k} = d\bar{z}_k$, exchanging the two bases of part (2), completing the proof.

The proposition makes the Hodge decomposition of a complex torus entirely computable: no harmonic analysis, no elliptic operators, no appeal to the general Hodge theorem is needed, because the harmonic representatives of H^1 are visibly the translation-invariant forms dz_k and $d\bar{z}_k$. This is the sense in which the complex torus is the simplest nontrivial example in Hodge theory, and why the whole subject of this book can be developed by explicit linear algebra where the general Kähler theory of [4] would require the harmonic projection. The data of the Hodge structure amounts to the position of a single g -dimensional subspace $H^{1,0} \subset H^1(X, \mathbb{C})$, a subspace transverse to its own conjugate ($H^{1,0} \oplus \overline{H^{1,0}} = H^1(X, \mathbb{C})$); recovering the complex structure on V from this subspace is the inverse construction, and a polarization (Chapter 3) is a further alternating form making the subspace “positive.”

The wedge-power structure of theorem 1.3.2 propagates the weight-one decomposition to every degree. Applying $\bigwedge^n$ to $H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$ splits $H^n(X, \mathbb{C})$ into pieces $H^{p,q} = \bigwedge^p H^{1,0} \otimes \bigwedge^q H^{0,1}$ with $p + q = n$, the Hodge decomposition of weight n . Weight two is the first place where this structure carries geometry the reader will use directly, because the class of a line bundle lives in H^2 ; the next result records the integral degree-two picture and isolates the piece that will become the Néron-Severi group.

Proposition 1.3.5: H^2 as Alternating Forms on the Lattice

Let $X = V/\Lambda$ be a complex torus. There is a canonical isomorphism

$$H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee \cong \{ E: \Lambda \times \Lambda \rightarrow \mathbb{Z} \mid E \text{ is } \mathbb{Z}\text{-bilinear and alternating} \}$$

identifying a degree-two class with an alternating integral form on Λ . Extending scalars to $\mathbb{C}$ and using definition 1.3.3, this decomposes as

$$H^2(X, \mathbb{C}) = H^{2,0} \oplus H^{1,1} \oplus H^{0,2}, \quad H^{p,q} = \bigwedge^p H^{1,0} \otimes \bigwedge^q H^{0,1}$$

with $\dim_{\mathbb{C}} H^{2,0} = \dim_{\mathbb{C}} H^{0,2} = \binom{g}{2}$ and $\dim_{\mathbb{C}} H^{1,1} = g^2$. A class $E \in H^2(X, \mathbb{R})$, viewed as an alternating $\mathbb{R}$ -bilinear form on V , lies in $H^{1,1}$ if and only if it satisfies the invariance $E(iv, iw) = E(v, w)$ for all $v, w \in V$.

Proof:

Step 1: The integral identification. By theorem 1.3.2, $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$. An element of $\bigwedge^2 \Lambda^\vee$ is an alternating $\mathbb{Z}$ -bilinear form on Λ : the pairing $\bigwedge^2 \Lambda^\vee \rightarrow \text{Hom}(\bigwedge^2 \Lambda, \mathbb{Z})$ sending $\varphi \wedge \psi$ to the form $(\lambda, \mu) \mapsto \varphi(\lambda)\psi(\mu) - \varphi(\mu)\psi(\lambda)$ is an isomorphism onto the alternating forms, because Λ is free of finite rank. This is the stated identification $H^2(X, \mathbb{Z}) \cong \{E: \Lambda \times \Lambda \rightarrow \mathbb{Z} \text{ alternating}\}$.

Step 2: The Hodge decomposition of H^2 . Wedging the decomposition $H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$ with itself, $\bigwedge^2 H^1(X, \mathbb{C}) = \bigwedge^2 H^{1,0} \oplus (H^{1,0} \otimes H^{0,1}) \oplus \bigwedge^2 H^{0,1}$, which are the summands $H^{2,0}$, $H^{1,1}$, $H^{0,2}$ respectively. Since $\dim_{\mathbb{C}} H^{1,0} = \dim_{\mathbb{C}} H^{0,1} = g$, the outer summands have dimension $\dim \bigwedge^2 \mathbb{C}^g = \binom{g}{2}$ and the middle summand has dimension $g \cdot g = g^2$. As a check, $2\binom{g}{2} + g^2 = g(g-1) + g^2 = 2g^2 - g = \binom{2g}{2}$, the rank of H^2 .

Step 3: The $(1, 1)$ condition. An alternating $\mathbb{R}$ -bilinear form E on V extends $\mathbb{C}$ -bilinearly to $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$, which decomposes as $V^{1,0} \oplus V^{0,1}$ under $J =$ multiplication by i (the $\pm i$ -eigenspaces of $J \otimes 1$). The form E , as an element of $\bigwedge^2 V_{\mathbb{C}}^*$, lies in the $(1, 1)$ -piece exactly when it pairs $V^{1,0}$ with $V^{0,1}$ and annihilates $\bigwedge^2 V^{1,0}$ and $\bigwedge^2 V^{0,1}$. A direct computation identifies this with the eigenvalue condition under J : E is of type $(1, 1)$ if and only if $E(Jv, Jw) = E(v, w)$ for all v, w , that is, $E(iv, iw) = E(v, w)$. Indeed, on the $(2, 0) \oplus (0, 2)$ part J acts by $E(Jv, Jw) = -E(v, w)$ (each of v, w contributing a factor i from the same isotropic subspace), while on the $(1, 1)$ part the two factors i and $-i$ cancel to give $E(Jv, Jw) = E(v, w)$; the condition $E(iv, iw) = E(v, w)$ therefore cuts out exactly $H^{1,1} \cap H^2(X, \mathbb{R})$, as we sought to show.

The last clause is the technical heart of Part I stated in advance. An integral class $E \in H^2(X, \mathbb{Z})$ that is of type $(1, 1)$ is an alternating integral form on Λ satisfying $E(iv, iw) = E(v, w)$, and by the correspondence $H \mapsto E = \text{Im} H$, with inverse $H(v, w) = E(iv, w) + iE(v, w)$, such a form is exactly the imaginary part of a Hermitian form H on V . These are the classes that arise as first Chern classes of line bundles: the group of them is the Néron-Severi group $\text{NS}(X)$, developed in Chapter 2, and the requirement that the Hermitian form H be positive definite is the Riemann relation that singles out the algebraic tori in Chapter 3. Thus the abstract statement “ $H^2(X, \mathbb{Z}) =$ alternating forms on Λ , with a distinguished $(1, 1)$ part” is the frame into which the entire theory of polarizations will be poured. In degree two the integral Lefschetz $(1, 1)$ -theorem holds on any complex torus: every integral $(1, 1)$ class in H^2 is algebraic, because the Appell-Humbert theorem of Chapter 2 will realize it as the first Chern class of an actual line bundle on X . (This is a statement about divisors only; in higher codimension the analogous integral Hodge conjecture can fail for abelian varieties, and nothing here bears on it.)

The general picture is best seen first in the case $g = 1$, where the lattice has rank 2, the cohomology is that of a 2-torus, and every subspace is one-dimensional.

Example 1.3.6: The Hodge Structure of an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$, the standard elliptic-curve lattice ([3]). Here $g = 1$, $V = \mathbb{C}$ with coordinate z , and the topological facts of proposition 1.3.1 and theorem 1.3.2 read:

- $H_1(E, \mathbb{Z}) \cong \Lambda = \mathbb{Z} \cdot 1 \oplus \mathbb{Z} \cdot \tau$, generated by the two loops $\gamma_1: t \mapsto t$ and $\gamma_\tau: t \mapsto t\tau$, $t \in [0, 1]$, the sides of the fundamental parallelogram.
- $H^1(E, \mathbb{Z}) \cong \Lambda^\vee = \mathbb{Z}\alpha \oplus \mathbb{Z}\beta$, where α, β is the dual basis: $\alpha(1) = 1$, $\alpha(\tau) = 0$, $\beta(1) = 0$, $\beta(\tau) = 1$.
- $H^2(E, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee = \mathbb{Z}(\alpha \wedge \beta)$, the fundamental class, and $H^n(E, \mathbb{Z}) = 0$ for $n > 2$.

The Hodge decomposition of definition 1.3.3 is

$$H^1(E, \mathbb{C}) = H^{1,0} \oplus H^{0,1} = \mathbb{C} \cdot dz \oplus \mathbb{C} \cdot d\bar{z}$$

a splitting of a 2-dimensional space into two lines. The single holomorphic 1-form dz spans $H^{1,0} = H^0(E, \Omega_E^1)$, consistent with $\dim H^0(E, \Omega_E^1) = g = 1$ (a genus-one curve has a one-dimensional space of holomorphic differentials), and $d\bar{z} = d\bar{z}$ spans $H^{0,1}$.

The relation between the two bases is recorded by the *periods*: the numbers 1 and τ that generate Λ also fix the position of the Hodge line $H^{1,0}$ inside $H^1(E, \mathbb{C})$. The form dz is a class in

$H^1(E, \mathbb{C}) = H^1(E, \mathbb{Z}) \otimes \mathbb{C}$; expressing it in the integral basis α, β amounts to integrating dz over the loops γ_1, γ_τ . The period integrals are

$$\int_{\gamma_1} dz = \int_0^1 dt = 1, \quad \int_{\gamma_\tau} dz = \int_0^1 \tau dt = \tau$$

so that, under the pairing of $H^1(E, \mathbb{C})$ with $H_1(E, \mathbb{Z})$, the class of dz is $\alpha + \tau \beta$. The row vector $(1 \ \tau)$ of its periods is the period matrix Π of E , the 1×2 matrix whose columns record the $\mathbb{Z}$ -basis $1, \tau$ of Λ in the $\mathbb{C}$ -basis $\{1\}$ of $V = \mathbb{C}$, and the ratio of periods $\tau = (\int_{\gamma_\tau} dz) / (\int_{\gamma_1} dz)$ recovers the point of the upper half-plane classifying E up to isomorphism ([3]). The positivity $\text{Im} \tau > 0$ is the $g = 1$ Riemann relation: it is equivalent to $\frac{i}{2} \int_E dz \wedge d\bar{z} > 0$, the statement that the natural Hermitian form is positive, which is the condition studied in general in Chapter 3.

The elliptic curve exhibits the entire structure of this section in miniature: the lattice is H_1 , its dual is H^1 , the top wedge is the fundamental class in H^2 , and the Hodge line $H^{1,0} = \mathbb{C} dz$ tilts inside $H^1(E, \mathbb{C})$ by an amount recorded in the period τ . Moving that line, that is, varying τ over the upper half-plane, varies the complex structure while leaving the lattice and its cohomology fixed; this is the first appearance of a period domain and of moduli, and in dimension g the Hodge line becomes a g -dimensional subspace whose variation is governed by the Siegel upper half-space of Chapter 3.

The unifying statement of the section is that the passage $X \mapsto (H^1(X, \mathbb{Z}), H^{1,0})$ is a functor from complex tori to weight-one Hodge structures, and that it loses nothing: a complex torus and its weight-one Hodge structure carry the same information, because the complex structure on V is recovered from the position of $H^{1,0} \subset H^1(X, \mathbb{C})$, and $\Lambda = H_1(X, \mathbb{Z})$ is recovered as the integral lattice. This is the precise form of the assertion in the chapter opening that a complex torus is governed by linear algebra, and it is the bridge on which the rest of the book is built. Everything added in Chapters 2 through 6 is extra structure on this Hodge structure: a line bundle is a class in the integral $(1, 1)$ part of H^2 ; a polarization is such a class whose Hermitian form is positive; the endomorphism algebra is the algebra of Hodge-structure endomorphisms.

1.4 Which Complex Tori Are Algebraic?

The preceding sections have made a complex torus $X = V/\Lambda$ (definition 1.1.2) into a thoroughly computable object: a compact complex Lie group whose morphisms are linear algebra over $\mathbb{Z}$ (proposition 1.1.7) and whose cohomology is the exterior algebra of a lattice (theorem 1.3.2). Nothing so far has required that X be anything more than a complex manifold. The question that organizes the rest of Part I is whether X is also an algebraic object, that is, whether it can be realized as a projective variety cut out by polynomial equations. This section makes the question precise, settles it completely in dimension one, and exhibits an explicit two-dimensional torus for which the answer is no.

The stake is the following. A complex torus is a quotient, and quotients are cheap: any lattice of rank $2g$ in $\mathbb{C}^g$ produces one, and the space of lattices is $2g^2$ -complex-dimensional. Projective varieties, by contrast, are constrained: their meromorphic functions form a field of transcendence degree equal to the dimension, and the ambient projective space forces relations among coordinates. To ask that a torus be projective is to ask that its meromorphic function field be as large as possible. The tension between the ease of writing down tori and the scarcity of projective ones is the phenomenon this section names, and the criterion that resolves it, the Riemann relations, is the subject of Chapter 3.

Definition 1.4.1: Abelian Variety (Provisional)

A complex torus $X = V/\Lambda$ is called *algebraic*, or an *abelian variety*, if there exists a holomorphic embedding $X \hookrightarrow \mathbb{P}^N$ into some complex projective space, that is, a holomorphic map that is an isomorphism onto a closed complex submanifold of $\mathbb{P}^N$.

This definition is provisional in two respects, both resolved later. It is stated in the analytic category, in terms of a holomorphic embedding into $\mathbb{P}^N$; that a compact complex submanifold of projective space is automatically the zero locus of homogeneous polynomials, hence a projective variety in the algebraic sense, is Chow's theorem, cited to [10]. And the criterion for such an embedding to exist is not yet visible from the definition. The refined definition, in terms of the existence of a polarization, is definition 3.2.1, and theorem 3.2.6 records the equivalence of the two formulations. The name *abelian variety* anticipates that the group structure carried by X (proposition 1.1.3) becomes, once X is projective, a morphism of varieties: an abelian variety is a projective group.

Why insist on projectivity rather than some weaker form of algebraicity? The answer is that projectivity is exactly the condition under which the group law and the geometry interact well. A projective variety is proper, so the image of any morphism out of it is closed; a connected group variety that is proper is automatically commutative and its group law is uniquely determined by the identity, facts developed scheme-theoretically in Part II. None of this holds for a general complex torus regarded only as a complex manifold. Projectivity is what promotes a compact complex Lie group to an object of algebraic geometry.

The one-dimensional case is settled by machinery already in hand, and settled affirmatively for every torus without exception.

Proposition 1.4.2: One-Dimensional Tori Are Elliptic Curves

Every one-dimensional complex torus $X = \mathbb{C}/\Lambda$ is an abelian variety. Explicitly, the map

$$\varphi: \mathbb{C}/\Lambda \longrightarrow \mathbb{P}^2, \quad z \longmapsto \begin{cases} [\wp(z) : \wp'(z) : 1] & z \notin \Lambda \\ [0 : 1 : 0] & z \in \Lambda \end{cases}$$

built from the Weierstrass function $\wp$ of Λ and its derivative is a holomorphic embedding whose image is the smooth plane cubic

$$Y^2 Z = 4X^3 - g_2(\Lambda) X Z^2 - g_3(\Lambda) Z^3 \quad (1.1)$$

with $g_2(\Lambda) = 60 \sum_{\lambda \neq 0} \lambda^{-4}$ and $g_3(\Lambda) = 140 \sum_{\lambda \neq 0} \lambda^{-6}$.

Proof:

The Weierstrass function $\wp$ of the lattice Λ is the Λ -periodic meromorphic function

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda \in \Lambda \setminus \{0\}} \left(\frac{1}{(z - \lambda)^2} - \frac{1}{\lambda^2} \right)$$

with a double pole at each lattice point and no others; its construction, convergence, and the differential equation

$$\wp'(z)^2 = 4\wp(z)^3 - g_2(\Lambda) \wp(z) - g_3(\Lambda) \quad (1.2)$$

are developed in full in [10] and applied to elliptic curves in [3]. Only (1.2) and the pole orders of $\wp$ and $\wp'$ are used here.

By (1.2), for $z \notin \Lambda$ the point $(\wp(z), \wp'(z))$ satisfies the affine equation $Y^2 = 4X^3 - g_2X - g_3$, so $\wp(z)$ lies on the projective curve (1.1). Both $\wp$ and $\wp'$ are Λ -periodic, so φ is well-defined on the quotient $\mathbb{C}/\Lambda$. At a lattice point $\wp$ has a double pole and $\wp'$ a triple pole, and the local expansions $\wp(z) \sim z^{-2}$, $\wp'(z) \sim -2z^{-3}$ near $z = 0$ give

$$[\wp(z) : \wp'(z) : 1] = [z^{-2} : -2z^{-3} : 1] = [z : -2 : z^3] \longrightarrow [0 : 1 : 0]$$

as $z \rightarrow 0$, so the assignment $\varphi(0) = [0 : 1 : 0]$ extends φ holomorphically across the lattice points. Away from Λ , φ is a composition of the meromorphic functions $\wp, \wp'$ with the standard chart of $\mathbb{P}^2$, hence holomorphic, so φ is holomorphic on all of $\mathbb{C}/\Lambda$.

The cubic (1.1) is smooth because the lattice discriminant $g_2^3 - 27g_3^2$ is nonzero for a genuine lattice ([10]), so the polynomial $4X^3 - g_2X - g_3$ has distinct roots. That φ is a biholomorphism onto this smooth cubic is the uniformization theorem, proved in full as the isomorphism of Riemann surfaces $\mathbb{C}/\Lambda \cong E(\mathbb{C})$ in [3]. Injectivity uses that $\wp$ has degree two and $\wp'$ is odd, so the fibers of $\wp$ are the pairs $\{a, -a\}$ separated by the value of $\wp'$; surjectivity uses that $\wp$ attains every value; and a bijective holomorphic map of compact Riemann surfaces is biholomorphic. A biholomorphism onto a smooth cubic in $\mathbb{P}^2$ is in particular a holomorphic embedding, so $X = \mathbb{C}/\Lambda$ is an abelian variety in the sense of definition 1.4.1, completing the proof.

The content of proposition 1.4.2 is that the two descriptions of an elliptic curve, the transcendental quotient $\mathbb{C}/\Lambda$ and the algebraic plane cubic (1.1), are the same object. The Weierstrass function is precisely the nonconstant meromorphic function whose existence witnesses algebraicity: it and its derivative generate the field of Λ -periodic meromorphic functions, and that field is $\mathbb{C}(X)$, of transcendence degree one. Every one-dimensional torus carries such a function, and so every one is projective. The purpose of the rest of Part I is to explain why this uniformity breaks down in higher dimension, and the first sign of the breakdown is that in dimension two or more, a torus may have no nonconstant meromorphic functions at all.

The right invariant to track is the size of the meromorphic function field. For a compact complex manifold X of dimension g , the field $\mathcal{M}(X)$ of meromorphic functions is a finitely generated extension of $\mathbb{C}$ whose transcendence degree, the *algebraic dimension* $a(X) := \text{trdeg}_{\mathbb{C}} \mathcal{M}(X)$, satisfies $0 \leq a(X) \leq g$. This bound and the finite generation are theorems of Siegel; the detailed development for complex tori, including the equivalences below, is in Birkenhake and Lange, *Complex Abelian Varieties*. The two extremes are the cases of interest. At the top, $a(X) = g$ holds exactly when X is an abelian variety: the function field is then as large as the dimension permits, the torus is projective, and $\mathcal{M}(X) = \mathbb{C}(X)$ is the field of rational functions of the projective variety. At the bottom, $a(X) = 0$ means the only meromorphic functions on X are the constants, so X admits no nonconstant holomorphic map to any projective space and is as far from algebraic as a compact complex manifold can be. For a torus, $a(X) = g$ is equivalent to the existence of a positive Hermitian form on V with imaginary part integral on Λ , which is the polarization condition of Chapter 3; the intermediate values $0 < a(X) < g$ occur and correspond to tori that are fibered over lower-dimensional abelian varieties. By proposition 1.4.2, $g = 1$ forces $a(X) = 1 = g$, so the phenomenon of a deficient function field is invisible until $g \geq 2$.

The following example makes the failure concrete. It exhibits a two-dimensional torus, given by an

explicit period matrix, that carries no nonconstant meromorphic functions, hence is not algebraic.

Example 1.4.3: A Generic Two-Dimensional Torus Is Not Algebraic

Fix $V = \mathbb{C}^2$ and let $\tau \in M_2(\mathbb{C})$ be a 2×2 complex matrix, chosen generically in a sense made precise below. Take the lattice

$$\Lambda = \tau\mathbb{Z}^2 + \mathbb{Z}^2 \subset \mathbb{C}^2$$

spanned by the columns of the period matrix $\Pi = (\tau \ I_2) \in M_{2 \times 4}(\mathbb{C})$, and set $X = \mathbb{C}^2/\Lambda$. For Λ to be a full lattice, the columns of Π must be $\mathbb{R}$ -linearly independent, which holds as long as $\text{Im}\tau$ is invertible; assume this. The claim is that for τ outside a countable union of proper analytic subsets of the space of such matrices, X admits no nonzero Hermitian form H on V whose imaginary part $E = \text{Im}H$ is integral on Λ , and consequently $a(X) = 0$: the only meromorphic functions on X are constants, and X is not an abelian variety.

The key computation is a count of linear constraints. By the analysis of Hermitian forms integral on a lattice, which the reader will meet in proposition 2.3.2, a Hermitian H on $V = \mathbb{C}^2$ with $E = \text{Im}H$ integral on Λ is the same datum as an alternating integral form $E: \Lambda \times \Lambda \rightarrow \mathbb{Z}$ satisfying the compatibility $E(iv, iw) = E(v, w)$ with the complex structure. An alternating integral form on the rank-four lattice Λ is specified by an alternating matrix $E \in M_4(\mathbb{Z})$, a choice of $\binom{4}{2} = 6$ integers. The compatibility condition $E(iv, iw) = E(v, w)$, written out in the period coordinates via Π , becomes the equation

$$\Pi E^{-1} \Pi^t = 0$$

on the entries of τ . Because E is alternating, E^{-1} is again skew, so the 2×2 complex matrix $\Pi E^{-1} \Pi^t$ is itself skew-symmetric: its diagonal vanishes identically and its two off-diagonal entries are negatives of one another. The equation is therefore the vanishing of a single complex number, one $\mathbb{C}$ -linear equation, hence two $\mathbb{R}$ -linear equations, in the six integer entries of E . Written out, the constraint is the vanishing of the single complex-linear form

$$-a - d\tau_{11} + b\tau_{12} - e\tau_{21} + f(\tau_{12}\tau_{21} - \tau_{11}\tau_{22}) + c\tau_{22}$$

in which a, b, c, d, e, f range over the six independent entries of E ; its real and imaginary parts are two $\mathbb{R}$ -linear equations whose coefficients are the real and imaginary parts of 1, the four entries of τ , and $\det \tau = \tau_{11}\tau_{22} - \tau_{12}\tau_{21}$. A system of two real equations on six real unknowns cannot cut it down to the origin; its real solution space is four-dimensional. The point is that for generic τ these coefficients are not $\mathbb{Q}$ -linear combinations of one another, so the four-dimensional real kernel is not defined over $\mathbb{Q}$ and meets the integer lattice $\mathbb{Z}^6$ only at the origin, forcing $E = 0$. The two demands, that E be integral (which confines it to $\mathbb{Z}^6$) and that it be compatible with the complex structure (which cuts out the τ -dependent real kernel above), are thus jointly satisfiable by a nonzero E only when τ lies in a countable union of proper analytic subsets; a generic τ avoids all of them.

Granting that $H = 0$ is the only integral Hermitian form, no positive such form exists, so X carries no polarization. Since X then admits no nonzero integral Hermitian form at all, its Néron-Severi group vanishes, and a complex torus whose line bundles are all topologically trivial has $a(X) = 0$. The full verification that the vanishing of the meromorphic function field follows from the absence of an integral Hermitian form, and the precise measure-theoretic statement of genericity, are carried out in Birkenhake and Lange, *Complex Abelian Varieties*; the mechanism recorded above

is the linear-algebraic heart of that argument. A concrete instance is the non-symmetric matrix

$$\tau = \begin{pmatrix} 2^{1/3} i & \sqrt{2} i \\ \sqrt{3} i & 3i \end{pmatrix}$$

for which $\text{Im}\tau$ is invertible, so Λ is a full lattice. Here the two real equations read $-a + (3 \cdot 2^{1/3} - \sqrt{6})f = 0$ and $\sqrt{2}b + 3c - 2^{1/3}d - \sqrt{3}e = 0$. The first forces $f = 0$ and hence $a = 0$, since $3 \cdot 2^{1/3} - \sqrt{6}$ is irrational; the second forces $b = c = d = e = 0$, since $1, \sqrt{2}, 2^{1/3}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$. Thus $E = 0$ is the only integral compatible form, so this particular X carries no polarization and has only constant meromorphic functions. That τ is non-symmetric is what forces the vanishing: a symmetric τ with $\text{Im}\tau$ positive definite lies in the Siegel upper half-space and always furnishes a principal polarization, as section 3.3 makes precise.

Two lessons stand behind this example. The first is that non-algebraic complex tori are not exotic: they are the generic case in every dimension $g \geq 2$, since the algebraicity conditions cut out a countable union of proper analytic subsets in the space of lattices, whose complement is everything. The second is that the obstruction is entirely linear algebraic, a question about whether a certain integral alternating form compatible with the complex structure exists, with no reference to the transcendental machinery of meromorphic functions once the equivalence $a(X) = g \iff$ polarization is granted. This is what makes the algebraicity question tractable: it reduces to the arithmetic of forms on Λ , which is exactly the reduction Chapter 3 carries out.

1.4.4 Foreshadowing: The Riemann Relations The condition separating the algebraic tori from the rest was named repeatedly above without being stated: a complex torus $X = V/\Lambda$ is an abelian variety precisely when V carries a positive definite Hermitian form H whose imaginary part $E = \text{Im}H$ takes integer values on Λ . This is the content of the *Riemann relations*, proved as theorem 3.1.4 in Chapter 3. The positivity of H is what produces enough sections of a line bundle to define a projective embedding, via the theta functions of Chapter 4; the integrality of $E = \text{Im}H$ on Λ is what makes the bundle descend to the quotient X . In dimension one every lattice admits such an H , which reproves proposition 1.4.2 from the general criterion; in dimension two or more the two demands, positivity and integrality, are jointly a strong constraint that example 1.4.3 shows a generic lattice fails. The organizing tension of Part I is exactly this gap: complex tori are effortless to construct and describe, yet only those satisfying the Riemann relations are algebraic, and the machinery of the next two chapters exists to characterize, and then to realize, that scarce algebraic subclass.

Line Bundles and the Appell-Humbert Theorem

Whether a complex torus embeds in projective space is a question about line bundles: an embedding into $\mathbb{P}^N$ pulls back the hyperplane bundle to an ample line bundle on X , and conversely a positive enough line bundle manufactures an embedding. Before one can speak of positivity, one needs to know what the line bundles on a complex torus are. This chapter answers that completely. The Appell-Humbert theorem describes every line bundle on $X = V/\Lambda$ by a startlingly small amount of data: a Hermitian form H on V whose imaginary part is integral on the lattice, together with a unitary semicharacter χ recording how the bundle twists around the lattice directions.

The route there runs through the exponential sequence, which splits the Picard group into a continuous part (the bundles that are topologically trivial, parametrized by characters of the lattice) and a discrete part (the Néron-Severi group, measuring the topological type through the first Chern class). Reading the first Chern class as an alternating integer-valued form on Λ turns the analytic classification into a problem in the arithmetic of bilinear forms, and it is that translation which, in Chapter 3, makes the algebraicity criterion computable. The chapter ends by realizing sections of a line bundle as classical theta functions, the holomorphic functions on V with prescribed quasi-periodicity that Chapter 4 studies in earnest.

2.1 The Picard Group and the Exponential Sequence

A complex torus $X = V/\Lambda$ is a compact complex Lie group, and section 1.3 exhausted the linear algebra of its cohomology: the integral cohomology is the exterior algebra $\bigwedge^\bullet \Lambda^\vee$, and the Hodge structure on H^1 is the explicit decomposition $V^* \oplus \bar{V}^*$. What that analysis did not touch is the geometry that makes a torus into a variety in the first place, namely its holomorphic line bundles.

A line bundle is what allows a compact complex manifold to be mapped to projective space: a choice of line bundle L together with a basis of its global sections $s_0, \dots, s_N$ produces the map $x \mapsto [s_0(x) : \dots : s_N(x)]$ into $\mathbb{P}^N$, and every holomorphic map to projective space arises this way. The question of which complex tori are algebraic, raised in section 1.4 and answered in Chapter 3, is at bottom the question of which tori carry enough line bundles to separate their points. This chapter classifies all of them.

This section sets up the bookkeeping. The holomorphic line bundles on any complex manifold form a group, the Picard group, and on a complex manifold there is one exact sequence of sheaves that controls it completely: the exponential sequence. Feeding it through the long exact sequence in cohomology splits the Picard group into a continuous part and a discrete part, and identifies each with a piece of the linear algebra already in hand. The continuous part turns out to be a complex torus in its own right, the discrete part a lattice of alternating forms. The next two sections make each part fully explicit; here the exponential sequence does the separating.

Definition 2.1.1: Picard Group

Let X be a complex manifold and let $\mathcal{O}_X^*$ denote the sheaf of nowhere-vanishing holomorphic functions on X , a sheaf of abelian groups under multiplication. The *Picard group* of X is

$$\mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^*)$$

the first sheaf cohomology of X with coefficients in $\mathcal{O}_X^*$. Its elements are the isomorphism classes of holomorphic line bundles on X , the group operation is tensor product $L \otimes L'$, the identity is the trivial bundle $\mathcal{O}_X$, and the inverse of L is the dual bundle $L^{-1} = \mathrm{Hom}(L, \mathcal{O}_X)$.

The identification of $H^1(X, \mathcal{O}_X^*)$ with isomorphism classes of line bundles is the standard Čech description: a line bundle is glued from trivial pieces $U_i \times \mathbb{C}$ by transition functions $g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$ satisfying the cocycle condition $g_{ij}g_{jk} = g_{ik}$, and two systems of transition functions define isomorphic bundles precisely when they differ by a coboundary. This is exactly the datum of a class in $H^1(X, \mathcal{O}_X^*)$, and tensor product of line bundles corresponds to multiplication of Čech cocycles, so the bijection is a group isomorphism. The full verification belongs to the general theory of line bundles on complex manifolds; see [10]. What matters here is that the Picard group is a cohomology group, because that is what makes it computable.

The reason line bundles govern maps to projective space is worth stating precisely, since it is the motivation for the entire chapter. On $\mathbb{P}^N$ there is the hyperplane bundle $\mathcal{O}(1)$, the Serre twisting sheaf, whose global sections are exactly the linear forms $X_0, \dots, X_N$. Pulling back along a holomorphic map $\varphi: X \rightarrow \mathbb{P}^N$ produces a line bundle $L = \varphi^*\mathcal{O}(1)$ on X together with the $N+1$ sections $s_i = \varphi^*X_i$, which have no common zero (the point $[X_0 : \dots : X_N]$ is always defined). Conversely a line bundle L and a base-point-free system of sections $s_0, \dots, s_N$ reconstruct φ . Thus classifying line bundles is the first step toward classifying the maps of X to projective space, and understanding $\mathrm{Pic}(X)$ is prerequisite to understanding when X is projective at all.

Example 2.1.2: The Picard Group of an Elliptic Curve

For the elliptic curve $E = \mathbb{C}/\Lambda$ with $g = 1$, the Picard group is understood through the degree homomorphism. A line bundle on a compact Riemann surface has a well-defined degree, the sum of the multiplicities of the zeros minus the poles of any meromorphic section, and this gives a surjective homomorphism $\deg: \mathrm{Pic}(E) \rightarrow \mathbb{Z}$. The kernel consists of the degree-zero line bundles,

and the Abel-Jacobi theorem for E ([3]) identifies it with E itself,

$$\mathrm{Pic}^0(E) \cong E \quad L \mapsto [\text{the point } p \text{ with } L \cong \mathcal{O}_E(p - 0)]$$

where $0 \in E$ is the origin. The degree sequence $0 \rightarrow \mathrm{Pic}^0(E) \rightarrow \mathrm{Pic}(E) \xrightarrow{\deg} \mathbb{Z} \rightarrow 0$ splits, since the class of $\mathcal{O}_E(0)$ maps to $1 \in \mathbb{Z}$, giving

$$\mathrm{Pic}(E) \cong \mathbb{Z} \oplus E$$

The discrete factor $\mathbb{Z}$ records the degree; the continuous factor E is the group of degree-zero bundles, itself an elliptic curve. This chapter recovers both factors for a general torus: this section identifies the continuous factor E as the dual torus, and section 2.3 replaces the discrete factor $\mathbb{Z}$ by the group of integral alternating forms.

The example exhibits, in the case $g = 1$, exactly the two-part structure this section extracts in general: a continuous group of “degree-zero” line bundles, which is again a torus, and a discrete group of numerical invariants. The mechanism that separates the two parts for an arbitrary complex torus, and indeed for any compact complex manifold, is the exponential sequence.

The exponential sequence relates the additive sheaf $\mathcal{O}_X$ to the multiplicative sheaf $\mathcal{O}_X^*$ through the exponential map $f \mapsto e^{2\pi i f}$. The factor $2\pi i$ is chosen so that the kernel is exactly the constant integer functions: $e^{2\pi i f} = 1$ holds precisely when f is an integer. This is the one short exact sequence of sheaves available on every complex manifold, and its long exact cohomology sequence is the engine of the whole classification.

Theorem 2.1.3: The Exponential Sheaf Sequence

Let X be a complex manifold. Write $\underline{\mathbb{Z}}$ for the constant sheaf $\mathbb{Z}$ on X . The sequence of sheaves of abelian groups

$$0 \rightarrow \underline{\mathbb{Z}} \rightarrow \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \rightarrow 0 \quad (2.1)$$

is exact, where the first map is the inclusion of locally constant integer functions and $\exp(f) = e^{2\pi i f}$. It induces a long exact sequence in cohomology

$$\cdots \rightarrow H^1(X, \underline{\mathbb{Z}}) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow H^1(X, \mathcal{O}_X^*) \xrightarrow{\delta} H^2(X, \underline{\mathbb{Z}}) \rightarrow H^2(X, \mathcal{O}_X) \rightarrow \cdots \quad (2.2)$$

in which the middle term $H^1(X, \mathcal{O}_X^*) = \mathrm{Pic}(X)$ is the Picard group and δ is the connecting homomorphism.

Proof:

Step 1: Exactness of the sheaf sequence (2.1). Exactness of a sequence of sheaves is checked on stalks, so fix a point $x \in X$ and work with germs of holomorphic functions at x . The inclusion $\underline{\mathbb{Z}} \rightarrow \mathcal{O}_X$ sends an integer n to the germ of the constant function n ; it is injective because a nonzero constant is a nonzero germ. The map $\exp$ sends a germ f to $e^{2\pi i f}$, which is a germ of a nowhere-vanishing holomorphic function, since $e^{2\pi i f}$ never takes the value zero. Composing, $\exp$ of an integer constant is $e^{2\pi i n} = 1$, so the image of $\underline{\mathbb{Z}}$ lies in the kernel of $\exp$.

Conversely, suppose $e^{2\pi i f} = 1$ as germs at x , with f holomorphic near x . Then $2\pi i f$ takes values in $2\pi i \mathbb{Z}$; since f is continuous and $\mathbb{Z}$ is discrete, f is locally constant near x with integer value, hence a germ from $\underline{\mathbb{Z}}$. Thus the kernel of $\exp$ is exactly the image of $\underline{\mathbb{Z}}$, which is exactness at $\mathcal{O}_X$.

Step 2: Surjectivity of $\exp$ on stalks. Surjectivity of a map of sheaves is also a stalk condition. Let u be the germ at x of a nowhere-vanishing holomorphic function. On a small enough simply connected neighborhood U of x , the function u has no zeros, so a holomorphic logarithm $\log u$ exists on U : since $u: U \rightarrow \mathbb{C}^*$ and U is simply connected, the holomorphic 1-form $\frac{du}{u}$ has a holomorphic primitive ℓ on U with $e^\ell = u$ after adjusting an additive constant. Setting $f = \frac{1}{2\pi i} \ell$ gives a holomorphic germ with $\exp(f) = e^{2\pi i f} = e^\ell = u$. Hence every germ of $\mathcal{O}_X^*$ lifts, and $\exp$ is surjective as a map of sheaves. This existence of a local holomorphic logarithm is the one nontrivial input, and it is a theorem of one complex variable; see [10]. This establishes the exactness of (2.1).

Step 3: The long exact sequence. A short exact sequence of sheaves of abelian groups on any topological space induces a long exact sequence in sheaf cohomology, with connecting homomorphisms $\delta: H^k(X, \mathcal{O}_X^*) \rightarrow H^{k+1}(X, \underline{\mathbb{Z}})$. This is the derived-functor long exact sequence of the cohomology functor $H^\bullet(X, -)$, proved in [5] for the cohomology of any left-exact functor applied to a short exact sequence. Applied to (2.1), it reads

$$\cdots \rightarrow H^k(X, \underline{\mathbb{Z}}) \rightarrow H^k(X, \mathcal{O}_X) \rightarrow H^k(X, \mathcal{O}_X^*) \xrightarrow{\delta} H^{k+1}(X, \underline{\mathbb{Z}}) \rightarrow \cdots$$

and the segment in degrees 1 and 2 is exactly (2.2), with $H^1(X, \mathcal{O}_X^*) = \text{Pic}(X)$ by definition 2.1.1, completing the proof.

The sheaf-cohomology group $H^k(X, \underline{\mathbb{Z}})$ of the constant sheaf agrees with the singular cohomology $H^k(X, \mathbb{Z})$ for a manifold, so the outer terms of (2.2) are the topological invariants computed in section 1.3, while the terms involving $\mathcal{O}_X$ are the holomorphic ones. The exponential sequence is thus the precise mechanism that couples topology to holomorphic geometry: a line bundle is a class in the middle, and reading it leftward recovers its holomorphic content while reading it rightward through δ recovers its topological content. Everything that follows is the extraction of these two readings.

The rightward reading is the more immediate one, because δ carries a line bundle to a class in $H^2(X, \mathbb{Z})$ that measures its topological type. This class is the first Chern class, and it is the primary discrete invariant of a line bundle.

Definition 2.1.4: First Chern Class

The *first Chern class* is the connecting homomorphism of the exponential sequence,

$$c_1: \text{Pic}(X) = H^1(X, \mathcal{O}_X^*) \xrightarrow{\delta} H^2(X, \mathbb{Z})$$

For a line bundle L , the class $c_1(L) \in H^2(X, \mathbb{Z})$ is a homomorphism of abelian groups: $c_1(L \otimes L') = c_1(L) + c_1(L')$, since δ is a homomorphism of cohomology groups.

The first Chern class is the topological shadow of a line bundle. Two line bundles with the same c_1 are topologically isomorphic as complex line bundles, differing only in their holomorphic structure;

conversely c_1 records exactly the underlying topological line bundle, which on a manifold is classified by $H^2(X, \mathbb{Z})$. The image of c_1 inside $H^2(X, \mathbb{Z})$ is the group of topological types that can be realized holomorphically, and identifying it is the subject of section 2.3, where it emerges as the group of alternating integral forms $E = \text{Im}H$. For the moment the object of interest is the kernel.

Proposition 2.1.5: The Degree-Zero Picard Group as a Complex Torus

Let $X = V/\Lambda$ be a complex torus of dimension g , and let $\text{Pic}^0(X) = \ker(c_1) \subseteq \text{Pic}(X)$ be the group of line bundles with trivial first Chern class. Then the exponential sequence induces an isomorphism

$$\text{Pic}^0(X) \cong H^1(X, \mathcal{O}_X) / \text{im}(H^1(X, \mathbb{Z}) \rightarrow H^1(X, \mathcal{O}_X)) \quad (2.3)$$

Moreover $H^1(X, \mathcal{O}_X) = H^{0,1} = \overline{V^*}$, the space of antilinear functionals on V , and the image of $H^1(X, \mathbb{Z}) \cong \Lambda^\vee$ is a full lattice in it. Consequently $\text{Pic}^0(X)$ is a complex torus of dimension g .

Proof:

Step 1: The kernel of c_1 as a cokernel. The relevant segment of the long exact sequence (2.2) is

$$H^1(X, \mathbb{Z}) \xrightarrow{\iota} H^1(X, \mathcal{O}_X) \xrightarrow{\exp_*} \text{Pic}(X) \xrightarrow{c_1} H^2(X, \mathbb{Z})$$

By exactness at $\text{Pic}(X)$, the kernel of c_1 equals the image of $\exp_*$. By exactness at $H^1(X, \mathcal{O}_X)$, the kernel of $\exp_*$ equals the image of ι . Hence $\exp_*$ induces an isomorphism from the cokernel of ι onto its image $\text{Pic}^0(X) = \ker(c_1)$,

$$\text{Pic}^0(X) \cong H^1(X, \mathcal{O}_X) / \text{im}(\iota)$$

which is (2.3).

Step 2: Identifying $H^1(X, \mathcal{O}_X)$. The Dolbeault isomorphism identifies $H^1(X, \mathcal{O}_X)$ with the Hodge piece $H^{0,1}(X)$ of $H^1(X, \mathbb{C})$. For a complex torus this piece was computed explicitly in proposition 1.3.4: the holomorphic cotangent space is $H^{1,0} = V^*$, spanned by $dz_1, \dots, dz_g$, and its conjugate is

$$H^1(X, \mathcal{O}_X) = H^{0,1} = \overline{V^*}$$

the space of $\mathbb{C}$ -antilinear functionals on V , spanned by $d\bar{z}_1, \dots, d\bar{z}_g$. This has complex dimension g .

Step 3: The image of $H^1(X, \mathbb{Z})$ is a full lattice. By theorem 1.3.2 the integral cohomology is $H^1(X, \mathbb{Z}) \cong \Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$, a free abelian group of rank $2g$. The map ι is the composite of the change of coefficients $H^1(X, \mathbb{Z}) \rightarrow H^1(X, \mathbb{C})$ with the projection $H^1(X, \mathbb{C}) = H^{1,0} \oplus H^{0,1} \rightarrow H^{0,1}$ onto the second summand. The change of coefficients is injective with image the real form $H^1(X, \mathbb{Z}) \otimes \mathbb{R} = H^1(X, \mathbb{R})$, a real subspace of $H^1(X, \mathbb{C})$ of real dimension $2g$ that meets neither $H^{1,0}$ nor $H^{0,1}$ except in 0: indeed $H^1(X, \mathbb{R})$ is the fixed space of complex conjugation, which interchanges $H^{1,0}$ and $H^{0,1}$, so a real class lying in $H^{0,1}$ would be conjugation-fixed and hence also in $H^{1,0}$, forcing it to be zero. It follows that the projection $H^1(X, \mathbb{R}) \rightarrow H^{0,1}$ is a real-linear isomorphism onto the underlying real space of $\overline{V^*}$, of real dimension $2g$. The image $\text{im}(\iota) = \iota(\Lambda^\vee)$ is therefore the image of the full lattice $\Lambda^\vee$ under a real-linear isomorphism, hence a full lattice in $H^{0,1} = \overline{V^*}$.

Step 4: The quotient is a complex torus. Combining the three steps,

$$\mathrm{Pic}^0(X) \cong \overline{V^*} / \iota(\Lambda^\vee)$$

the quotient of the g -dimensional complex vector space $\overline{V^*}$ by a full lattice. This is a complex torus of dimension g in the sense of definition 1.1.2, as we sought to show.

The proposition says that the degree-zero Picard group is not just a group but a complex torus of the same dimension as X , built from the antilinear dual $\overline{V^*}$ and the lattice $\Lambda^\vee$. This torus is the *dual torus* $\hat{X}$, the central construction of Chapter 5, where its geometry is developed in full and the pairing between X and $\hat{X}$ made explicit. The description as $\overline{V^*}/\Lambda^\vee$ is the analytic incarnation of the dual; the same object reappears in Part II as the identity component $\mathrm{Pic}_{X/k}^0$ of the Picard scheme. For now the point is that the continuous part of $\mathrm{Pic}(X)$ carries its own complex structure inherited from that of X .

There is a second description of $\mathrm{Pic}^0(X)$ that the Appell-Humbert theorem of section 2.2 will make precise. A degree-zero line bundle is, by (2.3), a class in $H^1(X, \mathcal{O}_X)$ modulo the integral lattice, and such a class is equivalent to a unitary character of the lattice Λ . Concretely, the isomorphism $\mathrm{Pic}^0(X) \cong \mathrm{Hom}(\Lambda, U(1))$ presents a degree-zero bundle as a homomorphism $\chi: \Lambda \rightarrow U(1)$, the group of characters of Λ . This is the semicharacter datum with $H = 0$ that theorem 2.2.8 attaches to a line bundle, and it is the concrete face of the abstract quotient (2.3).

Example 2.1.6: Degree-Zero Bundles on an Elliptic Curve

For $g = 1$ the proposition reads $\mathrm{Pic}^0(E) \cong \overline{V^*}/\Lambda^\vee$ with $V = \mathbb{C}$ one-dimensional, so $\overline{V^*} \cong \mathbb{C}$ and $\Lambda^\vee$ is a rank-2 lattice in it. The quotient is a one-dimensional complex torus, and it is isomorphic to E itself, recovering the identification $\mathrm{Pic}^0(E) \cong E$ of example 2.1.2. Under the character description, a degree-zero line bundle on $E = \mathbb{C}/\Lambda$ corresponds to a character $\chi: \Lambda \rightarrow U(1)$; writing $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$, such a character is determined by the two unit-modulus values $\chi(1)$ and $\chi(\tau)$, so

$$\mathrm{Pic}^0(E) \cong \mathrm{Hom}(\Lambda, U(1)) \cong U(1) \times U(1)$$

as real Lie groups, a real 2-torus carrying the complex structure that makes it isomorphic to E . The trivial character $\chi = 1$ is the trivial bundle $\mathcal{O}_E$, the origin of $\mathrm{Pic}^0(E)$.

The two examples bracket the general theory. The degree homomorphism of example 2.1.2 and the character description of example 2.1.6 are the $g = 1$ shadows of the two exact-sequence readings: the rightward reading through c_1 produces the discrete degree, and the leftward reading through $\exp_*$ produces the continuous group of characters. For general g the same split holds, packaged in the short exact sequence extracted from the exponential sequence.

The exponential sequence has done its structural work. By definition 2.1.4 the map c_1 is defined on all of $\mathrm{Pic}(X)$, and by proposition 2.1.5 its kernel is the complex torus $\mathrm{Pic}^0(X)$. The image of c_1 is a subgroup of $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$, and section 2.3 will identify it as the group of alternating integral forms arising from Hermitian data, the Néron-Severi group $\mathrm{NS}(X)$. The result is the exact sequence

$$0 \longrightarrow \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}(X) \xrightarrow{c_1} \mathrm{NS}(X) \longrightarrow 0 \quad (2.4)$$

which separates every line bundle into its continuous invariant, a point of the dual torus $\mathrm{Pic}^0(X)$, and its discrete invariant, an alternating integral form $c_1(L) \in \mathrm{NS}(X)$. The continuous invariant is the content of a character of Λ ; the discrete invariant is the topological type. This is the clean

separation the exponential sequence provides, and the two invariants are exactly what the next two sections compute: section 2.2 realizes both parts at once through the Appell-Humbert data (H, χ) , and section 2.3 pins down $\text{NS}(X)$ as Hermitian forms with integral imaginary part.

2.2 Factors of Automorphy and the Appell-Humbert Theorem

The exponential sequence of section 2.1 organizes $\text{Pic}(X)$ into a discrete part (the image of the first Chern class in $H^2(X, \mathbb{Z})$) and a continuous part ($\text{Pic}^0(X)$), but it does not yet exhibit a single line bundle concretely. To compute with a bundle one wants a formula: a rule for gluing the trivial bundle on V back into a bundle on the quotient $X = V/\Lambda$. Because V is contractible, every holomorphic line bundle on X pulls back to the trivial bundle on V , and the entire content of the bundle is packaged in how the two trivializations differ across the deck transformations by Λ . That packaging is a factor of automorphy. The main result of this section, the Appell-Humbert theorem, says that a factor of automorphy can always be normalized to a canonical form determined by a pair (H, χ) : a Hermitian form on V and a unitary semicharacter of Λ . This turns the classification of line bundles into a problem in linear algebra over the lattice, and it is the computational engine for everything that follows.

A holomorphic line bundle on X pulled back to V becomes $\pi^*L \cong V \times \mathbb{C}$, since V is contractible and Stein. The lattice Λ acts on $V \times \mathbb{C}$ by deck transformations covering its action on V , and any such action lifting translation must scale the fiber $\mathbb{C}$ linearly and holomorphically. This is exactly the data of a factor of automorphy.

Definition 2.2.1: Factor of Automorphy

A *factor of automorphy* for Λ acting on V is a holomorphic map

$$a: \Lambda \times V \rightarrow \mathbb{C}^*$$

satisfying the *cocycle condition*

$$a(\lambda + \mu, v) = a(\lambda, \mu + v) a(\mu, v) \quad \text{for all } \lambda, \mu \in \Lambda, v \in V$$

Given such an a , the group Λ acts on the trivial bundle $V \times \mathbb{C}$ by

$$\lambda \cdot (v, t) = (v + \lambda, a(\lambda, v)t)$$

and the quotient

$$L(a) := (V \times \mathbb{C})/\Lambda$$

is a holomorphic line bundle on $X = V/\Lambda$.

The cocycle condition is precisely what makes the formula $\lambda \cdot (v, t) = (v + \lambda, a(\lambda, v)t)$ a group action rather than a mere assignment: applying first μ and then λ sends (v, t) to $(v + \mu + \lambda, a(\lambda, \mu + v)a(\mu, v)t)$, and this must agree with applying $\lambda + \mu$ in one step, which sends (v, t) to $(v + \lambda + \mu, a(\lambda + \mu, v)t)$. Equating the fiber factors gives exactly $a(\lambda + \mu, v) = a(\lambda, \mu + v)a(\mu, v)$. Since Λ acts freely and properly discontinuously on V (hence on $V \times \mathbb{C}$), the quotient $L(a)$ is a manifold, and the projection $V \times \mathbb{C} \rightarrow V$ descends to a holomorphic line bundle $L(a) \rightarrow X$ whose fiber over $v + \Lambda$ is the $\mathbb{C}$ carried along v .

Two factors of automorphy can produce isomorphic bundles. An isomorphism of the trivialized bundles over V is a nowhere-vanishing holomorphic function $h: V \rightarrow \mathbb{C}^*$, and it intertwines the two Λ -actions exactly when the factors differ by the corresponding coboundary. This is the equivalence one must quotient by.

Definition 2.2.2: Equivalence of Factors and Coboundaries

A factor of automorphy a is a *coboundary* if there is a nowhere-vanishing holomorphic function $h: V \rightarrow \mathbb{C}^*$ with

$$a(\lambda, v) = \frac{h(v + \lambda)}{h(v)} \quad \text{for all } \lambda \in \Lambda, v \in V$$

Two factors a, a' are *equivalent* if a'/a is a coboundary. The factors of automorphy form an abelian group under pointwise multiplication (the cocycles Z^1), the coboundaries form a subgroup B^1 , and the quotient Z^1/B^1 is a group under multiplication.

The factor of automorphy is a group cocycle for Λ with coefficients in the multiplicative group $\mathcal{O}_V^*$ of nowhere-vanishing holomorphic functions on V , where Λ acts on $\mathcal{O}_V^*$ by translation. The cocycle condition of definition 2.2.1 is the 1-cocycle identity, and the coboundary relation of definition 2.2.2 is the image of the differential $\mathcal{O}_V^* \rightarrow Z^1$, $h \mapsto (\lambda \mapsto (v \mapsto h(v + \lambda)/h(v)))$. Thus $Z^1/B^1 = H^1(\Lambda, \mathcal{O}_V^*)$ in the sense of group cohomology ([5], group cohomology of a discrete group acting on a module). The construction $a \mapsto L(a)$ realizes this cohomology group as line bundles.

Proposition 2.2.3: Line Bundles as Factors of Automorphy

The assignment $a \mapsto L(a)$ induces an isomorphism of groups

$$Z^1/B^1 \xrightarrow{\sim} \text{Pic}(X), \quad [a] \mapsto [L(a)]$$

Every holomorphic line bundle on X arises from a factor of automorphy, and $L(a) \cong L(a')$ as line bundles if and only if a and a' are equivalent.

Proof:

Step 1: Every bundle arises from a factor. Let L be a holomorphic line bundle on X and $\pi: V \rightarrow X$ the projection. The pullback π^*L is a holomorphic line bundle on V . Since $V = \mathbb{C}^g$ is contractible and Stein, every holomorphic line bundle on V is trivial: $H^1(V, \mathcal{O}_V^*) = 0$, because the exponential sequence on V gives $H^1(V, \mathcal{O}_V^*) \rightarrow H^1(V, \mathcal{O}_V^*) \rightarrow H^2(V, \mathbb{Z})$ with both outer groups zero ([10], triviality of holomorphic line bundles on a contractible Stein manifold). Fix a trivialization $\varphi: \pi^*L \xrightarrow{\sim} V \times \mathbb{C}$.

The deck group Λ acts on π^*L (covering its translation action on V) with quotient L , since $\pi^*L = V \times_X L$. Transporting this action through φ gives an action on $V \times \mathbb{C}$ of the form $\lambda \cdot (v, t) = (v + \lambda, a(\lambda, v)t)$ for some holomorphic $a(\lambda, -): V \rightarrow \mathbb{C}^*$; that the action is by bundle automorphisms forces linearity in the fiber, and nowhere-vanishing since each λ acts invertibly. The associativity of the action is the cocycle condition. By construction $L(a) = (V \times \mathbb{C})/\Lambda \cong \pi^*L/\Lambda = L$.

Step 2: Equivalent factors give isomorphic bundles. Suppose $a' = a \cdot \delta h$ with $\delta h(\lambda, v) =$

$h(v + \lambda)/h(v)$ for a nowhere-vanishing holomorphic $h: V \rightarrow \mathbb{C}^*$. The map $V \times \mathbb{C} \rightarrow V \times \mathbb{C}$, $(v, t) \mapsto (v, h(v)t)$, is a holomorphic bundle automorphism over V , and it intertwines the a -action with the a' -action:

$$(v, t) \xrightarrow{\lambda} (v + \lambda, a(\lambda, v)t) \xrightarrow{h} (v + \lambda, h(v + \lambda)a(\lambda, v)t)$$

while

$$(v, t) \xrightarrow{h} (v, h(v)t) \xrightarrow{\lambda} (v + \lambda, a'(\lambda, v)h(v)t)$$

and $a'(\lambda, v)h(v) = a(\lambda, v)h(v + \lambda)$ by the coboundary relation. The two composites agree, so the automorphism descends to an isomorphism $L(a) \cong L(a')$.

Step 3: Isomorphic bundles give equivalent factors. Conversely, an isomorphism $L(a) \cong L(a')$ pulls back to a Λ -equivariant bundle isomorphism $V \times \mathbb{C} \rightarrow V \times \mathbb{C}$ over the identity of V , hence is multiplication by some nowhere-vanishing holomorphic $h: V \rightarrow \mathbb{C}^*$ in each fiber. Equivariance is precisely $a'(\lambda, v)h(v) = a(\lambda, v)h(v + \lambda)$, that is, $a'/a = \delta h$ is a coboundary. Steps 2 and 3 show $L(a) \cong L(a')$ iff $[a] = [a']$.

Step 4: The map is a homomorphism. If a, a' are factors, the tensor product $L(a) \otimes L(a')$ is $(V \times \mathbb{C})/\Lambda$ for the fiberwise-product action, whose factor is $a \cdot a'$; thus $L(aa') \cong L(a) \otimes L(a')$. The trivial bundle corresponds to the constant factor $a \equiv 1$. Therefore $[a] \mapsto [L(a)]$ is a group homomorphism, and by Steps 1 and 3 it is a bijection, completing the proof.

This proposition reduces the classification of line bundles to the classification of factors of automorphy up to coboundary. The identity $Z^1/B^1 \cong \text{Pic}(X)$ is not yet a computation: the space of holomorphic factors is enormous. The Appell-Humbert theorem accomplishes the reduction by showing that each equivalence class contains one, and only one, factor of a rigidly prescribed canonical shape. The shape involves a Hermitian form and a unitary semicharacter, which are introduced next.

The semicharacter is the discrete piece of the data. It records how the bundle winds around the lattice directions, and its defect from being an ordinary homomorphism is governed by the alternating form $E = \text{Im}H$ evaluated on lattice vectors. Recall from section 1.3 the conventions on Hermitian forms fixed for this book: $H: V \times V \rightarrow \mathbb{C}$ is linear in the first variable and conjugate-linear in the second, with $H(w, v) = \overline{H(v, w)}$, and $E := \text{Im}H$ is an alternating $\mathbb{R}$ -bilinear form satisfying $E(iv, iw) = E(v, w)$.

Definition 2.2.4: Semicharacter

Let H be a Hermitian form on V with $E = \text{Im}H$ integral on Λ , that is, $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$. A *semicharacter* for E is a map

$$\chi: \Lambda \rightarrow U(1) = \{z \in \mathbb{C} \mid |z| = 1\}$$

satisfying

$$\chi(\lambda + \mu) = \chi(\lambda) \chi(\mu) e^{i\pi E(\lambda, \mu)} \quad \text{for all } \lambda, \mu \in \Lambda$$

Since $E(\lambda, \mu) \in \mathbb{Z}$, the twisting factor is $e^{i\pi E(\lambda, \mu)} = (-1)^{E(\lambda, \mu)}$, so equivalently $\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu)(-1)^{E(\lambda, \mu)}$. A pair (H, χ) with H Hermitian, $E = \text{Im}H$ integral on Λ , and χ a semicharacter for E is called *Appell-Humbert data*.

A semicharacter is a homomorphism twisted by the sign cocycle $(\lambda, \mu) \mapsto (-1)^{E(\lambda, \mu)}$. When E vanishes on the lattice (in particular when $H = 0$) the defect disappears and χ is an ordinary group homomorphism $\Lambda \rightarrow U(1)$. In general the identity $\chi(\lambda + \mu)/[\chi(\lambda)\chi(\mu)] = (-1)^{E(\lambda, \mu)}$ measures how far χ is from being a character, and the discrepancy is symmetric in the sense that swapping λ, μ multiplies by $(-1)^{E(\mu, \lambda) - E(\lambda, \mu)} = (-1)^{-2E(\lambda, \mu)} = 1$, consistent with the commutativity of Λ . That E is integral on Λ is exactly the condition needed for the sign to be well-defined; it is the integrality that section 2.3 identifies with the first Chern class lying in $H^2(X, \mathbb{Z})$.

Given Appell-Humbert data (H, χ) , one now writes down the canonical factor and checks that it is a factor of automorphy. The Hermitian form supplies the continuous, curvature-carrying part of the factor, and the semicharacter supplies the discrete twisting; the two are locked together precisely by the semicharacter identity.

Definition 2.2.5: Appell-Humbert Bundle

Let (H, χ) be Appell-Humbert data. The *canonical factor of automorphy* associated to (H, χ) is

$$a_{(H, \chi)}(\lambda, v) = \chi(\lambda) \exp\left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right)$$

and the *Appell-Humbert bundle* is the line bundle

$$L(H, \chi) := L(a_{(H, \chi)}) = (V \times \mathbb{C})/\Lambda, \quad \lambda \cdot (v, t) = (v + \lambda, a_{(H, \chi)}(\lambda, v) t)$$

where $H(v, \lambda)$ is linear in v and $H(\lambda, \lambda) \in \mathbb{R}$ because H is Hermitian.

Proposition 2.2.6: The Canonical Factor is a Factor of Automorphy

For any Appell-Humbert data (H, χ) , the map $a_{(H, \chi)}$ is a factor of automorphy. Consequently $L(H, \chi)$ is a well-defined holomorphic line bundle on X .

Proof:

Write $a = a_{(H, \chi)}$ and $f(\lambda, v) = \exp(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda))$, so $a(\lambda, v) = \chi(\lambda) f(\lambda, v)$. For each fixed λ , the function $v \mapsto a(\lambda, v)$ is holomorphic and nowhere-vanishing: $H(v, \lambda)$ is $\mathbb{C}$ -linear in v , so $f(\lambda, -)$ is the exponential of a holomorphic function, and $\chi(\lambda) \in U(1)$ is a nonzero constant. It remains to verify the cocycle condition

$$a(\lambda + \mu, v) = a(\lambda, \mu + v) a(\mu, v)$$

Step 1: Expand the exponential part. Using bilinearity of H (linear in the first slot, conjugate-linear in the second) compute the exponent of the right-hand product $f(\lambda, \mu + v) f(\mu, v)$:

$$\pi H(\mu + v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda) + \pi H(v, \mu) + \frac{\pi}{2} H(\mu, \mu)$$

Since $H(\mu + v, \lambda) = H(\mu, \lambda) + H(v, \lambda)$, this exponent equals

$$\pi H(v, \lambda) + \pi H(v, \mu) + \pi H(\mu, \lambda) + \frac{\pi}{2} (H(\lambda, \lambda) + H(\mu, \mu))$$

Now the exponent of the left-hand factor $f(\lambda + \mu, v)$ is

$$\pi H(v, \lambda + \mu) + \frac{\pi}{2} H(\lambda + \mu, \lambda + \mu) = \pi H(v, \lambda) + \pi H(v, \mu) + \frac{\pi}{2} (H(\lambda, \lambda) + H(\lambda, \mu) + H(\mu, \lambda) + H(\mu, \mu))$$

using $H(v, \lambda + \mu) = H(v, \lambda) + H(v, \mu)$ (additivity in the second slot, which holds for a conjugate-linear argument since conjugation affects only scalar multiplication, not addition) and the expansion of $H(\lambda + \mu, \lambda + \mu)$.

Step 2: Compare the two exponents. Subtracting the exponent of $f(\lambda + \mu, v)$ from that of $f(\lambda, \mu + v)f(\mu, v)$, the terms $\pi H(v, \lambda) + \pi H(v, \mu)$ and $\frac{\pi}{2} (H(\lambda, \lambda) + H(\mu, \mu))$ cancel, leaving

$$\pi H(\mu, \lambda) - \frac{\pi}{2} (H(\lambda, \mu) + H(\mu, \lambda)) = \frac{\pi}{2} (H(\mu, \lambda) - H(\lambda, \mu))$$

By the Hermitian symmetry $H(\lambda, \mu) = \overline{H(\mu, \lambda)}$, the difference $H(\mu, \lambda) - H(\lambda, \mu) = H(\mu, \lambda) - \overline{H(\mu, \lambda)} = 2i \operatorname{Im} H(\mu, \lambda) = 2i E(\mu, \lambda)$. Therefore

$$\frac{f(\lambda, \mu + v) f(\mu, v)}{f(\lambda + \mu, v)} = \exp(i\pi E(\mu, \lambda))$$

Step 3: Absorb the sign into the semicharacter. Multiply through by the semicharacter values. The right-hand side of the cocycle is

$$a(\lambda, \mu + v) a(\mu, v) = \chi(\lambda) \chi(\mu) f(\lambda, \mu + v) f(\mu, v) = \chi(\lambda) \chi(\mu) e^{i\pi E(\mu, \lambda)} f(\lambda + \mu, v)$$

Because E is integral on Λ and alternating, $E(\mu, \lambda) = -E(\lambda, \mu)$ is an integer, so $e^{i\pi E(\mu, \lambda)} = (-1)^{E(\mu, \lambda)} = (-1)^{E(\lambda, \mu)} = e^{i\pi E(\lambda, \mu)}$. The semicharacter identity of definition 2.2.4 then gives

$$\chi(\lambda) \chi(\mu) e^{i\pi E(\lambda, \mu)} = \chi(\lambda + \mu)$$

so that $a(\lambda, \mu + v) a(\mu, v) = \chi(\lambda + \mu) f(\lambda + \mu, v) = a(\lambda + \mu, v)$, which is the cocycle condition. Hence a is a factor of automorphy, and $L(H, \chi)$ is a holomorphic line bundle on X , as we sought to show.

The proof shows exactly where the semicharacter condition comes from: it is forced. The exponential part f satisfies the cocycle condition only up to the sign $e^{i\pi E(\mu, \lambda)}$, and the twisting law $\chi(\lambda + \mu) = \chi(\lambda) \chi(\mu) (-1)^{E(\lambda, \mu)}$ is designed to cancel precisely that sign. This is why one cannot use an ordinary character: the Hermitian part alone fails the cocycle identity by a sign, and only a semicharacter repairs it. The integrality $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$ is what makes $(-1)^{E(\lambda, \mu)}$ meaningful and makes the two signs $e^{\pm i\pi E(\lambda, \mu)}$ coincide on the lattice.

Before the general theorem, the canonical factor is worth seeing on the elliptic curve, where every quantity is a single complex number.

Example 2.2.7: The Canonical Factor for an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$. Take the Hermitian form

$$H(v, w) = \frac{v \bar{w}}{\operatorname{Im} \tau}$$

which is positive definite, with imaginary part $E(v, w) = \operatorname{Im}(v \bar{w} / \operatorname{Im} \tau)$. On the ordered basis $(1, \tau)$

of Λ one computes $E(1, \tau) = \text{Im}(\bar{\tau}/\text{Im}\tau) = -1$ and $E(\tau, 1) = +1$, while $E(1, 1) = E(\tau, \tau) = 0$. Thus E takes integer values on Λ (it is ± 1 on the basis and extends $\mathbb{Z}$ -bilinearly), so H is integral. A semicharacter χ for E is determined by its values $\chi(1), \chi(\tau) \in U(1)$ on the basis together with the twisting rule extending it to all of Λ . Taking $\chi(1) = \chi(\tau) = 1$ and extending forces $\chi(m+n\tau) = (-1)^{mn}$, which satisfies the semicharacter identity: for $\lambda = m_1 + n_1\tau$ and $\mu = m_2 + n_2\tau$ one has $E(\lambda, \mu) = m_2n_1 - m_1n_2$, and $\chi(\lambda + \mu)/[\chi(\lambda)\chi(\mu)] = (-1)^{(m_1+m_2)(n_1+n_2) - m_1n_1 - m_2n_2} = (-1)^{m_1n_2 + m_2n_1} = (-1)^{E(\lambda, \mu)}$, the last equality because $m_1n_2 + m_2n_1$ and $m_2n_1 - m_1n_2$ differ by $2m_1n_2$. The canonical factor becomes, for $\lambda \in \Lambda$ and $v \in \mathbb{C}$,

$$a_{(H, \chi)}(\lambda, v) = \chi(\lambda) \exp\left(\frac{\pi v \bar{\lambda}}{\text{Im}\tau} + \frac{\pi}{2} \frac{|\lambda|^2}{\text{Im}\tau}\right)$$

This is the automorphy factor whose sections are the classical Jacobi theta functions, recovered in section 2.4; the exponential quadratic in λ is the source of the Gaussian factor in the theta series.

Everything is now in place for the central theorem of the chapter. It asserts that the map $(H, \chi) \mapsto L(H, \chi)$ is a bijection onto the full Picard group: every line bundle is canonically one of these, and no two distinct data give the same bundle. The proof is a normalization argument. Given an arbitrary factor of automorphy, one first adjusts it by a coboundary so that its class is represented by a factor whose logarithm is affine in v , then reads off (H, χ) from the coefficients. The technical heart is showing that every holomorphic factor is equivalent to one of this affine type, which rests on the vanishing $H^1(V, \mathcal{O}_V) = 0$ from the Stein property of V together with the Hodge-theoretic description of $\text{Pic}^0(X)$ established in section 2.1.

Theorem 2.2.8: Appell-Humbert Theorem

Let $X = V/\Lambda$ be a complex torus. The assignment

$$(H, \chi) \longmapsto L(H, \chi)$$

is a bijection from the set of Appell-Humbert data (pairs (H, χ) with H Hermitian, $E = \text{Im}H$ integral on Λ , and χ a semicharacter for E) onto $\text{Pic}(X)$. Every holomorphic line bundle on X is isomorphic to a unique $L(H, \chi)$.

Proof:

By proposition 2.2.3, $\text{Pic}(X) = Z^1/B^1$, and by proposition 2.2.6 each $L(H, \chi)$ is a line bundle. The proof establishes that every class in Z^1/B^1 contains exactly one canonical factor $a_{(H, \chi)}$.

Step 1: Passing to the logarithm. Let a be any factor of automorphy. Because each $a(\lambda, -): V \rightarrow \mathbb{C}^*$ is holomorphic on the simply connected V , it has a holomorphic logarithm: there exist holomorphic $g_\lambda: V \rightarrow \mathbb{C}$ with $a(\lambda, v) = \exp(2\pi i g_\lambda(v))$, unique up to an additive integer constant. The cocycle condition $a(\lambda + \mu, v) = a(\lambda, \mu + v)a(\mu, v)$ becomes, in the logarithm,

$$g_{\lambda+\mu}(v) \equiv g_\lambda(\mu + v) + g_\mu(v) \pmod{\mathbb{Z}}$$

This is a 1-cocycle for Λ with values in the holomorphic functions on V , twisted by $\mathbb{Z}$.

Step 2: Normalizing to an affine cocycle. The goal is to replace a by an equivalent factor whose logarithm $g_\lambda(v)$ is affine in v , that is, of the form $g_\lambda(v) = \ell(\lambda, v) + c(\lambda)$ with $\ell(\lambda, -): V \rightarrow \mathbb{C}$ a

$\mathbb{C}$ -linear functional depending on λ and $c(\lambda)$ a constant. Modifying a by a coboundary δh with $h = \exp(2\pi i k)$, $k: V \rightarrow \mathbb{C}$ holomorphic, changes each g_λ by $k(v + \lambda) - k(v)$; since V is Stein and simply connected, every holomorphic $k: V \rightarrow \mathbb{C}$ is admissible.

The reduction is carried out on the second-order derivatives of g_λ . Each g_λ is holomorphic on $V = \mathbb{C}^g$, so its holomorphic Hessian

$$B_\lambda(v) := \left(\frac{\partial^2 g_\lambda}{\partial v_j \partial v_k}(v) \right)_{1 \leq j, k \leq g}$$

is a holomorphic map $V \rightarrow \text{Sym}^2(\mathbb{C}^g)^\vee$, symmetric in (j, k) . Differentiating the Step 1 relation twice in v (the additive $\mathbb{Z}$ -constant and the first-order translation terms drop) gives

$$B_{\lambda+\mu}(v) = B_\lambda(v + \mu) + B_\mu(v)$$

so $\lambda \mapsto B_\lambda$ is a 1-cocycle for Λ valued in the sheaf $\mathcal{H} := \text{Hess}(\mathcal{O}_V)$ of holomorphic symmetric fields that arise as Hessians (equivalently the integrable ones, $\partial_l \gamma_{jk} = \partial_j \gamma_{lk}$). Two vanishings on the Stein contractible V combine to make this cocycle a coboundary. First, the operator Hess is surjective on global sections: it sits in the exact sheaf sequence $0 \rightarrow \mathcal{A} \rightarrow \mathcal{O}_V \xrightarrow{\text{Hess}} \mathcal{H} \rightarrow 0$, where $\mathcal{A}$ is the finite-dimensional space of affine functions (the kernel of Hess), and since $H^1(V, \mathcal{A}) = 0$ ($\mathcal{A}$ is a trivial finite-rank bundle and V is contractible) the global-section sequence $\mathcal{O}_V(V) \rightarrow \mathcal{H}(V) \rightarrow H^1(V, \mathcal{A}) = 0$ shows every global integrable field, in particular each B_λ , is the Hessian of a global holomorphic function. Second, the Λ -cocycle $\lambda \mapsto B_\lambda$ is a Λ -coboundary: because $V = \mathbb{C}^g$ is holomorphically contractible with $H^1(V, \mathcal{O}_V) = 0$ ([10]), the group cohomology $H^1(\Lambda, \mathcal{O}_V(V)) = 0$ for the translation action ([5]), and its image under the Hess-surjection kills the class of B_λ ; concretely, solving the coboundary equation in $\mathcal{O}_V(V)$ and applying Hess produces a global holomorphic $k: V \rightarrow \mathbb{C}$ with $\beta := \text{Hess}(k)$ satisfying $B_\lambda(v) = \beta(v + \lambda) - \beta(v)$ for every λ . Replacing g_λ by $g_\lambda - (k(\cdot + \lambda) - k)$ leaves a factor equivalent to a whose logarithm has Hessian $B_\lambda - (\beta(\cdot + \lambda) - \beta) = 0$ for every λ . A holomorphic function on V with identically vanishing Hessian is affine, so after this coboundary adjustment the representative has

$$g_\lambda(v) = \ell(\lambda, v) + c(\lambda)$$

with $c(\lambda) \in \mathbb{C}$ and $\ell(\lambda, -) = dg_\lambda$ a functional. Because g_λ is holomorphic, its differential $\ell(\lambda, -)$ is $\mathbb{C}$ -linear, so $\ell(\lambda, -) \in V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$. This $\mathbb{C}$ -linearity is used decisively in Step 4.

Step 3: The cocycle condition in affine form. Substituting $g_\lambda(v) = \ell(\lambda, v) + c(\lambda)$ into the Step 1 relation and using linearity of $\ell(\lambda, -)$,

$$\ell(\lambda + \mu, v) + c(\lambda + \mu) \equiv \ell(\lambda, v) + \ell(\lambda, \mu) + c(\lambda) + \ell(\mu, v) + c(\mu) \pmod{\mathbb{Z}}$$

Comparing the parts that are linear in v gives

$$\ell(\lambda + \mu, v) = \ell(\lambda, v) + \ell(\mu, v)$$

so $\lambda \mapsto \ell(\lambda, -)$ is additive in λ : it is a homomorphism $\Lambda \rightarrow \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$. Comparing the v -independent parts gives

$$c(\lambda + \mu) \equiv c(\lambda) + c(\mu) + \ell(\lambda, \mu) \pmod{\mathbb{Z}} \quad (*)$$

Step 4: Extracting the Hermitian form. Extend $\lambda \mapsto \ell(\lambda, -)$ from Λ to all of $V = \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$ by $\mathbb{R}$ -linearity in the first slot, writing $\ell'(u, w)$ for the value of the extended functional with first argument $u \in V$ evaluated at $w \in V$; thus $\ell'(u, -)$ is $\mathbb{C}$ -linear in w for each u , and ℓ' is $\mathbb{R}$ -bilinear. Define an $\mathbb{R}$ -bilinear alternating form on V by

$$E(v, w) := \ell'(w, v) - \ell'(v, w)$$

Antisymmetrizing Step 3's relation $(*)$ (writing $(*)$ with λ, μ swapped and subtracting) eliminates c , yielding $\ell(\mu, \lambda) - \ell(\lambda, \mu) \equiv 0 \pmod{\mathbb{Z}}$; since ℓ' extends ℓ , this reads

$$E(\lambda, \mu) \in \mathbb{Z} \quad \text{for } \lambda, \mu \in \Lambda$$

so E is an integer-valued alternating form on Λ , hence $\mathbb{R}$ -valued on all of V by $\mathbb{R}$ -bilinear extension.

It remains to see E is the imaginary part of a Hermitian form. Set

$$H(v, w) := E(iv, w) + iE(v, w)$$

following the correspondence fixed in section 1.3 ([4] for the general Hodge-structure statement). By the criterion recalled there, H is Hermitian, that is, $\mathbb{C}$ -linear in v with $H(w, v) = \overline{H(v, w)}$, if and only if E satisfies the compatibility $E(iv, iw) = E(v, w)$. This compatibility is where the holomorphicity of a enters, through the $\mathbb{C}$ -linearity of $\ell(\lambda, -)$ established in Step 2. Compute the defect $D(v, w) := E(iv, iw) - E(v, w)$ directly from the definition of E and the $\mathbb{C}$ -linearity of ℓ' in its second slot ($\ell'(u, iw) = i\ell'(u, w)$):

$$D(v, w) = [\ell'(iw, iv) - \ell'(iv, iw)] - [\ell'(w, v) - \ell'(v, w)]$$

Using $\ell'(iv, iw) = i\ell'(iv, w)$ and $\ell'(iw, iv) = i\ell'(iw, v)$, the bracketed difference $\ell'(iw, iv) - \ell'(iv, iw) = i(\ell'(iw, v) - \ell'(iv, w))$, so D is $\mathbb{C}$ -bilinear in the complexified arguments, that is, a holomorphic alternating 2-form: $D \in \bigwedge^2 V^*$, of pure Hodge type $(2, 0)$. On the other hand E is $\mathbb{R}$ -valued, so $D = E(i \cdot, i \cdot) - E$ is $\mathbb{R}$ -valued as well. A form that is simultaneously $\mathbb{C}$ -bilinear (type $(2, 0)$) and $\mathbb{R}$ -valued must vanish: its complex conjugate is both equal to itself (reality) and of type $(0, 2)$ (conjugating a $(2, 0)$ -form), and $(2, 0) \cap (0, 2) = 0$. Hence $D \equiv 0$, giving $E(iv, iw) = E(v, w)$. Thus H is a Hermitian form with $\text{Im}H = E$: indeed $\text{Im}H(v, w) = E(v, w)$ directly from the definition, and E is integral on Λ .

Step 5: Matching the linear part to H . The canonical factor $a_{(H,1)}$ (semicharacter $\equiv 1$) has logarithm whose v -linear part is $\frac{1}{2\pi i} \pi H(v, \lambda) = \frac{1}{2i} H(v, \lambda)$, a $\mathbb{C}$ -linear functional of v ; write $L(\lambda, v) := \frac{1}{2i} H(v, \lambda)$. Compare it with $\ell(\lambda, v)$, the $\mathbb{C}$ -linear part produced in Step 2. Both are $\mathbb{C}$ -linear in v , and both have the *same* alternating part under interchange of λ and v : indeed

$$L(\lambda, v) - L(v, \lambda) = \frac{1}{2i} (H(v, \lambda) - H(\lambda, v)) = \frac{1}{2i} (H(v, \lambda) - \overline{H(v, \lambda)}) = \text{Im}H(v, \lambda) = E(v, \lambda)$$

while $\ell(\lambda, v) - \ell(v, \lambda) = \ell'(\lambda, v) - \ell'(v, \lambda) = E(v, \lambda)$ by the definition of E in Step 4. Their difference $\ell(\lambda, v) - L(\lambda, v)$ is therefore symmetric in (λ, v) and $\mathbb{C}$ -linear in each, hence the restriction to $v \in \Lambda$ of a symmetric $\mathbb{C}$ -bilinear form $Q(u, v)$ on V . Setting $k(v) := \frac{1}{2} Q(v, v)$, a holomorphic quadratic, the coboundary δk has v -linear part $k(v + \lambda) - k(v) = Q(\lambda, v) + \frac{1}{2} Q(\lambda, \lambda)$, whose linear-in- v term is exactly $\ell(\lambda, v) - L(\lambda, v)$. Subtracting this coboundary replaces ℓ by L :

after one further coboundary adjustment the normalized logarithm has v -linear part $\frac{1}{2i}H(v, \lambda)$, so a and $a_{(H,1)}$ have identical v -dependence up to a coboundary, and their ratio is a factor of automorphy $\lambda \mapsto \chi(\lambda)$ with $\chi(\lambda) \in \mathbb{C}^*$ independent of v .

Step 6: Extracting the semicharacter. The residual $\chi = a/a_{(H,1)}$ inherits its cocycle law from (*). Its cocycle condition reads $\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu)e^{2\pi i \rho(\lambda, \mu)}$, where $\rho(\lambda, \mu)$ is the discrepancy between the constant part of the logarithm of a and that of $a_{(H,1)}$. The canonical factor is a genuine cocycle (proposition 2.2.6), so its constant part is exact; the constant part of a obeys (*), that is $c(\lambda + \mu) - c(\lambda) - c(\mu) \equiv \ell(\lambda, \mu) \pmod{\mathbb{Z}}$. Only the alternating part of $\ell(\lambda, \mu)$ survives in ρ (the symmetric part was absorbed into H and $a_{(H,1)}$ in Step 5), and that alternating part is

$$\frac{1}{2}(\ell(\lambda, \mu) - \ell(\mu, \lambda)) = \frac{1}{2}(\ell'(\lambda, \mu) - \ell'(\mu, \lambda)) = -\frac{1}{2}E(\lambda, \mu)$$

using $E(\lambda, \mu) = \ell'(\mu, \lambda) - \ell'(\lambda, \mu)$ from Step 4. Since $E(\lambda, \mu) \in \mathbb{Z}$, one has $e^{2\pi i(-\frac{1}{2}E(\lambda, \mu))} = e^{-i\pi E(\lambda, \mu)} = (-1)^{-E(\lambda, \mu)} = (-1)^{E(\lambda, \mu)} = e^{i\pi E(\lambda, \mu)}$, so

$$\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu)e^{i\pi E(\lambda, \mu)}$$

and χ is a semicharacter for E . Finally, $|\chi(\lambda)| = 1$: writing $\chi(\lambda) = |\chi(\lambda)|e^{i \arg \chi(\lambda)}$, the modulus $\lambda \mapsto \log |\chi(\lambda)|$ is an ordinary $\mathbb{R}$ -valued homomorphism on Λ (the sign twist $e^{i\pi E}$ is unimodular, so it drops out of the moduli), hence extends to an $\mathbb{R}$ -linear functional $u: V \rightarrow \mathbb{R}$. To remove the modulus by a coboundary one must use a *holomorphic* gauge, as definition 2.2.2 demands: let $\ell \in V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ be the unique $\mathbb{C}$ -linear functional with $\text{Re} \ell = u$, which exists because $l \mapsto \text{Re}(\ell(\cdot))$ is an $\mathbb{R}$ -linear isomorphism $V^* \xrightarrow{\sim} \text{Hom}_{\mathbb{R}}(V, \mathbb{R})$. Then $h(v) = e^{\ell(v)}$ is a nowhere-vanishing holomorphic function on V , its coboundary is $\delta h(\lambda, v) = h(v + \lambda)/h(v) = e^{\ell(\lambda)}$, and its modulus is $|e^{\ell(\lambda)}| = e^{\text{Re} \ell(\lambda)} = e^{u(\lambda)} = |\chi(\lambda)|$. Dividing the factor by this coboundary replaces χ by $\chi/|\chi|$ without changing the bundle and leaves $\chi: \Lambda \rightarrow U(1)$. Thus a is equivalent to $a_{(H, \chi)}$, proving *existence*: every line bundle is some $L(H, \chi)$.

Step 7: Uniqueness. Suppose $L(H, \chi) \cong L(H', \chi')$. By proposition 2.2.3 the factors $a_{(H, \chi)}$ and $a_{(H', \chi')}$ differ by a coboundary δh , $h = e^{2\pi i k}$. Taking logarithms, $\pi H(v, \lambda) + \frac{\pi}{2}H(\lambda, \lambda) - \pi H'(v, \lambda) - \frac{\pi}{2}H'(\lambda, \lambda) + \log \frac{\chi(\lambda)}{\chi'(\lambda)} \equiv 2\pi i(k(v + \lambda) - k(v)) \pmod{2\pi i\mathbb{Z}}$. The left side is affine in v (linear part $\pi(H - H')(v, \lambda)$), so $k(v + \lambda) - k(v)$ is affine in v for every λ . Applying the holomorphic Hessian $k'' := (\partial^2 k / \partial v_j \partial v_k)$, affineness of the lattice-difference means $k''(v + \lambda) = k''(v)$ for all $\lambda \in \Lambda$, so k'' is Λ -periodic and descends to a holomorphic $\text{Sym}^2 V^*$ -valued function on the compact torus X ; a holomorphic function on the compact connected X is constant, so k'' is a constant symmetric form and k is a polynomial of degree ≤ 2 . The v -linear part of $2\pi i \delta k$ is then a $\mathbb{C}$ -bilinear form $2\pi i Q(\lambda, v)$ *symmetric* in (λ, v) , whereas the v -linear part $\pi(H - H')(v, \lambda)$ of the left side is $\mathbb{C}$ -linear in v but conjugate-linear in λ . A $\mathbb{C}$ -bilinear form and a form conjugate-linear in λ agree for all λ only if both vanish, so $H - H' = 0$, that is, $H = H'$. With $H = H'$ the linear and quadratic parts cancel entirely, leaving $\chi(\lambda)/\chi'(\lambda) = e^{2\pi i(k(v + \lambda) - k(v))}$ independent of v and of modulus 1. Since $H = H'$, the difference $k(v + \lambda) - k(v)$ is v -independent, so k is affine, say $k(v) = \psi(v) + c$ with $\psi \in V^*$; then $\chi(\lambda)/\chi'(\lambda) = e^{2\pi i \psi(\lambda)}$. Both χ and χ' are semicharacters for the same E , so their ratio $\lambda \mapsto e^{2\pi i \psi(\lambda)}$ is a genuine character $\Lambda \rightarrow U(1)$; but a character valued in $U(1)$ of the form $e^{2\pi i \psi(\lambda)}$ with ψ a $\mathbb{C}$ -linear functional forces $\psi(\lambda) \in \mathbb{R}$ for all $\lambda \in \Lambda$, and a $\mathbb{C}$ -linear functional real-valued on the full lattice Λ (hence on $V = \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$) vanishes identically. Thus the ratio is trivial and $\chi = \chi'$. Hence the data (H, χ) is unique, completing the proof.

The Appell-Humbert theorem is the structural pillar of the analytic theory. It says that the infinite-dimensional-looking group $\text{Pic}(X)$ is parametrized by a pair of finite pieces of linear-algebraic data: a Hermitian form, which is a matrix of complex numbers, and a semicharacter, which is a finite amount of $U(1)$ -data on lattice generators. Every question about line bundles on X can be translated into this language: positivity of L becomes positivity of H (the polarization condition of section 3.1), the first Chern class becomes $\text{Im}H$ (section 2.3), and the sections of L become theta functions with the quasi-periodicity dictated by $a_{(H,\chi)}$ (section 2.4). The Hermitian form is the “continuous” invariant carrying the topological type, and the semicharacter is the “arithmetic” choice of a specific bundle within a fixed topological type.

Two immediate consequences pin down the pieces of $\text{Pic}(X)$ predicted by the exponential sequence. When $H = 0$, the semicharacter condition degenerates to the character condition $\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu)$, so $\chi \in \text{Hom}(\Lambda, U(1))$; these are exactly the topologically trivial bundles. When $H \neq 0$, the class $\text{Im}H$ is the first Chern class, and it ranges over the integral Hermitian forms. The theorem thus splits $\text{Pic}(X)$ into a continuous torus of characters and a discrete lattice of Hermitian forms.

Corollary 2.2.9: The Two Pieces of the Picard Group

Under the Appell-Humbert parametrization:

1. The bundles $L(0, \chi)$ with $\chi \in \text{Hom}(\Lambda, U(1))$ are exactly the elements of $\text{Pic}^0(X)$, and

$$\text{Pic}^0(X) \cong \text{Hom}(\Lambda, U(1))$$

2. There is an exact sequence

$$0 \rightarrow \text{Pic}^0(X) \rightarrow \text{Pic}(X) \xrightarrow{c_1} \text{NS}(X) \rightarrow 0, \quad c_1(L(H, \chi)) = H$$

where the quotient $\text{NS}(X)$ is the group of Hermitian forms H with $\text{Im}H$ integral on Λ that arise as first Chern classes, given its formal treatment as the Néron-Severi group in section 2.3.

Proof:

1. When $H = 0$ the factor $a_{(0,\chi)}(\lambda, v) = \chi(\lambda)$ is a constant unitary factor, whose bundle $L(0, \chi)$ is flat, hence topologically trivial and in $\text{Pic}^0(X)$. Conversely, by proposition 2.1.5 the group $\text{Pic}^0(X)$ is the image of $H^1(X, \mathcal{O}_X)$ in $\text{Pic}(X)$; by exactness of the exponential sequence theorem 2.1.3 this image is the kernel of c_1 , so $\text{Pic}^0(X)$ consists exactly of the classes with trivial first Chern class. By the theorem these are the $L(H, \chi)$ with $\text{Im}H = 0$, and a Hermitian form with $\text{Im}H = 0$ is 0. The map $\chi \mapsto L(0, \chi)$ is a homomorphism $\text{Hom}(\Lambda, U(1)) \rightarrow \text{Pic}^0(X)$ by proposition 2.2.10, injective and surjective by the theorem’s bijectivity restricted to $H = 0$.
2. The map c_1 sends $L(H, \chi)$ to H ; it is well-defined and surjective onto $\text{NS}(X)$ because every integral Hermitian H admits at least one semicharacter: fixing a $\mathbb{Z}$ -basis of Λ and any integral bilinear B with $B - B^t = E$ (for instance the strictly-upper-triangular part of E in that basis), the formula $\chi(\lambda) = (-1)^{B(\lambda, \lambda)}$ satisfies the semicharacter identity, since $B(\lambda + \mu, \lambda + \mu) - B(\lambda, \lambda) - B(\mu, \mu) = B(\lambda, \mu) + B(\mu, \lambda) \equiv B(\lambda, \mu) - B(\mu, \lambda) = E(\lambda, \mu) \pmod{2}$. Its kernel is the set of data with $H = 0$, which is $\text{Pic}^0(X)$ by part (1). Exactness

follows, completing the proof.

The group structure on Appell-Humbert data is the last piece needed to make the parametrization an isomorphism of groups rather than of sets. Tensoring line bundles multiplies factors of automorphy, and the canonical factors multiply by adding Hermitian forms and multiplying semicharacters, so the bijection of the theorem is a group isomorphism.

Proposition 2.2.10: Group Law on Appell-Humbert Data

The set of Appell-Humbert data is a group under

$$(H_1, \chi_1) \cdot (H_2, \chi_2) = (H_1 + H_2, \chi_1 \chi_2)$$

and $(H, \chi) \mapsto L(H, \chi)$ is an isomorphism of this group onto $\text{Pic}(X)$ (definition 2.1.1). Under it, tensor product of bundles corresponds to the sum $(H_1 + H_2, \chi_1 \chi_2)$, the trivial bundle to $(0, 1)$, and the inverse $L(H, \chi)^{-1}$ to $(-H, \chi^{-1})$.

Proof:

First, $(H_1 + H_2, \chi_1 \chi_2)$ is again valid data: $H_1 + H_2$ is Hermitian with $\text{Im}(H_1 + H_2) = E_1 + E_2$ integral on Λ , and $\chi_1 \chi_2$ is a semicharacter for $E_1 + E_2$ because

$$(\chi_1 \chi_2)(\lambda + \mu) = \chi_1(\lambda) \chi_1(\mu) e^{i\pi E_1(\lambda, \mu)} \chi_2(\lambda) \chi_2(\mu) e^{i\pi E_2(\lambda, \mu)} = (\chi_1 \chi_2)(\lambda) (\chi_1 \chi_2)(\mu) e^{i\pi(E_1 + E_2)(\lambda, \mu)}$$

The canonical factors multiply directly:

$$a_{(H_1, \chi_1)}(\lambda, v) a_{(H_2, \chi_2)}(\lambda, v) = \chi_1(\lambda) \chi_2(\lambda) \exp\left(\pi(H_1 + H_2)(v, \lambda) + \frac{\pi}{2}(H_1 + H_2)(\lambda, \lambda)\right) = a_{(H_1 + H_2, \chi_1 \chi_2)}(\lambda, v)$$

Since tensor product of line bundles multiplies factors of automorphy (proposition 2.2.3, Step 4), $L(H_1, \chi_1) \otimes L(H_2, \chi_2) \cong L(H_1 + H_2, \chi_1 \chi_2)$. The identity element is the trivial data $(0, 1)$, whose factor is $\equiv 1$ (the trivial bundle $\mathcal{O}_X$), and inverses are $(-H, \chi^{-1})$ by the same multiplication. The map is therefore a group homomorphism, and it is a bijection by theorem 2.2.8, hence a group isomorphism onto $\text{Pic}(X)$, as we sought to show.

The isomorphism identifies the flat bundles $L(0, \chi)$ with $\text{Hom}(\Lambda, U(1))$ as a subgroup. Here $\chi: \Lambda \rightarrow U(1)$ is a homomorphism of *abstract groups* from the lattice to the circle, not a homomorphism of tori; the group $\text{Hom}(\Lambda, U(1))$ of all such characters is itself a real torus of dimension $2g$, the dual real torus of Λ . The lattice-level maps recorded by $\text{Hom}(\Lambda, U(1))$ are the same kind of Λ -datum that underlies a homomorphism of complex tori (definition 1.1.6), where a holomorphic homomorphism $X \rightarrow X'$ is pinned down by a $\mathbb{Z}$ -linear map $\Lambda \rightarrow \Lambda'$; the exponential sequence theorem 2.1.3 exhibited $\text{Pic}^0(X)$ as $H^1(X, \mathcal{O}_X)/H^1(X, \mathbb{Z})$, and the present description $\text{Pic}^0(X) \cong \text{Hom}(\Lambda, U(1))$ is the same group read through characters.

The worked example that closes the section returns to the elliptic curve and recovers, in the (H, χ) language, the line bundles the reader already knows from the theory of divisors on a genus-one curve. On $E = \mathbb{C}/\Lambda$, line bundles are classified by degree and by a point of E (the divisor class of degree zero), and this classical picture matches the Appell-Humbert data exactly.

Example 2.2.11: Line Bundles on an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$, $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$. Here $g = 1$, and a Hermitian form on $V = \mathbb{C}$ is $H(v, w) = \alpha v\bar{w}$ for a real scalar $\alpha \in \mathbb{R}$, with $E = \text{Im}H$ given by $E(v, w) = \alpha \text{Im}(v\bar{w})$.

1. *The topologically trivial bundles.* Taking $H = 0$ forces $\chi \in \text{Hom}(\Lambda, U(1))$, a genuine character determined freely by $\chi(1), \chi(\tau) \in U(1)$. Thus

$$\text{Pic}^0(E) \cong \text{Hom}(\Lambda, U(1)) \cong U(1) \times U(1)$$

a real 2-torus. This matches the classical isomorphism $\text{Pic}^0(E) \cong E$: the degree-zero divisor classes on E form a group isomorphic to E itself, and $E = \mathbb{C}/\Lambda$ is precisely the 2-torus $U(1) \times U(1)$ as a real Lie group. A character χ with $\chi(1) = e^{2\pi i s}$, $\chi(\tau) = e^{2\pi i t}$ corresponds to the point $s + t\tau \in E$ (up to the normalization fixing the isomorphism), so the flat bundle $L(0, \chi)$ is the degree-zero bundle $\mathcal{O}_E(P - 0)$ associated to that point P .

2. *Integrality and the degree.* For $H \neq 0$, integrality of $E = \text{Im}H$ on Λ requires $E(1, \tau) \in \mathbb{Z}$. With $H(v, w) = \alpha v\bar{w}$ one has $E(1, \tau) = \alpha \text{Im}\bar{\tau} = -\alpha \text{Im}\tau$, so integrality forces $\alpha \text{Im}\tau \in \mathbb{Z}$, that is, $\alpha = d/\text{Im}\tau$ for the integer $d = E(\tau, 1) = \alpha \text{Im}\tau$. This integer d is the degree of the line bundle: the first Chern class $c_1(L(H, \chi)) = E$ evaluated on the fundamental class of E is $E(\tau, 1) = d$ (section 2.3 makes this identification precise). Positivity $H > 0$ (equivalently $\alpha > 0$) corresponds to $d > 0$, an ample bundle.

3. *The degree-one theta bundle.* The choice $d = 1$, that is,

$$H(v, w) = \frac{v\bar{w}}{\text{Im}\tau}, \quad E(\tau, 1) = 1$$

with any semicharacter χ for this E (for instance $\chi(m + n\tau) = (-1)^{mn}$, as in example 2.2.7), gives the positive line bundle of degree one. This is the principal polarization: as section 3.4 will show, $\dim H^0(E, L) = 1$, so it has a unique section up to scalar, whose zero divisor is a single point. That section is the classical Riemann theta function, and the bundle $L(H, \chi)$ is $\mathcal{O}_E(P)$ for the base point P determined by χ . Every positive line bundle on E is a translate-and-power of this one: $L(dH, \chi') \cong \mathcal{O}_E(D)$ for a divisor D of degree d , recovering the classical statement that $\text{Pic}^d(E) \cong E$ for each d .

The full picture is the exact sequence $0 \rightarrow \text{Pic}^0(E) \rightarrow \text{Pic}(E) \xrightarrow{\deg} \mathbb{Z} \rightarrow 0$ of corollary 2.2.9, with $\text{Pic}^0(E) \cong E$ the connected component and $\deg = c_1$ the topological degree, exactly reproducing the classification of line bundles on a genus-one curve by degree together with a point.

2.3 The Neron-Severi Group and the First Chern Class

The Appell-Humbert theorem describes a line bundle on $X = V/\Lambda$ by a pair (H, χ) , and the exponential sequence sorts these pairs into two registers: the semicharacter χ varies continuously and records the topologically trivial part, while the Hermitian form H is the discrete invariant, the one that survives passage to cohomology. That discrete invariant is the subject of this section. The Hermitian forms H that occur, namely those whose imaginary part is integer-valued on the lattice, assemble into a finitely generated abelian group, the Neron-Severi group, and the map sending a

bundle to its H is exactly the first Chern class. Everything that will distinguish an abelian variety from a general complex torus lives here: the question of projectivity, taken up in Chapter 3, becomes the question of whether this group of forms contains one that is positive definite.

The technical heart of the section is a translation. A Hermitian form carries a complex structure inside it, while the first Chern class, living in $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$, is by nature a real alternating form on the lattice. These two kinds of object are the same data seen from two sides, and the dictionary between them is the correspondence $H \leftrightarrow \text{Im}H$. Making that dictionary precise turns the analytic classification of line bundles into a problem about integral alternating forms, which is arithmetic the reader can compute with. The elliptic curve shows the mechanism in miniature: on $E = \mathbb{C}/\Lambda$ the Neron-Severi group is just $\mathbb{Z}$, and the integer attached to a line bundle is its degree.

The starting point is the observation, extracted from the Appell-Humbert theorem in section 2.2, that not every Hermitian form on V underlies a line bundle. The factor of automorphy is well defined on the lattice only when the semicharacter relation can be satisfied, and that relation forces $\text{Im}H$ to take integer values on Λ . The forms meeting this constraint are precisely the ones that survive as invariants of bundles, and they deserve a name.

Definition 2.3.1: Neron-Severi Group

Let $X = V/\Lambda$ be a complex torus. The *Neron-Severi group* of X is

$$\text{NS}(X) = \{ H : V \times V \rightarrow \mathbb{C} \text{ Hermitian} \mid \text{Im}H(\Lambda, \Lambda) \subseteq \mathbb{Z} \}$$

the group, under addition of forms, of Hermitian forms on V whose imaginary part is integer-valued on the lattice. Equivalently, by the Appell-Humbert theorem it is the image of the first Chern class

$$c_1 : \text{Pic}(X) \longrightarrow H^2(X, \mathbb{Z})$$

the group of line bundles modulo those that are topologically trivial.

Two descriptions of the same group are packaged here, and their agreement is the content of the section. The first is intrinsic linear algebra: a Hermitian form on V subject to one integrality constraint. The second is topological: the isomorphism classes of line bundles that remain once one forgets the continuous parameter χ , recorded by their first Chern class. The Appell-Humbert theorem 2.2.8 makes these coincide, since it identifies $\text{Pic}(X)$ with the group of pairs (H, χ) and identifies $\text{Pic}^0(X)$, the topologically trivial bundles, with the pairs $(0, \chi)$; the quotient is exactly the group of admissible H . That $\text{NS}(X)$ is a group under addition is immediate from the first description, and that it is free of finite rank will follow once it is realized inside $\bigwedge^2 \Lambda^\vee$, which is a free $\mathbb{Z}$ -module of rank $\binom{2g}{2}$.

The name is borrowed from algebraic geometry, where for a smooth projective variety the Neron-Severi group is the group of divisors modulo algebraic equivalence, a finitely generated abelian group by the theorem of the base. For a complex torus the analytic definition above reproduces that group, and Part II will recover it a third time as $\text{NS}(X) = \text{Pic}(X)/\text{Pic}^0(X)$ with Pic^0 the connected component of the Picard scheme. The three faces of an abelian variety each supply their own construction of the same lattice of forms.

To compute with $\text{NS}(X)$ one must resolve the tension already flagged: an element of $\text{NS}(X)$ is a Hermitian form, an object built from the complex structure on V , whereas its first Chern class is an alternating form on the underlying real lattice, an object that has forgotten the complex structure.

The bridge is the imaginary part. Passing from H to $E = \text{Im}H$ discards no information provided one remembers a single compatibility that E inherits from the complex structure, and the whole of H can be rebuilt from E by an explicit formula. This is the central mechanism of the section.

Proposition 2.3.2: Hermitian and Alternating Forms Correspond

Let V be a complex vector space and $\Lambda \subset V$ a full lattice. Sending a Hermitian form to its imaginary part,

$$H \mapsto E := \text{Im}H$$

is a bijection from the set of Hermitian forms H on V whose imaginary part is integer-valued on Λ onto the set of alternating $\mathbb{R}$ -bilinear forms $E: V \times V \rightarrow \mathbb{R}$ that are integer-valued on Λ and satisfy the compatibility

$$E(iv, iw) = E(v, w) \quad \text{for all } v, w \in V \quad (2.5)$$

The inverse sends such an E to the Hermitian form

$$H(v, w) = E(iv, w) + i E(v, w) \quad (2.6)$$

Under the correspondence, H is positive definite if and only if the symmetric form $(v, w) \mapsto E(iv, w)$ is positive definite.

Proof:

The argument has two directions, together with the verification of the reconstruction formula; the integrality condition rides along passively, since $E = \text{Im}H$ is integral on Λ exactly when the reconstructed H is admissible, so it suffices to establish the bijection between real forms and Hermitian forms and observe that it respects the integrality of E .

Step 1: The imaginary part of a Hermitian form is alternating and complex-compatible. Let H be Hermitian, so H is $\mathbb{C}$ -linear in the first variable, $\mathbb{C}$ -antilinear in the second, and $H(w, v) = \overline{H(v, w)}$. Write $E = \text{Im}H$ and $S = \text{Re}H$, so $H = S + iE$ with S, E real-valued. The Hermitian symmetry $H(w, v) = \overline{H(v, w)}$ gives $S(w, v) + iE(w, v) = S(v, w) - iE(v, w)$, whence S is symmetric and E is alternating. For the compatibility, use that H is $\mathbb{C}$ -linear in the first slot and $\mathbb{C}$ -antilinear in the second: $H(iv, iw) = i \cdot \bar{i} \cdot H(v, w) = i(-i)H(v, w) = H(v, w)$. Taking imaginary parts of $H(iv, iw) = H(v, w)$ yields (2.5). Since $E = \text{Im}H$ takes integer values on Λ by hypothesis, E lies in the target set.

Step 2: The reconstruction formula recovers H from E . Still with $H = S + iE$ Hermitian, the real and imaginary parts are not independent, and S is expressible through E alone. Using $\mathbb{C}$ -linearity in the first variable, $H(iv, w) = i H(v, w) = i(S(v, w) + iE(v, w)) = -E(v, w) + i S(v, w)$, while directly $H(iv, w) = S(iv, w) + iE(iv, w)$. Comparing imaginary parts of these two expressions gives

$$S(v, w) = E(iv, w) \quad (2.7)$$

Hence S is determined by E , and therefore so is H :

$$H(v, w) = S(v, w) + i E(v, w) = E(iv, w) + i E(v, w)$$

which is (2.6). So the map $H \mapsto \text{Im}H$ is injective.

Step 3: Every complex-compatible alternating form arises this way. Let E be a real alternating

form with $E(iv, iw) = E(v, w)$, and define H by the formula (2.6), $H(v, w) = E(iv, w) + iE(v, w)$. It remains to check that H is Hermitian; then $\text{Im}H = E$ by construction and surjectivity follows.

First, H is $\mathbb{R}$ -bilinear because E is. Since $\mathbb{R}$ -linearity is already in hand, $\mathbb{C}$ -linearity in the first variable reduces to the single identity $H(iv, w) = iH(v, w)$. Expanding the left side by the definition and using $E(i(iv), w) = E(-v, w) = -E(v, w)$,

$$H(iv, w) = E(i(iv), w) + iE(iv, w) = -E(v, w) + iE(iv, w)$$

while the right side is

$$iH(v, w) = i(E(iv, w) + iE(v, w)) = -E(v, w) + iE(iv, w)$$

and the two agree. For Hermitian symmetry, $H(w, v) = \overline{H(v, w)}$ splits into the real-part identity $E(iw, v) = E(iv, w)$ and the imaginary-part identity $E(w, v) = -E(v, w)$; the latter is the alternating property of E . For the former, the compatibility (2.5) applied to the pair (iw, v) gives $E(iw, v) = E(i \cdot iw, iv) = E(-w, iv) = -E(w, iv) = E(iv, w)$, the last step by the alternating property. So $H(w, v) = \overline{H(v, w)}$, and H is Hermitian. Antilinearity in the second variable follows from linearity in the first together with Hermitian symmetry.

Step 4: Integrality and positivity. If E is integer-valued on Λ then the reconstructed H has $\text{Im}H = E$ integral on Λ , so $H \in \text{NS}(X)$; conversely $\text{Im}H$ integral is the defining condition on E . This matches the two sets slot for slot. Finally, by (2.7) the real part of H is the symmetric form $S(v, w) = E(iv, w)$, and $H(v, v) = S(v, v) = E(iv, v)$ is real for every v , so $H > 0$ precisely when this symmetric form is positive definite, as we sought to show.

The correspondence is the organizing principle of the entire analytic theory. On one side sits a Hermitian form, the natural home for positivity and hence for the notion of an ample line bundle. On the other sits an integral alternating form on Λ , the natural home for cohomology and hence for the first Chern class and the intersection theory of divisors. The single compatibility (2.5) is precisely the condition that an alternating form, a priori insensitive to the complex structure, be compatible with multiplication by i ; it is the analytic shadow of the Hodge-theoretic statement that the class lies in $H^{1,1}$. Chapter 3 will read the positivity clause as the Riemann relations, and the entire question of whether X is projective will reduce to whether the alternating form E can be chosen so that $E(iv, w)$ is positive definite.

Before turning to the identification with the first Chern class, it is worth pinning down the compatibility condition on a concrete torus, where the abstract clause $E(iv, iw) = E(v, w)$ becomes a linear equation on the entries of a matrix.

Example 2.3.3: The Compatibility Condition on a Torus

Take $g = 1$, $V = \mathbb{C}$ with real basis $\{1, i\}$, and let E be a real alternating form on $\mathbb{C}$. An alternating form on a 2-dimensional real space is determined by a single number $E(1, i) =: m$, and every such form is automatically compatible with the complex structure:

$$E(i \cdot 1, i \cdot i) = E(i, -1) = -E(i, 1) = E(1, i)$$

so $E(iv, iw) = E(v, w)$ holds for all v, w by bilinearity. Writing $z = x + iy$ and $w = u + iv$ and expanding E from $E(1, i) = m$, $E(1, 1) = E(i, i) = 0$ gives $E(z, w) = m(xv - yu) = m\text{Im}(\bar{z}w)$. The reconstructed Hermitian form $H(z, w) = E(iz, w) + iE(z, w)$ then works out to

$$H(z, w) = -m \text{Re}(\bar{z}w) + im \text{Im}(\bar{z}w) = -m z\bar{w}$$

which is positive definite precisely when $m < 0$. So for $g = 1$ every integral alternating form on Λ is complex-compatible, and the Neron-Severi group is all of $\bigwedge^2 \Lambda^\vee \cong \mathbb{Z}$, with the positive classes forming one of the two half-lines.

For $g = 2$, an alternating form on $\mathbb{R}^4$ has $\binom{4}{2} = 6$ parameters, while the compatibility (2.5) imposes linear relations cutting the space down. In the real basis $\{e_1, ie_1, e_2, ie_2\}$, the condition $E(iv, iw) = E(v, w)$ leaves $E(e_1, ie_1)$ and $E(e_2, ie_2)$ free and imposes $E(e_1, e_2) = E(ie_1, ie_2)$ together with $E(e_1, ie_2) = -E(ie_1, e_2)$, leaving a 4-dimensional space of compatible forms, matching the real dimension $g^2 = 4$ of the space of Hermitian forms on $\mathbb{C}^2$.

The correspondence just established is stated for real vector spaces, but its content is arithmetic once restricted to the lattice: a complex compatible alternating form that is integer-valued on Λ is an element of $\bigwedge^2 \Lambda^\vee$, the group in which the first Chern class lives. What remains is to check that the alternating form the correspondence attaches to $L(H, \chi)$ is its first Chern class, and not some other invariant that happens to resemble one. This is a direct computation with the connecting homomorphism of the exponential sequence.

Proposition 2.3.4: The First Chern Class Is the Imaginary Part

Let $X = V/\Lambda$ and let $L = L(H, \chi)$ be the line bundle attached by the Appell-Humbert theorem theorem 2.2.8 to the data (H, χ) . Under the identification $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$ of theorem 1.3.2, the first Chern class of L is

$$c_1(L) = E = \text{Im}H$$

the alternating integral form $E(\lambda, \mu) = \text{Im}H(\lambda, \mu)$ on Λ . In particular c_1 depends only on H , not on the semicharacter χ , and the induced map $\text{NS}(X) \rightarrow \bigwedge^2 \Lambda^\vee$, $H \mapsto \text{Im}H$, identifies $\text{NS}(X)$ with the group of complex-compatible integral alternating forms on Λ .

Proof:

The first Chern class is the image of L under the connecting map $\delta: H^1(X, \mathcal{O}_X^*) \rightarrow H^2(X, \mathbb{Z})$ of the long exact sequence attached to the exponential sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 0$ (definition 2.1.4). Computing δ amounts to lifting the cocycle that represents L from $\mathcal{O}_X^*$ to $\mathcal{O}_X$ and measuring the failure of the lift to be a cocycle, which lands in the constant sheaf $\mathbb{Z}$. The computation is cleanest in the description of L by a factor of automorphy, where the cocycle is given by the canonical factor from section 2.2.

Step 1: From the factor of automorphy to a group 2-cocycle. The bundle $L(H, \chi)$ is $V \times \mathbb{C}/\Lambda$ with $\lambda \cdot (v, t) = (v + \lambda, a(\lambda, v)t)$, where

$$a(\lambda, v) = \chi(\lambda) \exp\left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right)$$

is the canonical factor of automorphy. Since V is contractible, the cohomology of $X = V/\Lambda$ with coefficients in a locally constant or holomorphic sheaf is computed by the group cohomology of Λ acting on functions on V ; this is the identification $H^\bullet(X, \mathcal{O}_X) \cong H^\bullet(\Lambda, \mathcal{O}(V))$ underlying theorem 1.3.2. The line bundle L is represented in $H^1(\Lambda, \mathcal{O}(V)^*)$ by the 1-cocycle $\lambda \mapsto a(\lambda, -)$, which satisfies the cocycle identity $a(\lambda + \mu, v) = a(\lambda, \mu + v) a(\mu, v)$ dictated by the action. Choose

a branch of the logarithm and lift a to a $\mathbb{C}$ -valued cochain $a(\lambda, v) = \exp(2\pi i g(\lambda, v))$ with

$$g(\lambda, v) = \frac{1}{2\pi i} \left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda) \right) + c(\lambda) = \frac{1}{2i} H(v, \lambda) + \frac{1}{4i} H(\lambda, \lambda) + c(\lambda)$$

where $c(\lambda) = \frac{1}{2\pi i} \log \chi(\lambda) \in \mathbb{R}$ is fixed modulo $\mathbb{Z}$ by $\chi(\lambda) = \exp(2\pi i c(\lambda))$.

Step 2: The connecting map is the coboundary of the lift. The connecting homomorphism δ sends the class of a to the class of the integer-valued 2-cocycle measuring the failure of the lift g to satisfy the same cocycle identity as a . Taking $\frac{1}{2\pi i} \log$ of the identity $a(\lambda + \mu, v) = a(\lambda, \mu + v) a(\mu, v)$ turns products into sums up to an integer, and that integer is

$$F(\lambda, \mu) = g(\lambda + \mu, v) - g(\lambda, \mu + v) - g(\mu, v)$$

which lies in $\mathbb{Z}$ and, as the computation now shows, is independent of v . Substitute the expression for g and collect by type of term. The terms linear in v contribute

$$\frac{1}{2i} (H(v, \lambda + \mu) - H(\mu + v, \lambda) - H(v, \mu)) = \frac{1}{2i} (-H(\mu, \lambda))$$

using additivity of H in each slot, so the v -dependence cancels. The quadratic terms $\frac{1}{4i} H(-, -)$ on the diagonal contribute

$$\frac{1}{4i} (H(\lambda + \mu, \lambda + \mu) - H(\lambda, \lambda) - H(\mu, \mu)) = \frac{1}{4i} (H(\lambda, \mu) + H(\mu, \lambda))$$

and the semicharacter relation $\chi(\lambda + \mu) = \chi(\lambda)\chi(\mu) e^{i\pi E(\lambda, \mu)}$ from definition 2.2.4 gives $c(\lambda + \mu) - c(\lambda) - c(\mu) = \frac{1}{2} E(\lambda, \mu)$ modulo $\mathbb{Z}$.

Step 3: Collecting the terms. Adding the three contributions,

$$F(\lambda, \mu) = -\frac{1}{2i} H(\mu, \lambda) + \frac{1}{4i} (H(\lambda, \mu) + H(\mu, \lambda)) + \frac{1}{2} E(\lambda, \mu)$$

modulo $\mathbb{Z}$. The two H -terms combine to $\frac{1}{4i} (H(\lambda, \mu) - H(\mu, \lambda)) = \frac{1}{4i} (H(\lambda, \mu) - \overline{H(\lambda, \mu)}) = \frac{1}{2} \text{Im} H(\lambda, \mu) = \frac{1}{2} E(\lambda, \mu)$, using Hermitian symmetry $H(\mu, \lambda) = \overline{H(\lambda, \mu)}$ and $\frac{1}{2i} (z - \bar{z}) = \text{Im} z$. Hence

$$F(\lambda, \mu) = \frac{1}{2} E(\lambda, \mu) + \frac{1}{2} E(\lambda, \mu) = E(\lambda, \mu)$$

which is integer-valued on Λ because E is integral there, so the congruence modulo $\mathbb{Z}$ is an equality. The 2-cocycle $F = E$ is alternating and $\mathbb{Z}$ -bilinear, hence a well-defined element of $\bigwedge^2 \Lambda^\vee \cong H^2(X, \mathbb{Z})$ (theorem 1.3.2). By construction it is $\delta(L) = c_1(L)$, so $c_1(L) = E = \text{Im} H$. The semicharacter entered only through the term $\frac{1}{2} E(\lambda, \mu)$, which combined with the Hermitian part to produce exactly E ; changing χ within its coset alters c by a genuine character, whose coboundary is a symmetric integer-valued cochain and so does not change the alternating class F . Thus $c_1(L)$ depends only on H , completing the proof.

The proposition closes the loop opened by the Appell-Humbert theorem. That theorem gave a bijection between line bundles and pairs (H, χ) ; the exponential sequence organized $\text{Pic}(X)$ into the extension $0 \rightarrow \text{Pic}^0(X) \rightarrow \text{Pic}(X) \xrightarrow{c_1} \text{NS}(X) \rightarrow 0$; and now the first Chern class in that sequence is seen to be nothing other than the map $H \mapsto \text{Im} H$ of proposition 2.3.2. The three descriptions of $\text{NS}(X)$, as admissible Hermitian forms, as the image of c_1 , and as complex-compatible integral

alternating forms on Λ , are now literally the same group, and one may compute in whichever is convenient. For questions of positivity one keeps H ; for questions of cohomology, intersection, and rank one passes to E .

Because $E = \text{Im}H$ is an integral alternating form on the rank- $2g$ lattice Λ , the group $\text{NS}(X)$ embeds in the free $\mathbb{Z}$ -module $\bigwedge^2 \Lambda^\vee$ of rank $\binom{2g}{2}$, and is therefore itself free of finite rank. Its rank is the invariant that measures how many independent line-bundle types a torus carries, and it earns a name.

Definition 2.3.5: Picard Number

The *Picard number* of a complex torus $X = V/\Lambda$ is the rank of its Neron-Severi group,

$$\rho(X) = \text{rank}_{\mathbb{Z}} \text{NS}(X)$$

the number of independent classes of line bundles modulo topological triviality.

Two bounds frame the Picard number. On one side, $\text{NS}(X)$ sits inside the real vector space of Hermitian forms on $V = \mathbb{C}^g$, which has real dimension g^2 ; since $\text{NS}(X)$ is a discrete subgroup of that space, its rank cannot exceed g^2 , giving $\rho(X) \leq g^2$. On the other side, $\rho(X) \geq 0$, with equality exactly when the only admissible Hermitian form is $H = 0$, which is the generic situation for a complex torus of dimension at least two and the reason most complex tori are not projective. The gap between these bounds is where the arithmetic of the individual torus lives: a very special torus, such as one with complex multiplication or a product of isogenous factors, pushes ρ toward the upper bound g^2 , while a very general torus sits at the floor $\rho = 0$. The elliptic curve, sitting at $g = 1$, has no room for either phenomenon: the bound reads $0 \leq \rho \leq 1$, and since an elliptic curve is projective it realizes $\rho = 1$.

Example 2.3.6: The Neron-Severi Group of an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$, as in example 1.1.4. Here $g = 1$, so $V = \mathbb{C}$ is a real 2-dimensional space and every alternating form on V is automatically complex-compatible, as computed in example 2.3.3. An alternating integral form on $\Lambda = \mathbb{Z}\langle 1, \tau \rangle$ is determined by its value on the basis pair, and every integer value occurs, so

$$\text{NS}(E) \cong \mathbb{Z}, \quad \rho(E) = 1$$

Under proposition 2.3.2 the positive generator is the class of the Hermitian form

$$H_1(z, w) = \frac{z\bar{w}}{\text{Im}\tau}$$

whose imaginary part $E_1 = \text{Im}H_1$ is the alternating form with $E_1(\tau, 1) = 1$; a general class is $H_m = mH_1$ for $m \in \mathbb{Z}$, positive definite precisely when $m > 0$. The integer m is the degree of the corresponding line bundle: $L(H_m, \chi)$ has $\deg L = m$, matching the classical fact from [3] that line bundles on an elliptic curve are classified up to Pic^0 by their degree. The positive generator is the class of the origin as a divisor, the bundle $\mathcal{O}_E(0)$ of degree 1, the principal polarization studied in Chapter 3.

The elliptic curve exhausts the possibilities for $g = 1$, but as soon as $g = 2$ the Picard number becomes a genuine invariant that separates one torus from another. A product of two elliptic curves already carries more line-bundle classes than either factor, because line bundles can be pulled back

from each factor and, when the factors are isogenous, a further class appears from the graph of the isogeny. This is the first place where the higher-dimensional theory departs qualitatively from the elliptic-curve model.

Example 2.3.7: The Neron-Severi Group of a Product $E \times E$

Let $E = \mathbb{C}/\Lambda$ be an elliptic curve and $X = E \times E$, a complex torus of dimension $g = 2$ with lattice $\Lambda \oplus \Lambda \subset \mathbb{C}^2$ and upper bound $\rho(X) \leq g^2 = 4$. Three line-bundle classes are visible at once.

1. The pullback $p_1^*\mathcal{O}_E(0)$ along the first projection, whose Chern class is the alternating form E_1 supported on the first factor, $E_1((\lambda, \lambda'), (\mu, \mu')) = E_0(\lambda, \mu)$ with E_0 the generator of $\text{NS}(E)$ from example 2.3.6.
2. The pullback $p_2^*\mathcal{O}_E(0)$ along the second projection, with Chern class $E_2((\lambda, \lambda'), (\mu, \mu')) = E_0(\lambda', \mu')$.
3. The class of the diagonal $\Delta \subset E \times E$, the image of E under $z \mapsto (z, z)$; its Chern class E_Δ is not a combination of E_1 and E_2 because it pairs the two factors, taking the value $E_0(\lambda, \mu')$ on the cross terms.

These three classes are $\mathbb{Z}$ -linearly independent in $\text{NS}(X)$, so

$$\rho(E \times E) \geq 3$$

Whether the rank is exactly 3 or larger is decided by the arithmetic of E . If E has no complex multiplication, so $\text{End}(E) = \mathbb{Z}$, then the only homomorphisms $E \rightarrow E$ are the integers $[n]$, and one checks that these three classes span: $\rho(E \times E) = 3$. If E has complex multiplication by an order in an imaginary quadratic field K , so $\text{End}(E) \otimes \mathbb{Q} = K$ is 2-dimensional over $\mathbb{Q}$, then each endomorphism $\alpha \in \text{End}(E)$ contributes the class of its graph $\Gamma_\alpha = \{(z, \alpha z)\}$, and these graphs supply a fourth independent class; the maximum $\rho(E \times E) = 4 = g^2$ is attained. The Picard number thus reads off the endomorphism algebra of E , a first instance of the principle, developed fully in Chapter 6, that the arithmetic of an abelian variety is encoded in its endomorphisms.

The two examples mark the two extremes the Picard number can occupy and show that the invariant is sensitive to arithmetic that is invisible at the level of the underlying real torus. What makes $\text{NS}(X)$ the arithmetic heart of the classification is that every question about the algebraicity of X passes through it. A complex torus is a compact complex manifold regardless of its Neron-Severi group, but it is an abelian variety, embeddable in projective space, only when $\text{NS}(X)$ contains a class that is not just nonzero but positive definite as a Hermitian form. The existence of such a class is a condition on the lattice Λ relative to the complex structure on V , and translating it into the language of the alternating form E produces the Riemann relations. That translation, and the definition of an abelian variety it makes possible, is the business of Chapter 3, for which the correspondence $H \leftrightarrow \text{Im}H$ of this section is the indispensable dictionary.

2.4 Sections and the Space $H^0(X, L)$

The Appell-Humbert theorem classifies every line bundle on $X = V/\Lambda$ by a pair (H, χ) , but a classification is not yet a geometry. What makes a line bundle useful is its space of global sections: a positive bundle carries enough sections to separate points and tangent vectors, and the map to

projective space assembled from a basis of those sections is what realizes X as a projective variety. This section identifies the global sections of $L(H, \chi)$ concretely. Because $L(H, \chi)$ is built from the trivial bundle $V \times \mathbb{C}$ by gluing along the factor of automorphy, a section is not an abstract cohomology class but a holomorphic function on V obeying a prescribed transformation law under the lattice. These functions are the theta functions, and the whole projective story of the next chapters rests on counting them.

Two facts organize the count. First, the transformation law is so rigid that a nonzero section can exist only when H is positive semidefinite: a Hermitian form that fails to be positive somewhere forces every candidate section to grow too slowly to survive on a compact torus, and Liouville's theorem kills it. Second, when H is strictly positive the sections form a finite-dimensional space whose dimension is exactly the Pfaffian of $\text{Im}H$, the product of the elementary divisors of the polarization. The first fact is proved here in full; the second is stated here and its proof, together with an explicit basis, is the business of section 3.4 and of Chapter 4.

Definition 2.4.1: Theta Function for a Line Bundle

Let $L = L(H, \chi)$ be the line bundle on $X = V/\Lambda$ associated by the Appell-Humbert construction (definition 2.2.5) to a datum (H, χ) , with canonical factor of automorphy

$$a_{(H, \chi)}(\lambda, v) = \chi(\lambda) \exp\left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right)$$

A *theta function* for (H, χ) is a holomorphic function $\theta: V \rightarrow \mathbb{C}$ satisfying the functional equation

$$\theta(v + \lambda) = a_{(H, \chi)}(\lambda, v) \theta(v) \quad \text{for all } v \in V, \lambda \in \Lambda \quad (2.8)$$

The complex vector space of all theta functions for (H, χ) is denoted $H^0(X, L)$.

The identification of this space with the sheaf-cohomology group $H^0(X, L)$ of global sections is a direct unwinding of the Appell-Humbert construction, and spelling it out fixes the dictionary used for the rest of the book. Recall from definition 2.2.5 that $L(H, \chi)$ is the quotient of $V \times \mathbb{C}$ by the Λ -action $\lambda \cdot (v, t) = (v + \lambda, a_{(H, \chi)}(\lambda, v) t)$. A global section of L is a holomorphic map $s: X \rightarrow L$ splitting the projection $L \rightarrow X$; pulling back along $V \rightarrow X$, it lifts to a holomorphic map $\tilde{s}: V \rightarrow V \times \mathbb{C}$ of the form $\tilde{s}(v) = (v, \theta(v))$ for a holomorphic $\theta: V \rightarrow \mathbb{C}$. Compatibility with the Λ -action, $\tilde{s}(v + \lambda) = \lambda \cdot \tilde{s}(v)$, reads $(v + \lambda, \theta(v + \lambda)) = (v + \lambda, a_{(H, \chi)}(\lambda, v) \theta(v))$, which is exactly (2.8). Thus a section is the same datum as a theta function, and the space of theta functions of definition 2.4.1 coincides with the sheaf-cohomology group $H^0(X, L)$ of global sections of L . The functional equation is the reason these functions are called *quasi-periodic*: θ is not Λ -periodic (that would make it a section of the trivial bundle), but it reproduces itself under lattice translation up to the explicit automorphy factor.

A theta function is a function on V , not on X , so only its zero set descends. The functional equation (2.8) shows $a_{(H, \chi)}(\lambda, v)$ never vanishes, so the zero locus $\{\theta = 0\} \subseteq V$ is Λ -invariant and descends to a well-defined divisor on X , the divisor of the section. When $H > 0$ this divisor is the geometric object, a theta divisor, that Chapter 4 studies in its own right. For now the point is structural: sections are quasi-periodic entire functions, and their count is controlled entirely by the growth that the factor of automorphy permits.

Before counting sections when they exist, the first question is when they exist at all. The functional equation constrains the modulus of θ against the Hermitian form H , and on a compact torus modulus constraints become boundedness constraints. The following growth argument extracts from a single

nonzero section the positivity of H .

Proposition 2.4.2: Nonpositive Bundles Have No Sections

Let $L = L(H, \chi)$ be the line bundle associated to a datum (H, χ) on $X = V/\Lambda$. If H is not positive semidefinite, then

$$H^0(X, L) = 0$$

Equivalently, if L admits a nonzero global section, then $H \geq 0$.

Proof:

Let $\theta \in H^0(X, L)$ be a theta function for (H, χ) , and consider the real-valued function

$$\varphi(v) = |\theta(v)| \exp\left(-\frac{\pi}{2} H(v, v)\right) \quad (2.9)$$

which is well-defined because $H(v, v) \in \mathbb{R}$ for a Hermitian form.

Step 1: φ descends to the compact torus. The claim is that φ is Λ -periodic. Fix $\lambda \in \Lambda$. Since $|\chi(\lambda)| = 1$ and $H(\lambda, \lambda) \in \mathbb{R}$, taking moduli in the functional equation (2.8) gives

$$|\theta(v + \lambda)| = |\theta(v)| \exp\left(\pi \operatorname{Re} H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right)$$

On the other side, expanding $H(v + \lambda, v + \lambda)$ by bilinearity and using $H(\lambda, v) = \overline{H(v, \lambda)}$ yields

$$H(v + \lambda, v + \lambda) = H(v, v) + 2 \operatorname{Re} H(v, \lambda) + H(\lambda, \lambda)$$

Substituting both computations into (2.9),

$$\varphi(v + \lambda) = |\theta(v)| \exp\left(\pi \operatorname{Re} H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right) \exp\left(-\frac{\pi}{2} H(v, v) - \pi \operatorname{Re} H(v, \lambda) - \frac{\pi}{2} H(\lambda, \lambda)\right)$$

The two terms involving $\operatorname{Re} H(v, \lambda)$ cancel, as do the two terms involving $H(\lambda, \lambda)$, leaving $\varphi(v + \lambda) = |\theta(v)| \exp\left(-\frac{\pi}{2} H(v, v)\right) = \varphi(v)$. Thus φ is Λ -periodic, so it descends to a continuous function on the compact space $X = V/\Lambda$ and is therefore bounded: there is a constant $M \geq 0$ with

$$|\theta(v)| \leq M \exp\left(\frac{\pi}{2} H(v, v)\right) \quad \text{for all } v \in V \quad (2.10)$$

Step 2: a negative direction forces θ to vanish. Suppose H is not positive semidefinite. Then there is a vector $v_0 \in V$ with $H(v_0, v_0) < 0$. Fix an arbitrary $w \in V$ and restrict θ to the complex line through w in the direction v_0 , setting

$$g(z) = \theta(w + zv_0) \quad (z \in \mathbb{C})$$

which is an entire function of z . Expanding the Hermitian form along this line, using linearity in the first slot and conjugate-linearity in the second,

$$H(w + zv_0, w + zv_0) = H(w, w) + 2 \operatorname{Re}(z H(v_0, w)) + |z|^2 H(v_0, v_0)$$

The bound (2.10) applied at $v = w + zv_0$ therefore gives

$$|g(z)| \leq M \exp\left(\frac{\pi}{2} H(w, w) + \pi \operatorname{Re}(z H(v_0, w)) + \frac{\pi}{2} |z|^2 H(v_0, v_0)\right)$$

As $|z| \rightarrow \infty$ the quadratic term $\frac{\pi}{2}|z|^2 H(v_0, v_0)$ tends to $-\infty$, since $H(v_0, v_0) < 0$, while the linear term $\pi \operatorname{Re}(z H(v_0, w))$ grows at most like $|z|$. The quadratic term dominates, so the exponent tends to $-\infty$ and $|g(z)| \rightarrow 0$ as $|z| \rightarrow \infty$. An entire function that tends to 0 at infinity is bounded, hence constant by Liouville's theorem, and its constant value must be 0. Thus $g \equiv 0$; in particular $g(0) = \theta(w) = 0$.

Since $w \in V$ was arbitrary, θ vanishes identically. Therefore $H^0(X, L) = 0$, as we sought to show.

The mechanism deserves emphasis. The factor of automorphy is designed precisely so that the Gaussian-modulated modulus $\varphi(v) = |\theta(v)| \exp(-\frac{\pi}{2} H(v, v))$ becomes periodic, converting the transcendental transformation law into a boundedness statement on a compact space. Once bounded, the section can grow no faster than $\exp(\frac{\pi}{2} H(v, v))$, and along any direction in which H is negative this is exponential decay, which Liouville cannot tolerate for a nonzero entire function. The positivity of H is thus not an extra hypothesis imposed by hand: it is forced by the mere existence of a section. This is the analytic shadow of a fact that will reappear algebraically in Part II, that an effective line bundle has nonnegative degree. Combined with the classification of theorem 2.2.8, it means that of all the line bundles on X , only those whose Hermitian form is positive semidefinite can contribute sections at all, and the interesting geometry lives among the strictly positive ones.

The converse question, how many sections a strictly positive bundle supports, has an exact answer in terms of the alternating form $E = \operatorname{Im} H$. When $H > 0$ the form E is nondegenerate and integral on Λ , and the elementary-divisor theorem produces a symplectic basis in which E has invariants $d_1 |d_2| \cdots |d_g|$, the type of the polarization. The dimension of the section space is the product of these invariants, equivalently the Pfaffian of E .

Proposition 2.4.3: Dimension of the Space of Sections

Let $L = L(H, \chi)$ with $H > 0$, and let $E = \operatorname{Im} H$, a nondegenerate alternating integral form on Λ with elementary divisors $d_1 |d_2| \cdots |d_g|$. Then $H^0(X, L)$ is finite-dimensional, with

$$\dim_{\mathbb{C}} H^0(X, L) = \operatorname{Pf}(E) = d_1 d_2 \cdots d_g = \sqrt{\det E}$$

In particular $\dim_{\mathbb{C}} H^0(X, L) \geq 1$, so every positive line bundle has a nonzero section.

The proof constructs an explicit basis of theta functions, indexed by a set of coset representatives whose cardinality is exactly $d_1 \cdots d_g$, and verifies that these functions span. That construction is the content of section 3.4, where the dimension formula is derived as the Riemann-Roch theorem for abelian varieties, and of Chapter 4, where the basis is written down in classical coordinates as the Riemann theta functions with characteristics. The deferral is deliberate: the counting argument is cleanest once the reduction of E to symplectic normal form by the elementary-divisor theorem ([7]) is available in the polarization language of definition 3.2.3 in Chapter 3, and the explicit Fourier-series form of the basis belongs with the classical theta function of Chapter 4. Stated here, the formula records the payoff of the growth estimate above: positivity is not only necessary for sections but produces precisely $\operatorname{Pf}(E)$ of them, and this integer is the projective dimension budget that the Lefschetz embedding theorem will spend.

The dimension count is already transparent in the one-dimensional case, where it recovers a computation the reader has met analytically and where the sections are the classical theta functions of Jacobi.

Example 2.4.4: Theta Functions on an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$ and $\tau \in \mathbb{H}$, the $g = 1$ case. A Hermitian form on $V = \mathbb{C}$ is $H(v, w) = h v \bar{w}$ for a real scalar h , and $H > 0$ means $h > 0$. The imaginary part $E = \text{Im}H$ is the alternating form determined by its value on the basis $(\tau, 1)$ of Λ ,

$$E(\tau, 1) = \text{Im}(h \tau \cdot \bar{1}) = h \text{Im}\tau$$

so E is integral on Λ precisely when $h \text{Im}\tau \in \mathbb{Z}$. Writing $d = h \text{Im}\tau = E(\tau, 1)$, this integer is the single elementary divisor, the type (d) of the polarization, and it is the degree of the corresponding line bundle. proposition 2.4.3 then gives

$$\dim_{\mathbb{C}} H^0(E, L) = \text{Pf}(E) = d$$

For the principal polarization $d = 1$, so $h = 1/\text{Im}\tau$ and there is, up to scalar, a unique theta function. It is the classical *Jacobi theta function*

$$\vartheta(z, \tau) = \sum_{n \in \mathbb{Z}} \exp(\pi i n^2 \tau + 2\pi i n z) \quad (2.11)$$

an entire function of z whose series converges because $\text{Im}\tau > 0$ forces $|\exp(\pi i n^2 \tau)| = \exp(-\pi n^2 \text{Im}\tau)$ to decay faster than any geometric rate. Its quasi-periodicity is the direct analogue of the functional equation (2.8): reindexing the series gives

$$\vartheta(z + 1, \tau) = \vartheta(z, \tau), \quad \vartheta(z + \tau, \tau) = \exp(-\pi i \tau - 2\pi i z) \vartheta(z, \tau)$$

The first identity is immediate since $\exp(2\pi i n) = 1$; for the second, the substitution $\pi i n^2 \tau + 2\pi i n \tau = \pi i \tau (n + 1)^2 - \pi i \tau$ followed by the reindexing $m = n + 1$ produces the stated factor. These two transformation laws are exactly the values of a factor of automorphy on the basis $\lambda = 1$ and $\lambda = \tau$ of Λ ; up to the coboundary that passes between the classical normalization and the canonical Appell-Humbert factor, ϑ is the section of the degree-one bundle predicted by proposition 2.4.3.

For a general positive type (d) with $d \geq 1$, the dimension is d , and a basis is furnished by the d theta functions with rational characteristics $\vartheta \begin{bmatrix} a/d \\ 0 \end{bmatrix} (z, \tau)$ for $a = 0, 1, \dots, d - 1$, translates of (2.11) that Chapter 4 introduces in general. The count $\dim H^0(E, L) = d = \deg L$ is the analytic form of the Riemann-Roch statement for elliptic curves, recovering from the present transcendental viewpoint the dimension the reader computed algebraically in [3].

The elliptic case makes the general pattern legible. A section of a line bundle is a concrete holomorphic function on the universal cover with a completely explicit transformation law under the deck group, and once that law is fixed the sections form a finite-dimensional space whose dimension is a discrete invariant of the bundle, its degree in dimension one and its Pfaffian in general. Two features of this dimension are what power everything downstream. First, it is positive exactly when the bundle is positive, so the analytic condition $H > 0$ that permits sections is the same condition that will permit a projective embedding. Second, it grows: replacing L by $L^{\otimes n}$ multiplies each elementary divisor by n and hence multiplies $\text{Pf}(E)$ by n^g , so high tensor powers of a positive bundle supply an abundance of sections.

Polarizations and the Riemann Relations

The classification of Chapter 2 reduces the algebraicity of a complex torus to a single question: among the Hermitian forms H with $\text{Im}H$ integral on Λ , is there one that is positive definite? This chapter shows that the answer decides everything. A positive such form is a *polarization*, and its existence is exactly the Riemann relations, the classical bilinear conditions on the period matrix that separate the projective tori from the transcendental ones. A complex torus that carries a polarization is an *abelian variety*; one that does not is forever analytic, with too few meromorphic functions to cut out anything in projective space.

With the definition in hand, the chapter develops the two invariants a polarization carries. Its type, a tuple of elementary divisors $(d_1, \dots, d_g)$, records the arithmetic of the form on the lattice and specializes to the notion of a principal polarization when the type is trivial; the associated period matrices, normalized by the type, sweep out the Siegel upper half-space and give the first picture of a moduli space of abelian varieties. Finally, Riemann-Roch computes the number of independent theta functions a polarization supports, $\dim H^0(X, L) = d_1 \cdots d_g$, the count that Chapter 4 will realize by writing the theta functions down explicitly and using them to embed the variety.

3.1 Positive Line Bundles and the Riemann Relations

The previous chapter classified every line bundle on a complex torus $X = V/\Lambda$ by a pair (H, χ) , where H is a Hermitian form on V whose imaginary part $E = \text{Im}H$ is integral on the lattice, and χ is a semicharacter for E . That classification is purely formal: it records which line bundles exist, but says nothing about which of them have sections, and therefore nothing about whether X can be embedded in projective space. The class $c_1(L(H, \chi)) = H$ in $\text{NS}(X)$ carries all the information relevant to this

question, and the answer turns entirely on one attribute of H that the Appell-Humbert data does not constrain, namely its *signature*. This section isolates the condition, positive definiteness of H , and proves that it is exactly the condition under which X is algebraic.

The statement that a positive Hermitian form with integral imaginary part exists is the Riemann relations. In the period-matrix language it becomes a pair of explicit conditions on the $g \times 2g$ matrix Π of periods, one an algebraic identity and one a positivity, and these are the conditions Riemann wrote down for the periods of an abelian integral. The equivalence of the intrinsic Hermitian statement with the coordinate one is a linear-algebra computation carried out in full below. The consequence, developed across the rest of this chapter, is that the tori $\mathbb{C}^g/\Lambda$ admitting such a form are precisely the abelian varieties, a vanishingly thin subset of all complex tori once $g \geq 2$.

Definition 3.1.1: Positive Line Bundle

Let $X = V/\Lambda$ be a complex torus and $L = L(H, \chi)$ a line bundle on X , with Hermitian form H and semicharacter χ as in the Appell-Humbert classification. The bundle L is *positive* if its Hermitian form H is positive definite, written $H > 0$; that is, $H(v, v) > 0$ for every nonzero $v \in V$.

Positivity depends only on H , not on the semicharacter χ , and $H = c_1(L)$ is the image of L in $\text{NS}(X)$ under the first Chern class. Positivity is therefore a property of the class of L in the Néron-Severi group $\text{NS}(X)$ (definition 2.3.1), unchanged by tensoring L with any bundle from $\text{Pic}^0(X)$, which alters χ but leaves H fixed. The definition is the analytic incarnation of ampleness: a line bundle on a projective variety is ample when a power of it embeds the variety in projective space, and the content of this chapter and the next is that $H > 0$ is precisely the analytic signature of that geometric condition. The positive-definiteness requirement is restrictive. A Hermitian form on $\mathbb{C}^g$ has a signature (p, q) with $p + q = g$, and only the extreme case $(g, 0)$ is positive; the intermediate signatures give line bundles that are nontrivial in $\text{NS}(X)$ but have no global sections, while $(0, g)$ gives an anti-ample class.

The prototype is the elliptic curve, where the condition is automatically satisfiable and one sees the whole mechanism in a single complex variable.

Example 3.1.2: Positive Bundles on an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$, $\tau \in \mathbb{H}$, so $V = \mathbb{C}$ and $g = 1$. A Hermitian form on $\mathbb{C}$ is $H(v, w) = c v \bar{w}$ for a real scalar c , and $H > 0$ means $c > 0$. Its imaginary part is $E(v, w) = c \text{Im}(v \bar{w})$, an alternating $\mathbb{R}$ -form on $\mathbb{C}$. Evaluate E on the basis $\lambda_1 = \tau$, $\lambda_2 = 1$ of Λ :

$$E(\tau, 1) = c \text{Im}(\tau \cdot 1) = c \text{Im}(\tau)$$

while $E(1, 1) = E(\tau, \tau) = 0$. Integrality of E on Λ demands $E(\tau, 1) \in \mathbb{Z}$, so $c \text{Im}\tau$ must be an integer. Taking the smallest positive value, $c = 1/\text{Im}\tau > 0$, gives a positive Hermitian form

$$H(v, w) = \frac{v \bar{w}}{\text{Im}\tau}$$

with $E(\tau, 1) = 1$ and matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ on the basis $(\tau, 1)$. Thus every elliptic curve carries a positive line bundle. This is the analytic counterpart of the ample bundle $\mathcal{O}_E(O)$ attached to the origin in E , whose sections are the classical theta functions of $\mathbb{C}/\Lambda$ ([3]); the constant $1/\text{Im}\tau$ is the reciprocal of the covolume of Λ , which is why it is exactly this normalization that makes E integral.

The example shows the two constraints acting together: positivity of H and integrality of $E = \operatorname{Im} H$ on Λ . A line bundle whose Hermitian form satisfies both is what makes X algebraic, and this pairing of conditions deserves a name.

Definition 3.1.3: Polarization (Analytic)

Let $X = V/\Lambda$ be a complex torus. A *polarization* of X is a positive-definite Hermitian form H on V whose imaginary part $E = \operatorname{Im} H$ takes integer values on Λ , i.e. $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$. Equivalently, a polarization is a class in $\operatorname{NS}(X)$ (definition 2.3.1) that is represented by a positive-definite Hermitian form; such a class is called a *positive class*. The torus X is *polarizable* if it admits a polarization.

A polarization is thus a point of $\operatorname{NS}(X)$ lying in the open cone of positive-definite forms, subject to the integrality that membership in $\operatorname{NS}(X)$ already imposes. By theorem 2.2.8, every such H is the Chern class of an actual line bundle $L(H, \chi)$ on X , and different choices of semicharacter χ produce line bundles differing by an element of $\operatorname{Pic}^0(X)$; the polarization is the common class, an object slightly coarser than a line bundle. Recording a polarization as the Hermitian form H rather than as the bundle L is the convention throughout: what matters for algebraicity is the positive class, and the semicharacter is auxiliary data that the theory of theta functions in the next chapter pins down when an explicit bundle is needed. The word “polarization” is chosen to evoke a rigidification: a bare complex torus has no distinguished cohomology class, while a polarized torus comes equipped with one, and it is this extra datum that anchors the transcendental object to projective geometry.

The central result of the section states that polarizability is the precise dividing line for algebraicity, and gives the condition two forms, one intrinsic and one in coordinates.

Theorem 3.1.4: Riemann Relations

Let $X = V/\Lambda$ be a complex torus of dimension g , with $V = \mathbb{C}^g$. The following are equivalent.

1. X admits a polarization: there is a positive-definite Hermitian form H on V whose imaginary part $E = \operatorname{Im} H$ is integral on Λ .
2. (*Period-matrix form.*) For a period matrix $\Pi \in M_{g \times 2g}(\mathbb{C})$ of X , there exists a nondegenerate alternating integral matrix $E \in M_{2g}(\mathbb{Z})$ such that

$$\Pi E^{-1} \Pi^t = 0 \quad \text{and} \quad i \Pi E^{-1} \bar{\Pi}^t > 0 \quad (3.1)$$

where the second inequality asserts that the Hermitian matrix $i \Pi E^{-1} \bar{\Pi}^t$ is positive definite.

When either condition holds, X is an abelian variety: it admits a holomorphic embedding into some projective space $\mathbb{P}^N$, and conversely every complex torus that so embeds satisfies these conditions.

Condition (1) is intrinsic to V and Λ ; condition (2) is its transcription into the periods, and the two Riemann relations (3.1) are the classical statement, an algebraic identity together with a positivity. The final sentence records the geometric payoff, which is the reason the conditions are studied at all: they are the exact criterion separating algebraic tori from transcendental ones. Before proving the theorem it is worth being clear about what each half of the equivalence contributes. The equivalence

of (1) and (2) is a self-contained linear-algebra computation, carried out below. The passage from (1) to a projective embedding is a genuine construction, requiring the sections of a positive line bundle to be produced explicitly; that construction is the theory of theta functions, and the resulting Lefschetz embedding theorem is proved in the next chapter. The reverse passage, from a projective embedding to a polarization, extracts the Hermitian form from the first Chern class of the very ample bundle furnished by the embedding.

Proof:

The proof has three parts: the equivalence of the two analytic conditions (1) and (2), and the two implications relating (1) to algebraicity. The core computation, the equivalence of (1) and (2), occupies the first two steps; the geometric implications are then addressed in turn.

Step 1: From a polarization to the period relations. Suppose H is a positive-definite Hermitian form on $V = \mathbb{C}^g$ with $E = \text{Im}H$ integral on Λ . Fix the standard $\mathbb{C}$ -basis of V , and let $\mathcal{H} \in M_g(\mathbb{C})$ be the matrix of H , so that $H(v, w) = v^t \mathcal{H} \bar{w}$; the Hermitian symmetry $H(w, v) = \overline{H(v, w)}$ is the condition $\mathcal{H}^t = \mathcal{H}$, and positive definiteness of H is positive definiteness of $\mathcal{H}$. Let $\lambda_1, \dots, \lambda_{2g}$ be the $\mathbb{Z}$ -basis of Λ that forms the columns of Π , and let $E \in M_{2g}(\mathbb{Z})$ be the integer matrix with entries $E_{jk} = E(\lambda_j, \lambda_k)$, alternating because E is, and nondegenerate because H is nondegenerate on the real span $V = \Lambda \otimes \mathbb{R}$.

Assemble the periods into the invertible big period matrix

$$\hat{\Pi} = \begin{pmatrix} \Pi \\ \bar{\Pi} \end{pmatrix} \in M_{2g}(\mathbb{C})$$

whose invertibility records that $\lambda_1, \dots, \lambda_{2g}$ is an $\mathbb{R}$ -basis of $V \cong \mathbb{R}^{2g}$. Writing $M = \Pi^t \mathcal{H} \bar{\Pi}$, the (j, k) entry of M is $\lambda_j^t \mathcal{H} \bar{\lambda}_k = H(\lambda_j, \lambda_k)$, so $E = \text{Im}M$ entrywise. Since $\bar{M} = \bar{\Pi}^t \mathcal{H} \Pi = \bar{\Pi}^t \mathcal{H}^t \Pi$, this reads

$$E = \frac{1}{2i} (M - \bar{M}) = \frac{1}{2i} (\Pi^t \mathcal{H} \bar{\Pi} - \bar{\Pi}^t \mathcal{H}^t \Pi) = \frac{1}{2i} \hat{\Pi}^t B \hat{\Pi}, \quad B = \begin{pmatrix} 0 & \mathcal{H} \\ -\bar{\mathcal{H}} & 0 \end{pmatrix} \quad (3.2)$$

the last equality being the block expansion $\hat{\Pi}^t B \hat{\Pi} = \Pi^t \mathcal{H} \bar{\Pi} - \bar{\Pi}^t \mathcal{H} \Pi$, valid because $\mathcal{H}^t = \bar{\mathcal{H}}$. The block matrix B is invertible, with

$$B^{-1} = \begin{pmatrix} 0 & -\bar{\mathcal{H}}^{-1} \\ \mathcal{H}^{-1} & 0 \end{pmatrix}$$

as one checks by multiplying out. Inverting (3.2) gives $E^{-1} = 2i \hat{\Pi}^{-1} B^{-1} (\hat{\Pi}^t)^{-1}$, hence

$$\hat{\Pi} E^{-1} \hat{\Pi}^t = 2i B^{-1} = 2i \begin{pmatrix} 0 & -\bar{\mathcal{H}}^{-1} \\ \mathcal{H}^{-1} & 0 \end{pmatrix} \quad (3.3)$$

Now read off the $g \times g$ blocks of the left-hand side. Since $\hat{\Pi} = \begin{pmatrix} \Pi \\ \bar{\Pi} \end{pmatrix}$, the top-left block of $\hat{\Pi} E^{-1} \hat{\Pi}^t$ is $\Pi E^{-1} \Pi^t$ and the top-right block is $\Pi E^{-1} \bar{\Pi}^t$. Comparing with the right-hand side of (3.3), whose top-left block is 0 and whose top-right block is $-2i \bar{\mathcal{H}}^{-1}$, gives

$$\Pi E^{-1} \Pi^t = 0 \quad \text{and} \quad i \Pi E^{-1} \bar{\Pi}^t = i(-2i \bar{\mathcal{H}}^{-1}) = 2 \bar{\mathcal{H}}^{-1}$$

Since $\mathcal{H}$ is positive definite, so is $\mathcal{H}^{-1}$, and so is its conjugate $\bar{\mathcal{H}}^{-1}$; hence $i \Pi E^{-1} \bar{\Pi}^t = 2 \bar{\mathcal{H}}^{-1} > 0$. Both relations of (3.1) hold, establishing (1) $\Rightarrow$ (2).

Step 2: From the period relations to a polarization. Conversely, suppose $E \in M_{2g}(\mathbb{Z})$ is nondegenerate alternating integral and the two relations (3.1) hold for Π . Define an $\mathbb{R}$ -bilinear alternating form on $V = \Lambda \otimes \mathbb{R}$ by declaring $E(\lambda_j, \lambda_k)$ to be the (j, k) entry of E and extending $\mathbb{R}$ -bilinearly; it is integral on Λ by construction. Set

$$H(v, w) := E(iv, w) + iE(v, w)$$

Two things must be checked: that H is a Hermitian form (which requires the compatibility $E(iv, iw) = E(v, w)$), and that it is positive definite. Both follow from (3.1).

For the compatibility, the relation $\Pi E^{-1} \Pi^t = 0$ is exactly the vanishing of the top-left block of $\hat{\Pi} E^{-1} \hat{\Pi}^t$; conjugating the whole matrix identity, the bottom-right block $\bar{\Pi} E^{-1} \bar{\Pi}^t$ vanishes as well. Writing $P = \Pi E^{-1} \bar{\Pi}^t$ for the top-right block and using that $\hat{\Pi} E^{-1} \hat{\Pi}^t$ is alternating (its bottom-left block is therefore $-P^t$), the big matrix is block-antidiagonal:

$$\hat{\Pi} E^{-1} \hat{\Pi}^t = \begin{pmatrix} 0 & P \\ -P^t & 0 \end{pmatrix} \quad (3.4)$$

This block-antidiagonality is equivalent to the compatibility $E(iv, iw) = E(v, w)$. Indeed, the computation of Step 1 is reversible: for any real alternating form E on $V = \mathbb{R}^{2g}$, let $\mathcal{E}$ denote its matrix in the standard real coordinates and J the matrix of multiplication by i ; then a direct expansion of $\hat{\Pi} E^{-1} \hat{\Pi}^t$ shows its top-left block vanishes if and only if $J^t \mathcal{E} J = \mathcal{E}$, which is the coordinate form of $E(iv, iw) = E(v, w)$. When $E = \text{Im} H$ for a Hermitian H this compatibility holds automatically, giving (3.4) with $P = -2i\bar{\mathcal{H}}^{-1}$ as in Step 1; conversely, (3.4) forces $J^t \mathcal{E} J = \mathcal{E}$, so the E furnished by the Riemann relations is compatible. Granting $E(iv, iw) = E(v, w)$, the form $H(v, w) = E(iv, w) + iE(v, w)$ is Hermitian. It is $\mathbb{R}$ -bilinear by construction, and $\mathbb{C}$ -linear in the first variable because

$$H(iv, w) = E(-v, w) + iE(iv, w) = -E(v, w) + iE(iv, w) = i(E(iv, w) + iE(v, w)) = iH(v, w)$$

Conjugate symmetry $H(w, v) = \overline{H(v, w)}$ splits into two pieces. On imaginary parts, $\text{Im} H(w, v) = E(w, v) = -E(v, w) = -\text{Im} H(v, w)$ because E is alternating. On real parts, $\text{Re} H(w, v) = E(iw, v)$ must equal $\text{Re} H(v, w) = E(iv, w)$, and this follows from the compatibility applied to the pair (iw, v) : $E(iw, v) = E(i \cdot iw, iv) = E(-w, iv) = -E(w, iv) = E(iv, w)$.

For positive definiteness, the second relation is decisive. The identity $i \Pi E^{-1} \bar{\Pi}^t = 2\bar{\mathcal{H}}^{-1}$ derived in Step 1 holds as a formal consequence of (3.2) and (3.3) once H is Hermitian, where now $\mathcal{H}$ is the matrix of the reconstructed $H(v, w) = v^t \mathcal{H} \bar{w}$; reading it backwards, the hypothesis $i \Pi E^{-1} \bar{\Pi}^t > 0$ forces $\bar{\mathcal{H}}^{-1} > 0$, hence $\mathcal{H} > 0$, hence $H > 0$. Thus H is a polarization, establishing (2) $\Rightarrow$ (1) and completing the equivalence of the two analytic conditions.

Step 3: A polarization yields a projective embedding. Suppose X carries a polarization H . By theorem 2.2.8 there is a semicharacter χ making $L = L(H, \chi)$ a line bundle with $c_1(L) = H$, and since $H > 0$ the bundle L is positive. The sections of a high tensor power of a positive line bundle are the Riemann theta functions, and they separate points and tangent directions on X . This is the Lefschetz embedding theorem, proved in Chapter 4: for a positive line bundle L on a complex torus, $L^{\otimes 3}$ is very ample, so the associated map $X \rightarrow \mathbb{P}^N$ sending a point to the ratios of a basis of $H^0(X, L^{\otimes 3})$ is a holomorphic embedding. The construction of the sections, and the verification that they separate points, is the content of the theta-function machinery of that

chapter, so the argument is deferred to it; what the present theorem contributes is the criterion, that a positive H exists, which is the input the embedding theorem consumes. Granting Chapter 4, a polarizable torus embeds in projective space and is therefore an abelian variety.

Step 4: A projective embedding yields a polarization. Suppose conversely that $X \hookrightarrow \mathbb{P}^N$ is a holomorphic embedding. The restriction to X of the hyperplane bundle $\mathcal{O}_{\mathbb{P}^N}(1)$ is a very ample line bundle L on X , and its first Chern class $c_1(L) \in H^2(X, \mathbb{Z})$ is the restriction of the Fubini-Study class, which is a positive $(1, 1)$ -form on $\mathbb{P}^N$. Under the identification of $\text{NS}(X)$ with Hermitian forms (definition 2.3.1 and proposition 2.3.2), a positive $(1, 1)$ -class corresponds to a positive-definite Hermitian form H , whose imaginary part $E = \text{Im}H = c_1(L)$ is the integral class; positivity of the Fubini-Study form restricts to positivity along the complex submanifold X , so $H > 0$, and $E = \text{Im}H$ is integral because $c_1(L)$ is an integral cohomology class. Hence H is a polarization, and X is polarizable. This closes the circle: polarizability and projective embeddability are the same condition, as we sought to show.

The theorem is the structural dividing line of the analytic theory. Every complex torus is a compact complex Lie group and a differentiable object of the utmost regularity, yet only the polarizable ones are algebraic, and (3.1) measures exactly the failure of the rest. The first relation $\Pi E^{-1} \Pi^t = 0$ is a system of $\binom{g}{2}$ complex equations on the entries of Π , cutting out a proper analytic subvariety of the space of all period matrices; the positivity is an open condition on the complement. For $g = 1$ the first relation is vacuous, one antisymmetric equation among a single pair of indices contributes nothing, which is why every one-dimensional torus is an elliptic curve; for $g \geq 2$ the relations are nontrivial constraints, and a torus with a generic period matrix satisfies none of them. The precise mechanism of this genericity is examined in the worked contrast at the end of the section.

The passage from the intrinsic Hermitian condition to the two period relations is worth isolating as a statement in its own right, because it is the form in which the Riemann relations are used in computation. The proposition below is the coordinate incarnation of theorem 3.1.4, extracted from the proof and recorded with the explicit dictionary between the Hermitian matrix and the periods.

Proposition 3.1.5: The Period-Matrix Riemann Relations

Let $X = V/\Lambda$ have period matrix $\Pi \in M_{g \times 2g}(\mathbb{C})$ with respect to a $\mathbb{Z}$ -basis $\lambda_1, \dots, \lambda_{2g}$ of Λ , and let $E \in M_{2g}(\mathbb{Z})$ be a nondegenerate alternating integral matrix. Then E is the imaginary part on Λ of a Hermitian form H on V if and only if

$$\Pi E^{-1} \Pi^t = 0$$

and in that case H is positive definite if and only if

$$i \Pi E^{-1} \bar{\Pi}^t > 0$$

The Hermitian matrix is recovered from the periods by $\mathcal{H} = 2(\overline{i \Pi E^{-1} \bar{\Pi}^t})^{-1}$, where $H(v, w) = v^t \mathcal{H} \bar{w}$.

Proof:

The first relation is equivalent to the existence of a compatible H , and the second to its positivity; both equivalences were established in Steps 1 and 2 of theorem 3.1.4, and it remains only to record the reconstruction formula. From the identity $i \Pi E^{-1} \bar{\Pi}^t = 2 \mathcal{H}^{-1}$ obtained in (3.3), solve

for $\mathcal{H}$: taking inverses gives $\bar{\mathcal{H}} = 2(i \Pi E^{-1} \bar{\Pi}^t)^{-1}$, and conjugating gives

$$\mathcal{H} = 2 \overline{(i \Pi E^{-1} \bar{\Pi}^t)^{-1}} = 2 (\overline{i \Pi E^{-1} \bar{\Pi}^t})^{-1}$$

which is the stated formula, completing the proof.

The utility of the proposition is that it turns the search for a polarization into linear algebra over $\mathbb{Z}$ followed by a positivity check over $\mathbb{R}$. One writes down Π , chooses a candidate integer alternating matrix E (for instance the standard symplectic form on the basis at hand), tests the algebraic identity $\Pi E^{-1} \bar{\Pi}^t = 0$, and if it holds, forms the Hermitian matrix $i \Pi E^{-1} \bar{\Pi}^t$ and checks that it is positive definite. The reconstruction formula shows that no information is lost in passing between H and the pair (Π, E) : the Hermitian form is a rational, indeed polynomial-in-the-inverse, function of the periods and the polarizing form. This is the calculation performed for the elliptic curve next, and it is the calculation that fails, provably, for the transcendental torus discussed afterward.

Example 3.1.6: The Riemann Relations for an Elliptic Curve

Take $g = 1$, $V = \mathbb{C}$, and $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$ with $\tau \in \mathbb{H}$, so that the period matrix on the basis $\lambda_1 = \tau$, $\lambda_2 = 1$ is

$$\Pi = \begin{pmatrix} \tau & 1 \end{pmatrix} \in M_{1 \times 2}(\mathbb{C})$$

Choose the standard symplectic integer matrix

$$E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad E^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

nondegenerate and alternating over $\mathbb{Z}$. Test the first Riemann relation:

$$\Pi E^{-1} \bar{\Pi}^t = \begin{pmatrix} \tau & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \bar{\tau} \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -\tau \end{pmatrix} \begin{pmatrix} \bar{\tau} \\ 1 \end{pmatrix} = \tau - \tau = 0$$

so the algebraic relation holds identically, for every τ ; this is the reason there is no obstruction in dimension one. Now the positivity. Compute

$$i \Pi E^{-1} \bar{\Pi}^t = i \begin{pmatrix} 1 & -\tau \end{pmatrix} \begin{pmatrix} \bar{\tau} \\ 1 \end{pmatrix} = i(\bar{\tau} - \tau) = i(-2i \operatorname{Im} \tau) = 2 \operatorname{Im} \tau$$

which is a positive real number because $\tau \in \mathbb{H}$. Both Riemann relations hold, so E is the imaginary part of a positive Hermitian form, and $X = \mathbb{C}/\Lambda$ is an abelian variety. The reconstruction formula gives $\mathcal{H} = 2 \overline{(2 \operatorname{Im} \tau)}^{-1} = 1/\operatorname{Im} \tau$, so

$$H(v, w) = \frac{v \bar{w}}{\operatorname{Im} \tau}$$

recovering the positive form found by hand in example 3.1.2. Every elliptic curve is therefore an abelian variety, and the polarization it carries is principal (its type is (1), since E has determinant 1). This is the $g = 1$ instance of the general theory, and it succeeds for every τ precisely because the single algebraic relation degenerates to the tautology $\tau - \tau = 0$.

The contrast with higher dimension is the point of the whole development. When $g \geq 2$ the first Riemann relation is a nonempty system, and for a lattice with generic periods it has no integral solution at all. This is not an abstract possibility but an explicit phenomenon, already exhibited in section 1.4: the two-dimensional torus of example 1.4.3, built from a period matrix whose entries are algebraically independent transcendentals, admits no nonzero integral Hermitian form, hence no polarization, hence no projective embedding, and in fact has no nonconstant meromorphic functions

whatsoever. Set against example 3.1.6, the two examples bracket the theorem exactly. The elliptic curve is the case where the algebraic relation is vacuous and a polarization always exists; the generic torus of example 1.4.3 is the case where the relation is a genuine constraint and no polarization exists. The Riemann relations are the wall between them, the condition an integral alternating form must satisfy against the periods before it can be the shadow of a positive line bundle, and they are the reason that among all complex tori of dimension $g \geq 2$, the abelian varieties form a countable union of proper subvarieties of the moduli of tori rather than the whole space. Everything that follows in this book concerns the tori on the algebraic side of that wall.

3.2 The Definition of an Abelian Variety

The Riemann relations of section 3.1 single out, among all complex tori, exactly those that carry a positive Hermitian form integral on the lattice. This condition is the analytic gate to projective space: a torus on the positive side of the gate embeds as a closed algebraic subvariety of some $\mathbb{P}^N$, while a torus on the wrong side supports too few meromorphic functions to embed anywhere. The term *abelian variety* names the objects on the good side. This section fixes that definition, records the two numerical invariants a polarization carries (its type and, in the extremal case, its principality), and states the equivalence with projectivity whose forward half is the Lefschetz embedding theorem of Chapter 4.

A caution on terminology precedes the definition. Chapter 1 introduced a *provisional* notion (definition 1.4.1): a complex torus that happens to admit some holomorphic embedding into projective space. That definition is geometrically transparent but analytically opaque, since it quantifies over embeddings one cannot see. The polarization reformulates the same class of tori in terms of a single piece of linear-algebraic data on V and Λ , a positive Hermitian form, and it is this reformulation that makes the class computable. The two descriptions agree, and theorem 3.1.4 is precisely the bridge between them.

Definition 3.2.1: Abelian Variety

An *abelian variety* (over $\mathbb{C}$) is a complex torus $X = V/\Lambda$ that admits a polarization, that is, a positive definite Hermitian form $H > 0$ on V whose imaginary part $E = \text{Im}H$ takes integer values on Λ (definition 3.1.3). A *polarized abelian variety* is a pair (X, H) consisting of an abelian variety together with a choice of such a form.

This refines definition 1.4.1. The provisional definition asked for a holomorphic embedding $X \hookrightarrow \mathbb{P}^N$; definition 3.2.1 asks instead for a polarization. That the two requirements pick out the same tori is the content of the Riemann relations together with the embedding theorem: theorem 3.1.4 supplies the polarization from the embedding by pulling back the Fubini-Study form, and the Lefschetz theorem of Chapter 4 supplies the embedding from the polarization by constructing enough theta functions. From now on “abelian variety” means definition 3.2.1, and the provisional notion is retired.

The definition is a condition on X alone, not on a chosen H : an abelian variety is a torus for which *at least one* polarization exists, and different polarizations on the same torus are distinct geometric structures that cut out different projective embeddings. This is why the polarized pair (X, H) is recorded as a separate object. The bare torus underlying an abelian variety may carry many inequivalent polarizations, and the moduli problem of section 3.3 is sensitive to the choice.

Example 3.2.2: The Elliptic Curve as an Abelian Variety

Take $g = 1$, so $X = E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ and $\tau \in \mathbb{H}$. A Hermitian form on $V = \mathbb{C}$ is $H(v, w) = cv\bar{w}$ for a real scalar c , and $H > 0$ means $c > 0$. Its imaginary part is $E(v, w) = \operatorname{Im}H(v, w) = c\operatorname{Im}(v\bar{w})$, an alternating $\mathbb{R}$ -form on $\mathbb{C}$. Integrality on Λ requires $E(1, \tau) \in \mathbb{Z}$ and $E(\tau, 1) \in \mathbb{Z}$; since $E(1, \tau) = c\operatorname{Im}(\bar{\tau}) = -c\operatorname{Im}\tau$ and $E(\tau, 1) = c\operatorname{Im}\tau$, and the remaining pairings $E(1, 1) = E(\tau, \tau) = 0$ vanish, integrality amounts to the single condition $c\operatorname{Im}\tau \in \mathbb{Z}$. Choosing $c = (\operatorname{Im}\tau)^{-1}$ gives $E(\tau, 1) = 1$, a positive Hermitian form integral on Λ . Every one-dimensional complex torus therefore admits a polarization, so every elliptic curve is an abelian variety, matching proposition 1.4.2: in dimension one the gate is always open.

The one-dimensional case makes the structure visible in miniature: a polarization is a positive multiple of the natural symplectic form on $\mathbb{C}$, normalized so that the fundamental parallelogram has integer area against E . In higher dimensions the same picture holds pointwise on V , but the integrality of E on a rank- $2g$ lattice is a strong constraint that most tori fail, which is why the class of abelian varieties is a strict subclass of all complex tori for $g \geq 2$ (as the transcendental torus of example 1.4.3 showed).

A polarization carries more information than the mere fact of its existence. The alternating integral form $E = \operatorname{Im}H$ has, like any nondegenerate alternating form over $\mathbb{Z}$, a normal form under change of symplectic basis, and the diagonal of that normal form is an invariant of E that measures how far the polarization is from being “as tight as possible.” The elementary-divisor theorem supplies the normal form, and its diagonal is the type.

Definition 3.2.3: Type of a Polarization

Let H be a polarization of $X = V/\Lambda$, with $E = \operatorname{Im}H$ the associated nondegenerate alternating form, integral on Λ . By the elementary-divisor theorem ([7]) there is a *symplectic basis* $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ of Λ in which the Gram matrix of E has the block form

$$\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}, \quad D = \operatorname{diag}(d_1, \dots, d_g)$$

with positive integers $d_1 | d_2 | \dots | d_g$. The tuple $(d_1, \dots, d_g)$ is the *type* of the polarization; it is independent of the chosen basis.

The type reads off directly from the elementary divisors of the alternating form E on the free $\mathbb{Z}$ -module Λ . Concretely, E is given by an antisymmetric integer matrix once a $\mathbb{Z}$ -basis of Λ is fixed; Smith normal form for antisymmetric integer matrices, which is the symplectic refinement of the elementary-divisor theorem, brings this matrix to the block form above by an integral symplectic change of basis. The invariance of $(d_1, \dots, d_g)$ is the invariance of elementary divisors: $d_1 \cdots d_k$ is the greatest common divisor of the $2k \times 2k$ Pfaffian minors of E , hence a function of E and Λ alone, not of the basis realizing the normal form. Because $H > 0$ forces E to be nondegenerate, all $d_i \geq 1$, and $\det E = (d_1 \cdots d_g)^2$, so the Pfaffian $\operatorname{Pf}(E) = d_1 \cdots d_g$ is the product of the type, a number that will reappear in section 3.4 as the dimension of the space of theta functions.

The extremal case is the one where the type is as small as it can be.

Definition 3.2.4: Principal Polarization

A polarization H of X is *principal* if its type is $(1, \dots, 1)$, that is, if the alternating form $E = \text{Im}H$ admits a symplectic basis of Λ in which its Gram matrix is $\begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$. A *principally polarized abelian variety* (a *ppav*) is a pair (X, H) with H a principal polarization.

A principal polarization is one for which E is *unimodular* on Λ , that is, $\det E = 1$ and $\text{Pf}(E) = 1$. Equivalently, E induces an isomorphism $\Lambda \xrightarrow{\sim} \Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ rather than an injection with finite cokernel; the cokernel $\Lambda^\vee/E(\Lambda)$ has order $(d_1 \cdots d_g)^2 = \det E$ in general and is trivial exactly for a principal polarization. Principal polarizations are the ones that behave most simply under duality and are the source of the canonical examples: the theta divisor of a ppav is unique up to translation (section 3.4 gives $\dim H^0(X, L) = 1$ in this case), and the Jacobian of a curve is canonically a ppav. Not every abelian variety admits a principal polarization, but every abelian variety is isogenous to one that does, a point that belongs to the later theory of isogenies.

Example 3.2.5: Types on the Square Torus

Let $g = 1$ and $X = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}i)$, the square torus. The polarization $H(v, w) = v\bar{w}$ has $E(v, w) = \text{Im}(v\bar{w})$. Take the symplectic basis $\lambda_1 = i, \mu_1 = 1$; then $E(\lambda_1, \mu_1) = E(i, 1) = \text{Im}(i \cdot 1) = \text{Im}(i) = 1$, while $E(\lambda_1, \lambda_1) = E(\mu_1, \mu_1) = 0$, so the Gram matrix in the ordered basis (λ_1, μ_1) is $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and the type is (1): this is a principal polarization. Scaling the form to $H'(v, w) = 3v\bar{w}$ multiplies E by 3, giving Gram matrix $\begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$ and type (3). The bundle L' with $c_1(L') = E' = 3E$ is still positive and still integral, hence still a polarization, but a non-principal one; by the Riemann-Roch count of section 3.4 it will carry $\text{Pf}(E') = 3$ independent theta functions rather than one. In dimension one the type is a single integer, the degree of the associated line bundle.

For $g > 1$ the type records g separate divisibilities. A two-dimensional abelian variety can carry a polarization of type (1, 1) (principal), or of type (1, d) with $d > 1$, or of type (d_1, d_2) with $1 < d_1 | d_2$; these are distinct geometric structures on the same or different tori, and the moduli of section 3.3 are organized type by type. The type is thus the first place where the higher-dimensional theory diverges from the elliptic-curve case, where a polarization is determined up to scale by a single degree.

With the definition and its invariants in place, the equivalence that motivates the whole chapter can be stated cleanly. It ties three conditions together: the analytic condition (a polarization exists), the Riemann-relation condition on the period matrix, and the geometric condition (the torus is projective).

Theorem 3.2.6: Polarizable if and only if Projective

Let $X = V/\Lambda$ be a complex torus. The following are equivalent.

1. X is an abelian variety, that is, X admits a polarization $H > 0$ with $E = \text{Im}H$ integral on Λ .
2. The period matrix Π of X satisfies the Riemann relations: there is a nondegenerate alternating integral matrix $E \in M_{2g}(\mathbb{Z})$ with $\Pi E^{-1} \Pi^t = 0$ and $i \Pi E^{-1} \bar{\Pi}^t > 0$.
3. X admits a closed holomorphic embedding into some projective space $\mathbb{P}^N$, and its image is a smooth projective algebraic variety.

Proof:

The equivalence of (1) and (2) is theorem 3.1.4: statement (2) is the coordinate incarnation of the positive integral Hermitian form in (1), obtained by writing H against the period matrix Π , and proposition 3.1.5 records the translation in both directions. Nothing further is needed here.

Projective implies polarizable, (3) $\Rightarrow$ (1). Suppose $X \hookrightarrow \mathbb{P}^N$ is a closed holomorphic embedding. The Fubini-Study Kähler form ω_{FS} on $\mathbb{P}^N$ restricts to a closed positive $(1, 1)$ -form on X whose cohomology class lies in $H^2(X, \mathbb{Z})$, being the restriction of the integral hyperplane class. Under the identification $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$ of proposition 1.3.5, this class is an alternating integral form E on Λ ; positivity of ω_{FS} makes the associated Hermitian form $H(v, w) = E(iv, w) + iE(v, w)$ positive definite, and the compatibility $E(iv, iw) = E(v, w)$ is automatic because ω_{FS} is of type $(1, 1)$ for the complex structure. Thus H is a polarization and X is an abelian variety.

Polarizable implies projective, (1) $\Rightarrow$ (3). This is the Lefschetz embedding theorem, proved in full in Chapter 4. Given a polarization H , the associated positive line bundle $L = L(H, \chi)$ (definition 3.1.1) has, for $n \geq 3$, enough global theta functions that the map $X \rightarrow \mathbb{P}^N$ sending x to the ratios of a basis of $H^0(X, L^{\otimes n})$ is a closed holomorphic embedding; that $L^{\otimes 3}$ is already very ample is the precise statement. The construction of the theta functions and the verification that they separate points and tangent vectors is carried out in Chapter 4, so the embedding is deferred to there rather than reproduced now. Chow's theorem then upgrades the analytic image to a projective algebraic variety. Combining the two implications with the equivalence of (1) and (2) completes the cycle, as we sought to show.

The theorem is the analytic definition of an abelian variety made precise and useful. Condition (1) is what one checks in practice, since it is a finite linear-algebra problem on V and Λ ; condition (2) is what one computes from an explicit period matrix; and condition (3) is what the name “variety” promises. The only piece not proved here is the passage from a polarization to an actual embedding, which requires the theta functions of Chapter 4 and is the reason polarizations were worth isolating in the first place. Everything downstream, the moduli space of section 3.3 and the Riemann-Roch count of section 3.4, works with condition (1) and never needs to see the embedding directly.

The class of abelian varieties is closed under the operations one expects of a good class of geometric objects, and the simplest of these is the product. Products let one build higher-dimensional examples from lower ones and, in the principally polarized case, preserve the principality.

Example 3.2.7: Products of Principally Polarized Abelian Varieties

Let (X_1, H_1) and (X_2, H_2) be abelian varieties with $X_j = V_j/\Lambda_j$ and polarizations H_j . The product torus is $X_1 \times X_2 = (V_1 \oplus V_2)/(\Lambda_1 \oplus \Lambda_2)$, and the form

$$H(v_1 + v_2, w_1 + w_2) = H_1(v_1, w_1) + H_2(v_2, w_2)$$

is Hermitian on $V_1 \oplus V_2$. It is positive definite because each H_j is, and a sum of positive forms on complementary summands is positive on the direct sum. Its imaginary part $E = E_1 \oplus E_2$ is integral on $\Lambda_1 \oplus \Lambda_2$ because each E_j is integral on Λ_j and the cross terms vanish. Thus H is a polarization and $X_1 \times X_2$ is an abelian variety.

Suppose now that both H_1 and H_2 are principal. Choose symplectic bases of Λ_1 and Λ_2 in which E_1 and E_2 have Gram matrices $\begin{pmatrix} 0 & I_{g_1} \\ -I_{g_1} & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & I_{g_2} \\ -I_{g_2} & 0 \end{pmatrix}$. Concatenating the two λ -blocks and

the two μ -blocks produces a symplectic basis of $\Lambda_1 \oplus \Lambda_2$ in which $E = E_1 \oplus E_2$ has Gram matrix $\begin{pmatrix} 0 & I_{g_1+g_2} \\ -I_{g_1+g_2} & 0 \end{pmatrix}$, so the type of H is $(1, \dots, 1)$. A product of principally polarized abelian varieties is therefore principally polarized. In particular $E \times E$, with the product of the standard principal polarization on each factor, is a principally polarized abelian surface.

The product construction shows that principal polarizations are stable under the most basic building operation, but it also shows that not every ppav is indecomposable: $E \times E$ is a principally polarized surface that splits as a product. The indecomposable principally polarized abelian surfaces, the ones that do not split as a product of two elliptic curves, are the Jacobians of genus-two curves, and this is the first instance of the single most important source of principally polarized abelian varieties in the subject.

Example 3.2.8: The Canonical Polarization of a Jacobian

Let C be a smooth projective curve of genus $g \geq 1$ over $\mathbb{C}$. Its Jacobian is the complex torus

$$\text{Jac}(C) = H^0(C, \Omega_C^1)^* / H_1(C, \mathbb{Z})$$

of dimension g , where $H_1(C, \mathbb{Z})$ embeds in the dual of the space of holomorphic 1-forms by integration, $\gamma \mapsto (\omega \mapsto \int_\gamma \omega)$. The cup-product intersection pairing on $H_1(C, \mathbb{Z})$ is a nondegenerate alternating integral form, and with respect to a symplectic basis of 1-cycles (an a_i, b_i basis with $a_i \cdot b_j = \delta_{ij}$ and $a_i \cdot a_j = b_i \cdot b_j = 0$) its Gram matrix is exactly $\begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$. This form is the imaginary part of a positive Hermitian form on $H^0(C, \Omega_C^1)^*$, the one coming from the positivity of the Hodge inner product $\frac{i}{2} \int_C \omega \wedge \bar{\eta}$; the positivity is the Riemann bilinear relations for the curve C . The resulting polarization is principal by construction, since the intersection form is unimodular, and it depends on nothing but C . Every curve thus manufactures a principally polarized abelian variety of dimension g , canonically and with no choices.

The Jacobian construction is the reason principal polarizations are central rather than exceptional: they are the polarizations that arise from geometry without any auxiliary data, and the map sending a curve to its Jacobian is the bridge between the theory of curves and the theory of abelian varieties. That this map is close to injective, so that a curve can be recovered from its principally polarized Jacobian, is the Torelli theorem, and the full algebraic construction of the Jacobian together with its canonical polarization occupies Chapter 12. The analytic version sketched here already shows the mechanism: the intersection form on $H_1(C, \mathbb{Z})$ is unimodular for topological reasons, and the Hodge metric supplies the positivity, so the two ingredients of a principal polarization are both present the moment one writes down the Jacobian.

The word “abelian variety” now has a precise analytic meaning. It is a complex torus carrying a positive integral Hermitian form, equivalently a torus whose period matrix satisfies the Riemann relations, equivalently a projective complex torus. All three descriptions are of the same objects, and the equivalence is the analytic heart of Part I. The definition is, however, tied to $\mathbb{C}$ at every step: it invokes a Hermitian form on a complex vector space, a period matrix, and holomorphic embeddings, none of which survive the passage to a general field. Part II gives the field-independent algebraic definition, an abelian variety as a complete connected group variety over an arbitrary field, in which projectivity is a theorem rather than a hypothesis and no metric or transcendental input appears. That the two definitions agree over $\mathbb{C}$, so that the analytic abelian varieties of this chapter are exactly the complex points of the algebraic ones, is the comparison theorem taken up in Chapter 11, where the analytic period data of Part I is matched against the algebraic structure of Part II.

3.3 The Period Matrix, the Siegel Upper Half-Space, and Moduli

The Riemann relations decide, for a single torus, whether it is an abelian variety. Once that question is settled, a second one takes its place: the enumeration and organization of all abelian varieties of a given dimension. For an elliptic curve the answer is already familiar from [3]. A lattice in $\mathbb{C}$ normalizes to $\mathbb{Z}\tau + \mathbb{Z}$ with τ in the upper half-plane, two such lattices give isomorphic curves exactly when their parameters are related by $\mathrm{SL}_2(\mathbb{Z})$, and the quotient $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ is the moduli space of elliptic curves, uniformized by the j -invariant. This section carries out the same program in dimension g . The role of the upper half-plane is played by the Siegel upper half-space $\mathfrak{H}_g$, the role of $\mathrm{SL}_2(\mathbb{Z})$ by the symplectic group $\mathrm{Sp}(2g, \mathbb{Z})$, and the quotient $\mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z})$ is the first picture of the moduli space of principally polarized abelian varieties of dimension g .

The mechanism is a normalization of the period matrix. A principal polarization supplies a symplectic basis of the lattice (definition 3.2.3), and in a complex basis adapted to it the period matrix collapses to the compact form (τI_g) for a single $g \times g$ matrix τ . The Riemann relations, which in a general basis are a pair of bilinear conditions on the full period matrix, become the two transparent conditions that τ is symmetric and has positive definite imaginary part. That is the definition of $\mathfrak{H}_g$, so every principally polarized abelian variety is a point of $\mathfrak{H}_g$, and the ambiguity in the construction (the freedom in the choice of symplectic basis) is precisely the $\mathrm{Sp}(2g, \mathbb{Z})$ -action.

Definition 3.3.1: Siegel Upper Half-Space

The *Siegel upper half-space* of genus g is

$$\mathfrak{H}_g = \{ \tau \in M_g(\mathbb{C}) \mid \tau = \tau^t, \operatorname{Im} \tau > 0 \}$$

the set of symmetric complex $g \times g$ matrices whose imaginary part $\operatorname{Im} \tau = \frac{1}{2i}(\tau - \bar{\tau})$ is a positive definite real symmetric matrix. It is an open subset of the $\mathbb{C}$ -vector space $\operatorname{Sym}^2 \mathbb{C}^g$ of symmetric $g \times g$ matrices, hence a complex manifold of dimension

$$\dim_{\mathbb{C}} \mathfrak{H}_g = \frac{g(g+1)}{2}$$

the number of entries on and above the diagonal of a symmetric matrix.

The condition $\tau = \tau^t$ cuts the symmetric matrices out of all of $M_g(\mathbb{C})$, a linear condition leaving the $g(g+1)/2$ independent entries. The condition $\operatorname{Im} \tau > 0$ is open and nonempty, so it selects an open region rather than a lower-dimensional slice, and the dimension is unchanged. For $g = 1$ a 1×1 matrix is automatically symmetric and $\operatorname{Im} \tau > 0$ is the single inequality defining the upper half-plane, so $\mathfrak{H}_1 = \mathbb{H}$ and $\dim_{\mathbb{C}} \mathfrak{H}_1 = 1$; the Siegel upper half-space is the exact multivariable analogue of the domain that uniformizes elliptic curves. The positivity of $\operatorname{Im} \tau$ is the shadow of the positivity of the polarization: it is what will make the Hermitian form H come out positive definite, and it is what makes $\mathfrak{H}_g$ a bounded symmetric domain rather than all of $\operatorname{Sym}^2 \mathbb{C}^g$.

Example 3.3.2: The Siegel Upper Half-Space in Genus Two

For $g = 2$ a point of $\mathfrak{H}_2$ is a symmetric matrix

$$\tau = \begin{pmatrix} \tau_{11} & \tau_{12} \\ \tau_{12} & \tau_{22} \end{pmatrix} \quad \tau_{ij} \in \mathbb{C}$$

with three free entries $\tau_{11}, \tau_{12}, \tau_{22}$, so $\dim_{\mathbb{C}} \mathfrak{H}_2 = 3 = \frac{2 \cdot 3}{2}$. Writing $\tau_{ij} = x_{ij} + iy_{ij}$, the positivity condition $\text{Im} \tau > 0$ reads

$$y_{11} > 0 \quad \text{and} \quad y_{11}y_{22} - y_{12}^2 > 0$$

the leading-principal-minor (Sylvester) criterion for the symmetric real matrix $\text{Im} \tau = \begin{pmatrix} y_{11} & y_{12} \\ y_{12} & y_{22} \end{pmatrix}$ to be positive definite. The diagonal points $\tau = \text{diag}(\tau_1, \tau_2)$ with $\tau_1, \tau_2 \in \mathbb{H}$ form a copy of $\mathbb{H} \times \mathbb{H}$ inside $\mathfrak{H}_2$; these are the period matrices of the products $E_1 \times E_2$ of two elliptic curves, and the off-diagonal entry τ_{12} measures how far a principally polarized abelian surface is from splitting as such a product.

The example makes the parameter count concrete and displays the smallest case where the Siegel domain is more than a half-plane: a principally polarized abelian surface depends on three complex parameters, of which two record a pair of elliptic curves and the third records how they are glued. To turn a torus into such a point, one must first put its period matrix into normal form.

Recall the setup from section 3.1. A period matrix for $X = V/\Lambda$ is a matrix $\Pi \in M_{g \times 2g}(\mathbb{C})$ whose columns are the coordinates, in a fixed $\mathbb{C}$ -basis of V , of a $\mathbb{Z}$ -basis $\lambda_1, \dots, \lambda_{2g}$ of Λ . A principal polarization (definition 3.2.4) is a positive Hermitian form H whose imaginary part $E = \text{Im} H$ is a unimodular integral alternating form on Λ , and by definition 3.2.3 (with type $(1, \dots, 1)$) there is a symplectic basis $\mu_1, \dots, \mu_g, \nu_1, \dots, \nu_g$ of Λ in which E has matrix $\begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$. The next result says that using this basis, and one adapted change of complex coordinates, brings Π to the form $(\tau \ I_g)$ with $\tau \in \mathfrak{H}_g$, and that the Riemann relations become exactly the two conditions defining $\mathfrak{H}_g$.

Proposition 3.3.3: Normalized Period Matrix of a Principally Polarized Abelian Variety

Let $X = V/\Lambda$ be a principally polarized abelian variety of dimension g , with polarization H and $E = \text{Im} H$. Choose a symplectic basis $\mu_1, \dots, \mu_g, \nu_1, \dots, \nu_g$ of Λ for E , so that

$$E(\mu_i, \mu_j) = 0 \quad E(\nu_i, \nu_j) = 0 \quad E(\mu_i, \nu_j) = \delta_{ij}$$

Then $\nu_1, \dots, \nu_g$ is a $\mathbb{C}$ -basis of V . Expressing Λ in this basis, the period matrix takes the form $\Pi = (\tau \ I_g)$, where $\tau \in M_g(\mathbb{C})$ is the matrix whose j -th column is μ_j written in the basis $\nu_1, \dots, \nu_g$. Moreover $\tau \in \mathfrak{H}_g$, that is

$$\tau = \tau^t \quad \text{and} \quad \text{Im} \tau > 0$$

and these two conditions are precisely the Riemann relations for the period matrix $(\tau \ I_g)$ with respect to E . The lattice is $\Lambda = \tau \mathbb{Z}^g + \mathbb{Z}^g$.

Proof:

The argument has three steps: first that the ν_j form a complex basis, then the normalization of the period matrix, and finally the translation of the two Riemann relations into symmetry and positivity of τ .

Step 1: The vectors $\nu_1, \dots, \nu_g$ are a $\mathbb{C}$ -basis of V . Since Λ has rank $2g$ over $\mathbb{Z}$ and $V = \Lambda \otimes_{\mathbb{Z}} \mathbb{R}$ has real dimension $2g$, the $2g$ vectors μ_i, ν_j form an $\mathbb{R}$ -basis of V . It suffices to show that $\nu_1, \dots, \nu_g$ are linearly independent over $\mathbb{C}$, for then they span the g -dimensional complex space V . Suppose $\sum_j c_j \nu_j = 0$ with $c_j \in \mathbb{C}$; write $c_j = a_j + ib_j$ with $a_j, b_j \in \mathbb{R}$. Then

$$\sum_j a_j \nu_j + \sum_j b_j (i\nu_j) = 0$$

The ν_j are Lagrangian for E , so $E(\nu_j, \nu_k) = 0$ for all j, k ; and by the identity $\operatorname{Re} H(v, w) = E(iv, w)$ of section 3.1, together with the symmetry $\operatorname{Re} H(v, w) = \operatorname{Re} H(w, v)$,

$$E(i\nu_j, \nu_k) = \operatorname{Re} H(\nu_j, \nu_k) = \operatorname{Re} H(\nu_k, \nu_j) = E(i\nu_k, \nu_j)$$

The matrix S with entries $S_{jk} = E(i\nu_j, \nu_k) = \operatorname{Re} H(\nu_j, \nu_k)$ is therefore real symmetric, and it is positive definite because H is positive definite and $\operatorname{Re} H$ is its real part, whose restriction to the real span of the ν_j is positive definite (a positive definite Hermitian form has positive definite real part on any real subspace on which it is defined). Pairing the displayed relation against each ν_k under E gives $\sum_j b_j S_{jk} = 0$ for all k , hence $b = 0$ since S is invertible; the relation then reads $\sum_j a_j \nu_j = 0$ with real a_j , and pairing against $i\nu_k$ under E gives $\sum_j a_j S_{kj} = 0$, whence $a = 0$. Thus $c = 0$ and the ν_j are $\mathbb{C}$ -linearly independent, as we sought to show.

Step 2: The period matrix is (τI_g) . Adopt $\nu_1, \dots, \nu_g$ as the $\mathbb{C}$ -basis of V . In these coordinates ν_j is the j -th standard basis vector, so the block of the period matrix recording $\nu_1, \dots, \nu_g$ is the identity I_g . Let $\tau \in M_g(\mathbb{C})$ be the matrix whose j -th column is the coordinate vector of μ_j in the basis $\nu_1, \dots, \nu_g$, so that $\mu_j = \sum_k \tau_{kj} \nu_k$. Ordering the lattice basis as $\mu_1, \dots, \mu_g, \nu_1, \dots, \nu_g$, the period matrix is

$$\Pi = (\tau I_g)$$

and the lattice is $\Lambda = \tau \mathbb{Z}^g + \mathbb{Z}^g$, the $\mathbb{Z}$ -span of the columns of τ together with the standard basis vectors. This is the claimed normal form.

Step 3: The Riemann relations become $\tau = \tau^t$ and $\operatorname{Im} \tau > 0$. In the symplectic basis E has matrix $J = \begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$, with inverse $J^{-1} = \begin{pmatrix} 0 & -I_g \\ I_g & 0 \end{pmatrix}$. The Riemann relations for the period matrix $\Pi = (\tau I_g)$ with respect to E (proposition 3.1.5) are

$$\Pi J^{-1} \Pi^t = 0 \quad \text{and} \quad i \Pi J^{-1} \bar{\Pi}^t > 0$$

Compute the first product. Writing $\Pi = (\tau I_g)$ in blocks,

$$\Pi J^{-1} = (\tau I_g) \begin{pmatrix} 0 & -I_g \\ I_g & 0 \end{pmatrix} = (I_g \quad -\tau)$$

so that

$$\Pi J^{-1} \Pi^t = (I_g \quad -\tau) \begin{pmatrix} \tau^t \\ I_g \end{pmatrix} = \tau^t - \tau$$

The first Riemann relation $\Pi J^{-1} \Pi^t = 0$ is therefore equivalent to $\tau = \tau^t$: the matrix τ is symmetric. For the second, using $\bar{\Pi} = (\bar{\tau} \ I_g)$,

$$i \Pi J^{-1} \bar{\Pi}^t = i (I_g - \tau) \begin{pmatrix} \bar{\tau}^t \\ I_g \end{pmatrix} = i(\bar{\tau}^t - \tau)$$

Since τ is symmetric, $\bar{\tau}^t = \bar{\tau}$, and

$$i(\bar{\tau} - \tau) = i(-2i \operatorname{Im} \tau) = 2 \operatorname{Im} \tau$$

The second Riemann relation $i \Pi J^{-1} \bar{\Pi}^t > 0$ is thus equivalent to $\operatorname{Im} \tau > 0$. Both relations hold because H is a polarization, so τ is symmetric with positive definite imaginary part, that is $\tau \in \mathfrak{H}_g$, completing the proof.

The proposition is the structural heart of the section. It converts a principally polarized abelian variety, an object presented by a full $g \times 2g$ complex period matrix subject to two matrix identities, into a single point τ of the domain $\mathfrak{H}_g$, with the two identities absorbed into the definition of the domain itself. Nothing has been lost: from τ one recovers the lattice $\Lambda = \tau \mathbb{Z}^g + \mathbb{Z}^g$, hence the torus $X = \mathbb{C}^g / \Lambda$, and the Hermitian form $H(v, w) = v^t (\operatorname{Im} \tau)^{-1} \bar{w}$ whose imaginary part is unimodular integral on Λ and which is positive definite exactly because $\operatorname{Im} \tau > 0$; this is the polarization back again. So the map from (symplectic basis)-marked principally polarized abelian varieties to $\mathfrak{H}_g$ is a bijection. The freedom in the marking is what remains to be quotiented out.

Example 3.3.4: The Product of Two Elliptic Curves as a Period Matrix

Let $E_1 = \mathbb{C}/(\mathbb{Z}\tau_1 + \mathbb{Z})$ and $E_2 = \mathbb{C}/(\mathbb{Z}\tau_2 + \mathbb{Z})$ with $\tau_1, \tau_2 \in \mathbb{H}$, each carrying its principal polarization from [3]. The product $X = E_1 \times E_2$ is the torus $\mathbb{C}^2/\Lambda$ with lattice generated by the columns of

$$\Pi = \begin{pmatrix} \tau_1 & 0 & 1 & 0 \\ 0 & \tau_2 & 0 & 1 \end{pmatrix}$$

whose product polarization has $E = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$ in the evident symplectic basis $\mu_1 = (\tau_1, 0)$, $\mu_2 = (0, \tau_2)$, $\nu_1 = (1, 0)$, $\nu_2 = (0, 1)$. The block of ν -periods is already I_2 , so no change of coordinates is needed, and the associated Siegel point is

$$\tau = \begin{pmatrix} \tau_1 & 0 \\ 0 & \tau_2 \end{pmatrix} \in \mathfrak{H}_2$$

a diagonal matrix. It is symmetric, and $\operatorname{Im} \tau = \operatorname{diag}(\operatorname{Im} \tau_1, \operatorname{Im} \tau_2) > 0$ since $\tau_1, \tau_2 \in \mathbb{H}$; both Riemann relations reduce to their $g = 1$ instances on each factor. This is the diagonal locus $\mathbb{H} \times \mathbb{H} \subset \mathfrak{H}_2$ of example 3.3.2: products of elliptic curves are exactly the principally polarized abelian surfaces whose Siegel point is diagonal.

The normal form is not unique, because the symplectic basis is not unique. Two symplectic bases of Λ for the same E differ by an integral matrix that preserves the form J , and such matrices are exactly the integral symplectic group. Changing the symplectic basis changes the normalized τ , and the resulting equivalence relation on $\mathfrak{H}_g$ is what identifies period matrices that describe isomorphic abelian varieties. The next definition records the group and its action.

Definition 3.3.5: The Symplectic Group and its Action on $\mathfrak{H}_g$

The *integral symplectic group* is

$$\mathrm{Sp}(2g, \mathbb{Z}) = \{M \in \mathrm{GL}_{2g}(\mathbb{Z}) \mid M^t J M = J\} \quad J = \begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$$

the group of integer matrices preserving the standard alternating form J . Writing $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ in $g \times g$ blocks, the group acts on the Siegel upper half-space $\mathfrak{H}_g$ by

$$M \cdot \tau = (A\tau + B)(C\tau + D)^{-1}$$

the *Siegel modular transformation* associated to M .

The block condition $M^t J M = J$ unpacks into $A^t C = C^t A$, $B^t D = D^t B$, and $A^t D - C^t B = I_g$; these are the defining relations of the symplectic group in block form, and for $g = 1$ they collapse to the single relation $ad - bc = 1$, so $\mathrm{Sp}(2, \mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})$. That the formula $M \cdot \tau = (A\tau + B)(C\tau + D)^{-1}$ lands back in $\mathfrak{H}_g$, that $C\tau + D$ is invertible for $\tau \in \mathfrak{H}_g$, and that the assignment is a group action are the content of the following verification, which also identifies the action with change of symplectic basis.

Proposition 3.3.6: The Siegel Action is Well-Defined and Realizes Change of Symplectic Basis

For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathrm{Sp}(2g, \mathbb{Z})$ and $\tau \in \mathfrak{H}_g$, the matrix $C\tau + D$ is invertible, the point $M \cdot \tau = (A\tau + B)(C\tau + D)^{-1}$ lies in $\mathfrak{H}_g$, and $\tau \mapsto M \cdot \tau$ is an action of $\mathrm{Sp}(2g, \mathbb{Z})$ on $\mathfrak{H}_g$. Two points $\tau, \tau' \in \mathfrak{H}_g$ arise from the same principally polarized abelian variety, by two choices of symplectic basis, if and only if $\tau' = M \cdot \tau$ for some $M \in \mathrm{Sp}(2g, \mathbb{Z})$.

Proof:

Step 1: $C\tau + D$ is invertible and $M \cdot \tau \in \mathfrak{H}_g$. Write $P = A\tau + B$ and $Q = C\tau + D$. The block relations $A^t D - C^t B = I_g$, $A^t C = C^t A$, and $B^t D = D^t B$ from definition 3.3.5, together with $\tau = \tau^t$ and the reality of A, B, C, D , give the identity

$$\bar{Q}^t P - \bar{P}^t Q = \tau - \bar{\tau} = 2i \operatorname{Im} \tau \quad (3.5)$$

obtained by expanding both products and cancelling the terms $\bar{\tau}(C^t A - A^t C)\tau = 0$ and $D^t B - B^t D = 0$, while the remaining terms contribute $-\bar{\tau}(A^t D - C^t B) + (D^t A - B^t C)\tau = -\bar{\tau} + \tau$ using $D^t A - B^t C = (A^t D - C^t B)^t = I_g$. If $Qx = 0$ for some $x \in \mathbb{C}^g$, then multiplying the identity on the left by $\bar{x}^t$ and on the right by x makes the left-hand side vanish (both $\bar{x}^t \bar{Q}^t = \overline{Qx}^t = 0$ and $Qx = 0$), so $2i \bar{x}^t (\operatorname{Im} \tau) x = 0$; since $\operatorname{Im} \tau > 0$ this forces $x = 0$. Hence $Q = C\tau + D$ is invertible, and $M \cdot \tau = PQ^{-1}$ is defined.

It remains to check $W := M \cdot \tau = PQ^{-1} \in \mathfrak{H}_g$. For symmetry, the relations give $P^t Q = Q^t P$: expanding, $P^t Q - Q^t P = \tau(A^t C - C^t A)\tau + \tau(A^t D - C^t B) - (D^t A - B^t C)\tau + (B^t D - D^t B) = \tau - \tau = 0$, using $A^t C = C^t A$, $B^t D = D^t B$, and $A^t D - C^t B = (D^t A - B^t C)^t = I_g$ with τ symmetric. Hence $W^t = (Q^t)^{-1} P^t = (Q^t)^{-1} (P^t Q) Q^{-1} = (Q^t)^{-1} (Q^t P) Q^{-1} = PQ^{-1} = W$, so W is symmetric. For positivity, symmetry of W gives $\bar{W} = (\bar{Q}^t)^{-1} \bar{P}^t$, and using (3.5),

$$\bar{Q}^t (W - \bar{W}) Q = \bar{Q}^t P - \bar{P}^t Q = 2i \operatorname{Im} \tau$$

so that $\text{Im}W = (\bar{Q}^t)^{-1}(\text{Im}\tau)Q^{-1}$. This is the congruence of the positive definite matrix $\text{Im}\tau$ by the invertible matrix Q^{-1} , hence positive definite. Therefore $W = M \cdot \tau$ is symmetric with $\text{Im}W > 0$, that is $M \cdot \tau \in \mathfrak{H}_g$.

Step 2: The assignment is a group action. The identity matrix I_{2g} has $C = 0$, $D = I_g$, so $I_{2g} \cdot \tau = \tau$. For the composite of $M_1 = \begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix}$ and $M_2 = \begin{pmatrix} A_2 & B_2 \\ C_2 & D_2 \end{pmatrix}$, the automorphy factor satisfies the cocycle relation

$$C_{21}\tau + D_{21} = (C_2(M_1 \cdot \tau) + D_2)(C_1\tau + D_1)$$

where $\begin{pmatrix} A_{21} & B_{21} \\ C_{21} & D_{21} \end{pmatrix} = M_2 M_1$, obtained by multiplying out the bottom block row of $M_2 M_1$ and substituting $M_1 \cdot \tau = (A_1\tau + B_1)(C_1\tau + D_1)^{-1}$. Dividing the top block row identity by this factor gives $(M_2 M_1) \cdot \tau = M_2 \cdot (M_1 \cdot \tau)$, so the assignment is a left action of $\text{Sp}(2g, \mathbb{Z})$ on $\mathfrak{H}_g$.

Step 3: The action realizes change of symplectic basis. Let $\Pi = (\tau I_g)$ be a normalized period matrix as in proposition 3.3.3, with symplectic basis columns $\mu_1, \dots, \mu_g, \nu_1, \dots, \nu_g$. A second $\mathbb{Z}$ -basis of Λ is $\Pi' = \Pi N$ for a unique $N \in \text{GL}_{2g}(\mathbb{Z})$, and Π' is again a symplectic basis for E if and only if N preserves the matrix J of E , that is $N^t J N = J$, so $N \in \text{Sp}(2g, \mathbb{Z})$. The transformed period matrix $\Pi N = (\tau I_g)N$ need not be in normal form; but both principally polarized abelian varieties it presents are the same (X, H) , since Π and ΠN span the same lattice and carry the same E . Renormalizing ΠN by the recipe of proposition 3.3.3 returns a point $\tau' \in \mathfrak{H}_g$ that is, by construction, obtained from τ by an integral symplectic change of the coordinates (μ, ν) ; the Siegel transformation of definition 3.3.5, applied to the symplectic matrix $M = N^t \in \text{Sp}(2g, \mathbb{Z})$, produces exactly this $\tau' = M \cdot \tau$ (both are the normalized period point of the same polarized torus in the new basis, and the normalization is unique by proposition 3.3.3). Conversely every $M \in \text{Sp}(2g, \mathbb{Z})$ arises from such a change of basis. Thus two points $\tau, \tau' \in \mathfrak{H}_g$ present the same principally polarized abelian variety, through different symplectic bases, if and only if $\tau' = M \cdot \tau$ for some $M \in \text{Sp}(2g, \mathbb{Z})$, as we sought to show.

The proposition completes the dictionary. A principally polarized abelian variety, once a symplectic basis is chosen, is a point of $\mathfrak{H}_g$; forgetting the choice of basis quotients by $\text{Sp}(2g, \mathbb{Z})$; and points in the same orbit describe isomorphic polarized varieties. Everything needed for the moduli statement is now in place. The one point deserving emphasis is what “isomorphic” means here: an isomorphism of principally polarized abelian varieties is a biholomorphism $X \rightarrow X'$ carrying the polarization of X to that of X' , equivalently a $\mathbb{C}$ -linear isomorphism $V \rightarrow V'$ carrying Λ to Λ' and H to H' . Under normalization this is exactly a change of symplectic basis, so the orbit space is the set of isomorphism classes.

Theorem 3.3.7: Moduli of Principally Polarized Abelian Varieties

The map sending a principally polarized abelian variety of dimension g to the $\mathrm{Sp}(2g, \mathbb{Z})$ -orbit of its normalized Siegel point induces a bijection

$$\left\{ \begin{array}{c} \text{principally polarized abelian} \\ \text{varieties of dimension } g \end{array} \right\} / \cong \xrightarrow{\sim} \mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z})$$

between isomorphism classes of g -dimensional principally polarized abelian varieties and the orbit space of the Siegel upper half-space under the integral symplectic group. This orbit space is a complex analytic space of dimension

$$\dim_{\mathbb{C}} \mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z}) = \dim_{\mathbb{C}} \mathfrak{H}_g = \frac{g(g+1)}{2}$$

the dimension of the moduli space $\mathcal{A}_g$ of principally polarized abelian varieties of dimension g .

Proof:

By proposition 3.3.3, every principally polarized abelian variety of dimension g , together with a choice of symplectic basis, determines a point $\tau \in \mathfrak{H}_g$, and conversely every $\tau \in \mathfrak{H}_g$ determines such data: the torus $X_\tau = \mathbb{C}^g / (\tau \mathbb{Z}^g + \mathbb{Z}^g)$ with the Hermitian form $H_\tau(v, w) = v^t (\mathrm{Im} \tau)^{-1} \bar{w}$. This form is positive definite because $\mathrm{Im} \tau > 0$, and its imaginary part $E_\tau = \mathrm{Im} H_\tau$ has, in the basis given by the columns of $\Pi = (\tau \ I_g)$, the standard symplectic matrix J (this is the content of Step 3 of proposition 3.3.3 read backwards), so E_τ is a unimodular integral alternating form and H_τ is a principal polarization. Thus $\tau \mapsto (X_\tau, H_\tau, \text{basis})$ is a two-sided inverse to normalization, giving a bijection between $\mathfrak{H}_g$ and basis-marked principally polarized abelian varieties.

By proposition 3.3.6, two points of $\mathfrak{H}_g$ give isomorphic principally polarized abelian varieties (forgetting the basis) if and only if they lie in the same $\mathrm{Sp}(2g, \mathbb{Z})$ -orbit. Passing to orbits on the left and to isomorphism classes on the right, the bijection descends to the stated bijection between isomorphism classes and $\mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z})$.

For the dimension, the action of $\mathrm{Sp}(2g, \mathbb{Z})$ on $\mathfrak{H}_g$ is properly discontinuous (the group is discrete and acts by biholomorphisms of a bounded symmetric domain), so the quotient inherits the structure of a complex analytic space of the same dimension as $\mathfrak{H}_g$ (a quotient of a complex manifold by a properly discontinuous action has the same dimension everywhere; the fixed loci are lower-dimensional suborbifolds, not lower-dimensional components). Hence $\dim_{\mathbb{C}} \mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z}) = \dim_{\mathbb{C}} \mathfrak{H}_g = g(g+1)/2$ by definition 3.3.1, as we sought to show.

This is the first sight of the moduli space of abelian varieties, denoted $\mathcal{A}_g$: as a set, the isomorphism classes of g -dimensional principally polarized abelian varieties; as a space, the quotient $\mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z})$, of dimension $g(g+1)/2$. Two cautions fix the level of the claim. First, the theorem describes $\mathcal{A}_g$ as an analytic quotient and as a set of isomorphism classes; it is a *coarse* description, in that points of the quotient are in bijection with isomorphism classes but the quotient does not by itself carry a universal family of abelian varieties over it. The obstruction is exactly the extra automorphisms that some polarized varieties possess (every abelian variety has the automorphism $[-1]$, and special ones have more), which is why $\mathcal{A}_g$ carries the structure of an orbifold, or in the algebraic category a Deligne-Mumford stack, rather than a smooth manifold or variety. Developing $\mathcal{A}_g$ as a genuine moduli *scheme*, and its compactifications and Siegel modular forms, is outside this book's scope; it is

pointed to in Chapter 14 and taken up in the companion volume `NateShimuraVarieties`. Second, the quotient here is constructed purely analytically, from period matrices and the symplectic action; the arithmetic of the space, its special points and their fields of definition, is developed later. The special points where X has complex multiplication are the CM points, treated in Chapter 6, and they are the seeds from which the arithmetic theory of $\mathcal{A}_g$ and of Shimura varieties grows, developed in the companion volume `NateShimuraVarieties`.

The genus-one case is the sanity check, and it recovers a construction the reader has already met in full.

Example 3.3.8: Genus One: the Modular Curve and the j -Line

Take $g = 1$. A 1×1 matrix $\tau = (\tau_{11})$ is automatically symmetric, and $\text{Im}\tau > 0$ is the single condition $\text{Im}\tau_{11} > 0$, so

$$\mathfrak{H}_1 = \mathbb{H} = \{\tau \in \mathbb{C} \mid \text{Im}\tau > 0\}$$

is the upper half-plane, of dimension $\frac{1+2}{2} = 1$. The symplectic group is $\text{Sp}(2, \mathbb{Z}) = \text{SL}_2(\mathbb{Z})$, since for a 2×2 integer matrix the condition $M^t J M = J$ with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is the single equation $\det M = 1$. The Siegel action specializes, for $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, to

$$M \cdot \tau = \frac{a\tau + b}{c\tau + d}$$

the classical action of the modular group on $\mathbb{H}$ by Möbius transformations. Theorem 3.3.7 then reads: principally polarized abelian varieties of dimension 1, up to isomorphism, correspond to $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$. A 1-dimensional abelian variety is an elliptic curve, and its unique principal polarization is the one furnished by a point (the class of a degree-one divisor); so this is exactly the statement, proved analytically in [3], that $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ is the moduli space of elliptic curves, uniformized by the j -invariant $j: \text{SL}_2(\mathbb{Z}) \backslash \mathbb{H} \xrightarrow{\sim} \mathbb{C}$. Thus $\mathcal{A}_1$ is the affine j -line, and $\mathfrak{H}_g / \text{Sp}(2g, \mathbb{Z})$ is its higher-dimensional replacement, with the j -invariant giving way to Siegel modular forms. The two elliptic points $\tau = i$ and $\tau = e^{2\pi i/3}$, where the curve has extra automorphisms, are the $g = 1$ instances of the orbifold points that prevent $\mathcal{A}_g$ from being a fine moduli space.

3.4 Riemann-Roch on an Abelian Variety

A polarization is a positive line bundle in the sense of definition 3.1.1, and a positive line bundle is supposed to have sections. Chapter 2 already isolated the space that carries them: for $L = L(H, \chi)$ the space $H^0(X, L)$ consists of the holomorphic functions $\theta: V \rightarrow \mathbb{C}$ transforming by the canonical factor of automorphy, the *theta functions* of definition 2.4.1, and proposition 2.4.3 announced that this space is finite-dimensional with dimension the Pfaffian $\text{Pf}(\text{Im}H)$ when $H > 0$. This section proves that count. It is the analytic Riemann-Roch theorem for an abelian variety: a single integer, read off the type of the polarization, controls how many independent theta functions the bundle supports, and hence how large a projective space the variety can embed into once those functions are written down.

The number that emerges is $d_1 d_2 \cdots d_g$, the product of the elementary divisors of the polarization from definition 3.2.3. Two features make this the correct generalization of the elliptic-curve count $\dim H^0(E, L) = \deg L$. First, all higher cohomology vanishes, so the dimension equals the Euler characteristic and the answer is a topological invariant of the bundle. Second, in the principal case

$(1, \dots, 1)$ the count collapses to $\dim H^0(X, L) = 1$: a principal polarization has, up to translation, exactly one effective divisor, the theta divisor that will organize the entire geometry of Jacobians in Chapter 12.

The strategy for the dimension count is a Fourier decomposition. A theta function is quasiperiodic for the full lattice Λ ; but Λ contains a sublattice on which the factor of automorphy is trivial up to an ordinary character, and along that sublattice a theta function is periodic. Expanding in the resulting Fourier series turns the transformation law under the remaining lattice directions into a finite system of linear recursions on the Fourier coefficients, and counting the free coefficients gives $d_1 \cdots d_g$. The positivity $H > 0$ enters exactly where it must: it is what forces the series to converge, so that every formal solution of the recursion is an actual holomorphic function.

3.4.1 Convention: reduction to a normal form Throughout this section $L = L(H, \chi)$ with $H > 0$ and $E = \text{Im}H$ of type $(d_1, \dots, d_g)$, $d_1 \mid \cdots \mid d_g$, as in definition 3.2.3. The dimension $\dim H^0(X, L)$ is unchanged by an isomorphism of line bundles, so the argument may be run in any $\mathbb{Z}$ -basis of Λ adapted to E ; the symplectic basis of definition 3.2.3 is the convenient one. Moreover the count below is carried out for a fixed but arbitrary semicharacter χ and returns $d_1 \cdots d_g$ no matter which χ is chosen, so the answer in fact depends only on H ; that the underlying bundles $L(H, \chi) \otimes L_0$ with $L_0 \in \text{Pic}^0(X)$ are related by translation, which explains this χ -independence conceptually, is established in proposition 3.4.5.

The reduction recorded above is what lets the proof choose coordinates freely. The dimension is an invariant of H alone, so the semicharacter χ may be normalized and the lattice put in symplectic form without loss; only the positive Hermitian form H , through its elementary divisors, survives into the final count.

Theorem 3.4.2: Riemann-Roch on an Abelian Variety

Let $X = V/\Lambda$ be a complex torus and let $L = L(H, \chi)$ be a line bundle with $H > 0$. Let $E = \text{Im}H$, of type $(d_1, \dots, d_g)$. Then

$$\dim_{\mathbb{C}} H^0(X, L) = \text{Pf}(E) = d_1 d_2 \cdots d_g = \sqrt{\det E}$$

In particular $H^0(X, L) \neq 0$, so a polarization always has a nonzero global section.

Proof:

By box 3.4.1 the dimension depends only on H , and by definition 2.4.1 an element of $H^0(X, L)$ is a holomorphic function $\theta: V \rightarrow \mathbb{C}$ satisfying the quasiperiodicity

$$\theta(v + \lambda) = a_{(H, \chi)}(\lambda, v) \theta(v) = \chi(\lambda) \exp\left(\pi H(v, \lambda) + \frac{\pi}{2} H(\lambda, \lambda)\right) \theta(v) \quad (3.6)$$

for every $\lambda \in \Lambda$. The proof decomposes Λ into a sublattice on which θ is periodic and a complementary one that acts by a shift, expands θ in a Fourier series along the first, and counts the coefficients that the second permits.

Step 1: A symplectic splitting of the lattice. By definition 3.2.3 choose a $\mathbb{Z}$ -basis $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ of Λ in which E has the matrix $\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$ with $D = \text{diag}(d_1, \dots, d_g)$; that is, $E(\lambda_j, \mu_k) = d_j \delta_{jk}$

and $E(\lambda_j, \lambda_k) = E(\mu_j, \mu_k) = 0$. Set

$$\Lambda_1 = \sum_j \mathbb{Z} \lambda_j, \quad \Lambda_2 = \sum_j \mathbb{Z} \mu_j$$

so that $\Lambda = \Lambda_1 \oplus \Lambda_2$ and E vanishes on each of Λ_1, Λ_2 separately. Because Λ is a full lattice, $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ is an $\mathbb{R}$ -basis of V ; in particular $\lambda_1, \dots, \lambda_g$ are $\mathbb{R}$ -linearly independent. They are moreover $\mathbb{C}$ -linearly independent. If not, the real subspace $V_1 = \sum_j \mathbb{R} \lambda_j$ would contain a complex line, so there would be a nonzero $w \in V_1$ with $iw \in V_1$; but E vanishes on Λ_1 , hence on its real span V_1 , and then $H(w, w) = E(iw, w) = 0$ with $w \neq 0$, contradicting $H > 0$. Thus $\lambda_1, \dots, \lambda_g$ are $\mathbb{C}$ -linearly independent, and by the dimension count they form a $\mathbb{C}$ -basis of V . Use it to identify $V \cong \mathbb{C}^g$ and write $v = \sum_j z_j \lambda_j$ with coordinates $z = (z_1, \dots, z_g)$.

Step 2: Periodicity along Λ_1 and the Fourier series. The idea is to strip the transformation law (3.6) down to its core content by multiplying θ by a fixed nowhere-zero *holomorphic* exponential, the standard passage to a *classical theta* normalization. The prefactor must stay holomorphic, so it cannot be the non-holomorphic $e^{-\pi H(v, v)/2}$; instead it is built from the $\mathbb{C}$ -bilinear form that agrees with H along Λ_1 . Because $E = \text{Im} H$ vanishes on Λ_1 (Step 1), the restriction of H to the real span $V_1 = \sum_j \mathbb{R} \lambda_j$ is real symmetric, so the numbers $H(\lambda_j, \lambda_k)$ are real and symmetric in j, k . Let B be the symmetric $\mathbb{C}$ -bilinear form on $V \cong \mathbb{C}^g$ with $B(\lambda_j, \lambda_k) = H(\lambda_j, \lambda_k)$ on the $\mathbb{C}$ -basis of Step 1, extended $\mathbb{C}$ -bilinearly; in the coordinates z it is the holomorphic quadratic $B(v, v) = \sum_{j,k} H(\lambda_j, \lambda_k) z_j z_k$. Set

$$\theta'(v) = \exp\left(-\frac{\pi}{2} B(v, v)\right) \theta(v)$$

a *holomorphic* function, since both factors are. The prefactor is nowhere zero, so $\theta \mapsto \theta'$ is a bijection between holomorphic θ obeying (3.6) and holomorphic θ' satisfying the transformed law, and it preserves linear independence, so no dimension is lost by working with θ' . Substituting $\theta = e^{\pi B(v, v)/2} \theta'$ into (3.6) and using $B(v + \lambda, v + \lambda) = B(v, v) + 2B(v, \lambda) + B(\lambda, \lambda)$ (valid because B is $\mathbb{C}$ -bilinear and symmetric) gives, for every $\lambda \in \Lambda$,

$$\theta'(v + \lambda) = \chi(\lambda) \exp\left(\pi(H(v, \lambda) - B(v, \lambda)) + \frac{\pi}{2}(H(\lambda, \lambda) - B(\lambda, \lambda))\right) \theta'(v) \quad (3.7)$$

Now take $\lambda = \lambda_j \in \Lambda_1$. Here B was defined to agree with H on the $\mathbb{C}$ -basis, and both $v \mapsto H(v, \lambda_j)$ and $v \mapsto B(v, \lambda_j)$ are the $\mathbb{C}$ -linear functional with the same values $H(\lambda_k, \lambda_j)$ on λ_k , so $H(v, \lambda_j) = B(v, \lambda_j)$ identically, and likewise $H(\lambda_j, \lambda_j) = B(\lambda_j, \lambda_j)$. Every exponent in (3.7) therefore cancels, leaving

$$\theta'(v + \lambda_j) = \chi(\lambda_j) \theta'(v)$$

genuine periodicity up to the fixed *unitary constant* $\chi(\lambda_j) \in U(1)$, with θ' still holomorphic. Writing $\chi(\lambda_j) = e^{2\pi i \varphi_j}$ and setting $\psi(v) = \exp(-2\pi i \sum_j \varphi_j z_j) \theta'(v)$, still holomorphic since the new exponent is $\mathbb{C}$ -linear in z , absorbs the constant: under $z_j \mapsto z_j + 1$ the linear exponential contributes $e^{-2\pi i \varphi_j}$, which cancels $\chi(\lambda_j)$, so $\psi(v + \lambda_j) = \psi(v)$ for every j . Thus ψ is holomorphic and $\mathbb{Z}^g$ -periodic in z , and it admits a Fourier expansion

$$\psi(v) = \sum_{m \in \mathbb{Z}^g} c_m e^{2\pi i \langle m, z \rangle}, \quad \langle m, z \rangle = \sum_j m_j z_j \quad (3.8)$$

with coefficients $c_m \in \mathbb{C}$, absolutely and uniformly convergent on compact sets because ψ is holomorphic and 1-periodic in each coordinate z_j . The original theta is recovered as $\theta(v) = \exp(\frac{\pi}{2} B(v, v) + 2\pi i \sum_j \varphi_j z_j) \psi(v)$ through this fixed holomorphic prefactor.

Step 3: The recursion imposed by Λ_2 . It remains to impose (3.6) for $\lambda = \mu_k \in \Lambda_2$. Through the fixed holomorphic prefactor relating θ and ψ recorded at the end of Step 2, that quasiperiodicity of θ under μ_k is equivalent to a transformation law for ψ of the same shape: $\psi(v + \mu_k)$ equals $\psi(v)$ times a nowhere-zero holomorphic exponential in z . In the $\mathbb{C}$ -basis $\lambda_1, \dots, \lambda_g$ of Step 1 write $\mu_k = \sum_j \tau_{jk} \lambda_j$, so that $\tau = (\tau_{jk}) \in M_g(\mathbb{C})$ has μ_k for its k -th column and the period matrix in the ordered basis $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ is $\Pi = (I_g \ \tau)$.

Two properties of τ are needed, and both come directly from the Riemann relations of proposition 3.1.5 for this Π and this E , whose matrix in the present basis is $\mathcal{E} = \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$ with inverse $\mathcal{E}^{-1} = \begin{pmatrix} 0 & -D^{-1} \\ D^{-1} & 0 \end{pmatrix}$. This is *not* the principal normalization of proposition 3.3.3: there the $\mathbb{C}$ -basis is the second lattice half and $D = I_g$, whereas here the $\mathbb{C}$ -basis is the first half $\lambda_1, \dots, \lambda_g$ and $D = \text{diag}(d_1, \dots, d_g)$, so the symmetry and positivity of that proposition do not transfer and must be recomputed. Blockwise $\Pi \mathcal{E}^{-1} = (\tau D^{-1} \ -D^{-1})$, so the first Riemann relation reads

$$\Pi \mathcal{E}^{-1} \Pi^t = \tau D^{-1} - D^{-1} \tau^t = 0,$$

that is $D\tau = (D\tau)^t$: the matrix $D\tau$ is symmetric, though τ itself is symmetric only in the principal case $d_1 = \dots = d_g$. Using $\tau D^{-1} = D^{-1} \tau^t$, the second relation is

$$i \Pi \mathcal{E}^{-1} \bar{\Pi}^t = i(\tau D^{-1} - D^{-1} \bar{\tau}^t) = i D^{-1}(\tau - \bar{\tau})^t = -2 D^{-1}(\text{Im} \tau)^t = -2 \text{Im} \tau D^{-1},$$

the last equality because $D \text{Im} \tau = \text{Im}(D\tau)$ is symmetric. Since $H > 0$ this Hermitian matrix is positive definite, so the real symmetric matrix

$$Y := -D \text{Im} \tau = -\text{Im}(D\tau)$$

is positive definite; equivalently $\text{Im}(D\tau) < 0$. This is the correct form of the positivity in the present swapped, type- $(d_1, \dots, d_g)$ configuration, and $Y > 0$ is the sole consequence of $H > 0$ that the convergence argument will use.

Translation by μ_k is the shift $z_j \mapsto z_j + \tau_{jk}$. Substituting (3.8) into the transformation law for ψ and matching the coefficient of $e^{2\pi i \langle m, z \rangle}$ on both sides yields, for each k , a relation of the form

$$c_{m+d_k e_k} = w_k(m) c_m, \quad w_k(m) = C_k \exp\left(-2\pi i \sum_j (m_j + d_k \delta_{jk}) \tau_{jk}\right), \quad (3.9)$$

where e_k is the k -th standard basis vector, $C_k \in \mathbb{C}^*$ is a constant of modulus $|C_k| = \exp(2\pi \sum_j \varphi_j (\text{Im} \tau)_{jk} - \pi d_k (\text{Im} \tau)_{kk})$ built from $\chi(\mu_k) = e^{2\pi i \varphi_k}$ and the diagonal shift, and the index moves by exactly d_k in the k -th slot because $\mathcal{H} \text{Im} \tau = -D$ (the matrix form of the second relation, with $\mathcal{H}_{pq} = H(\lambda_p, \lambda_q)$ the real symmetric Gram matrix of Step 2) forces the linear part of the multiplier to be $e^{2\pi i d_k z_k}$, while $E(\lambda_k, \mu_k) = d_k$ governs the quasiperiod in that direction. Thus each coefficient $c_{m+d_k e_k}$ is determined, through the nonzero multiplier $w_k(m)$, by c_m . Consistency of the g recursions among themselves is automatic: the μ_k commute and $E(\mu_j, \mu_k) = 0$, so the shifts in different coordinates compose without contradiction.

Step 4: Counting free coefficients. The recursion (3.9) partitions the index set $\mathbb{Z}^g$ into orbits under the translation lattice $D\mathbb{Z}^g = \bigoplus_j d_j \mathbb{Z}$. On each orbit every coefficient is determined by any one of them via the nonzero multipliers, so an orbit contributes at most one free parameter: the value of c_m at a single chosen representative. The number of orbits is the index

$$[\mathbb{Z}^g : D\mathbb{Z}^g] = d_1 d_2 \cdots d_g$$

Hence the space of *formal* solutions of the system (3.9) has dimension at most $d_1 \cdots d_g$: a solution is a choice of c_m on one representative per orbit, freely, with all other coefficients then forced.

Step 5: Convergence, from positivity. A formal solution defines an actual holomorphic ψ , and hence through the fixed holomorphic prefactor an actual theta function θ , precisely when the series (3.8) converges. From the explicit multiplier (3.9) its modulus is

$$\log |w_k(m)| = 2\pi \sum_j (m_j + \varphi_j) (\operatorname{Im} \tau)_{jk} + \pi d_k (\operatorname{Im} \tau)_{kk},$$

linear in m ; telescoping the N steps that move m to $m + Nd_k e_k$ sums $\log |w_k(m + \ell d_k e_k)|$ over $\ell = 0, \dots, N-1$, and the ℓ -dependence through $(\operatorname{Im} \tau)_{kk}$ contributes the quadratic term $2\pi (\operatorname{Im} \tau)_{kk} d_k \binom{N}{2} = \pi d_k (\operatorname{Im} \tau)_{kk} (N^2 - N)$, giving

$$|c_{m+Nd_k e_k}| = |c_m| \exp\left(\pi d_k (\operatorname{Im} \tau)_{kk} N^2 + O(N)\right) = |c_m| \exp\left(-\pi Y_{kk} N^2 + O(N)\right),$$

where $Y = -D \operatorname{Im} \tau$ is the positive-definite matrix of Step 3 and the coefficient of $-N^2$ is $\pi Y_{kk} = -\pi d_k (\operatorname{Im} \tau)_{kk} > 0$ exactly because $Y > 0$ has positive diagonal, itself the consequence of $H > 0$. The same estimate holds in every coordinate direction; for a general shift $m \mapsto m + Dn$, $n \in \mathbb{Z}^g$, the same telescoping produces $\exp(-\pi n^t Y n + O(|n|))$, a genuine Gaussian because $Y > 0$, with the off-diagonal entries of Y contributing the cross terms. Thus each coefficient decays faster than any geometric rate as its index leaves a bounded region, and the series converges absolutely and locally uniformly to a holomorphic function on all of V . This is where the hypothesis $H > 0$ is used, and used essentially: without it Y would fail to be positive definite, the quadratic form in the exponent would change sign, the series would diverge, and the formal solutions would not be functions. Every one of the $d_1 \cdots d_g$ free choices therefore produces a genuine element of $H^0(X, L)$, and distinct choices give linearly independent functions since they differ already in their Fourier coefficients.

Combining Steps 4 and 5, the space $H^0(X, L)$ has dimension exactly $d_1 d_2 \cdots d_g$. Finally, for a nondegenerate alternating integral matrix in the symplectic normal form $\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$ the Pfaffian is $\operatorname{Pf}(E) = d_1 \cdots d_g$ and $\det E = (d_1 \cdots d_g)^2$, so $\operatorname{Pf}(E) = \sqrt{\det E}$, giving

$$\dim_{\mathbb{C}} H^0(X, L) = \operatorname{Pf}(E) = d_1 d_2 \cdots d_g = \sqrt{\det E}$$

as we sought to show.

The theorem is the analytic heart of the projective geometry of abelian varieties. It says that positivity is not a qualitative condition for algebraicity alone but a quantitative one: the polarization prescribes the exact dimension of the linear system of theta functions, and that dimension is a product of the elementary divisors, computable from the period matrix by a determinant. The proof also explains the mechanism. The lattice splits into a periodic half and a translating half; Fourier analysis handles the periodic half and turns the translating half into a finite orbit count, and positivity is precisely the input that promotes formal solutions into convergent, holomorphic ones. Nothing here yet writes a single theta function down. The count is an existence-and-dimension statement; Chapter 4 supplies the explicit basis, the Riemann theta series indexed by the $d_1 \cdots d_g$ orbit representatives, and uses it to construct the projective embedding whose target dimension this count determines.

One loose end remains before the count deserves the name Riemann-Roch. The classical theorem computes an Euler characteristic, an alternating sum of cohomology dimensions, and it is only

because the higher terms vanish that the alternating sum reduces to $\dim H^0$. The next result records that vanishing.

Proposition 3.4.3: Vanishing of Higher Cohomology

Let $X = V/\Lambda$ be a complex torus and $L = L(H, \chi)$ with $H > 0$. Then

$$H^i(X, L) = 0 \quad \text{for all } i > 0$$

Proof Key ideas:

The vanishing is standard, and the general- i statement rests on the theory of the cohomology of line bundles on a complex torus computed through the Dolbeault complex or through the theta-group action; the full computation of all $H^i(X, L)$ for $L(H, \chi)$ is carried out in Mumford, *Abelian Varieties*, and is cited here rather than reproduced because it repeats the harmonic-oscillator style Fourier analysis of theorem 3.4.2 once for each cohomological degree without adding a new idea. The mechanism is this. On a complex torus $H^i(X, L) = H_{\bar{\partial}}^{0,i}(X, L)$ is computed by the $\bar{\partial}$ -complex with coefficients in L . Diagonalizing the Hermitian form H and running the Fourier expansion of theorem 3.4.2 in each of the g complex directions separately, the $\bar{\partial}$ -Laplacian acting on L -valued $(0, i)$ -forms decouples into a sum of g one-dimensional operators. Because $H > 0$, each of these one-dimensional operators is strictly positive on the antiholomorphic directions, so a harmonic L -valued $(0, i)$ -form with $i > 0$ must vanish; only the $(0, 0)$ -part supports harmonic representatives, and those are exactly the holomorphic sections counted in theorem 3.4.2. Positivity is again the load-bearing hypothesis: it is what makes every higher $\bar{\partial}$ -cohomology group vanish, completing the argument.

The vanishing is the second half of what makes the dimension count a true Riemann-Roch theorem. With no higher cohomology to correct for, the Euler characteristic of the bundle reduces to the single number computed above, and that number is therefore a topological invariant, insensitive to the analytic choices in the construction of L .

Corollary 3.4.4: Euler Characteristic of a Positive Bundle

Let L be a line bundle with first Chern class $H > 0$ of type $(d_1, \dots, d_g)$, and set $E = \text{Im} H$. Then the Euler characteristic of L , written with the standard symbol $\chi(-)$ for an alternating sum of cohomology dimensions, is

$$\chi(L) = \sum_{i \geq 0} (-1)^i \dim H^i(X, L) = \text{Pf}(E) = d_1 d_2 \cdots d_g$$

Proof:

By proposition 3.4.3 every term with $i > 0$ vanishes, so $\chi(L) = \dim H^0(X, L)$, which equals $\text{Pf}(E) = d_1 \cdots d_g$ by theorem 3.4.2, completing the proof.

That the Euler characteristic equals $\text{Pf}(E) = \frac{1}{g!} \int_X c_1(L)^g$ is the abelian-variety incarnation of Hirzebruch-Riemann-Roch: the topological expression $\int_X e^{c_1(L)} \text{td}(X)$ collapses, because the tangent bundle of a torus is trivial and so its Todd class is 1, to the top self-intersection $\frac{1}{g!} \int_X c_1(L)^g$, and

that integral is exactly the Pfaffian of $E = c_1(L)$. The analytic count of theorem 3.4.2 and the topological count of Hirzebruch thus agree, each confirming the other.

The most consequential special case is the smallest one. When the type is trivial the product of elementary divisors is 1, so the polarization supports a one-dimensional space of theta functions, a single section up to scale, and hence a single effective divisor. This is the defining feature of a principal polarization and the reason such polarizations dominate the theory.

Proposition 3.4.5: Principal Polarizations Have a Unique Theta Divisor

Let $L = L(H, \chi)$ define a principal polarization on X , so that $H > 0$ has type $(1, \dots, 1)$ in the sense of definition 3.2.4. Then

$$\dim_{\mathbb{C}} H^0(X, L) = 1$$

Consequently L has, up to scaling, a unique nonzero global section θ , and the associated effective divisor $\Theta = \{\theta = 0\}$ is the unique effective divisor in the linear equivalence class of L . A different choice of the semicharacter χ , or of a bundle in $\text{Pic}^0(X)$ to twist by, replaces Θ by a translate; so a principal polarization determines a divisor Θ uniquely up to translation.

Proof:

By definition 3.2.4 a principal polarization has type $(1, \dots, 1)$, so $d_1 = \dots = d_g = 1$ and theorem 3.4.2 gives $\dim_{\mathbb{C}} H^0(X, L) = 1 \cdot 1 \cdots 1 = 1$. A one-dimensional space of sections has a nonzero element θ unique up to a scalar; its divisor $\Theta = \{\theta = 0\}$ is therefore well defined independently of the scalar. Any effective divisor D linearly equivalent to L is the zero divisor of some global section of L , hence of a scalar multiple of θ , so $D = \Theta$; thus Θ is the only effective divisor in the class.

For the last statement, the polarization is the class $H \in \text{NS}(X)$, and by the Appell-Humbert classification the line bundles with first Chern class H are the twists $L(H, \chi) \otimes L_0$ with $L_0 \in \text{Pic}^0(X)$; each is again positive of the same type, so each has a one-dimensional space of sections and a single theta divisor. Two such bundles differ by an element of $\text{Pic}^0(X) = \{L(0, \chi') : \chi' \in \text{Hom}(\Lambda, U(1))\}$, and it suffices to realize the passage from L to $L \otimes L(0, \chi')$ as a translation. This is a direct computation with the canonical factor. For $a \in V$ write $t_a : X \rightarrow X$ for translation by the image of a , lifted to $v \mapsto v + a$ on V ; then $t_a^* L(H, \chi)$ has factor of automorphy

$$a_{(H, \chi)}(\lambda, v + a) = a_{(H, \chi)}(\lambda, v) e^{\pi H(a, \lambda)}$$

since H is linear in its first slot. The extra factor $e^{\pi H(a, \lambda)}$ differs by a coboundary from the constant $U(1)$ -valued character $\lambda \mapsto e^{2\pi i E(a, \lambda)}$. Indeed the $\mathbb{R}$ -linear functional $\ell(v) = \pi E(ia, v) - \pi i E(a, v)$ is in fact $\mathbb{C}$ -linear, since $E(iu, iw) = E(u, w)$ gives $\ell(iv) = i \ell(v)$; so $h(v) = e^{\ell(v)}$ is a nowhere-zero holomorphic function on V , and its coboundary $h(v + \lambda)/h(v) = e^{\ell(\lambda)}$ satisfies

$$e^{\pi H(a, \lambda)} = e^{2\pi i E(a, \lambda)} e^{\ell(\lambda)}$$

using $H(a, \lambda) = E(ia, \lambda) + i E(a, \lambda)$. Multiplying the factor of $t_a^* L(H, \chi)$ by this coboundary replaces $e^{\pi H(a, \lambda)}$ by the constant character $e^{2\pi i E(a, \lambda)}$ without changing the isomorphism class, so, in $\text{Pic}(X)$,

$$t_a^* L(H, \chi) \cong L(H, \chi) \otimes L(0, \chi_a), \quad \chi_a(\lambda) = e^{2\pi i E(a, \lambda)}$$

Now $H > 0$ makes $E = \text{Im} H$ nondegenerate on V , so the $\mathbb{R}$ -linear map $V \rightarrow \text{Hom}(\Lambda, \mathbb{R})$, $a \mapsto E(a, -)$, is an isomorphism; composing with the projection $\text{Hom}(\Lambda, \mathbb{R}) \twoheadrightarrow \text{Hom}(\Lambda, \mathbb{R})/\text{Hom}(\Lambda, \mathbb{Z}) =$

$\mathrm{Hom}(\Lambda, U(1))$ shows that $a \mapsto \chi_a$ maps V onto $\mathrm{Hom}(\Lambda, U(1))$. Thus every $\chi' \in \mathrm{Hom}(\Lambda, U(1))$ equals χ_a for some a , and $L \otimes L(0, \chi') \cong t_a^* L$: tensoring by the degree-zero bundle really is pulling back along a translation, and it therefore carries the theta divisor Θ of L to its translate $t_a^{-1}(\Theta)$. Hence the theta divisors of the various bundles carrying the principal polarization are translates of one another, and Θ is determined by H up to translation, completing the proof.

The uniqueness is what gives a principally polarized abelian variety its distinguished geometry. An ample class in general has a positive-dimensional linear system, a whole family of divisors moving inside it; the principal case has a single divisor, rigid up to translation, and that divisor carries the polarization as its own class. The theta divisor Θ is thus an intrinsic feature of the polarized variety rather than an auxiliary choice, and it is the object through which Jacobians of curves acquire their canonical structure. A curve of genus g produces a principally polarized abelian variety of dimension g , its Jacobian, and the theta divisor is the image of the symmetric product of the curve; the Torelli theorem of Chapter 12 recovers the curve from the pair $(\mathrm{Jac} C, \Theta)$, which is possible only because Θ is canonical, which is possible only because $\dim H^0 = 1$.

Before turning to consequences, it is worth grounding the whole apparatus in the one dimension where every quantity is already familiar. In genus one the type is a single integer, the Pfaffian is that integer, and the count reproduces the Riemann-Roch dimension that elliptic-curve theory supplies by divisors.

Example 3.4.6: Riemann-Roch for Elliptic Curves

Let $g = 1$, so $E = \mathbb{C}/\Lambda$ is an elliptic curve and $V = \mathbb{C}$. A Hermitian form on $\mathbb{C}$ is $H(v, w) = h v \bar{w}$ for a real scalar h , and $H > 0$ means $h > 0$. Its imaginary part $E = \mathrm{Im} H$ is a single alternating integral form on the rank-two lattice Λ , and its type is the single elementary divisor $d_1 = E(\lambda_1, \lambda_2)$ for an oriented $\mathbb{Z}$ -basis λ_1, λ_2 of Λ ; this integer is positive because $H > 0$. The Pfaffian of a 2×2 alternating matrix $\begin{pmatrix} 0 & d_1 \\ -d_1 & 0 \end{pmatrix}$ is d_1 , so theorem 3.4.2 reads

$$\dim_{\mathbb{C}} H^0(E, L) = \mathrm{Pf}(E) = d_1$$

This matches the classical elliptic-curve Riemann-Roch. A line bundle L on E of degree $\deg L = n \geq 1$ has $\dim H^0(E, L) = n$ by the Riemann-Roch theorem for curves of genus one ([3], where the divisor form $\ell(D) = \deg D$ for $\deg D \geq 1$ is proved). The bridge between the two counts is that for a line bundle on an elliptic curve the degree equals the single elementary divisor of its first Chern class,

$$\deg L = \int_E c_1(L) = E(\lambda_1, \lambda_2) = d_1$$

the middle equality being the evaluation of the fundamental class against $c_1(L) = \mathrm{Im} H$ under $H^2(E, \mathbb{Z}) \cong \mathbb{Z}$. So the analytic formula $\dim H^0 = d_1$ and the classical formula $\dim H^0 = \deg L$ are the same statement, and the higher-dimensional Pfaffian $d_1 \cdots d_g$ is the natural replacement for $\deg L$ once $g > 1$, where a line bundle no longer has a single degree but a whole tuple of elementary divisors.

The elliptic-curve case fixes the reading of the general theorem. For $g = 1$ there is one elementary divisor, it is the degree, and $\dim H^0$ is that degree; a principal polarization is a degree-one bundle, with a one-dimensional space of sections and the single point of its divisor serving as the origin. For $g > 1$ the polarization no longer reduces to a number but to the tuple $(d_1, \dots, d_g)$, and the theorem asserts that their product, not their sum or their maximum, counts the theta functions. That product is exactly the input Chapter 4 requires: it is the number of coordinates in the projective embedding a polarization furnishes, the $d_1 \cdots d_g$ Riemann theta functions written out explicitly and used to map

X into $\mathbb{P}^{d_1 \cdots d_g - 1}$ when the polarization is base-point free. Among all polarizations the principal ones sit at the base of this count, with $\dim H^0 = 1$ and a single canonical theta divisor, and they are the case the theory returns to again and again because Jacobians of curves realize them: every curve manufactures a principally polarized abelian variety, and the rigidity of its lone theta divisor is what lets the geometry of the curve be read off from it.

Theta Functions

Chapter 3 counted the sections of a positive line bundle without writing a single one down: the Riemann-Roch theorem gave $\dim H^0(X, L) = d_1 \cdots d_g$ by a Fourier argument that produced the dimension and nothing more. This chapter supplies the functions themselves. They are the theta functions, holomorphic functions on V with the prescribed quasi-periodicity of the Appell-Humbert factor, and once they are in hand the projective geometry of an abelian variety becomes explicit: a basis of theta functions is exactly a system of homogeneous coordinates, and the map it defines is the embedding whose existence Chapter 3 could only assert.

Three threads run through the chapter. The first is the theta group, the finite Heisenberg group $\mathcal{G}(L)$ of symmetries of L , whose unique irreducible representation is the section space $H^0(X, L)$; this recovers the dimension count of Chapter 3 as a fact about the Heisenberg group and fixes the natural basis. The second is the classical Riemann theta function and its translates with characteristics, the concrete series that realize that basis and converge precisely because the period matrix lies in the Siegel upper half-space. The third is the payoff: the Lefschetz theorem, that the third power of a polarization is very ample, so every polarizable torus embeds in projective space and the provisional definition of an abelian variety from Chapter 1 is at last discharged. The chapter closes with the theta divisor, the zero locus of the single section of a principal polarization, the geometric object through which Jacobians will later carry their structure.

4.1 Canonical Factors and the Theta Group

The dimension count of theorem 3.4.2 answered the question “how many sections does a positive line bundle have?” with the number $d_1 \cdots d_g$, the product of the elementary divisors of the polarization. That count was extracted by a direct convergence estimate on quasi-periodic functions, and it left the space $H^0(X, L)$ looking like an accident of Fourier analysis: a vector space of the right dimension, with no structure singling out one basis over another. This section supplies the missing structure.

Attached to every line bundle L on $X = V/\Lambda$ is a finite abelian group $K(L)$, and above it a nonabelian group $\mathcal{G}(L)$ of symmetries of L , the *theta group*. The group $\mathcal{G}(L)$ acts on $H^0(X, L)$, and when L is a polarization this action is irreducible; the dimension of the unique such irreducible representation recovers $d_1 \cdots d_g$ without a single convergence estimate. The theta group is the intrinsic symmetry of a polarized abelian variety, and its commutator is the germ of the Weil pairing that organizes Chapter 5.

Everything rests on a lattice larger than Λ , on which the alternating form $E = \text{Im}H$ is still integral. Recall from definition 3.1.3 that a line bundle $L = L(H, \chi)$ carries the Hermitian form H and its imaginary part E , an alternating $\mathbb{R}$ -form that is integral on Λ : $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$. The next definition asks for the largest sublattice of V on which E remains integral against all of Λ .

Definition 4.1.1: The Lattice $\Lambda(L)$

Let $X = V/\Lambda$ and let $L = L(H, \chi)$ be a line bundle with $E = \text{Im}H$ integral on Λ . Define

$$\Lambda(L) := \{ v \in V : E(v, \lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda \}$$

the set of vectors on which E pairs integrally with the whole lattice.

Because $E(\Lambda, \Lambda) \subseteq \mathbb{Z}$, every lattice vector lies in $\Lambda(L)$, so $\Lambda \subseteq \Lambda(L)$. The set $\Lambda(L)$ is a subgroup of V : it is the preimage of the integral lattice $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ under the $\mathbb{R}$ -linear map $v \mapsto E(v, -)|_\Lambda$ from V to $\text{Hom}_\mathbb{R}(\Lambda, \mathbb{R}) = \text{Hom}_\mathbb{R}(V, \mathbb{R})$. When E is nondegenerate this map is an isomorphism of $\mathbb{R}$ -vector spaces, so $\Lambda(L)$ is the preimage of a lattice under a linear isomorphism, hence itself a full lattice in V containing Λ with finite index. The quotient measures how far E is from being unimodular on Λ .

Definition 4.1.2: The Group $K(L)$

With notation as in definition 4.1.1, define the finite abelian group

$$K(L) := \Lambda(L)/\Lambda$$

the cokernel of the inclusion $\Lambda \hookrightarrow \Lambda(L)$.

The group $K(L)$ is the analytic incarnation of the kernel of the polarization isogeny φ_L that Chapter 5 will construct: a point $\bar{v} \in K(L)$ is exactly a translation of X under which L is preserved up to isomorphism. For now it is a finite abelian group whose order is dictated entirely by the type of the polarization, as the following computation shows. The proof is a direct application of the symplectic normal form for E recorded in definition 3.2.3.

The order of $K(L)$ is the payload of the next result, and it will be reproved representation-theoretically at the end of the section. The motivating point is that $|K(L)|$ is a perfect square, $(d_1 \cdots d_g)^2$, whose square root is the dimension of $H^0(X, L)$; the theta group will explain why.

Proposition 4.1.3: The Order of $K(L)$

Let $L = L(H, \chi)$ with $H > 0$ a polarization of type $(d_1, \dots, d_g)$, so $E = \text{Im} H$ is nondegenerate integral alternating on Λ . Then

$$K(L) \cong \left(\bigoplus_{i=1}^g \mathbb{Z}/d_i \mathbb{Z} \right)^2$$

and in particular

$$|K(L)| = (d_1 d_2 \cdots d_g)^2 = \det E$$

Proof:

By definition 3.2.3 there is a symplectic basis $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ of Λ in which E has matrix

$$\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}, \quad D = \text{diag}(d_1, \dots, d_g), \quad d_1 | d_2 | \cdots | d_g$$

that is, $E(\lambda_i, \mu_j) = d_i \delta_{ij}$ and $E(\lambda_i, \lambda_j) = E(\mu_i, \mu_j) = 0$. Since Λ has real rank $2g$ and spans V over $\mathbb{R}$, the vectors $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ form an $\mathbb{R}$ -basis of V , so every $v \in V$ has unique real coordinates against them.

Step 1: Coordinatize $\Lambda(L)$. Write a general $v \in V$ in the symplectic basis as $v = \sum_i x_i \lambda_i + \sum_i y_i \mu_i$ with $x_i, y_i \in \mathbb{R}$. From the matrix above, $E(\lambda_i, \lambda_j) = 0$ and $E(\mu_i, \lambda_j) = -E(\lambda_j, \mu_i) = -d_i \delta_{ij}$, while $E(\lambda_i, \mu_j) = d_i \delta_{ij}$ and $E(\mu_i, \mu_j) = 0$. Pairing v against the basis vectors of Λ ,

$$E(v, \lambda_j) = \sum_i y_i E(\mu_i, \lambda_j) = -d_j y_j, \quad E(v, \mu_j) = \sum_i x_i E(\lambda_i, \mu_j) = d_j x_j$$

Thus $v \in \Lambda(L)$ if and only if $d_j x_j \in \mathbb{Z}$ and $d_j y_j \in \mathbb{Z}$ for every j , that is, if and only if $x_j \in \frac{1}{d_j} \mathbb{Z}$ and $y_j \in \frac{1}{d_j} \mathbb{Z}$ for all j . Hence

$$\Lambda(L) = \sum_{i=1}^g \frac{1}{d_i} \mathbb{Z} \lambda_i + \sum_{i=1}^g \frac{1}{d_i} \mathbb{Z} \mu_i$$

Step 2: Take the quotient. The lattice $\Lambda = \sum_i \mathbb{Z} \lambda_i + \sum_i \mathbb{Z} \mu_i$ sits inside $\Lambda(L)$ coordinatewise, and the quotient splits as a direct sum over the $2g$ coordinate directions:

$$K(L) = \Lambda(L)/\Lambda \cong \bigoplus_{i=1}^g \left(\frac{1}{d_i} \mathbb{Z}/\mathbb{Z} \right) \oplus \bigoplus_{i=1}^g \left(\frac{1}{d_i} \mathbb{Z}/\mathbb{Z} \right)$$

Each factor $\frac{1}{d_i} \mathbb{Z}/\mathbb{Z}$ is cyclic of order d_i , isomorphic to $\mathbb{Z}/d_i \mathbb{Z}$ via multiplication by d_i . Grouping the two copies gives $K(L) \cong \left(\bigoplus_i \mathbb{Z}/d_i \mathbb{Z} \right)^2$, as claimed.

Step 3: Compute the order. The order is the product of the orders of the cyclic factors:

$$|K(L)| = \prod_{i=1}^g d_i \cdot \prod_{i=1}^g d_i = (d_1 \cdots d_g)^2$$

Finally, in the symplectic basis $\det E = \det \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix} = \det(D)^2 = (d_1 \cdots d_g)^2$, and the determinant of the Gram matrix of an alternating form is a basis-independent invariant up to the square of the change-of-basis determinant, which is 1 for a change between $\mathbb{Z}$ -bases of Λ . Hence $|K(L)| = \det E$, as we sought to show.

The proof exhibits $K(L)$ as $2g$ cyclic pieces, one $\mathbb{Z}/d_i$ for each “ λ -direction” and one for each “ μ -direction” of the symplectic basis. The square in $(d_1 \cdots d_g)^2$ is not cosmetic: it is the statement that $K(L)$ pairs with itself. Half of $K(L)$ comes from the λ -directions and half from the μ -directions, and the alternating form E pairs the two halves nondegenerately, exactly as the standard symplectic form pairs the two Lagrangian halves of a symplectic vector space. This self-pairing is the commutator of the theta group, developed below; the appearance of $\sqrt{|K(L)|} = d_1 \cdots d_g$ as $\dim H^0(X, L)$ is then the statement that $H^0(X, L)$ is a Lagrangian-sized representation of $K(L)$.

Example 4.1.4: $K(L)$ for a Principal Polarization

A principal polarization has type $(1, \dots, 1)$, so $E = \text{Im} H$ is *unimodular* on Λ : its Gram matrix is the standard symplectic matrix $\begin{pmatrix} 0 & I_g \\ -I_g & 0 \end{pmatrix}$, with $\det E = 1$. Following definition 4.1.1, the condition $E(v, \lambda) \in \mathbb{Z}$ for all $\lambda \in \Lambda$ then already forces $v \in \Lambda$, because the coordinate map $v \mapsto E(v, -)|_\Lambda$ carries Λ isomorphically onto $\Lambda^\vee$. Thus $\Lambda(L) = \Lambda$ and $K(L) = 0$ is trivial, with $|K(L)| = 1 = 1^2$. By theorem 3.4.2 the corresponding space of sections is one-dimensional: a principally polarized abelian variety has, up to scalar, a single theta function, whose divisor is the theta divisor. The higher-type case $|K(L)| = (d_1 \cdots d_g)^2 > 1$ measures precisely the failure of this uniqueness.

With $K(L)$ in hand, the nonabelian group sitting above it comes into view. The line bundle L has automorphisms, and it also has “partial symmetries”: for a translation $t_x: X \rightarrow X$ by a point $x = \bar{v} \in X$, one can ask whether the pulled-back bundle $t_x^* L$ is isomorphic to L . The set of x for which this holds is exactly $K(L)$ (this is the analytic content of definition 4.1.2, and will be identified with $\ker \varphi_L$ in Chapter 5). The theta group collects, for each such x , the isomorphisms $t_x^* L \xrightarrow{\sim} L$ themselves, not just the fact that one exists. Because the automorphisms of a line bundle over a compact connected complex manifold are the nonzero scalars $\mathbb{C}^*$, choosing such an isomorphism is a $\mathbb{C}^*$ -torsor of choices, and the resulting group is a central extension of $K(L)$ by $\mathbb{C}^*$.

Definition 4.1.5: The Theta Group

Let L be a line bundle on $X = V/\Lambda$. The *theta group* (or *Heisenberg group*) of L is

$$\mathcal{G}(L) := \{ (x, \psi) : x \in K(L), \psi: t_x^* L \xrightarrow{\sim} L \text{ an isomorphism of line bundles} \}$$

with multiplication

$$(x, \psi) \cdot (x', \psi') = (x + x', \psi \circ t_x^* \psi')$$

where $t_x^* \psi': t_{x+x'}^* L = t_x^*(t_{x'}^* L) \xrightarrow{\sim} t_x^* L$ is the pullback of ψ' along t_x , so that the composite lands in L .

Equivalently, and this is the description used in the sequel, $\mathcal{G}(L)$ is the group of pairs (x, ψ) acting on the total space of L by a bundle map covering the translation t_x of the base. The composition law is exactly the composition of such covering maps: to translate by $x + x'$ and identify the result with L , first translate by x' and identify (via ψ'), then translate by x and identify (via ψ), pulling ψ' back along t_x so the two identifications compose. This makes $\mathcal{G}(L)$ literally the symmetry group of

the pair (X, L) over the translations that fix the isomorphism class of L .

Proposition 4.1.6: The Theta Group is a Central Extension

Let L be a line bundle on X . The map sending a nonzero scalar $c \in \mathbb{C}^*$ to the automorphism $(0, c) \in \mathcal{G}(L)$ (translation by 0, identified with L via multiplication by c) and the map $(x, \psi) \mapsto x$ fit into a short exact sequence of groups

$$1 \longrightarrow \mathbb{C}^* \longrightarrow \mathcal{G}(L) \longrightarrow K(L) \longrightarrow 0 \quad (4.1)$$

in which the image of $\mathbb{C}^*$ is central. Equivalently, $\mathcal{G}(L)$ acts $\mathbb{C}$ -linearly on $H^0(X, L)$, the central $\mathbb{C}^*$ acting by scalar multiplication.

Proof:

Step 1: Exactness. The second map $\pi: (x, \psi) \mapsto x$ is a homomorphism by the multiplication law, and it is surjective because every $x \in K(L)$ admits an isomorphism $t_x^* L \cong L$ by the defining property of $K(L)$ (definition 4.1.2). Its kernel consists of pairs $(0, \psi)$ with $\psi: L \xrightarrow{\sim} L$ an automorphism of L . The torus X is compact and connected (proposition 1.1.3), so by the maximum principle its global holomorphic functions are the constants; hence a line-bundle automorphism is multiplication by a nowhere-zero global function, that is, by a nonzero constant $c \in \mathbb{C}^*$. This identifies $\ker \pi$ with $\mathbb{C}^*$ and gives (4.1).

Step 2: Centrality. An element $(0, c)$ with $c \in \mathbb{C}^*$ acts on the fibres of L by the scalar c and covers the identity translation. For any (x, ψ) ,

$$(0, c) \cdot (x, \psi) = (x, c \cdot \psi), \quad (x, \psi) \cdot (0, c) = (x, \psi \circ t_x^*(c)) = (x, c \cdot \psi)$$

because pulling back the scalar c along any translation returns the same scalar c (scalars are global constants), and scalar multiplication commutes with the linear isomorphism ψ . Hence the two products agree and $\mathbb{C}^*$ is central.

Step 3: The action on sections. A pair $(x, \psi) \in \mathcal{G}(L)$ acts on $s \in H^0(X, L)$ by

$$(x, \psi) \cdot s := \psi \circ t_x^* s \circ t_{-x} = \psi(s \circ t_{-x})$$

translating the section by $-x$ and reidentifying via ψ ; this is again a global section of L , and the composition law of definition 4.1.5 makes it a left action. The central $(0, c)$ sends s to $c s$, scalar multiplication. This is the promised linear representation, completing the proof.

The exact sequence (4.1) is the structural heart of the section. It says that $\mathcal{G}(L)$ is not much larger than the finite group $K(L)$: it is $K(L)$ with a copy of $\mathbb{C}^*$ threaded through its center, exactly the pattern of the Heisenberg group $\begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix}$, whose center is the scalars along the top-right entry and whose quotient by the center is the abelian $(*, *)$. The nonabelianness of $\mathcal{G}(L)$ is concentrated in a single invariant, the commutator, which because $\mathbb{C}^*$ is central and $K(L)$ is abelian descends to a pairing on $K(L)$ itself.

To see this, notice that for $(x, \psi), (y, \varphi) \in \mathcal{G}(L)$ the commutator $(x, \psi)(y, \varphi)(x, \psi)^{-1}(y, \varphi)^{-1}$ projects under π to $x + y - x - y = 0$ in the abelian group $K(L)$, hence lies in the central $\mathbb{C}^*$. It is therefore a scalar, and because $\mathbb{C}^*$ is central the scalar is unchanged under replacing (x, ψ) by $(x, c\psi)$ or (y, φ) by $(y, c'\varphi)$: the commutator depends only on the images $x, y \in K(L)$. This produces a well-defined

pairing $K(L) \times K(L) \rightarrow \mathbb{C}^*$, whose value the next result computes explicitly from the alternating form E .

Proposition 4.1.7: The Commutator Pairing

Let $L = L(H, \chi)$ with $E = \text{Im}H$. The commutator in $\mathcal{G}(L)$ induces a map

$$e^L: K(L) \times K(L) \rightarrow \mathbb{C}^*, \quad e^L(\bar{v}, \bar{w}) = (w, \varphi)(v, \psi)(w, \varphi)^{-1}(v, \psi)^{-1}$$

where $(v, \psi), (w, \varphi) \in \mathcal{G}(L)$ are any lifts of $\bar{v}, \bar{w} \in K(L)$; the value depends only on $\bar{v}, \bar{w}$, and it is given by the formula

$$e^L(\bar{v}, \bar{w}) = \exp(2\pi i E(v, w)) \quad (4.2)$$

for any representatives $v, w \in \Lambda(L)$ of $\bar{v}, \bar{w} \in K(L) = \Lambda(L)/\Lambda$. The pairing e^L is alternating and nondegenerate when $H > 0$.

Proof:

Step 1: Well-definedness of the formula on $K(L)$. The right side of (4.2) must not depend on the choice of representatives $v \in \bar{v}$ and $w \in \bar{w}$ modulo Λ . Replace v by $v + \lambda$ with $\lambda \in \Lambda$; then

$$E(v + \lambda, w) = E(v, w) + E(\lambda, w)$$

Now $w \in \Lambda(L)$, so by definition 4.1.1 $E(w, \lambda) \in \mathbb{Z}$ for all $\lambda \in \Lambda$, and $E(\lambda, w) = -E(w, \lambda) \in \mathbb{Z}$ since E is alternating. Thus $E(v + \lambda, w) \equiv E(v, w) \pmod{\mathbb{Z}}$ and $\exp(2\pi i E(v + \lambda, w)) = \exp(2\pi i E(v, w))$. Symmetrically the value is unchanged when w is altered by a lattice vector. This is precisely the point where the enlargement to $\Lambda(L)$ is used: the inclusion $E(\Lambda(L), \Lambda) \subseteq \mathbb{Z}$ is what makes e^L well-defined on the quotient $K(L)$.

Step 2: The commutator equals the formula. Realize $\mathcal{G}(L)$ through the canonical factor of automorphy of definition 2.2.5. A section of L is a holomorphic $\theta: V \rightarrow \mathbb{C}$ with $\theta(u + \lambda) = a_{(H, \chi)}(\lambda, u) \theta(u)$, where $a_{(H, \chi)}(\lambda, u) = \chi(\lambda) \exp(\pi H(u, \lambda) + \frac{\pi}{2} H(\lambda, \lambda))$. For $v \in \Lambda(L)$, the translation operator on functions

$$(U_v \theta)(u) := \exp\left(\pi H(u, v) + \frac{\pi}{2} H(v, v)\right)^{-1} \theta(u + v)$$

carries sections of L to sections of L ; the prefactor is the value at v of the canonical automorphy factor, chosen so that $U_v \theta$ again satisfies the transformation law for Λ . To verify this, compute $(U_v \theta)(u + \lambda)$ from the definitions: pushing $\theta(u + v + \lambda) = a_{(H, \chi)}(\lambda, u + v) \theta(u + v)$ through the prefactor, the ratio $(U_v \theta)(u + \lambda) / (a_{(H, \chi)}(\lambda, u) (U_v \theta)(u))$ collapses, after the $H(\cdot, \lambda)$ and $H(\cdot, v)$ terms cancel in the exponent, to $\exp(\pi(H(v, \lambda) - H(\lambda, v))) = \exp(2\pi i E(v, \lambda))$. This equals 1 for every $\lambda \in \Lambda$ precisely when $E(v, \lambda) \in \mathbb{Z}$ for all λ , that is, exactly when $v \in \Lambda(L)$ (definition 4.1.1); this is why the enlargement to $\Lambda(L)$ is the exact condition for U_v to preserve the transformation law. Indeed U_v is a concrete lift of translation by $\bar{v}$ to a symmetry of L , hence an element of $\mathcal{G}(L)$ over $\bar{v} \in K(L)$. Composing,

$$(U_v U_w \theta)(u) = \exp\left(-\pi H(u, v) - \frac{\pi}{2} H(v, v) - \pi H(u + v, w) - \frac{\pi}{2} H(w, w)\right) \theta(u + v + w)$$

and the same expression with v, w interchanged for $(U_w U_v \theta)(u)$. Subtracting the two exponents, the terms $-\frac{\pi}{2} H(v, v)$ and $-\frac{\pi}{2} H(w, w)$ cancel, and expanding $H(u + v, w) = H(u, w) + H(v, w)$ and

$H(u+w, v) = H(u, v) + H(w, v)$ shows all terms linear in u cancel as well, leaving only the constant $\pi(H(w, v) - H(v, w))$. Using $H(v, w) - H(w, v) = H(v, w) - \overline{H(v, w)} = 2i \operatorname{Im} H(v, w) = 2i E(v, w)$, this constant is $-2\pi i E(v, w)$, so $(U_v U_w \theta)(u) = e^{-2\pi i E(v, w)} (U_w U_v \theta)(u)$, that is,

$$U_w U_v = \exp(2\pi i E(v, w)) U_v U_w$$

Hence the group commutator $U_w U_v U_w^{-1} U_v^{-1}$ is the scalar $\exp(2\pi i E(v, w))$, and this is the value of $e^L(\bar{v}, \bar{w})$ in the ordering convention of (4.2). (Reversing the order replaces $E(v, w)$ by $E(w, v) = -E(v, w)$, consistent with the alternating property.)

Step 3: Alternating and nondegenerate. The form is alternating because E is: $e^L(\bar{v}, \bar{v}) = \exp(2\pi i E(v, v)) = \exp(0) = 1$. For nondegeneracy, suppose $\bar{w} \in K(L)$ satisfies $e^L(\bar{v}, \bar{w}) = 1$ for all $\bar{v} \in K(L)$, that is, $E(v, w) \in \mathbb{Z}$ for all $v \in \Lambda(L)$. Work in the symplectic coordinates of proposition 4.1.3, where $\Lambda(L) = \sum_i \frac{1}{d_i} \mathbb{Z} \lambda_i + \sum_i \frac{1}{d_i} \mathbb{Z} \mu_i$ and $w = \sum_i a_i \lambda_i + \sum_i b_i \mu_i$ with $a_i, b_i \in \frac{1}{d_i} \mathbb{Z}$. Testing against $v = \frac{1}{d_j} \lambda_j \in \Lambda(L)$ gives $E(v, w) = b_j \in \mathbb{Z}$, and testing against $v = \frac{1}{d_j} \mu_j$ gives $E(v, w) = -a_j \in \mathbb{Z}$. Thus every $a_j, b_j \in \mathbb{Z}$, so $w \in \Lambda$ and $\bar{w} = 0$. Hence e^L is nondegenerate, completing the proof.

The pairing e^L is the arithmetic shadow of the entire theta-group construction. It is a nondegenerate alternating pairing on a finite abelian group, and such a pairing on $K(L) \cong (\bigoplus \mathbb{Z}/d_i)^2$ realizes the group as an orthogonal sum of hyperbolic planes $\mathbb{Z}/d_i \oplus \mathbb{Z}/d_i$, the two summands of proposition 4.1.3 pairing against each other. Restricted to the n -torsion this pairing is the *Weil pairing* $e_n: X[n] \times X[n] \rightarrow \mu_n$ that Chapter 5 constructs analytically and Part II reconstructs over an arbitrary field as a duality of finite group schemes; the commutator of the theta group is where it is born. The nondegeneracy just proved is what forces $H^0(X, L)$ to be irreducible under $\mathcal{G}(L)$, by the following classical fact about Heisenberg groups.

A finite Heisenberg group, a central extension of a finite abelian group K by $\mathbb{C}^*$ whose commutator pairing is nondegenerate, has a unique irreducible representation, up to isomorphism, on which the center acts by scalars. This is the Stone-von Neumann theorem in its finite form. The uniqueness is a general fact about the Heisenberg group and its proof, a counting argument via the characters of a maximal isotropic subgroup, is standard; the representation-theoretic details are given in Mumford, *Abelian Varieties*, and are cited here rather than reproved (the mechanism is the same as for the infinite Heisenberg group in quantum mechanics). What matters for this book is the payoff: the space of sections of a polarization is that unique representation.

Theorem 4.1.8: Stone-von Neumann for the Theta Group

Let $L = L(H, \chi)$ with $H > 0$ a polarization of type $(d_1, \dots, d_g)$. Then $\mathcal{G}(L)$ has, up to isomorphism, a unique irreducible representation on which the central subgroup $\mathbb{C}^*$ acts by the identity character (scalar multiplication). Its dimension is

$$\sqrt{|K(L)|} = d_1 d_2 \cdots d_g$$

and the representation of $\mathcal{G}(L)$ on $H^0(X, L)$ from proposition 4.1.6 is isomorphic to it. In particular

$$\dim_{\mathbb{C}} H^0(X, L) = d_1 d_2 \cdots d_g$$

reproving theorem 3.4.2 representation-theoretically.

Proof:

Step 1: Uniqueness and dimension of the abstract representation. By proposition 4.1.7 the extension (4.1) has nondegenerate commutator pairing e^L on $K(L)$, so $\mathcal{G}(L)$ is a finite Heisenberg group over $K(L)$. The Stone-von Neumann theorem for such a group provides a unique irreducible representation W with $\mathbb{C}^*$ acting by the identity character; its dimension is $\sqrt{|K(L)|}$. Concretely, choose a maximal isotropic subgroup $M \subseteq K(L)$ for e^L ; nondegeneracy gives $|M|^2 = |K(L)|$. The preimage $\tilde{M} := \pi^{-1}(M) \subseteq \mathcal{G}(L)$ is abelian (isotropic subgroups lift to abelian subgroups, since the commutator of two lifts is e^L evaluated on their images, which is 1 on M), so a choice of character of $\tilde{M}$ extending the identity character of $\mathbb{C}^*$ induces a representation $W = \text{Ind}_{\tilde{M}}^{\mathcal{G}(L)}$ of dimension $[K(L) : M] = |K(L)|/|M| = |M| = \sqrt{|K(L)|}$. By proposition 4.1.3, $\sqrt{|K(L)|} = \sqrt{(d_1 \cdots d_g)^2} = d_1 \cdots d_g$. The uniqueness is the content cited to Mumford above.

Step 2: $H^0(X, L)$ carries this representation. By proposition 4.1.6, $\mathcal{G}(L)$ acts on $V_0 := H^0(X, L)$ with $\mathbb{C}^*$ acting by scalars, that is, V_0 is a representation of $\mathcal{G}(L)$ on which the center acts by the identity character. Any such representation is a direct sum of copies of the unique irreducible W (all irreducibles with a fixed nontrivial central character are isomorphic to W , again by Stone-von Neumann). To conclude $V_0 \cong W$ it suffices to show $V_0 \neq 0$ and that the multiplicity is one.

Step 3: Nonvanishing and multiplicity one. That $V_0 = H^0(X, L) \neq 0$ for $H > 0$ is proposition 2.4.3: a positive line bundle has nonzero sections. Write $V_0 \cong W^{\oplus m}$ for some $m \geq 1$. To pin down m , restrict to the maximal abelian $\tilde{M} = \pi^{-1}(M)$ of Step 1, realized by the commuting operators U_v of proposition 4.1.7 for v in the lattice $\Lambda(M)$ (with $\Lambda \subseteq \Lambda(M) \subseteq \Lambda(L)$ the sublattice corresponding to M). The joint eigenspaces of this commuting family on V_0 each have dimension m , since on W each eigencharacter of $\tilde{M}$ occurs exactly once. But a joint eigenfunction is a holomorphic $\theta: V \rightarrow \mathbb{C}$ satisfying a full system of quasi-periodicity relations for the lattice $\Lambda(M)$, whose Fourier coefficients are then determined up to a single overall scalar; hence every joint eigenspace is at most one-dimensional. Comparing gives $m = 1$. (This last step is the abstract shadow of the explicit basis constructed in the next section, where the joint eigenfunctions are exhibited as the theta functions with characteristics.) Hence $V_0 \cong W$ and

$$\dim_{\mathbb{C}} H^0(X, L) = \dim_{\mathbb{C}} W = d_1 \cdots d_g$$

This recovers the value of theorem 3.4.2 with no convergence estimate, completing the proof.

The theorem reverses the logic of Chapter 3. There the dimension $d_1 \cdots d_g$ was computed by counting lattice points and estimating a theta series; here it is forced by a symmetry principle. The space $H^0(X, L)$ is the unique irreducible representation of $\mathcal{G}(L)$ with the given central character, and its dimension is pinned by the order of $K(L)$ alone. This is why the basis of theta functions constructed in the next section is not arbitrary: the theta group acts on $H^0(X, L)$, its maximal isotropic subgroups give simultaneous eigenbases, and the functions with characteristics $a \in \bigoplus_i (\frac{1}{d_i} \mathbb{Z})/\mathbb{Z}$ are exactly the eigenvectors of one such isotropic system. The theta group is the intrinsic symmetry that makes the natural basis of theta functions *canonical*, and its commutator e^L is the germ of the Weil pairing constructed analytically in Chapter 5.

The whole construction is transparent in dimension one, where it reduces to the finite Heisenberg groups underlying the classical Jacobi theta functions.

Example 4.1.9: The Theta Group of an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$, and let L be a line bundle of degree $d \geq 1$, so $E = \text{Im}H$ is the alternating form on Λ with $E(1, \tau) = d$ (the polarization of type (d) , matching [3]). There is a single elementary divisor $d_1 = d$, so $g = 1$ and the type is (d) .

1. *Principal case* $d = 1$. Here $E(1, \tau) = 1$, so E is unimodular and $\Lambda(L) = \Lambda$, giving $K(L) = 0$ (example 4.1.4). The theta group is just its center $\mathbb{C}^*$, and $H^0(E, L)$ is one-dimensional: the unique classical theta function $\vartheta(z, \tau)$ up to scalar. This is the analytic source of the single-point theta divisor on an elliptic curve.
2. *Degree* d . By definition 4.1.1, $v \in \Lambda(L)$ iff $E(v, 1) \in \mathbb{Z}$ and $E(v, \tau) \in \mathbb{Z}$. Writing $v = x + y\tau$ with $x, y \in \mathbb{R}$ and using $E(1, \tau) = d$, these conditions read $-dy \in \mathbb{Z}$ and $dx \in \mathbb{Z}$, so

$$\Lambda(L) = \frac{1}{d}\mathbb{Z} + \frac{1}{d}\mathbb{Z}\tau, \quad K(L) = \Lambda(L)/\Lambda \cong \frac{1}{d}\mathbb{Z}/\mathbb{Z} \oplus \frac{1}{d}\mathbb{Z}/\mathbb{Z} \cong (\mathbb{Z}/d)^2$$

consistent with proposition 4.1.3: $|K(L)| = d^2 = \det E$. The commutator pairing (4.2) on the generators $\frac{1}{d}$ and $\frac{\tau}{d}$ is

$$e^L\left(\frac{1}{d}, \frac{\tau}{d}\right) = \exp\left(2\pi i E\left(\frac{1}{d}, \frac{\tau}{d}\right)\right) = \exp\left(2\pi i \cdot \frac{1}{d^2} E(1, \tau)\right) = \exp\left(\frac{2\pi i}{d}\right)$$

a primitive d -th root of unity. Thus $\mathcal{G}(L)$ is a central extension of $(\mathbb{Z}/d)^2$ by $\mathbb{C}^*$, generated over the center by the two lifted translation operators $S := U_{1/d}$ and $T := U_{\tau/d}$ of proposition 4.1.7, translation by $\frac{1}{d}$ and by $\frac{\tau}{d}$ together with their canonical automorphy prefactors. Their commutator is the root of unity just computed,

$$TST^{-1}S^{-1} = e^L\left(\frac{1}{d}, \frac{\tau}{d}\right) = e^{2\pi i/d} =: \zeta_d$$

so S and T satisfy $TS = \zeta_d ST$. These are the operators under which the d classical Jacobi theta functions with characteristics transform among themselves; S is the shift by a d -torsion translation and T the quasi-periodic “modulation” encoded by the automorphy factor.

3. *Stone-von Neumann*. By theorem 4.1.8 the unique irreducible representation of $\mathcal{G}(L)$ with the standard central character has dimension $\sqrt{d^2} = d$, and $H^0(E, L)$ is that representation. Hence $\dim H^0(E, L) = d = \deg L$, recovering the Riemann-Roch count example 3.4.6 for the elliptic curve, now as the dimension of the standard representation of the finite Heisenberg group $H(\mathbb{Z}/d)$.

The pairing $\zeta_d = e^{2\pi i/d}$ read off in part (2) is the Weil pairing e_d on $E[d]$, and the operators S, T are the theta-group symmetries that permute the d Jacobi theta functions. This is the $g = 1$ prototype of the entire section: the theta group is a finite Heisenberg group, its commutator is the Weil pairing, and the sections of L form its Schrödinger representation.

4.2 The Classical Riemann Theta Function

The previous section identified the section space $H^0(X, L)$ of a polarization abstractly, as the unique irreducible representation of the theta group, and read off its dimension from the order of $K(L)$.

What that argument does not supply is a single function. This section writes one down. For a principally polarized abelian variety in normalized coordinates the period matrix is $(\tau \ I_g)$ with τ in the Siegel upper half-space (proposition 3.3.3), and there is a completely explicit holomorphic function on $\mathbb{C}^g$, given by a Fourier series indexed by the integer lattice, whose quasi-periodicity under the lattice $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$ is exactly the transformation law of a section of the principal polarization. It is the Riemann theta function, and it is the direct generalization to g variables of the Jacobi theta series the reader met in the elliptic case (example 2.4.4).

The construction begins from the coordinates furnished by the normalization. A principally polarized abelian variety of dimension g is, by proposition 3.3.3, the torus $X = \mathbb{C}^g/\Lambda$ with $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$ for a matrix $\tau \in \mathfrak{H}_g$ (definition 3.3.1); the lattice is spanned by the columns of τ together with the standard basis vectors of $\mathbb{Z}^g$. A section of the principal polarization is a holomorphic function on $\mathbb{C}^g$ reproducing itself, up to the canonical factor of automorphy, under translation by each of these $2g$ generators. Riemann's insight is that the two families of generators play asymmetric roles: translation by the integer vectors $\mathbb{Z}^g$ should leave the function unchanged, which forces a Fourier expansion in $e^{2\pi i n^t z}$, and translation by the columns of τ then dictates a Gaussian weight $e^{\pi i n^t \tau n}$ on the Fourier coefficients. The result is a single series.

Definition 4.2.1: The Riemann Theta Function

Let $\tau \in \mathfrak{H}_g$ be a point of the Siegel upper half-space (definition 3.3.1). The *Riemann theta function* associated to τ is the function $\theta(-, \tau): \mathbb{C}^g \rightarrow \mathbb{C}$ defined by the series

$$\theta(z, \tau) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i n^t \tau n + 2\pi i n^t z) \quad (4.3)$$

where $z \in \mathbb{C}^g$ is written as a column vector, n ranges over the integer lattice $\mathbb{Z}^g$, and $n^t \tau n$ and $n^t z$ are the usual matrix products (so $n^t \tau n = \sum_{j,k} n_j \tau_{jk} n_k$ is a scalar). The convergence of the series is established in proposition 4.2.2.

The series (4.3) is the canonical section of the principal polarization on the torus with period matrix $(\tau \ I_g)$, the explicit holomorphic function whose existence proposition 2.4.3 and the Stone-von Neumann argument of theorem 4.1.8 could guarantee but not exhibit. Two structural features are already visible in the formula. The summation over $\mathbb{Z}^g$ builds in periodicity under the integer part of the lattice: replacing z by $z + m$ for $m \in \mathbb{Z}^g$ multiplies each term by $e^{2\pi i n^t m} = 1$, so θ is literally $\mathbb{Z}^g$ -periodic, and this is why θ is a Fourier series in the first place. The Gaussian factor $e^{\pi i n^t \tau n}$, which decays because $\text{Im} \tau > 0$, is what makes the series converge and what governs its behaviour under the other half of the lattice, the columns of τ . The quasi-periodicity computed below shows that these two behaviours are exactly the values of the canonical factor of automorphy on a symplectic basis, so θ descends to a section of the line bundle it names.

Before any of that, the series must be shown to define a function at all. The decay of the Gaussian weight is the entire content of the convergence, and it is worth extracting the estimate in full, because the same estimate reappears every time a theta series is differentiated, translated, or specialized.

Proposition 4.2.2: Absolute and Uniform Convergence of the Theta Series

For each fixed $\tau \in \mathfrak{H}_g$ the series (4.3) converges absolutely, and uniformly on every compact subset of $\mathbb{C}^g$. Consequently $\theta(-, \tau)$ is a holomorphic function on $\mathbb{C}^g$. Moreover the convergence is uniform in τ as well, on compact subsets of $\mathfrak{H}_g \times \mathbb{C}^g$, so θ is holomorphic jointly in (z, τ) on $\mathfrak{H}_g \times \mathbb{C}^g$.

Proof:

The strategy is to bound the modulus of the general term of (4.3) by a Gaussian in n and apply the Weierstrass M -test.

Step 1: The modulus of a term. Fix $n \in \mathbb{Z}^g$. Since the term is $\exp(\pi i n^t \tau n + 2\pi i n^t z)$, its modulus is the exponential of the real part of the exponent. Writing Im for the imaginary part of a complex number,

$$|\exp(\pi i n^t \tau n + 2\pi i n^t z)| = \exp(\text{Re}(\pi i n^t \tau n) + \text{Re}(2\pi i n^t z)) = \exp(-\pi n^t (\text{Im} \tau) n - 2\pi n^t \text{Im} z)$$

where the first term uses $\text{Re}(i w) = -\text{Im} w$ applied to the complex number $w = n^t \tau n$, giving $\text{Re}(\pi i n^t \tau n) = -\pi \text{Im}(n^t \tau n) = -\pi n^t (\text{Im} \tau) n$ since n is a real vector; and the second uses $\text{Re}(2\pi i n^t z) = -2\pi \text{Im}(n^t z) = -2\pi n^t \text{Im} z$.

Step 2: A Gaussian majorant on a compact set. Let $K \subset \mathbb{C}^g$ be compact. Since $\text{Im} \tau$ is a positive definite real symmetric matrix (definition 3.3.1), its smallest eigenvalue $c > 0$ gives the bound $n^t (\text{Im} \tau) n \geq c |n|^2$ for every $n \in \mathbb{R}^g$, where $|n|$ is the Euclidean norm. And since K is compact there is a radius $R \geq 0$ with $|\text{Im} z| \leq R$ for all $z \in K$, so by Cauchy-Schwarz $|n^t \text{Im} z| \leq |n| R$. Combining these with Step 1,

$$|\exp(\pi i n^t \tau n + 2\pi i n^t z)| \leq \exp(-\pi c |n|^2 + 2\pi R |n|) \quad \text{for all } z \in K \quad (4.4)$$

The right-hand side is independent of z , and completing the square in the exponent, $-\pi c |n|^2 + 2\pi R |n| = -\pi c (|n| - R/c)^2 + \pi R^2/c$, shows it is bounded above by $\exp(\pi R^2/c) \exp(-\frac{\pi c}{4} |n|^2)$ once $|n| \geq 2R/c$, since then $(|n| - R/c)^2 \geq |n|^2/4$. Thus, discarding the finitely many n with $|n| < 2R/c$ from the tail, the general term is dominated by a constant multiple of $\exp(-\frac{\pi c}{4} |n|^2)$.

Step 3: Summability of the majorant. The series $\sum_{n \in \mathbb{Z}^g} \exp(-\frac{\pi c}{4} |n|^2)$ converges: it factors as

$$\sum_{n \in \mathbb{Z}^g} \exp\left(-\frac{\pi c}{4} |n|^2\right) = \prod_{j=1}^g \left(\sum_{n_j \in \mathbb{Z}} \exp\left(-\frac{\pi c}{4} n_j^2\right) \right) = \left(\sum_{m \in \mathbb{Z}} e^{-\pi c m^2/4} \right)^g$$

and each one-dimensional factor is a convergent series, being dominated by the convergent geometric series $\sum_m e^{-(\pi c/4)|m|}$ for $|m| \geq 1$ (as $m^2 \geq |m|$). By the Weierstrass M -test the majorant from (4.4) makes the series (4.3) converge absolutely and uniformly on K . A uniform limit of holomorphic functions is holomorphic, and each partial sum is holomorphic in z , so $\theta(-, \tau)$ is holomorphic on $\mathbb{C}^g$.

Step 4: Joint holomorphy. Let $K' \subset \mathfrak{H}_g$ be compact. On K' the smallest eigenvalue of $\text{Im} \tau$ attains a positive minimum $c' > 0$ by continuity and compactness, so the constant c in Step 2 may be taken to be c' uniformly over $\tau \in K'$; the bound (4.4) then holds with c' in place of c for all $(\tau, z) \in K' \times K$, and Step 3 gives a single summable majorant independent of (τ, z) . Hence the series converges uniformly on $K' \times K$. Each term is holomorphic in (z, τ) jointly, so the sum is holomorphic on $\mathfrak{H}_g \times \mathbb{C}^g$, completing the proof.

The estimate (4.4) isolates the mechanism: positivity of $\text{Im}\tau$ turns the oscillatory factor $e^{\pi i n^t \tau n}$ into a genuine Gaussian $e^{-\pi n^t (\text{Im}\tau) n}$, whose super-exponential decay in $|n|$ overwhelms the at-most-exponential growth $e^{2\pi R|n|}$ coming from the compact set of z -values. This is the analytic reason the Siegel condition $\text{Im}\tau > 0$ is not optional: outside $\mathfrak{H}_g$ the weight fails to decay in some direction and the series diverges, so the theta function exists precisely on the domain that parametrizes principally polarized abelian varieties. The joint holomorphy in (z, τ) is what allows θ to be viewed as a single holomorphic object over the whole family, the germ of a section of a line bundle on the universal abelian variety over $\mathfrak{H}_g$; that global viewpoint belongs to the theory of Siegel modular forms, but the convergence estimate is already its foundation.

With θ established as a holomorphic function, its defining property is the transformation law under $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$. The two halves of the lattice behave differently, and the computation is a reindexing of the series in each case.

Proposition 4.2.3: Quasi-Periodicity of the Riemann Theta Function

Let $\tau \in \mathfrak{H}_g$ and let $\theta = \theta(-, \tau)$ be the Riemann theta function of definition 4.2.1. For every $m \in \mathbb{Z}^g$ and every $z \in \mathbb{C}^g$, the periodicity relation

$$\theta(z + m, \tau) = \theta(z, \tau) \quad (4.5)$$

holds under the integer part of the lattice, and the quasi-periodicity relation

$$\theta(z + \tau m, \tau) = \exp(-\pi i m^t \tau m - 2\pi i m^t z) \theta(z, \tau) \quad (4.6)$$

holds under the columns of τ .

Proof:

Both identities are proved by manipulating the absolutely convergent series (4.3), which may be reindexed freely.

Step 1: Periodicity under $\mathbb{Z}^g$. For $m \in \mathbb{Z}^g$, substituting $z + m$ for z in (4.3) multiplies the n -th term by $\exp(2\pi i n^t m)$. Since $n, m \in \mathbb{Z}^g$, the inner product $n^t m$ is an integer, so $\exp(2\pi i n^t m) = 1$ and every term is unchanged. Hence

$$\theta(z + m, \tau) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i n^t \tau n + 2\pi i n^t (z + m)) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i n^t \tau n + 2\pi i n^t z) = \theta(z, \tau)$$

which is (4.5).

Step 2: The exponent under a τ -translation. For $m \in \mathbb{Z}^g$, substitute $z + \tau m$ for z in (4.3). The n -th exponent becomes

$$\pi i n^t \tau n + 2\pi i n^t (z + \tau m) = \pi i n^t \tau n + 2\pi i n^t z + 2\pi i n^t \tau m$$

The idea is to absorb the cross term $2\pi i n^t \tau m$ into the quadratic form by completing the square in n . Because τ is symmetric (definition 3.3.1), $n^t \tau m = m^t \tau n$, and

$$\pi i n^t \tau n + 2\pi i n^t \tau m = \pi i (n + m)^t \tau (n + m) - \pi i m^t \tau m$$

as one checks by expanding $(n + m)^t \tau (n + m) = n^t \tau n + 2n^t \tau m + m^t \tau m$ using the symmetry of τ . Substituting, the n -th exponent equals

$$\pi i (n + m)^t \tau (n + m) + 2\pi i n^t z - \pi i m^t \tau m$$

Step 3: Reindexing. Set $k = n + m$, a bijection of $\mathbb{Z}^g$ with itself. Then $n = k - m$, so $2\pi i n^t z = 2\pi i k^t z - 2\pi i m^t z$, and the n -th exponent becomes

$$\pi i k^t \tau k + 2\pi i k^t z - \pi i m^t \tau m - 2\pi i m^t z$$

The last two terms are independent of the summation index k and factor out of the sum. Therefore

$$\theta(z + \tau m, \tau) = \exp(-\pi i m^t \tau m - 2\pi i m^t z) \sum_{k \in \mathbb{Z}^g} \exp(\pi i k^t \tau k + 2\pi i k^t z) = \exp(-\pi i m^t \tau m - 2\pi i m^t z) \theta(z, \tau)$$

which is (4.6), completing the proof.

The two identities are precisely the values of the canonical Appell-Humbert factor of automorphy on the symplectic basis of $\Lambda = \tau\mathbb{Z}^g + \mathbb{Z}^g$. The columns of I_g , the integer generators, act trivially by (4.5), reflecting the vanishing of the factor's exponent on the isotropic sublattice $\mathbb{Z}^g$; the columns of τ act through the nowhere-vanishing multiplier $\exp(-\pi i m^t \tau m - 2\pi i m^t z)$ of (4.6). Up to the holomorphic coboundary passing between this classical normalization and the canonical factor $a_{(H, \chi)}(\lambda, v)$ of definition 2.2.5, that multiplier is the value of the factor of automorphy of the principal polarization, whose Hermitian form is $H(v, w) = v^t (\text{Im} \tau)^{-1} \bar{w}$ in the normalized coordinates, on the lattice vector $\lambda = \tau m$. Thus θ satisfies a functional equation of the shape (2.8) defining a section of L , and it is nonzero, so it spans the one-dimensional space $H^0(X, L)$ predicted by proposition 3.4.5 for a principal polarization. The existence that Chapter 3 could only infer from a dimension count is now witnessed by an explicit series.

For polarizations of higher type, and for organizing the whole space of sections, one needs not a single theta function but a finite family of them, obtained by shifting the summation lattice and the argument by constant vectors. These shifts are the characteristics, and they produce a two-parameter family of theta functions whose members transform by the same law up to explicit unimodular constants.

Definition 4.2.4: Theta Function with Characteristics

Let $\tau \in \mathfrak{H}_g$ and let $a, b \in \mathbb{R}^g$. The *theta function with characteristic* $\begin{bmatrix} a \\ b \end{bmatrix}$ is

$$\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z, \tau) = \sum_{n \in \mathbb{Z}^g} \exp\left(\pi i (n + a)^t \tau (n + a) + 2\pi i (n + a)^t (z + b)\right) \quad (4.7)$$

for $z \in \mathbb{C}^g$. The vectors a and b are the *characteristics*; a shifts the summation lattice and b shifts the argument. The plain Riemann theta function of definition 4.2.1 is the case $a = b = 0$, that is $\theta(z, \tau) = \theta\left[\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right](z, \tau)$.

The characteristic version is not a new kind of object but a translate of the old one, and recording the exact relation makes both its convergence and its transformation law immediate. Expanding the exponent in (4.7) and comparing with (4.3) yields the identity

$$\theta\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right](z, \tau) = \exp\left(\pi i a^t \tau a + 2\pi i a^t (z + b)\right) \theta(z + \tau a + b, \tau) \quad (4.8)$$

valid for all $a, b \in \mathbb{R}^g$ and $z \in \mathbb{C}^g$: substituting $z + \tau a + b$ into (4.3) and using the symmetry of τ to write $\pi i n^t \tau n + 2\pi i n^t \tau a = \pi i (n + a)^t \tau (n + a) - \pi i a^t \tau a$ reproduces the summand of (4.7) after

multiplying by the displayed prefactor. Since θ is holomorphic and converges absolutely and uniformly on compacta (proposition 4.2.2), so does $\theta[\frac{a}{b}]$, being θ evaluated at the translated argument $z + \tau a + b$ times a holomorphic prefactor. The characteristics thus produce a whole family of holomorphic sections without any new convergence analysis, and their transformation law follows from that of θ by tracking the two constant shifts.

Proposition 4.2.5: Quasi-Periodicity of Theta with Characteristics

Let $\tau \in \mathfrak{H}_g$ and $a, b \in \mathbb{R}^g$. For every $m \in \mathbb{Z}^g$ and $z \in \mathbb{C}^g$, translation by the integer part of the lattice gives

$$\theta[\frac{a}{b}](z + m, \tau) = \exp(2\pi i a^t m) \theta[\frac{a}{b}](z, \tau) \quad (4.9)$$

and translation by the columns of τ gives

$$\theta[\frac{a}{b}](z + \tau m, \tau) = \exp(-2\pi i b^t m) \exp(-\pi i m^t \tau m - 2\pi i m^t z) \theta[\frac{a}{b}](z, \tau) \quad (4.10)$$

Proof:

Step 1: Translation by an integer vector. Substituting $z + m$ for z in (4.7) multiplies the n -th term by $\exp(2\pi i (n + a)^t m)$. Splitting $(n + a)^t m = n^t m + a^t m$ and using $n^t m \in \mathbb{Z}$, this factor equals $\exp(2\pi i a^t m)$, which is independent of n and pulls out of the sum. This gives (4.9).

Step 2: Translation by a τ -column, via the translate identity. Apply (4.8) at the argument $z + \tau m$:

$$\theta[\frac{a}{b}](z + \tau m, \tau) = \exp(\pi i a^t \tau a + 2\pi i a^t (z + \tau m + b)) \theta(z + \tau m + \tau a + b, \tau)$$

The plain theta on the right is quasi-periodic under τm : writing $w = z + \tau a + b$ and applying (4.6) from proposition 4.2.3 at w ,

$$\theta(w + \tau m, \tau) = \exp(-\pi i m^t \tau m - 2\pi i m^t w) \theta(w, \tau)$$

with $m^t w = m^t z + m^t \tau a + m^t b$. Substituting and collecting exponents, the total prefactor multiplying $\theta(z + \tau a + b, \tau)$ is the exponential of

$$\pi i a^t \tau a + 2\pi i a^t z + 2\pi i a^t \tau m + 2\pi i a^t b - \pi i m^t \tau m - 2\pi i m^t z - 2\pi i m^t \tau a - 2\pi i m^t b$$

The cross terms $2\pi i a^t \tau m$ and $-2\pi i m^t \tau a$ cancel by the symmetry $a^t \tau m = m^t \tau a$.

Step 3: Recognizing the characteristic theta. By (4.8) again, $\theta(z + \tau a + b, \tau) = \exp(-\pi i a^t \tau a - 2\pi i a^t (z + b)) \theta[\frac{a}{b}](z, \tau)$. Multiplying this into the prefactor from Step 2, the terms $\pi i a^t \tau a$, $2\pi i a^t z$, and $2\pi i a^t b$ cancel against their negatives, leaving the exponential of

$$-\pi i m^t \tau m - 2\pi i m^t z - 2\pi i m^t b$$

Since $m^t b = b^t m$, this is exactly $\exp(-2\pi i b^t m) \exp(-\pi i m^t \tau m - 2\pi i m^t z)$, and the whole expression equals the right-hand side of (4.10) times $\theta[\frac{a}{b}](z, \tau)$, as we sought to show.

The two constants $e^{2\pi i a^t m}$ and $e^{-2\pi i b^t m}$ are unimodular multipliers that alter the factor of automorphy only by a character of the lattice, so every $\theta[\frac{a}{b}]$ is a section of a line bundle differing from the principal one by a degree-zero twist, that is by an element of $\text{Pic}^0(X)$. When a and b range over

the finite set of rational representatives $a \in \bigoplus_i (\frac{1}{d_i} \mathbb{Z}) / \mathbb{Z}$ and $b = 0$, the multiplier $e^{2\pi i a^t m}$ tracks exactly the character by which the theta group permutes the sections, and the $d_1 \cdots d_g$ resulting functions transform under a common factor of automorphy, so they span a single space $H^0(X, L)$; that they form a basis is the subject of section 4.3. For the principal polarization the only relevant characteristics are the half-integer ones $a, b \in \frac{1}{2} \mathbb{Z}^g$, which produce the sixteen classical theta constants in genus two and, in every genus, the even and odd theta functions whose zeros locate the theta divisor of section 4.5.

The construction closes the circle with the elliptic case, where $g = 1$ and the whole apparatus collapses to a single classical series the reader has already computed.

Example 4.2.6: Recovering the Jacobi Theta Function

Take $g = 1$. The Siegel upper half-space is $\mathfrak{H}_1 = \mathbb{H}$ (example 3.3.8), so $\tau \in \mathbb{H}$ is a single complex number with $\text{Im}\tau > 0$, and $z \in \mathbb{C}$. A column vector $n \in \mathbb{Z}^1$ is an integer, and the matrix products degenerate to $n^t \tau n = n^2 \tau$ and $n^t z = nz$. The defining series (4.3) becomes

$$\theta(z, \tau) = \sum_{n \in \mathbb{Z}} \exp(\pi i n^2 \tau + 2\pi i n z)$$

which is exactly the Jacobi theta function $\vartheta(z, \tau)$ of (2.11) met in example 2.4.4. The Riemann theta function is thus the g -variable generalization of ϑ , and the two notations agree in genus one.

The quasi-periodicity specializes to match as well. For $m \in \mathbb{Z}$, (4.5) reads $\vartheta(z + m, \tau) = \vartheta(z, \tau)$, and for $m = 1$, (4.6) reads

$$\vartheta(z + \tau, \tau) = \exp(-\pi i \tau - 2\pi i z) \vartheta(z, \tau)$$

These are precisely the two transformation laws recorded in example 2.4.4 for the section of the degree-one line bundle on $E = \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$. There the laws were derived by an ad hoc reindexing of the one-variable series and identified as the values of a factor of automorphy on the basis $\lambda = 1$ and $\lambda = \tau$ of Λ ; here they fall out as the $g = 1$ instances of proposition 4.2.3. For a positive polarization of type (d) with $d \geq 1$, the d sections promised in example 2.4.4 are the characteristic thetas $\theta \left[\begin{smallmatrix} a/d \\ 0 \end{smallmatrix} \right] (z, \tau)$ for $a = 0, 1, \dots, d-1$ of definition 4.2.4, and (4.9) shows they are permuted up to the root-of-unity multipliers $e^{2\pi i am/d}$, the genus-one shadow of the theta-group action of section 4.1.

The genus-one specialization confirms what the general estimate already implied: the Riemann theta function is not a new invention but the resolution of an existence claim. Chapter 3 proved, by counting orbits and summing a Gaussian, that a principal polarization has a one-dimensional space of sections, without producing a section; the elliptic case then exhibited the single section as Jacobi's series. The Riemann theta function does the same in every dimension at once, turning the abstract guarantee of proposition 2.4.3 and theorem 4.1.8 into a concrete holomorphic function on $\mathbb{C}^g$, computable term by term, whose zeros and whose translates carry all the projective geometry that the remainder of the chapter extracts.

4.3 The Space of Theta Functions and Its Dimension

Two threads have been left deliberately open. In section 2.4 the dimension formula $\dim_{\mathbb{C}} H^0(X, L) = \text{Pf}(\text{Im}H)$ was stated as proposition 2.4.3 and its proof postponed; in section 3.4 the Riemann-Roch theorem 3.4.2 counted the sections of a positive line bundle abstractly, as $d_1 \cdots d_g$, without producing a single one of them. This section closes both threads at once, and it does so in the strongest possible form: not by recounting the dimension, but by exhibiting an explicit basis. The theta functions with characteristics from section 4.2, indexed by a finite set of rational shift vectors, turn out to be exactly $d_1 \cdots d_g$ in number and linearly independent, and they span. Once a basis is in hand the sections of L become concrete coordinate functions, and the ratio of any two of them is a genuine meromorphic function on X . That observation is the germ of the projective embedding developed in section 4.4: a basis of $H^0(X, L)$ assembles into a map $X \rightarrow \mathbb{P}^{d_1 \cdots d_g - 1}$, and the whole enterprise of realizing an abelian variety inside projective space reduces to understanding this one map.

Throughout, $X = V/\Lambda$ with $V = \mathbb{C}^g$, and $L = L(H, \chi)$ is a positive line bundle, $H > 0$, with $E = \text{Im}H$ integral on Λ of type $(d_1, \dots, d_g)$. By choosing a symplectic basis realizing this type (definition 3.2.3), the lattice is normalized so that $\Lambda = \tau\mathbb{Z}^g + D\mathbb{Z}^g$ with $\tau \in \mathfrak{H}_g$ and $D = \text{diag}(d_1, \dots, d_g)$, and the sections of L are the holomorphic $\theta: V \rightarrow \mathbb{C}$ obeying the quasi-periodicity attached to the canonical factor of automorphy of $L(H, \chi)$. The classical theta function $\theta(z, \tau)$ of definition 4.2.1 and its variants with characteristics $\theta[\begin{smallmatrix} a \\ 0 \end{smallmatrix}]$ of definition 4.2.4 are exactly the holomorphic functions with this behavior, and the transformation law of proposition 4.2.5 is precisely the quasi-periodicity a section must satisfy.

The counting begins with the shift index set. A theta function with characteristic a transforms under the sublattice $D\mathbb{Z}^g$ by the automorphy factor of L , but its behavior under the finer lattice $\mathbb{Z}^g$ picks out which coset of a modulo $\mathbb{Z}^g$ one has chosen. The distinct sections are therefore indexed by the shift vectors a modulo $\mathbb{Z}^g$ that are compatible with the type, and these form a finite abelian group of order $d_1 \cdots d_g$.

Definition 4.3.1: The Characteristic Group

For a positive line bundle L of type $(d_1, \dots, d_g)$ on $X = V/\Lambda$, the *characteristic group* is the finite abelian group

$$\mathfrak{a}(L) := \left(\bigoplus_{i=1}^g \frac{1}{d_i} \mathbb{Z} \right) / \mathbb{Z}^g \cong \bigoplus_{i=1}^g \mathbb{Z}/d_i \mathbb{Z}$$

of order $d_1 \cdots d_g$. For $a \in \mathfrak{a}(L)$, represented by a vector $a \in \bigoplus_i \frac{1}{d_i} \mathbb{Z}$, write $\theta_a := \theta[\begin{smallmatrix} a \\ 0 \end{smallmatrix}]$ for the associated theta function with characteristic of definition 4.2.4.

The group $\mathfrak{a}(L)$ is a set of representatives for one of the two Lagrangian pieces of $K(L)$. Recall from proposition 4.1.3 that the finite group $K(L) = \Lambda(L)/\Lambda$ has order $(d_1 \cdots d_g)^2$ and, in the symplectic normalization, decomposes as a product of two isotropic subgroups each isomorphic to $\bigoplus_i \mathbb{Z}/d_i \mathbb{Z}$. The characteristics in $\mathfrak{a}(L)$ label the translates t_a^* under one such subgroup, and the square root $\sqrt{|K(L)|} = d_1 \cdots d_g$ is exactly the order of $\mathfrak{a}(L)$. This is the first sign that the theta functions θ_a will match the Stone-von Neumann count of theorem 4.1.8.

The main result asserts that these $d_1 \cdots d_g$ functions are a basis. Linear independence is the transparent half and follows by comparing Fourier expansions; spanning is the substantive half, and there are two routes to it. One matches an arbitrary section's Fourier coefficients against the θ_a

directly; the other invokes the representation theory of the theta group, where the θ_a are visibly a basis because they diagonalize the action of a maximal abelian subgroup. Both are given, the Fourier argument in full as the primary proof and the representation-theoretic viewpoint as the conceptual companion.

Theorem 4.3.2: Basis of Theta Functions

Let $X = V/\Lambda$ and let $L = L(H, \chi)$ be a positive line bundle with $H > 0$ of type $(d_1, \dots, d_g)$, in the symplectic normalization $\Lambda = \tau\mathbb{Z}^g + D\mathbb{Z}^g$ with $\tau \in \mathfrak{H}_g$ and $D = \text{diag}(d_1, \dots, d_g)$. Then the $d_1 \cdots d_g$ theta functions

$$\{\theta_a \mid a \in \mathfrak{a}(L)\}, \quad \theta_a = \theta\left[\begin{smallmatrix} a \\ 0 \end{smallmatrix}\right](\cdot, \tau)$$

form a $\mathbb{C}$ -basis of $H^0(X, L)$. In particular $\dim_{\mathbb{C}} H^0(X, L) = d_1 \cdots d_g$, confirming proposition 2.4.3 and theorem 3.4.2 with an explicit basis.

Proof:

After the symplectic normalization the sections of L are, by proposition 4.2.5, precisely the holomorphic functions $\theta: \mathbb{C}^g \rightarrow \mathbb{C}$ satisfying the two quasi-periodicity laws

$$\theta(z + Dm) = \theta(z), \quad \theta(z + \tau m) = \exp(-\pi i m^t \tau m - 2\pi i m^t z) \theta(z) \quad (m \in \mathbb{Z}^g) \quad (4.11)$$

where the first line uses that $D\mathbb{Z}^g \subseteq \Lambda$ carries the trivial part of the automorphy factor and the second reproduces the transformation of proposition 4.2.3 on the τ -periods. Denote by $\mathcal{T}$ the $\mathbb{C}$ -vector space of all such θ ; the claim is that $\{\theta_a\}_{a \in \mathfrak{a}(L)}$ is a basis of $\mathcal{T} = H^0(X, L)$.

Step 1: Fourier expansion along the first period. The first law in (4.11) makes θ periodic under $D\mathbb{Z}^g$, so it admits a Fourier expansion in the variable z against the dual lattice of $D\mathbb{Z}^g$. Writing $D^{-1} = \text{diag}(d_1^{-1}, \dots, d_g^{-1})$, the characters of $\mathbb{C}^g/D\mathbb{Z}^g$ are $z \mapsto \exp(2\pi i r^t D^{-1} z)$ for $r \in \mathbb{Z}^g$, so

$$\theta(z) = \sum_{r \in \mathbb{Z}^g} c_r \exp(2\pi i r^t D^{-1} z)$$

for uniquely determined coefficients $c_r \in \mathbb{C}$. The task is to see how the second law in (4.11) constrains the c_r .

Step 2: The recursion on Fourier coefficients. Substitute the expansion into the second law. On the one hand,

$$\theta(z + \tau m) = \sum_{r \in \mathbb{Z}^g} c_r \exp(2\pi i r^t D^{-1} z) \exp(2\pi i r^t D^{-1} \tau m)$$

On the other hand the right-hand side of the law multiplies $\theta(z)$ by $\exp(-\pi i m^t \tau m - 2\pi i m^t z)$. Matching the two forces, for every frequency index, an identity between the coefficient of a given exponential $\exp(2\pi i r^t D^{-1} z)$ on each side. On the right the factor $\exp(-2\pi i m^t z) = \exp(-2\pi i (Dm)^t D^{-1} z)$ shifts the frequency index by Dm ; comparing the coefficient of $\exp(2\pi i r^t D^{-1} z)$ gives

$$c_r \exp(2\pi i r^t D^{-1} \tau m) = c_{r+Dm} \exp(-\pi i m^t \tau m) \quad (4.12)$$

The content of (4.12) is that c_r and c_{r+Dm} determine one another for every $m \in \mathbb{Z}^g$: the coefficients are governed by the coset of r in $\mathbb{Z}^g/D\mathbb{Z}^g$.

Step 3: Each coset has one free coefficient. The cosets $\mathbb{Z}^g/D\mathbb{Z}^g$ are in bijection with $\mathfrak{a}(L)$ via $r \mapsto D^{-1}r \bmod \mathbb{Z}^g$, since $D^{-1}\mathbb{Z}^g/\mathbb{Z}^g = \bigoplus_i (\frac{1}{d_i}\mathbb{Z})/\mathbb{Z}$. Fix a coset and a representative r_0 ; (4.12) then expresses every c_{r_0+Dm} as c_{r_0} times an explicit exponential factor, so the entire family of coefficients over the coset is determined by the single value c_{r_0} . Consistency of the recursion (that traversing the coset by different paths gives the same answer) is exactly the cocycle identity built into the automorphy factor, and holds because τ is symmetric: the accumulated phase for the shift $m + m'$ agrees with composing the shifts by m and m' , using $m^t \tau m' = m'^t \tau m$. Consequently $\dim_{\mathbb{C}} \mathcal{T}$ equals the number of cosets, namely $|\mathbb{Z}^g/D\mathbb{Z}^g| = d_1 \cdots d_g$.

Step 4: The θ_a realize the free coefficients. It remains to identify a section supported on each coset with a theta function with characteristic. Expanding the definition of definition 4.2.4 in the normalization at hand,

$$\theta_a(z) = \theta\left[\begin{smallmatrix} a \\ 0 \end{smallmatrix}\right](z, \tau) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i (n+a)^t \tau (n+a) + 2\pi i (n+a)^t z)$$

which, upon collecting the coefficient of $\exp(2\pi i (n+a)^t z)$, is a Fourier series whose nonzero frequencies $D^{-1}r = n+a$ run exactly over the coset $a + \mathbb{Z}^g$, that is, over the single coset of $\mathbb{Z}^g/D\mathbb{Z}^g$ corresponding to $a \in \mathfrak{a}(L)$. Thus θ_a is a nonzero element of $\mathcal{T}$ whose Fourier support meets only the coset labelled by a . Since distinct $a \in \mathfrak{a}(L)$ give disjoint Fourier supports, the θ_a are linearly independent: a vanishing combination $\sum_a \lambda_a \theta_a = 0$ has, on each coset, only the term $\lambda_a \theta_a$ contributing, and each θ_a has a nonzero coefficient there (the $n=0$ term $\exp(\pi i a^t \tau a)$ is nonzero), forcing every $\lambda_a = 0$.

Step 5: Independence plus the dimension count gives a basis. By Step 3, $\dim_{\mathbb{C}} \mathcal{T} = d_1 \cdots d_g = |\mathfrak{a}(L)|$; by Step 4 the $|\mathfrak{a}(L)|$ functions θ_a are linearly independent elements of $\mathcal{T}$. A linearly independent set whose size equals the dimension of the space spans it, so $\{\theta_a\}_{a \in \mathfrak{a}(L)}$ is a basis of $\mathcal{T} = H^0(X, L)$. The equality $\dim_{\mathbb{C}} H^0(X, L) = d_1 \cdots d_g$ recovers proposition 2.4.3 and theorem 3.4.2, now with the basis displayed explicitly, completing the proof.

The argument does more than reprove the dimension formula: it pins down which functions the abstract count was counting. Every section of L is a superposition $\sum_a \lambda_a \theta_a$, and the recursion (4.12) shows that a section is nothing but a choice of one Fourier coefficient per coset of $\mathbb{Z}^g/D\mathbb{Z}^g$, with all remaining coefficients propagated by the quasi-periodicity. The theta functions with characteristics are the sections that concentrate on a single coset, and the transparency of the answer, one basis vector per element of $\mathfrak{a}(L)$, is what makes theta functions the computational engine of the whole theory. For a principal polarization $D = I_g$, the group $\mathfrak{a}(L)$ is trivial and the basis is a single function, recovering $\dim H^0(X, L) = 1$ from proposition 3.4.5: there is one theta function, unique up to scale, and its zero locus is the theta divisor.

The Fourier computation is self-contained, but the structural reason the count comes out to $d_1 \cdots d_g$

rather than to $(d_1 \cdots d_g)^2 = |K(L)|$ is the theta group. The theta group $\mathcal{G}(L)$ of theorem 4.1.8 acts on $H^0(X, L)$ as its unique irreducible representation with the central $\mathbb{C}^*$ acting by scalars, and that representation has dimension $\sqrt{|K(L)|} = d_1 \cdots d_g$. The characteristics $\mathfrak{a}(L)$ are a maximal abelian subgroup of $K(L)$, lifting to a maximal abelian subgroup of $\mathcal{G}(L)$, and the θ_a are precisely the simultaneous eigenvectors for its action.

4.3.3 Remark: The Theta Basis as a Weight Basis The representation-theoretic reading of theorem 4.3.2 is that $\{\theta_a\}$ is a weight basis. Under the commutator pairing e^L of proposition 4.1.7, the finite group $K(L)$ splits as a product $K_1 \times K_2$ of two maximal isotropic subgroups, dual to one another under e^L ; here K_1 is the image of the characteristics $\mathfrak{a}(L)$ and K_2 its complement. The subgroup K_1 lifts to a maximal abelian subgroup $A \subset \mathcal{G}(L)$, and on the irreducible representation $H^0(X, L)$ the group A acts by a direct sum of $|K_1| = d_1 \cdots d_g$ distinct characters, one for each element of the dual group $K_2 \cong \widehat{K_1}$. The θ_a are exactly the corresponding character eigenlines: θ_a is the eigenvector on which A acts through the character indexed by a . This is why the count is $\sqrt{|K(L)|}$ and not $|K(L)|$: Stone-von Neumann forces the irreducible to have dimension equal to the square root of the order of the finite group, and the Fourier computation above is that theorem made explicit for the Heisenberg group of a polarized torus.

The two proofs are complementary. The Fourier argument is elementary and produces the basis by hand; the Stone-von Neumann argument explains, without computation, why a basis of that size had to exist and why the natural indexing set is a group. Both will be used downstream: the explicit functions when computing embeddings and theta relations, the group-theoretic picture when the theta group acts on families and its action must be tracked functorially.

With a basis fixed, the sections of L become homogeneous coordinates. Choosing an ordering $a_0, a_1, \dots, a_N$ of $\mathfrak{a}(L)$ with $N = d_1 \cdots d_g - 1$, the tuple $(\theta_{a_0}, \dots, \theta_{a_N})$ sends a point of X to a point of projective space, provided the sections do not all vanish there. This is the theta map, the concrete device that turns the abstract line bundle L into a map to $\mathbb{P}^N$.

Proposition 4.3.4: The Theta Map

Let $L = L(H, \chi)$ be positive of type $(d_1, \dots, d_g)$, with basis $\{\theta_a\}_{a \in \mathfrak{a}(L)}$ of $H^0(X, L)$ from theorem 4.3.2, and fix an ordering $a_0, \dots, a_N$ of $\mathfrak{a}(L)$, $N = d_1 \cdots d_g - 1$. On the open set

$$U := \{x \in X \mid \theta_{a_j}(x) \neq 0 \text{ for some } j\}$$

where the sections have no common zero, the assignment

$$\varphi_{|L|}: U \rightarrow \mathbb{P}^N, \quad x \mapsto [\theta_{a_0}(x) : \theta_{a_1}(x) : \cdots : \theta_{a_N}(x)]$$

is a well-defined holomorphic map, independent of the choice of basis up to the induced projective linear automorphism of $\mathbb{P}^N$.

Proof:

The functions θ_a are not functions on X but sections of L : each satisfies the quasi-periodicity (4.11), so under $z \mapsto z + \lambda$ every θ_a is multiplied by the *same* automorphy factor $a_{(H, \chi)}(\lambda, z)$, which is independent of a . Consequently the ratio $\theta_{a_j}(z + \lambda)/\theta_{a_k}(z + \lambda) = \theta_{a_j}(z)/\theta_{a_k}(z)$ is

Λ -invariant, so the homogeneous tuple $[\theta_{a_0}(z) : \cdots : \theta_{a_N}(z)]$ depends only on the class of z in X . This is the precise sense in which a common automorphy factor produces a well-defined point of projective space: rescaling all coordinates by a common nonzero factor is the identity in $\mathbb{P}^N$.

On U at least one coordinate is nonzero, so the tuple defines an actual point of $\mathbb{P}^N$, and in a neighborhood where $\theta_{a_k} \neq 0$ the map is given in the affine chart by the holomorphic functions $\theta_{a_j}/\theta_{a_k}$; hence $\varphi_{|L|}$ is holomorphic on U . Finally, a different basis $\{\theta'_b\}$ of the same space $H^0(X, L)$ is related to $\{\theta_a\}$ by an invertible matrix $M \in \mathrm{GL}_{N+1}(\mathbb{C})$, and replacing the coordinates by M applied to them changes $\varphi_{|L|}$ by the projective linear automorphism of $\mathbb{P}^N$ induced by M , as we sought to show.

The map $\varphi_{|L|}$ is the object the next section studies in depth. Two features are already visible. First, whether $\varphi_{|L|}$ extends to all of X , and whether it is injective and an immersion, depends on the base locus and on separation properties of the sections, which is why L itself need not embed but a suitable power does. Second, the target dimension $N = d_1 \cdots d_g - 1$ is dictated entirely by the type: a principal polarization gives $N = 0$ and a constant map, a type $(1, \dots, 1, 2)$ polarization gives a map to $\mathbb{P}^1$, and only once the type is large enough do the sections separate points. The systematic way to guarantee enough sections is to pass to a power $L^{\otimes n}$, whose type is $(nd_1, \dots, nd_g)$ and whose section space has dimension $n^g d_1 \cdots d_g$; this is the mechanism behind the Lefschetz embedding theorem of section 4.4.

The collection of hypersurfaces cut out by the sections of L , as the section ranges over $H^0(X, L)$, is a classical object worth naming: the complete linear system of L . It is the intrinsic, basis-free version of the theta map, and it is the language in which the classification of maps to projective space is usually phrased.

Definition 4.3.5: The Complete Linear System of a Line Bundle

Let L be a line bundle on X with $H^0(X, L) \neq 0$. The *complete linear system* of L is the projective space

$$|L| := \mathbb{P}(H^0(X, L)) = (H^0(X, L) \setminus \{0\})/\mathbb{C}^*$$

of nonzero sections up to scalar. A point of $|L|$ is the divisor $\mathrm{div}(s)$ of a nonzero section s , so $|L|$ is identified with the set of effective divisors $D \geq 0$ on X with $\mathcal{O}_X(D) \cong L$. The *base locus* of $|L|$ is the common zero set $\mathrm{Bl}(L) := \bigcap_{s \in H^0(X, L)} \mathrm{div}(s)$; when L is positive of type $(d_1, \dots, d_g)$, the theta functions $\{\theta_a\}_{a \in \mathfrak{a}(L)}$ of theorem 4.3.2 furnish a set of projective coordinates on $|L| \cong \mathbb{P}^{d_1 \cdots d_g - 1}$.

The complete linear system reframes the theta map coordinate-free. A choice of basis of $H^0(X, L)$ identifies $|L|$ with $\mathbb{P}^N$ and, dually, sends a point $x \in X$ off the base locus to the hyperplane of sections vanishing at x ; this is the map $\varphi_{|L|}$ of proposition 4.3.4 read through the isomorphism $|L| \cong \mathbb{P}^N$. Points of the base locus are exactly the points where $\varphi_{|L|}$ is undefined, so $\varphi_{|L|}$ is a morphism on all of X precisely when $|L|$ is base-point-free. For a positive line bundle of type $(d_1, \dots, d_g)$ the dimension of $|L|$ is $d_1 \cdots d_g - 1$, and the effective divisors it parametrizes are the translates and combinations of the theta divisor; the principal case, $\dim |L| = 0$, is the statement that the theta divisor is rigid, a single effective divisor with no linear freedom.

The genus-two case is the smallest one where the type varies nontrivially. In dimension one the

type is a single integer $d_1 = \deg L$ and the section count is that integer; in dimension two the type is a pair (d_1, d_2) with $d_1 \mid d_2$, and the section count $d_1 d_2$ already exhibits the full phenomenon that distinct polarizations of the same underlying torus give projective spaces of different dimensions.

Example 4.3.6: Theta Bases in Genus Two

Let $X = V/\Lambda$ be a two-dimensional complex torus, $g = 2$, carrying a positive line bundle $L = L(H, \chi)$ with $E = \text{Im} H$ of type (d_1, d_2) , $d_1 \mid d_2$. theorem 4.3.2 gives

$$\dim_{\mathbb{C}} H^0(X, L) = d_1 d_2, \quad \varphi_{|L|}: X \dashrightarrow \mathbb{P}^{d_1 d_2 - 1}$$

and the three smallest types illustrate the range of behavior.

1. *Principal, type (1, 1).* Here $\mathfrak{a}(L) = (\frac{1}{1}\mathbb{Z} \oplus \frac{1}{1}\mathbb{Z})/\mathbb{Z}^2$ is trivial, so $\dim H^0(X, L) = 1$: there is a single theta function $\theta = \theta[\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}](z, \tau)$, unique up to scale, and $\varphi_{|L|}$ maps X to $\mathbb{P}^0$, a point. The linear system $|L|$ is a single point, and its unique effective divisor is the theta divisor $\Theta = \text{div}(\theta)$, a curve inside X . This is the case relevant to Jacobians of genus-two curves: the Jacobian is principally polarized, and when X is such a Jacobian Θ is a smooth genus-two curve, the image of the curve under Abel-Jacobi.
2. *Type (1, 2).* Now $\mathfrak{a}(L) = (\mathbb{Z} \oplus \frac{1}{2}\mathbb{Z})/\mathbb{Z}^2 \cong \mathbb{Z}/2$, of order 2, so $\dim H^0(X, L) = 2$. A basis is $\{\theta_{(0,0)}, \theta_{(0,1/2)}\}$, the two theta functions with characteristics $a = (0, 0)$ and $a = (0, \frac{1}{2})$, and $\varphi_{|L|}: X \dashrightarrow \mathbb{P}^1$ is a map to the projective line. The two sections have a common zero locus in general, so $\varphi_{|L|}$ is only a rational map, undefined on the base locus.
3. *Type (2, 2).* Here $\mathfrak{a}(L) = (\frac{1}{2}\mathbb{Z} \oplus \frac{1}{2}\mathbb{Z})/\mathbb{Z}^2 \cong (\mathbb{Z}/2)^2$, of order 4, so $\dim H^0(X, L) = 4$ with basis

$$\{ \theta_{(0,0)}, \theta_{(1/2,0)}, \theta_{(0,1/2)}, \theta_{(1/2,1/2)} \}$$

the four theta functions with half-integer characteristics, and $\varphi_{|L|}: X \dashrightarrow \mathbb{P}^3$. This type is realized by $L_0^{\otimes 2}$ for a principal L_0 , and the four sections are the classical second-order theta functions; the resulting map factors through $X/\{\pm 1\}$ and realizes the Kummer surface inside $\mathbb{P}^3$ as a quartic.

The pattern is uniform: the characteristic group $\mathfrak{a}(L) \cong \mathbb{Z}/d_1 \oplus \mathbb{Z}/d_2$ indexes the basis, its order $d_1 d_2$ is the dimension of $H^0(X, L)$, and the target of the theta map is $\mathbb{P}^{d_1 d_2 - 1}$. Passing from a principal polarization to its square multiplies each d_i by 2 and takes the section count from 1 to 4, which is exactly the jump from a point to $\mathbb{P}^3$ that makes the theta map informative.

The genus-two computation makes the abstract dimension of theorem 3.4.2 tangible: the number $\text{Pf}(E) = d_1 \cdots d_g$ is not a bookkeeping figure but the count of coordinate functions available for mapping X to projective space, and the theta functions with characteristics are those coordinates written out. The principal case gives too few sections to embed, which is why a Jacobian is not cut out by its own theta divisor but by a power of it; the type (2, 2) case gives just enough sections to separate the generic point from its negative, which is the Kummer construction. The general principle, that enough sections force an embedding, is the content of the Lefschetz theorem, and theorem 4.3.2 supplies the explicit sections on which its proof runs.

4.4 Projective Embeddings: The Lefschetz Theorem

Every thread of Part I has been converging on a single promise. The Riemann relations (theorem 3.1.4) identified the complex tori that ought to be projective as exactly the polarizable ones; the definition of an abelian variety (definition 3.2.1) took the existence of a polarization as the working hypothesis; and theorem 3.2.6 then asserted the equivalence “polarizable if and only if projective” with one implication, the passage from a polarization to an actual embedding, left as a debt to be paid in this chapter. The theta functions built in the previous three sections are the currency. This section pays the debt: given a polarization L , the sections of $L^{\otimes 3}$ are numerous enough, and separate points and tangent directions well enough, that the map they define embeds X as a closed submanifold of projective space. The result is due to Lefschetz, and it is the analytic keystone of the entire theory, since it turns the linear-algebraic condition of positivity into a projective variety.

The strategy is classical and rests on a single geometric device: the translates of the theta divisor. A polarization L has, by Riemann-Roch (theorem 3.4.2), a computable and positive number of global sections, and each nonzero section vanishes on a divisor. Translating that divisor around the torus produces a family of divisors that, taken together, cover X and separate its points, because two distinct points of an abelian variety can always be pried apart by a translate of a positive divisor. Passing from L to $L^{\otimes 2}$ turns “covered” into “base-point-free”, and passing to $L^{\otimes 3}$ turns “separated” into “embedded”. The exponents 2 and 3 are not incidental: they are the minimal powers for which the translation argument produces, respectively, a section avoiding a given point and a pair of sections separating any two points together with any tangent direction.

Throughout, $L = L(H, \chi)$ is a polarization of $X = V/\Lambda$, so $H > 0$ and $E = \text{Im}H$ is integral on Λ ; write $(d_1, \dots, d_g)$ for its type (definition 3.2.3) and $d := d_1 \cdots d_g = \text{Pf}(E)$ for the product, which by theorem 3.4.2 equals $\dim_{\mathbb{C}} H^0(X, L)$. For a point $a \in X$, let $t_a: X \rightarrow X$ denote translation by a , and recall that t_a^*L is again a line bundle with the same first Chern class E as L , so that t_a^*L and L define the same polarization and differ only by an element of $\text{Pic}^0(X)$. A section $s \in H^0(X, L)$ pulled back along t_a is a section of t_a^*L ; its divisor is the translate $D_s - a$ of the divisor D_s of s . This interplay between translation and the theta divisor is the mechanism the whole argument exploits.

The first step is to remove base points. A *base point* of a line bundle M is a point at which every global section vanishes; the bundle is *base-point-free* when it has none, equivalently when the sections define an everywhere-regular map $\varphi_{|M|}: X \rightarrow \mathbb{P}(H^0(X, M)^*)$. A single polarization L need not be base-point-free, for a principal polarization has $\dim H^0(X, L) = 1$ (proposition 3.4.5) and its one section vanishes on the theta divisor, which is then a common (indeed the only possible) base locus. The remedy is to square. The sections of $L^{\otimes 2}$ include all products $s_a \cdot s_b$ of translated sections, and these products have enough freedom to avoid any prescribed point.

Proposition 4.4.1: $L^{\otimes 2}$ Is Base-Point-Free

Let $L = L(H, \chi)$ be a polarization of the abelian variety $X = V/\Lambda$. Then the line bundle $L^{\otimes 2}$ is base-point-free: for every point $x_0 \in X$ there is a global section $s \in H^0(X, L^{\otimes 2})$ with $s(x_0) \neq 0$. Equivalently, the theta functions of $L^{\otimes 2}$ have no common zero.

Proof:

Step 1: Translated sections of L multiply into sections of $L^{\otimes 2}$. Fix a nonzero section $s_0 \in H^0(X, L)$; such a section exists because $\dim H^0(X, L) = d \geq 1$ by theorem 3.4.2, the type having

all $d_i \geq 1$. Let $D_0 = \text{div}(s_0)$ be its divisor, an effective divisor on X with $\mathcal{O}_X(D_0) \cong L$. For $a \in X$, represented by $\alpha \in V$, the translate t_a^*L is computed from the Appell-Humbert data (theorem 2.2.8, definition 2.2.5): translating the canonical factor of automorphy by α multiplies the semicharacter χ by the character $\lambda \mapsto e^{2\pi i E(\alpha, \lambda)}$, so

$$t_a^*L \cong L(H, \chi_\alpha), \quad \chi_\alpha(\lambda) = \chi(\lambda) e^{2\pi i E(\alpha, \lambda)}$$

with the same Hermitian form H , hence the same Chern class E ; in particular t_a^*L is again a polarization of the same type as L , with $\dim H^0(X, t_a^*L) = d$. Applying this with α and with $-\alpha$ and multiplying the data (proposition 2.2.10),

$$t_a^*L \otimes t_{-a}^*L \cong L(2H, \chi_\alpha \chi_{-\alpha}) = L(2H, \chi^2) = L^{\otimes 2}$$

where the middle equality holds because $\chi_\alpha(\lambda) \chi_{-\alpha}(\lambda) = \chi(\lambda)^2 e^{2\pi i E(\alpha, \lambda)} e^{-2\pi i E(\alpha, \lambda)} = \chi(\lambda)^2$, the two exponential factors cancelling. Taking the specific pulled-back sections $s' := t_a^*s_0 \in H^0(X, t_a^*L)$ and $s'' := t_{-a}^*s_0 \in H^0(X, t_{-a}^*L)$, both nonzero, their product $s' \cdot s''$ is a nonzero section of $L^{\otimes 2}$ whose divisor is $(D_0 - a) + (D_0 + a)$, the sum of two translates of D_0 . (A generic nonzero section of t_a^*L need not have divisor $D_0 - a$ when $d > 1$; the pullbacks $t_a^*s_0, t_{-a}^*s_0$ are chosen precisely to realize these two translates.)

Step 2: Choose the translates to miss a prescribed point. Fix $x_0 \in X$; the goal is a section of $L^{\otimes 2}$ nonvanishing at x_0 , equivalently a pair of translates of D_0 neither of which passes through x_0 . A translate $D_0 + a$ contains x_0 if and only if $x_0 \in \text{supp}(D_0 + a)$, that is, if and only if $a \in x_0 - \text{supp}(D_0)$. Since D_0 is an effective divisor on the g -dimensional torus X , its support is a hypersurface, an analytic subset of complex codimension one, so $x_0 - \text{supp}(D_0)$ is a proper analytic subset of X ; its complement is nonempty and dense. Pick any a in that complement. Then $x_0 \notin D_0 + a$, and also $x_0 \notin D_0 - a$ unless a lies in the reflected set $\text{supp}(D_0) - x_0$, likewise a proper analytic subset. Because X is not the union of two proper analytic subsets, some a avoids both, and for that a the section $s' \cdot s''$ from Step 1 vanishes at neither of the two components of its divisor at x_0 , hence

$$(s' \cdot s'')(x_0) \neq 0$$

exhibiting a section of $L^{\otimes 2}$ nonzero at x_0 .

Since $x_0 \in X$ was arbitrary, no point is a base point of $L^{\otimes 2}$, so $L^{\otimes 2}$ is base-point-free, completing the proof.

The proof isolates the one property of abelian varieties that makes the whole embedding theory work: an effective divisor can be translated, and its translates sweep out the torus. On a general projective variety a positive line bundle carries no such homogeneity, and base-point-freeness of powers is a subtler matter; on an abelian variety the group law supplies exactly the room needed. The section $s' \cdot s''$ produced above is, in the language of the previous section, a product of two theta functions with shifted arguments, and this is the concrete form the translates take once a Riemann theta function (definition 4.2.1) is chosen: the addition formula for theta functions expresses such products in terms of the theta basis of $H^0(X, L^{\otimes 2})$ (theorem 4.3.2), so that Step 1 is an identity among explicit holomorphic functions on V .

With base points gone after squaring, the map $\varphi_{|L^{\otimes 2}|}$ is everywhere regular, but it need not yet be injective, nor an immersion. To force injectivity and the immersion property one more tensor power

is needed, and the extra room it provides is what separates points and tangent vectors.

The heart of the theorem is the separation argument. A regular map $\varphi_{|M|}: X \rightarrow \mathbb{P}^N$ defined by a base-point-free bundle M is a closed embedding precisely when it is injective and its differential is injective at every point, that is, when the sections of M *separate points* (for distinct $x \neq y$ there is a section vanishing at x but not at y) and *separate tangent vectors* (for each x and each nonzero tangent vector $v \in T_x X$ there is a section vanishing at x whose differential is nonzero on v). The next result records that $L^{\otimes 3}$, and more generally every $L^{\otimes n}$ with $n \geq 3$, achieves both.

Theorem 4.4.2: Lefschetz Embedding Theorem

Let $L = L(H, \chi)$ be a polarization of the abelian variety $X = V/\Lambda$, of type $(d_1, \dots, d_g)$ and with $d = d_1 \cdots d_g$. For every integer $n \geq 3$ the line bundle $L^{\otimes n}$ is very ample: the map

$$\varphi_{|L^{\otimes n}|}: X \longrightarrow \mathbb{P}(H^0(X, L^{\otimes n})^*) = \mathbb{P}^N, \quad N = n^g d - 1$$

associated to a basis of theta functions of $L^{\otimes n}$ is a closed holomorphic embedding of X onto a smooth projective subvariety. In particular $\varphi_{|L^{\otimes 3}|}: X \hookrightarrow \mathbb{P}^N$ with $N = 3^g d - 1$ is a closed holomorphic embedding.

Proof:

The dimension count is immediate: $L^{\otimes n}$ has Chern class nE , hence is a polarization of type $(nd_1, \dots, nd_g)$, so by theorem 3.4.2 its space of sections has dimension $\dim H^0(X, L^{\otimes n}) = \text{Pf}(nE) = (nd_1)(nd_2) \cdots (nd_g) = n^g d$, and the target is $\mathbb{P}^N$ with $N = n^g d - 1$. It remains to show, for $n \geq 3$, that $L^{\otimes n}$ is base-point-free, separates points, and separates tangent vectors. The cases $n = 3$ and $n = 4$ are treated directly below; the case $n \geq 5$ then follows because $L^{\otimes n} = L^{\otimes 3} \otimes L^{\otimes(n-3)}$ and tensoring a very ample bundle with a base-point-free bundle keeps it very ample. Here $L^{\otimes(n-3)}$ is base-point-free for $n \geq 5$: any power $L^{\otimes m}$ with $m \geq 2$ is a tensor product of copies of $L^{\otimes 2}$ and $L^{\otimes 3}$, hence base-point-free by proposition 4.4.1 and Step 1 below, and $n-3 \geq 2$ precisely when $n \geq 5$. This reduction does *not* apply at $n = 4$, where $L^{\otimes(n-3)} = L$ has a base point in the principal case, so $n = 4$ must be, and is, handled on its own. So fix first $n = 3$; the modifications for $n = 4$ are indicated at the end of each step.

Step 1: Base-point-freeness. The naive factorization $L^{\otimes 3} = L^{\otimes 2} \otimes L$ does not help, because L itself may have a base point: in the principal case $\dim H^0(X, L) = 1$ (proposition 3.4.5) and its single section vanishes on the theta divisor, so no section of L avoids a point x_0 lying there. The correct device is again to spread the theta divisor by translation, but now using *three* translates so that the $\text{Pic}^0(X)$ -twists cancel. Fix a nonzero section $s_0 \in H^0(X, L)$ with effective divisor D_0 . For $a, b, c \in X$ with $a + b + c = 0$ the Appell-Humbert computation of Step 1 of proposition 4.4.1, applied three times, gives

$$t_a^* L \otimes t_b^* L \otimes t_c^* L \cong L^{\otimes 3}$$

with the three character twists $e^{2\pi i E(\alpha, \cdot)}$, $e^{2\pi i E(\beta, \cdot)}$, $e^{2\pi i E(\gamma, \cdot)}$ multiplying to $e^{2\pi i E(\alpha + \beta + \gamma, \cdot)} = 1$ on Λ since $\alpha + \beta + \gamma \in \Lambda$ and E is integral there (the same identity is spelled out in Step 2 below). Following the convention fixed above, the divisor of $t_a^* s_0$ is the translate $D_0 - a$, so the product of the three pulled-back sections $t_a^* s_0 \cdot t_b^* s_0 \cdot t_c^* s_0$ is a section of $L^{\otimes 3}$ with divisor $(D_0 - a) + (D_0 - b) + (D_0 - c)$. Given x_0 , choose a in the complement of the proper analytic subset $\text{supp}(D_0) - x_0$, then b likewise avoiding $\text{supp}(D_0) - x_0$, and set $c = -a - b$, adjusting

b within its open dense set of admissible values so that c too avoids $\text{supp}(D_0) - x_0$; this is possible because each constraint excludes only a proper analytic subset. Then none of the three translates passes through x_0 , so the section is nonzero at x_0 . Thus $L^{\otimes 3}$ is base-point-free and $\varphi|_{L^{\otimes 3}}$ is a well-defined holomorphic map on all of X . For $n = 4$ the same reasoning applies with $L^{\otimes 4} = L^{\otimes 2} \otimes L^{\otimes 2}$: a product of two sections of $L^{\otimes 2}$ each nonvanishing at x_0 (proposition 4.4.1) is a section of $L^{\otimes 4}$ nonzero at x_0 , so $L^{\otimes 4}$ is base-point-free as well.

Step 2: Separation of points. Let $x \neq y$ be distinct points of X ; the goal is a section of $L^{\otimes 3}$ vanishing at x but not at y . As in proposition 4.4.1, fix a nonzero section $s_0 \in H^0(X, L)$ with effective divisor D_0 , and consider divisors of the form

$$D_a := (D_0 - a) + (D_0 - b) + (D_0 - c)$$

with $a + b + c = 0$ in X , which is the divisor of a section of $L^{\otimes 3}$: indeed the three-fold analogue of the identity used above,

$$t_a^* L \otimes t_b^* L \otimes t_c^* L \cong L^{\otimes 3} \quad \text{whenever } a + b + c = 0,$$

holds by the same Appell-Humbert computation as in Step 1 of proposition 4.4.1. Representing a, b, c by $\alpha, \beta, \gamma \in V$ with $\mu := \alpha + \beta + \gamma \in \Lambda$, the left side is $L(3H, \chi_\alpha \chi_\beta \chi_\gamma)$, and $\chi_\alpha \chi_\beta \chi_\gamma(\lambda) = \chi(\lambda)^3 e^{2\pi i E(\mu, \lambda)} = \chi(\lambda)^3$, since $E(\mu, \lambda) \in \mathbb{Z}$ by integrality of E on Λ ; thus the product is $L(3H, \chi^3) = L^{\otimes 3}$. A product of nonzero sections of the three translated bundles is then a section $s_{a,b,c}$ of $L^{\otimes 3}$ with $\text{div}(s_{a,b,c}) = D_a$. Now x lies on D_a if and only if x lies on one of the three translates, that is, if and only if one of a, b, c lies in $\text{supp}(D_0) - x$; and y avoids D_a if and only if none of a, b, c lies in $\text{supp}(D_0) - y$. The task is to choose $a \in \text{supp}(D_0) - x$ with $a \notin \text{supp}(D_0) - y$, then b and $c = -a - b$ generic avoiding $\text{supp}(D_0) - y$; the resulting section vanishes at x but not at y . Both are proper analytic subsets of X , so once a valid a is fixed the genericity of b (hence of $c = -a - b$) is again the codimension-one argument of Step 1. The one point that a single divisor D_0 does not settle is whether the hypersurface $\text{supp}(D_0) - x$ is contained in $\text{supp}(D_0) - y$: this fails exactly when $\text{supp}(D_0)$ is invariant under the translation $x - y$, and a positive divisor can carry a nonzero finite (torsion) translation stabilizer, so for such a pair no translate of D_0 separates x from y . That the sections of $L^{\otimes 3}$ nonetheless separate every pair $x \neq y$ is the transversality input supplied by the theta-group argument deferred to Mumford below, which produces enough sections beyond the products of translates of a single D_0 to cover these exceptional pairs. For $n = 4$ the same reduction runs with four translates: $D_a := (D_0 - a) + (D_0 - b) + (D_0 - c) + (D_0 - e)$ with $a + b + c + e = 0$ is the divisor of a section of $L^{\otimes 4}$ by the same Appell-Humbert computation, and the choice of $a \in \text{supp}(D_0) - x$ with $a \notin \text{supp}(D_0) - y$ and the remaining b, c generic with $e = -a - b - c$ all avoiding $\text{supp}(D_0) - y$ is governed by the same deferred input.

Step 3: Separation of tangent vectors. Fix $x \in X$ and a nonzero tangent vector $v \in T_x X$; the goal is a section s of $L^{\otimes 3}$ with $s(x) = 0$ but $ds_x(v) \neq 0$, so that $\varphi|_{L^{\otimes 3}}$ is an immersion at x . Work again with $D_a = (D_0 - a) + (D_0 - b) + (D_0 - c)$, $a + b + c = 0$. Choose a so that $D_0 - a$ passes through x as a *smooth* point of that single translate and with tangent hyperplane not annihilating v : concretely, $x \in \text{supp}(D_0 - a)$ means $a \in \text{supp}(D_0) - x$, and as a ranges over this hypersurface the point of D_0 through which the translate meets x ranges over all of $\text{supp}(D_0)$, in particular over its smooth locus, where the tangent hyperplane to $D_0 - a$ at x is a translate of the tangent hyperplane to D_0 and varies. Here D_0 is a hypersurface and not a union of translation-invariant subtori, since a positive divisor cannot be invariant under a positive-dimensional subgroup, as

its self-intersection would then vanish, whereas $(D_0^g) = g!d > 0$. This last self-intersection is computed on an abelian variety, where $\mathrm{td}(X) = 1$, from $(D_0^g) = \int_X c_1(L)^g$ together with the degree- g term of $\int_X e^{c_1(L)}$, namely $\int_X \frac{1}{g!} c_1(L)^g = \chi(L) = \mathrm{Pf}(E) = d$; hence $\int_X c_1(L)^g = g!d$, the factor $\mathrm{Pf}(E) = d$ being the section count of theorem 3.4.2. Consequently the Gauss map of D_0 is nonconstant on its smooth locus, so these tangent hyperplanes are not all equal, and the fact that some such a places a smooth branch through x with tangent hyperplane missing the prescribed v is the transversality input deferred to Mumford below. Fix such an a , so that $D_0 - a$ vanishes to order exactly one at x with $d(\text{local equation})_x(v) \neq 0$. Then choose b and $c = -a - b$ generic so that neither $D_0 - b$ nor $D_0 - c$ passes through x , which is possible by the codimension-one argument of Step 1. The section $s_{a,b,c}$ with divisor D_a then vanishes at x only along the single smooth branch $D_0 - a$, so near x it is a unit multiple of the local equation of $D_0 - a$, whence

$$s_{a,b,c}(x) = 0, \quad d(s_{a,b,c})_x(v) \neq 0$$

separating the tangent vector v . The requirement $n \geq 3$ enters exactly here: three translates give one degree of freedom to place a smooth branch through x tangent to a chosen hyperplane and two more to push the remaining branches off x , whereas two translates suffice only for point separation and one for neither. For $n = 4$ one uses four translates $D_a = (D_0 - a) + (D_0 - b) + (D_0 - c) + (D_0 - e)$ with $a + b + c + e = 0$: place the single smooth branch $D_0 - a$ through x with tangent hyperplane not annihilating v exactly as above, and push the three remaining branches $D_0 - b$, $D_0 - c$, $D_0 - e = D_0 + a + b + c$ off x , which the extra degree of freedom only makes easier.

By Steps 1 to 3 the sections of $L^{\otimes 3}$ are base-point-free and separate both points and tangent vectors, so $\varphi|_{L^{\otimes 3}}: X \rightarrow \mathbb{P}^N$ is an injective holomorphic immersion. Since X is compact, an injective immersion is a homeomorphism onto its image and hence a closed holomorphic embedding, and by Chow's theorem ([10]) the image is a smooth projective algebraic variety. The step-by-step modifications establish the same conclusion verbatim for $n = 4$, and the reduction at the start extends it from $n = 3, 4$ to all $n \geq 5$, proving very ampleness for every $n \geq 3$, as we sought to show.

The separation arguments above are self-contained modulo one standard input used in Steps 2 and 3: that the sections of $L^{\otimes 3}$ have enough independent behavior to separate the exceptional point pairs left over when a single theta divisor has a torsion translation stabilizer, and to isolate any tangent direction. A fully detailed verification, tracking the theta functions explicitly and confirming that $\varphi|_{L^{\otimes 3}}$ is injective with everywhere-injective differential, is carried out in Mumford, *Abelian Varieties*; the argument there is representation-theoretic, using the theta group of section 4.1 to show that the sections of $L^{\otimes n}$ for $n \geq 3$ generate enough jets at every point, and it is included there because this transversality claim is the one geometric fact whose full proof is longer than the linear-algebraic core of the theorem. The mechanism recorded above, translating the theta divisor to place branches and tangent hyperplanes at will, is the geometric heart of that argument.

The theorem is the analytic realization of an abelian variety as a projective variety, and it does more than assert projectivity: it names an explicit embedding. The coordinates of $\varphi|_{L^{\otimes 3}}$ are the $3^g d$ theta functions of theorem 4.3.2 for the bundle $L^{\otimes 3}$, so the embedded torus is cut out by explicit theta relations, the quadratic and higher identities among these functions. For a principal polarization, $d = 1$ and the embedding lands in $\mathbb{P}^{3^g - 1}$; for $g = 1$ this is $\mathbb{P}^2$, recovering the plane cubic, as the example below makes precise. The bound $n \geq 3$ is sharp in general: L itself and $L^{\otimes 2}$ can fail to be very ample, the former having only a single section, hence a base point, when $d = 1$ and the latter failing to separate certain points on some abelian varieties, so cubing is the universal guarantee.

The theorem discharges the outstanding debt of section 3.2 in one stroke.

Corollary 4.4.3: Every Polarizable Complex Torus Is Projective

Let $X = V/\Lambda$ be a complex torus admitting a polarization, that is, an abelian variety in the sense of definition 3.2.1. Then X admits a closed holomorphic embedding into projective space, and its image is a smooth projective algebraic variety. This supplies the missing implication of theorem 3.2.6, whose equivalence is thereby complete: a complex torus is polarizable if and only if it is projective.

Proof:

Let $H > 0$ be a polarization of X with $E = \text{Im}H$ integral on Λ , and let $L = L(H, \chi)$ be an associated positive line bundle (definition 3.1.1), which exists by the Appell-Humbert theorem for any semicharacter χ of E . By theorem 4.4.2 the bundle $L^{\otimes 3}$ is very ample, so $\varphi_{|L^{\otimes 3}|} : X \hookrightarrow \mathbb{P}^N$ is a closed holomorphic embedding onto a smooth projective variety. This establishes the forward implication “polarizable $\Rightarrow$ projective” of theorem 3.2.6, whose converse was proved there. Polarizability and projectivity are therefore equivalent, completing the proof.

This corollary is the payoff of Part I. Chapter 3 proposed the polarization as the algebraicity criterion and proved everything except that a polarization actually forces projectivity; that gap, flagged in definition 3.2.1 and left open in theorem 3.2.6, is now closed. The word “abelian variety” has, over $\mathbb{C}$, a fully justified meaning: a polarizable complex torus is a projective algebraic group, and the passage from the transcendental quotient $\mathbb{C}^g/\Lambda$ to a system of homogeneous polynomial equations in $\mathbb{P}^N$ is effected concretely by theta functions. Nothing in Part I now rests on an unproved forward reference; the analytic theory is complete, and Part II will recover the same class of objects over an arbitrary field by algebraic means.

The embedding furnished by the theorem is a construction worth naming, both because it is canonical up to the choice of a basis of sections and because it is the object that later chapters manipulate.

Definition 4.4.4: Theta Embedding

Let $L = L(H, \chi)$ be a polarization of the abelian variety X of type $(d_1, \dots, d_g)$, and let $\theta_0, \dots, \theta_N$ with $N = 3^g d_1 \cdots d_g - 1$ be a basis of the space $H^0(X, L^{\otimes 3})$ of theta functions of $L^{\otimes 3}$ (theorem 4.3.2). The *theta embedding* associated to L is the closed holomorphic embedding

$$\varphi_{|L^{\otimes 3}|} : X \longrightarrow \mathbb{P}^N, \quad x \longmapsto [\theta_0(x) : \cdots : \theta_N(x)]$$

of theorem 4.4.2. Its image is the abelian variety X presented as a smooth projective subvariety of $\mathbb{P}^N$, cut out by the theta relations among $\theta_0, \dots, \theta_N$.

The theta embedding depends on the polarization L and, through the choice of basis, on a coordinate system in $H^0(X, L^{\otimes 3})$, but two choices of basis differ by a projective linear change of coordinates, so the embedded variety is well defined up to the action of PGL_{N+1} . Different polarizations on the same torus give distinct embeddings, into projective spaces of different dimensions when the types differ, which is why the polarization, and not the bare torus, is the datum that a projective realization remembers. For a principally polarized abelian variety the theta functions with characteristics of theorem 4.3.2 give the coordinates explicitly, and the embedding is the one classically written in terms of the Riemann theta function $\theta[\frac{a}{0}]$ with characteristics $a \in (\frac{1}{3}\mathbb{Z}^g)/\mathbb{Z}^g$.

The definition becomes concrete in dimension one, where the theta embedding is an object the reader has already met from an entirely different route.

Example 4.4.5: The Plane Cubic as a Theta Embedding

Take $g = 1$ and $X = E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$. The principal polarization has $H(v, w) = (\operatorname{Im}\tau)^{-1}v\bar{w}$ and type (1), so $d = d_1 = 1$ and the associated line bundle $L = L(H, \chi)$ has $\dim H^0(E, L) = \operatorname{Pf}(E) = 1$ by theorem 3.4.2: a single theta function, the classical Jacobi theta $\theta(z, \tau)$ of theorem 3.1.4 and theorem 4.3.2, spanning the sections of the degree-one bundle L . This one section vanishes to order one at a single point, so L itself is far from very ample, and the theorem's insistence on cubing is visible already here: $\varphi|_L$ is not even defined, having a base point at the zero of θ .

Cube the polarization. The bundle $L^{\otimes 3}$ has Chern class $3E$, type (3), and by theorem 3.4.2

$$\dim_{\mathbb{C}} H^0(E, L^{\otimes 3}) = \operatorname{Pf}(3E) = 3$$

so $L^{\otimes 3}$ has degree three and three independent theta functions. By theorem 4.3.2 a basis is furnished by the three theta functions with characteristics

$$\theta_j(z) = \theta\left[\begin{smallmatrix} j/3 \\ 0 \end{smallmatrix}\right](z, \tau) = \sum_{n \in \mathbb{Z}} \exp\left(\pi i\left(n + \frac{j}{3}\right)^2 \tau + 2\pi i\left(n + \frac{j}{3}\right)z\right), \quad j = 0, 1, 2$$

and the theta embedding is

$$\varphi|_{L^{\otimes 3}}: E \longrightarrow \mathbb{P}^2, \quad z \longmapsto [\theta_0(z) : \theta_1(z) : \theta_2(z)]$$

a map of the elliptic curve into $\mathbb{P}^2 = \mathbb{P}^{3^1 \cdot 1 - 1}$. By theorem 4.4.2 this is a closed holomorphic embedding, and its image is a smooth plane curve of degree three, since a degree-three line bundle on a genus-one curve embeds it as a cubic: the image meets a generic line in $\deg L^{\otimes 3} = 3$ points.

This recovers, from the theta viewpoint, the Weierstrass embedding of proposition 1.4.2. There the embedding $E \hookrightarrow \mathbb{P}^2$ was built from the pair $(\wp, \wp')$, realizing E as the cubic $Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3$; here it is built from the triple $(\theta_0, \theta_1, \theta_2)$ of theta functions with characteristics. The two embeddings target the same $\mathbb{P}^2$ and their images are projectively equivalent smooth plane cubics, because both are the complete linear system of a degree-three line bundle on E and any two such systems differ by a projective linear change of coordinates. Concretely, the Weierstrass coordinates are recovered from the theta functions through the classical identity $\wp(z) = -\frac{d^2}{dz^2} \log \theta(z, \tau) + c$ for a constant c (and its z -derivative for $\wp'$), so the point $[\wp(z) : \wp'(z) : 1]$ and the point $[\theta_0(z) : \theta_1(z) : \theta_2(z)]$ are carried to one another by a fixed element of $\operatorname{PGL}_3(\mathbb{C})$. The classical fact that an elliptic curve is a plane cubic and the general Lefschetz theorem for abelian varieties are, in dimension one, the same statement seen through two lenses, the Weierstrass functions and the theta functions, and proposition 1.4.2 is the $g = 1$ instance of theorem 4.4.2. For $g = 1$ the analytic theory of Part I thus reduces, at its final step, to the theory of [3].

4.5 The Theta Divisor

The preceding sections turned a polarization into projective geometry: a positive line bundle L has a space of global sections built from theta functions, and a high tensor power of L embeds the abelian variety into projective space (theorem 4.3.2, theorem 4.4.2). When the polarization is principal, that space of sections collapses to a single line. A nonzero section then has a zero locus that is canonical up to scaling, and this zero locus, the theta divisor, is the geometric object in which a principally polarized abelian variety records its polarization. It is a divisor rather than a cohomology class, a subvariety one can point at, and it will turn out to carry enough information to reconstruct a curve from its Jacobian. This section defines the theta divisor, identifies its class and its self-intersection, and pins down the sense in which it is unique.

The starting point is the dimension count from Riemann-Roch. For a principal polarization the space of sections is one-dimensional (proposition 3.4.5), so up to a scalar there is exactly one theta function, and its zero locus is a well-defined effective divisor.

Definition 4.5.1: Theta Divisor

Let $X = V/\Lambda$ be an abelian variety with a principal polarization, and let $L = L(H, \chi)$ be a line bundle with $H > 0$ of type $(1, \dots, 1)$ representing it. By proposition 3.4.5, $\dim_{\mathbb{C}} H^0(X, L) = 1$; let θ be a nonzero section, unique up to a nonzero scalar. The *theta divisor* of the polarization is the effective divisor

$$\Theta := \operatorname{div}(\theta)$$

on X , the zero locus of θ counted with multiplicity. Since θ is unique up to a scalar and scaling does not change its zero locus, Θ is independent of the choice of θ .

The definition is legitimate precisely because the polarization is principal. For a polarization of type $(d_1, \dots, d_g)$ with $d_1 \cdots d_g > 1$ there is a whole projective space $\mathbb{P}(H^0(X, L))$ of sections, each with its own zero locus, and no single divisor is singled out. Principality is what forces $\dim H^0(X, L) = 1$, and hence what makes “the” theta divisor a meaningful phrase. The section θ is a theta function in the sense of definition 2.4.1: a holomorphic $\theta: V \rightarrow \mathbb{C}$ transforming by the canonical factor of automorphy, so its zero set is a Λ -invariant analytic hypersurface in V descending to the codimension-one subvariety $\Theta \subset X$.

The theta divisor is not a bookkeeping device attached to L ; it recovers L completely. The next result says that the line bundle $\mathcal{O}_X(\Theta)$ associated to the divisor is L itself, so Θ carries the full polarization, and it computes the one numerical invariant a principal polarization has: the self-intersection number $(\Theta^g) = g!$.

Proposition 4.5.2: The Theta Divisor Defines the Polarization

With notation as in definition 4.5.1, the divisor Θ satisfies the following.

1. There is a canonical isomorphism $\mathcal{O}_X(\Theta) \cong L$; in particular $c_1(\mathcal{O}_X(\Theta)) = c_1(L) = H$, so $\mathcal{O}_X(\Theta)$ represents the given principal polarization.
2. $\mathcal{O}_X(\Theta)$ is ample, and Θ is an ample divisor.
3. The self-intersection number is

$$(\Theta^g) = g!$$

Proof:

We treat the three assertions in turn.

1. *The line bundle of the divisor.* By construction $\theta \in H^0(X, L)$ is a global section of L whose divisor of zeros is $\Theta = \text{div}(\theta)$. On any complex manifold, a nonzero global section s of a holomorphic line bundle M gives a canonical isomorphism $\mathcal{O}_X(\text{div}(s)) \cong M$: locally s is a holomorphic function whose divisor is $\text{div}(s)$, and comparing the transition data of M with those of the divisor line bundle of $\text{div}(s)$ shows they agree (this is the standard correspondence between effective divisors with a chosen defining section and line bundles with a section, see [10]). Applying this with $M = L$ and $s = \theta$ gives $\mathcal{O}_X(\Theta) \cong L$. Taking first Chern classes and invoking $c_1(L(H, \chi)) = H$ (proposition 2.3.4) yields $c_1(\mathcal{O}_X(\Theta)) = H$, so $\mathcal{O}_X(\Theta)$ represents the principal polarization.
2. *Ampleness.* Since $H > 0$, the class L is a polarization (definition 3.1.3), and by the Lefschetz embedding theorem (theorem 4.4.2) some power $L^{\otimes n}$ is very ample; hence L is ample. By part (1), $\mathcal{O}_X(\Theta) \cong L$ is ample, and an ample line bundle admitting the effective section θ has ample associated divisor Θ .
3. *The self-intersection number.* The self-intersection (Θ^g) is the top power of the first Chern class,

$$(\Theta^g) = \int_X c_1(\mathcal{O}_X(\Theta))^g = \int_X c_1(L)^g$$

Under the identification $H^2(X, \mathbb{Z}) \cong \bigwedge^2 \Lambda^\vee$ of theorem 1.3.2, $c_1(L) = E = \text{Im}H$ is the alternating form of the polarization (proposition 2.3.4), and the top self-intersection of a line bundle on a g -dimensional complex torus is computed by the Pfaffian of E times $g!$. Concretely, choose a symplectic basis $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ of Λ in which E has matrix $\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$ with $D = \text{diag}(d_1, \dots, d_g)$ (definition 3.2.3), and let $x_1, \dots, x_g, y_1, \dots, y_g$ be the dual real coordinates on V/Λ . Then

$$E = \sum_{i=1}^g d_i dx_i \wedge dy_i$$

as an element of $\bigwedge^2 \Lambda^\vee$, and its g -th exterior power is

$$E^{\wedge g} = g! (d_1 d_2 \cdots d_g) dx_1 \wedge dy_1 \wedge \cdots \wedge dx_g \wedge dy_g$$

The fundamental class $[X]$ pairs with the volume form $dx_1 \wedge dy_1 \wedge \cdots \wedge dx_g \wedge dy_g$ to give 1 (this monomial is a generator of $H^{2g}(X, \mathbb{Z}) \cong \bigwedge^{2g} \Lambda^\vee \cong \mathbb{Z}$), so

$$(\Theta^g) = \int_X E^{\wedge g} = g! (d_1 d_2 \cdots d_g) = g! \operatorname{Pf}(E)$$

For the principal polarization all $d_i = 1$, so $\operatorname{Pf}(E) = 1$ and $(\Theta^g) = g!$, as we sought to show.

The three statements fit together as one picture. The class of Θ is H , so no information is lost in passing from the polarization L to the geometric divisor Θ ; the ampleness realizes Θ as a projective hypersurface, cut out inside the embedding of theorem 4.4.2; and the number $g!$ is the degree of X under the map defined by a suitable power of $\mathcal{O}_X(\Theta)$, the sole discrete invariant a principal polarization carries. The count $(\Theta^g) = g! \cdot \operatorname{Pf}(E)$ for a general type is exactly $g!$ times the Riemann-Roch dimension $\operatorname{Pf}(E) = d_1 \cdots d_g$ of theorem 3.4.2, which for the principal case is the statement $\dim H^0 = 1$ made numerical. That the self-intersection is $g!$ rather than 1 is the familiar phenomenon that a hypersurface of “degree one” in the polarization still meets g generic translates of itself in $g!$ points, the number of ways to order the g transverse sheets.

Nothing so far has fixed Θ as a subset of X , only its class and its numerical invariants. The freedom in the construction is the freedom in the line bundle representing the polarization: the semicharacter χ can be altered, or L tensored by a bundle in $\operatorname{Pic}^0(X)$, without changing the class H . Each such change moves Θ by a translation. The next result makes this precise, so that “the” theta divisor is well-defined up to translation and no more.

Proposition 4.5.3: Uniqueness up to Translation

Let X be a principally polarized abelian variety with polarization class $H > 0$ of type $(1, \dots, 1)$. The theta divisor of definition 4.5.1 is unique up to translation by a point of X : if Θ and Θ' are theta divisors attached to two line bundles $L(H, \chi)$ and $L(H, \chi')$ representing the same polarization H , then $\Theta' = t_a^* \Theta$ for some $a \in X$, where $t_a: X \rightarrow X$ is translation by a . Conversely, every translate $t_a^* \Theta$ is the theta divisor of a line bundle representing H .

Proof:

Step 1: Two representatives differ by a bundle in $\operatorname{Pic}^0(X)$. Two line bundles $L = L(H, \chi)$ and $L' = L(H, \chi')$ with the same Hermitian form H have the same first Chern class $c_1 = H$ (proposition 2.3.4), so $L' \otimes L^{-1}$ lies in the kernel of $c_1: \operatorname{Pic}(X) \rightarrow \operatorname{NS}(X)$, which is $\operatorname{Pic}^0(X)$ (theorem 2.2.8). Thus $L' = L \otimes P$ for a unique $P = L(0, \chi' \chi^{-1}) \in \operatorname{Pic}^0(X)$.

Step 2: Tensoring by $\operatorname{Pic}^0(X)$ is translation of the bundle. The polarization isogeny $\varphi_L: X \rightarrow \operatorname{Pic}^0(X)$ sending $x \mapsto t_x^* L \otimes L^{-1}$ is, for a principal polarization, an isomorphism (this is the analytic statement that $H > 0$ of type $(1, \dots, 1)$ makes φ_L bijective; the map φ_L is developed in Chapter 5, but its bijectivity in the principal case is the content of $\dim H^0 = 1$ we already have). Hence there is a unique $a \in X$ with $P = \varphi_L(a) = t_a^* L \otimes L^{-1}$, so

$$L' = L \otimes P = t_a^* L$$

as line bundles.

Step 3: Translation of bundles is translation of divisors. A section θ of L pulls back under t_a to a section $t_a^* \theta$ of $t_a^* L = L'$, and pullback of a section pulls back its divisor: $\operatorname{div}(t_a^* \theta) = t_a^* \operatorname{div}(\theta) = t_a^* \Theta$,

which is the translate of Θ by $-a$. Since $\dim H^0(X, L') = 1$, the section $t_a^*\theta$ is, up to scalar, the unique defining section of L' , so $\Theta' = \operatorname{div}(t_a^*\theta) = t_a^*\Theta$. This proves Θ and Θ' are translates.

Step 4: Converse. Conversely, given any $a \in X$, the bundle $t_a^*L \cong L \otimes \varphi_L(a)$ has the same class H and its section has divisor $t_a^*\Theta$; setting $P = \varphi_L(a)$ and χ' the semicharacter of $L \otimes P$ exhibits $t_a^*\Theta$ as a theta divisor for H . Every translate thus arises, completing the proof.

The translation ambiguity is not a defect to be removed but the actual state of affairs: a principal polarization determines a divisor class, and a divisor within that class only after a choice of origin-compatible line bundle. What the proposition guarantees is that the ambiguity is exactly a torsor under X acting by translation, so any geometric statement about Θ that is translation-invariant, its singularities, its intersection numbers, the pair (X, Θ) up to isomorphism, is well-defined independently of the choice. This is the sense in which one speaks of the theta divisor of a principally polarized abelian variety without further qualification. There is, however, a canonical way to reduce the ambiguity from a full translate to a sign, by choosing the representative that is symmetric under $x \mapsto -x$.

Definition 4.5.4: Symmetric Theta Divisor

Let $[-1]: X \rightarrow X$ denote the inversion $x \mapsto -x$ of the group X . A theta divisor Θ is *symmetric* if

$$[-1]^*\Theta = \Theta$$

as divisors on X . Equivalently, Θ is symmetric when the semicharacter χ of the representing bundle $L(H, \chi)$ is normalized so that $[-1]^*L \cong L$ and the section θ is even or odd, $\theta(-v) = \pm \theta(v)$.

A symmetric theta divisor exists because $[-1]^*L$ again represents H (inversion is a group automorphism, so it preserves the Hermitian form H), so by proposition 4.5.3 the divisor $[-1]^*\Theta$ is a translate of Θ ; translating by a suitable half-period makes the two coincide. Among the translates of a theta divisor there are exactly $|X[2]| = 2^{2g}$ symmetric ones, the orbit of 2-torsion, so symmetry cuts the translation ambiguity down to a finite set fixed by inversion rather than eliminating it entirely. The value of the normalization is that it fixes a canonical point structure: a symmetric Θ is invariant under $[-1]$, so it interacts cleanly with the 2-torsion $X[2]$ and with the even/odd theta functions that furnish the classical theta constants, and it is the representative one uses when the geometry of Θ near the origin, its multiplicity at 2-torsion points, matters. The role of Θ as the geometric heart of the polarization is sharpest in the symmetric normalization, where the divisor is genuinely a canonical feature of the pointed group $(X, 0)$ and not only of X up to translation.

The whole construction is transparent in dimension one, where the theta divisor is a single point of an elliptic curve and the count $(\Theta^g) = g!$ reads $(\Theta^1) = 1 = 1!$.

Example 4.5.5: The Theta Divisor of an Elliptic Curve

Let $X = E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$, a one-dimensional complex torus. Here $g = 1$ and any polarization is automatically principal: the alternating form E has a single elementary divisor d_1 , and rescaling H by $1/d_1$ makes it of type (1), so the type is (1). Take $L = L(H, \chi)$ with H the principal polarization normalized so that $E = \operatorname{Im} H$ is the standard alternating form on Λ with $E(1, \tau) = 1$.

By proposition 3.4.5, $\dim_{\mathbb{C}} H^0(E, L) = 1$, and the unique section (up to scalar) is the classical

Jacobi theta function

$$\theta(z, \tau) = \sum_{n \in \mathbb{Z}} \exp(\pi i n^2 \tau + 2\pi i n z)$$

of example 4.2.6. Its quasi-periodicity, $\theta(z + 1) = \theta(z)$ and $\theta(z + \tau) = \exp(-\pi i \tau - 2\pi i z)\theta(z)$, shows that the zero set is Λ -invariant, so it descends to a divisor on E . The number of zeros of θ in a fundamental parallelogram is computed by the argument principle: integrating $\frac{1}{2\pi i} d \log \theta$ around the boundary and using the quasi-periodicity, the four sides contribute a net winding of 1, so θ has exactly one zero (counted with multiplicity) modulo Λ . That zero is the half-period $z_0 = \frac{1+\tau}{2}$, the unique odd 2-torsion point where θ vanishes, and

$$\Theta = [z_0]$$

a single point of E with multiplicity one, an effective divisor of degree 1.

The class of Θ is $c_1(\mathcal{O}_E(\Theta)) = H$, matching $\dim H^0(E, \mathcal{O}_E(\Theta)) = 1$ from Riemann-Roch on the elliptic curve (example 3.4.6). The self-intersection number in dimension one is just the degree of the divisor,

$$(\Theta^1) = \deg \Theta = 1 = 1!$$

in agreement with proposition 4.5.2. The symmetric normalization (definition 4.5.4) is the choice of $z_0 \in E[2]$: since θ with characteristics can be shifted so that its zero sits at any of the four 2-torsion points, the symmetric theta divisors on E are exactly the four points of $E[2] = \frac{1}{2}\Lambda/\Lambda$, on which $[-1]$ acts trivially.

The elliptic case is the whole subject in miniature: a principally polarized abelian variety of dimension one is a pair $(E, [z_0])$, a curve together with a marked point, and the theta divisor is that point. This is why an elliptic curve is so often presented as a curve with a chosen origin. In higher dimensions the theta divisor stops being a point and becomes a genuine hypersurface of dimension $g - 1$, and its geometry, in particular its singular locus, becomes an object of study in its own right, but the principle is the same: the divisor is where the abelian variety stores its polarization in visible geometric form. The definition of a principal polarization (definition 3.2.4) is exactly the condition $\dim H^0 = 1$ that makes this storage possible, and it is the class of principally polarized abelian varieties, those carrying a theta divisor, that the moduli space $\mathfrak{H}_g / \mathrm{Sp}(2g, \mathbb{Z})$ of theorem 3.3.7 parametrizes.

The theta divisor is the point at which this book's two later strands converge. Chapter 12 realizes it concretely for the Jacobian $\mathrm{Jac}(C)$ of a smooth projective curve C of genus g : the image of the $(g - 1)$ -fold symmetric product of C under the Abel-Jacobi map is a theta divisor for the canonical principal polarization, so every curve manufactures not just a principally polarized abelian variety but an explicit theta divisor inside it. The reason this matters is the Torelli theorem, also taken up in Chapter 12: the pair $(\mathrm{Jac}(C), \Theta)$ determines the curve C up to isomorphism. The polarized Jacobian remembers its curve, and it remembers it through the geometry of the theta divisor, so that a hypersurface in an abelian variety encodes the entire projective curve that gave rise to it.

5

The Dual Torus, Isogenies, and Poincaré Reducibility

The first four chapters studied a single abelian variety and its line bundles. This chapter turns to the maps between abelian varieties and to the dual object that organizes them. The degree-zero line bundles on X , which Chapter 2 exhibited as a complex torus in their own right, are assembled here into the dual abelian variety $\hat{X}$: the same $\overline{V}^*/\Lambda^\vee$ met before, now regarded as a variety that varies functorially with X and satisfies $\hat{\hat{X}} = X$. A polarization, previously a positive Hermitian form, reappears as an actual homomorphism $\varphi_L: X \rightarrow \hat{X}$, and the type of the polarization becomes the degree of that isogeny.

Isogenies are the right notion of equivalence for the arithmetic of abelian varieties: a surjection with finite kernel, invertible up to that kernel. The torsion subgroups $X[n]$, the Tate-module shadows that Part II will exploit, and the Weil pairing that a polarization places on them are all introduced here in their transparent analytic form. The chapter ends with Poincaré complete reducibility, the statement that a polarization splits any abelian subvariety off up to isogeny, so that every abelian variety is isogenous to a product of simple ones. This is what reduces the study of endomorphism algebras, the subject of Chapter 6, to the case of a simple variety, whose endomorphism algebra is a division algebra.

5.1 The Dual Complex Torus

The degree-zero Picard group $\text{Pic}^0(X)$ of a complex torus was already computed in Chapter 2: the exponential sequence exhibited it as the quotient $\overline{V}^*/\iota(\Lambda^\vee)$ of the antilinear dual by an integral lattice, hence as a complex torus of the same dimension g as X (proposition 2.1.5). At that point $\text{Pic}^0(X)$ was a group of isomorphism classes of line bundles that happened to carry a complex

structure. This chapter promotes it to a geometric object in its own right, the *dual torus* $\hat{X}$, and organizes the constructions that make $X \mapsto \hat{X}$ a duality: a contravariant functor on complex tori, biduality $\hat{\hat{X}} = X$, and (in the next sections) the isogeny $X \rightarrow \hat{X}$ attached to a polarization. The dual torus is the target into which a polarization maps X , so pinning it down precisely is the prerequisite for everything about polarizations that follows.

The definition below builds $\hat{X}$ directly from the linear algebra of V and Λ , without reference to line bundles, and then proposition 5.1.3 reconciles this construction with the Picard-group description. Working from the linear algebra first is what makes the functoriality and biduality transparent: both are statements about antilinear duals of vector spaces and their integral lattices, and neither is obvious from the definition of $\text{Pic}^0(X)$ as a set of bundle classes.

The right ambient space is the space of antilinear functionals on V . Recall from the notational conventions of Chapter 1 that $\bar{V}^* = \text{Hom}_{\bar{\mathbb{C}}}(V, \mathbb{C})$ denotes the $\mathbb{C}$ -antilinear functionals $\ell: V \rightarrow \mathbb{C}$, those satisfying $\ell(cv) = \bar{c}\ell(v)$ for $c \in \mathbb{C}$; this is a g -dimensional complex vector space, the conjugate of the ordinary dual V^* .

Definition 5.1.1: Dual Complex Torus

Let $X = V/\Lambda$ be a complex torus of dimension g . Set

$$\hat{V} := \bar{V}^* = \text{Hom}_{\bar{\mathbb{C}}}(V, \mathbb{C})$$

the space of $\mathbb{C}$ -antilinear functionals on V , a complex vector space of dimension g , and let

$$\hat{\Lambda} := \{ \ell \in \hat{V} : \text{Im} \ell(\lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda \}$$

be the subgroup of antilinear functionals whose imaginary part is integer-valued on Λ . Then $\hat{\Lambda}$ is a full lattice in $\hat{V}$, and the *dual complex torus* of X is the quotient

$$\hat{X} := \hat{V} / \hat{\Lambda}$$

a complex torus of dimension g .

The definition packages the parameter space of degree-zero line bundles as a variety. A degree-zero bundle on X is, by proposition 2.1.5 and the character description that followed it, a unitary character $\chi: \Lambda \rightarrow U(1)$; writing $\chi(\lambda) = e^{2\pi i \varphi(\lambda)}$ presents such a character by a real-valued phase φ on Λ modulo $\mathbb{Z}$, and the antilinear functional ℓ with $\text{Im} \ell = \varphi$ (up to the lattice) is exactly the data $\hat{X}$ records. The condition $\text{Im} \ell(\lambda) \in \mathbb{Z}$ is the condition that the phase ℓ be trivial modulo the integral lattice, so that ℓ and ℓ' define the same character precisely when $\ell - \ell' \in \hat{\Lambda}$. Thus $\hat{X}$ is the moduli space of degree-zero line bundles, made into a complex torus by giving its universal cover the antilinear-dual complex structure. That $\hat{\Lambda}$ is a genuine lattice, and $\hat{X}$ a genuine torus, is the content of the following verification, carried out before the reconciliation with $\text{Pic}^0(X)$ so that the definition stands on the linear algebra alone.

Proof of the lattice claim in definition 5.1.1:

Only the assertion that $\hat{\Lambda}$ is a full lattice in $\hat{V}$ requires proof; granting it, $\hat{X} = \hat{V} / \hat{\Lambda}$ is a complex torus of dimension g by definition 1.1.2.

Step 1: The imaginary-part pairing. Sending an antilinear functional to the imaginary part of its restriction to V gives a map

$$\Phi: \hat{V} \longrightarrow \text{Hom}_{\mathbb{R}}(V, \mathbb{R}), \quad \ell \longmapsto \text{Im}\ell$$

This map is $\mathbb{R}$ -linear. It is a bijection: an antilinear functional ℓ is determined by $\text{Im}\ell$, because $\text{Re}\ell(v) = -\text{Im}\ell(iv)$. Indeed $\ell(iv) = \bar{i}\ell(v) = -i\ell(v)$ by antilinearity, so $\text{Im}\ell(iv) = \text{Im}(-i\ell(v)) = -\text{Re}\ell(v)$, which recovers the real part from the imaginary part. Conversely, given any $\mathbb{R}$ -linear $u: V \rightarrow \mathbb{R}$, the formula $\ell(v) := -u(iv) + iu(v)$ defines an antilinear functional with $\text{Im}\ell = u$: additivity is clear, and $\ell(iv) = -u(-v) + iu(iv) = u(v) + iu(iv) = -i(-u(iv) + iu(v)) = -i\ell(v)$, so ℓ is antilinear. Hence Φ is an $\mathbb{R}$ -linear isomorphism from the $2g$ -dimensional real space $\hat{V}$ onto the $2g$ -dimensional real space $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$.

Step 2: Transporting the integral lattice. Under Φ the defining condition of $\hat{\Lambda}$ reads $\Phi(\ell)(\lambda) = \text{Im}\ell(\lambda) \in \mathbb{Z}$ for all $\lambda \in \Lambda$, so

$$\Phi(\hat{\Lambda}) = \{ u \in \text{Hom}_{\mathbb{R}}(V, \mathbb{R}) : u(\Lambda) \subseteq \mathbb{Z} \} = \text{Hom}(\Lambda, \mathbb{Z}) = \Lambda^{\vee}$$

the dual lattice, where the middle equality uses that an $\mathbb{R}$ -linear functional integral on Λ is determined by its values on a $\mathbb{Z}$ -basis of Λ , which is an $\mathbb{R}$ -basis of V (the lattice is full). Now $\Lambda^{\vee}$ is a full lattice in $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$: it is the free abelian group on the basis dual to a $\mathbb{Z}$ -basis of Λ , and that basis is an $\mathbb{R}$ -basis of $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$. Since Φ is an $\mathbb{R}$ -linear isomorphism, $\hat{\Lambda} = \Phi^{-1}(\Lambda^{\vee})$ is a full lattice in $\hat{V}$, as we sought to show.

The proof isolates the mechanism that makes the dual work: the $\mathbb{R}$ -linear isomorphism $\ell \mapsto \text{Im}\ell$ between the antilinear dual $\hat{V}$ and the ordinary real dual $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$, under which the integral lattice $\hat{\Lambda}$ corresponds to the dual lattice $\Lambda^{\vee}$. This is the same identification that appeared inside proposition 2.1.5, where $\iota(\Lambda^{\vee})$ was the image of $\Lambda^{\vee}$ in $H^{0,1} = \overline{V}^*$; the present formulation makes the lattice intrinsic to the antilinear dual rather than an image under a cohomological projection. The two descriptions of $\hat{\Lambda}$, the imaginary-part condition and the dual lattice $\Lambda^{\vee}$, are the two faces of the dual torus that the next proposition matches.

The first concrete instance is $g = 1$, worked out here before the general reconciliation so that the abstract definition has an example within reach.

Example 5.1.2: The Dual of an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$, $\tau \in \mathbb{H}$, as in example 1.1.4. Here $V = \mathbb{C}$, and an antilinear functional $\ell: \mathbb{C} \rightarrow \mathbb{C}$ has the form $\ell(v) = \bar{v}w$ for a unique $w \in \mathbb{C}$ (antilinearity forces ℓ to be a conjugate-linear multiple), so $\hat{V} = \overline{\mathbb{C}}^* \cong \mathbb{C}$ via $\ell \leftrightarrow w$. The dual lattice is

$$\hat{\Lambda} = \{ \ell_w : \text{Im}(\bar{\lambda}w) \in \mathbb{Z} \text{ for } \lambda \in \{1, \tau\} \}$$

Writing the two integrality conditions $\text{Im}(w) \in \mathbb{Z}$ and $\text{Im}(\bar{\tau}w) \in \mathbb{Z}$, these cut out a rank-2 lattice in the w -plane, real-linearly isomorphic to $\Lambda^{\vee}$ under $w \mapsto (\text{Im}w, \text{Im}\bar{\tau}w)$. The quotient $\hat{E} = \hat{V}/\hat{\Lambda}$ is therefore a one-dimensional complex torus.

Every one-dimensional complex torus is an elliptic curve (proposition 1.4.2), so $\hat{E}$ is an elliptic curve. It is in fact isomorphic to E : by proposition 5.1.3 below, $\hat{E} \cong \text{Pic}^0(E)$, and example 2.1.2 identified $\text{Pic}^0(E) \cong E$ through the Abel-Jacobi theorem, sending a degree-zero bundle $\mathcal{O}_E(p - 0)$

to the point p . Thus

$$\hat{E} \cong \text{Pic}^0(E) \cong E$$

and the elliptic curve is *self-dual*. This self-duality is the $g = 1$ shadow of the principal polarization: the isomorphism $E \xrightarrow{\sim} \hat{E}$ is the polarization isogeny φ_L of section 5.3 for $L = \mathcal{O}_E(0)$, which is an isomorphism exactly because the natural polarization of an elliptic curve is principal. For general g an abelian variety need not be self-dual; the elliptic curve is the special case where the canonical map $X \rightarrow \hat{X}$ happens to be an isomorphism.

The construction of $\hat{X}$ from the antilinear dual and the construction of $\text{Pic}^0(X)$ from the exponential sequence produce the same complex torus. Establishing this canonically, and not merely up to a non-specified isomorphism, is what turns $\hat{X}$ from a piece of linear algebra into the geometric parameter space of degree-zero bundles.

Proposition 5.1.3: The Dual Torus Is the Degree-Zero Picard Group

Let $X = V/\Lambda$ be a complex torus. There is a canonical isomorphism of complex tori

$$\hat{X} \xrightarrow{\sim} \text{Pic}^0(X)$$

identifying $\hat{X} = \hat{V}/\hat{\Lambda}$ with the quotient $\overline{V^*}/\iota(\Lambda^\vee)$ of proposition 2.1.5. Explicitly, the class of an antilinear functional $\ell \in \hat{V}$ maps to the degree-zero line bundle $L(0, \chi_\ell)$ with unitary semicharacter

$$\chi_\ell(\lambda) = e^{2\pi i \text{Im} \ell(\lambda)} = e^{-\pi i (\ell(\lambda) - \overline{\ell(\lambda)})}$$

for $\lambda \in \Lambda$.

Proof:

Step 1: The two universal covers agree. The antilinear dual $\hat{V} = \overline{V^*}$ is by definition the space $\overline{V^*}$ of $\mathbb{C}$ -antilinear functionals appearing in proposition 2.1.5, where it was identified with $H^{0,1} = H^1(X, \mathcal{O}_X)$. So the universal covers of $\hat{X}$ and of the torus $\overline{V^*}/\iota(\Lambda^\vee)$ representing $\text{Pic}^0(X)$ are literally the same complex vector space; there is nothing to choose in identifying them.

Step 2: The two lattices agree. It remains to check $\hat{\Lambda} = \iota(\Lambda^\vee)$ inside $\hat{V} = \overline{V^*}$. In proposition 2.1.5 the lattice $\iota(\Lambda^\vee)$ was the image of $H^1(X, \mathbb{Z}) \cong \Lambda^\vee$ under the projection $H^1(X, \mathbb{R}) \rightarrow H^{0,1}$, which is a real-linear isomorphism onto $\overline{V^*}$. Concretely, a class in $H^1(X, \mathbb{Z}) = \text{Hom}(\Lambda, \mathbb{Z})$ is an integral $\mathbb{R}$ -linear functional $u: V \rightarrow \mathbb{R}$ with $u(\Lambda) \subseteq \mathbb{Z}$, and its image in $H^{0,1} = \overline{V^*}$ is the antilinear functional ℓ with $\text{Im} \ell = u$, by the computation of the projection in the proof of proposition 2.1.5. Thus $\iota(\Lambda^\vee)$ is exactly the set of $\ell \in \hat{V}$ with $\text{Im} \ell$ integer-valued on Λ , which is $\hat{\Lambda}$ by definition 5.1.1. The two lattices coincide, so the identity map on $\hat{V}$ descends to an isomorphism of complex tori $\hat{X} = \hat{V}/\hat{\Lambda} \xrightarrow{\sim} \overline{V^*}/\iota(\Lambda^\vee) = \text{Pic}^0(X)$.

Step 3: The semicharacter description. Under the character description following proposition 2.1.5, the class of $\ell \in H^1(X, \mathcal{O}_X) = \overline{V^*}$ modulo $\iota(\Lambda^\vee)$ corresponds to the unitary character $\chi_\ell: \Lambda \rightarrow U(1)$ obtained by exponentiating: a class in $H^1(X, \mathcal{O}_X)/\iota(\Lambda^\vee)$ is sent by $\exp_*$ to the

line bundle whose semicharacter records the phase $\text{Im} \ell$ on Λ , that is,

$$\chi_\ell(\lambda) = e^{2\pi i \text{Im} \ell(\lambda)}$$

Since $\text{Im} \ell(\lambda) = \frac{1}{2i}(\ell(\lambda) - \overline{\ell(\lambda)})$, this equals $e^{-\pi i (\ell(\lambda) - \overline{\ell(\lambda)})}$. Because $H = 0$, the semicharacter identity of definition 2.2.4 degenerates to $\chi_\ell(\lambda + \mu) = \chi_\ell(\lambda)\chi_\ell(\mu)$: χ_ℓ is an ordinary character, matching the description $\text{Pic}^0(X) \cong \text{Hom}(\Lambda, U(1))$. The bundle is $L(0, \chi_\ell)$ in the Appell-Humbert notation of theorem 2.2.8, and $\ell \mapsto L(0, \chi_\ell)$ is the canonical isomorphism, completing the proof.

The proposition is the identification that gives the dual torus its meaning. On the left, $\hat{X}$ is a complex torus manufactured from antilinear functionals, an object of linear algebra; on the right, $\text{Pic}^0(X)$ is the group of degree-zero line bundles on X , an object of geometry. That they are canonically the same complex torus means every degree-zero line bundle on X is a point of $\hat{X}$, and conversely, in a way compatible with the group law and the complex structure. The reconciliation of the two descriptions of $\hat{\Lambda}$, the imaginary-part-integrality condition of definition 5.1.1 and the image $\iota(\Lambda^\vee)$ of the dual lattice from proposition 2.1.5, is the concrete content: both describe the same rank- $2g$ subgroup of $\hat{V}$, and the semicharacter χ_ℓ makes the correspondence explicit at the level of line bundles. From here on $\hat{X}$ and $\text{Pic}^0(X)$ are used interchangeably, and the Poincaré bundle of the algebraic theory (Part II) will make the point of $\hat{X}$ tautologically a bundle on X .

A duality is not a construction but a functor, and the value of $\hat{X}$ is that $X \mapsto \hat{X}$ reverses arrows. A homomorphism $f: X \rightarrow Y$ pulls back line bundles on Y to line bundles on X , and this pullback carries degree-zero bundles to degree-zero bundles, hence induces a homomorphism $\hat{Y} \rightarrow \hat{X}$ in the opposite direction. The linear-algebraic incarnation of this pullback is the antilinear transpose, which is what makes the induced map manifestly a homomorphism of complex tori.

Proposition 5.1.4: Functoriality of the Dual

Let $f: X \rightarrow Y$ be a homomorphism of complex tori, induced (proposition 1.1.7) by the $\mathbb{C}$ -linear map $F: V \rightarrow W$ with $F(\Lambda_X) \subseteq \Lambda_Y$, where $X = V/\Lambda_X$ and $Y = W/\Lambda_Y$. Then:

1. The antilinear transpose $\hat{F}: \hat{W} \rightarrow \hat{V}$, $\ell \mapsto \ell \circ F$, is $\mathbb{C}$ -linear and satisfies $\hat{F}(\hat{\Lambda}_Y) \subseteq \hat{\Lambda}_X$, so it descends to a homomorphism of complex tori

$$\hat{f}: \hat{Y} \longrightarrow \hat{X}$$

called the *dual homomorphism*. Under proposition 5.1.3, $\hat{f}$ is the pullback of line bundles $f^*: \text{Pic}^0(Y) \rightarrow \text{Pic}^0(X)$.

2. The assignment $f \mapsto \hat{f}$ is contravariant and functorial: $\widehat{\text{id}_X} = \text{id}_{\hat{X}}$, and for a second homomorphism $h: Y \rightarrow Z$ one has $\widehat{h \circ f} = \hat{f} \circ \hat{h}$.

Proof:

1. *The transpose and its lattice condition.* For $\ell \in \hat{W} = \text{Hom}_{\overline{\mathbb{C}}}(W, \mathbb{C})$, the composite $\ell \circ F: V \rightarrow \mathbb{C}$ is antilinear, being the composite of the $\mathbb{C}$ -linear F with the antilinear ℓ , so $\hat{F}(\ell) = \ell \circ F \in \hat{V}$. The map $\hat{F}$ is $\mathbb{C}$ -linear in ℓ : scalar multiplication and addition of functionals are computed pointwise, and $(c\ell) \circ F = c(\ell \circ F)$ for $c \in \mathbb{C}$ because $c\ell$ scales

values, so $\hat{F}(c\ell) = c\hat{F}(\ell)$. To check the lattice condition, let $\ell \in \hat{\Lambda}_Y$, so $\text{Im}\ell(\mu) \in \mathbb{Z}$ for all $\mu \in \Lambda_Y$. For $\lambda \in \Lambda_X$ one has $F(\lambda) \in \Lambda_Y$ by hypothesis, hence

$$\text{Im}(\hat{F}(\ell)(\lambda)) = \text{Im}(\ell(F(\lambda))) \in \mathbb{Z}$$

so $\hat{F}(\ell) \in \hat{\Lambda}_X$. Thus $\hat{F}(\hat{\Lambda}_Y) \subseteq \hat{\Lambda}_X$, and by proposition 1.1.7 the $\mathbb{C}$ -linear $\hat{F}$ carrying $\hat{\Lambda}_Y$ into $\hat{\Lambda}_X$ descends to a homomorphism $\hat{f}: \hat{Y} \rightarrow \hat{X}$ of complex tori.

That $\hat{f}$ is the pullback f^* is a computation with semicharacters. By proposition 5.1.3 a point $\ell \in \hat{W}$ represents the bundle $L(0, \chi_\ell)$ on Y with $\chi_\ell(\mu) = e^{2\pi i \text{Im}\ell(\mu)}$. Its pullback $f^*L(0, \chi_\ell)$ is the bundle on X obtained by pulling back the factor of automorphy along F , whose semicharacter is $\lambda \mapsto \chi_\ell(F(\lambda)) = e^{2\pi i \text{Im}\ell(F(\lambda))} = e^{2\pi i \text{Im}(\hat{F}(\ell))(\lambda)} = \chi_{\hat{F}(\ell)}(\lambda)$. This is the semicharacter attached to $\hat{F}(\ell)$, so $f^*L(0, \chi_\ell) = L(0, \chi_{\hat{F}(\ell)})$, which is exactly the bundle represented by $\hat{f}(\ell)$. Hence $\hat{f} = f^*$ under the identification of proposition 5.1.3.

2. *Contravariance and functoriality.* The identity id_X is induced by id_V , whose antilinear transpose is $\text{id}_{\hat{V}}$, so $\widehat{\text{id}_X} = \text{id}_{\hat{X}}$. If $h: Y \rightarrow Z$ is induced by the $\mathbb{C}$ -linear $G: W \rightarrow U$ with $G(\Lambda_Y) \subseteq \Lambda_Z$, then $h \circ f$ is induced by $G \circ F$, and for $\ell \in \hat{U}$,

$$\widehat{G \circ F}(\ell) = \ell \circ (G \circ F) = (\ell \circ G) \circ F = \hat{F}(\hat{G}(\ell))$$

so $\widehat{h \circ f} = \hat{f} \circ \hat{h}$ on universal covers, and hence on the tori. The functor $X \mapsto \hat{X}$, $f \mapsto \hat{f}$ is therefore contravariant, as we sought to show.

The proposition records that duality is a genuine contravariant functor on the category of complex tori, computed on universal covers by the antilinear transpose $F \mapsto \hat{F} = (-) \circ F$. Two descriptions of $\hat{f}$ have been matched: the linear-algebraic one, $\ell \mapsto \ell \circ F$, which is manifestly a homomorphism of tori, and the geometric one, pullback of degree-zero line bundles along f , which is manifestly functorial. The equality of the two is what allows the dual to be computed either way, a convenience used repeatedly in the sequel. Contravariance is the reason a polarization, which will be a self-map up to the identification $X \cong \hat{X}$, produces a symmetric object: dualizing reverses the direction of $\varphi_L: X \rightarrow \hat{X}$, and the Rosati involution of Chapter 6 is built from exactly this reversal.

Applying the dual twice should return to the original torus, and it does, canonically. Biduality is the statement that promotes $X \mapsto \hat{X}$ from a functor to a duality: not only is $\hat{\hat{X}}$ isomorphic to X , the isomorphism is natural, given by a single formula independent of any choice of basis or lattice.

Theorem 5.1.5: Biduality

Let $X = V/\Lambda$ be a complex torus. The map

$$\kappa_X: X \longrightarrow \hat{\hat{X}}, \quad v + \Lambda \longmapsto (\ell \mapsto \overline{\ell(v)})$$

is a canonical isomorphism of complex tori. It is natural in X : for every homomorphism $f: X \rightarrow Y$ one has $\hat{\hat{f}} \circ \kappa_X = \kappa_Y \circ f$. Consequently $\hat{\hat{X}} = X$ canonically.

Proof:

Step 1: The formula defines a bidual element. Write $\hat{V} = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ and $\hat{\hat{V}} = \text{Hom}_{\mathbb{C}}(\hat{V}, \mathbb{C})$. For fixed $v \in V$, consider the assignment $K(v): \ell \mapsto \overline{\ell(v)}$ on $\hat{V}$. It is a functional on $\hat{V}$, and it is antilinear in ℓ : for $c \in \mathbb{C}$, $(c\ell)(v) = c\ell(v)$, so $K(v)(c\ell) = \overline{c\ell(v)} = \overline{c}\overline{\ell(v)} = \overline{c}K(v)(\ell)$. Hence $K(v) \in \hat{\hat{V}}$. Moreover $v \mapsto K(v)$ is $\mathbb{C}$ -linear: ℓ is antilinear, so $\ell(cv) = \overline{c}\ell(v)$ and $K(cv)(\ell) = \overline{\ell(cv)} = \overline{\overline{c}\ell(v)} = c\overline{\ell(v)} = cK(v)(\ell)$, and additivity in v is immediate. Thus $K: V \rightarrow \hat{\hat{V}}$ is a $\mathbb{C}$ -linear map.

Step 2: K is an isomorphism of vector spaces. Both V and $\hat{\hat{V}}$ have complex dimension g , so it suffices to check that K is injective. If $K(v) = 0$, then $\overline{\ell(v)} = 0$, hence $\ell(v) = 0$, for every antilinear functional ℓ on V . But the antilinear functionals separate points of V : if $v \neq 0$, extend v to a $\mathbb{C}$ -basis and take the antilinear functional dual to v in that basis, which does not vanish at v . Therefore $v = 0$, so K is injective, hence a $\mathbb{C}$ -linear isomorphism $V \xrightarrow{\sim} \hat{\hat{V}}$.

Step 3: K matches the lattices. It remains to show $K(\Lambda) = \hat{\hat{\Lambda}}$, where $\hat{\hat{\Lambda}} = \{\Xi \in \hat{\hat{V}} : \text{Im}\Xi(\ell) \in \mathbb{Z} \text{ for all } \ell \in \hat{\Lambda}\}$ is the lattice defining $\hat{\hat{X}}$ by definition 5.1.1 applied to $\hat{X}$. First, $K(\Lambda) \subseteq \hat{\hat{\Lambda}}$: for $\lambda \in \Lambda$ and $\ell \in \hat{\Lambda}$,

$$\text{Im}(K(\lambda)(\ell)) = \text{Im}(\overline{\ell(\lambda)}) = -\text{Im}\ell(\lambda) \in \mathbb{Z}$$

since $\text{Im}\ell(\lambda) \in \mathbb{Z}$ by definition of $\hat{\Lambda}$. Thus $K(\lambda) \in \hat{\hat{\Lambda}}$. For the reverse inclusion, both $K(\Lambda)$ and $\hat{\hat{\Lambda}}$ are full lattices in $\hat{\hat{V}}$ (the first because K is an $\mathbb{R}$ -linear isomorphism carrying the full lattice Λ , the second by definition 5.1.1), and $K(\Lambda) \subseteq \hat{\hat{\Lambda}}$; a containment of full lattices is an equality once the index is 1. The index is 1 because dualizing preserves lattice covolume up to the fixed pairing: taking the imaginary-part pairing Φ of Step 1 of the proof of definition 5.1.1, $\hat{\Lambda}$ is $\mathbb{R}$ -linearly identified with $\Lambda^\vee$, and $\hat{\hat{\Lambda}}$ with $(\Lambda^\vee)^\vee$, and the canonical identification $(\Lambda^\vee)^\vee = \Lambda$ of the double $\mathbb{Z}$ -dual of a lattice sends λ to evaluation at λ , which is exactly $K(\lambda)$ under Φ . Hence $K(\Lambda) = \hat{\hat{\Lambda}}$, and K descends to an isomorphism of complex tori $\kappa_X: X \xrightarrow{\sim} \hat{\hat{X}}$.

Step 4: Naturality. Let $f: X \rightarrow Y$ be induced by $F: V \rightarrow W$. Then $\hat{f}$ is induced by $\hat{F} = (-) \circ F$ (proposition 5.1.4), and $\hat{\hat{f}}$ by $\hat{\hat{F}} = (-) \circ \hat{F}$. For $v \in V$ and $\ell \in \hat{W}$,

$$(\hat{\hat{F}}(K_X(v)))(\ell) = K_X(v)(\hat{F}(\ell)) = K_X(v)(\ell \circ F) = \overline{(\ell \circ F)(v)} = \overline{\ell(F(v))} = K_Y(F(v))(\ell)$$

so $\hat{\hat{F}} \circ K_X = K_Y \circ F$ on universal covers, hence $\hat{\hat{f}} \circ \kappa_X = \kappa_Y \circ f$ on the tori. The isomorphism κ_X is therefore natural in X , completing the proof.

Biduality closes the construction: the passage to the dual torus is an involution on complex tori, canonically returning X from $\hat{\hat{X}}$. The conjugation in the formula $v \mapsto (\ell \mapsto \overline{\ell(v)})$ is forced by the two layers of antilinearity, each dual being built from conjugate-linear functionals, and it is what makes κ_X a $\mathbb{C}$ -linear rather than merely $\mathbb{R}$ -linear isomorphism. Naturality is the substantive part: it means the identification $X = \hat{\hat{X}}$ is compatible with all homomorphisms simultaneously, so one may treat a complex torus and its bidual as the same object without ever tracking the isomorphism. This is what

licenses the standard abuses of the sequel, writing $\hat{X} = X$, regarding the dual of $\hat{f}: \hat{Y} \rightarrow \hat{X}$ as f again, and treating the polarization isogeny $\varphi_L: X \rightarrow \hat{X}$ as a map whose own dual $\hat{\varphi}_L: \hat{X} = X \rightarrow \hat{X}$ can be compared with it (proposition 5.3.6).

The dual torus is the second complex torus canonically attached to X , after X itself, and it is the object toward which the theory now turns. The functor $X \mapsto \hat{X}$ converts the degree-zero Picard group, a group of line bundles, into a variety that varies functorially and contravariantly with X , and biduality makes that variety carry X back. A polarization is the datum of a homomorphism $X \rightarrow \hat{X}$ with a positivity property, so with the dual torus in hand the polarization can finally be read as a map, the isogeny φ_L constructed in section 5.3; the intervening section 5.2 develops the isogenies and torsion in which φ_L takes its values.

5.2 Isogenies, Torsion, and the Weil Pairing

The dual torus of section 5.1 records how line bundles on X vary; the present section records how abelian varieties map to one another. The morphisms that matter for the classification of abelian varieties are not the isomorphisms, of which there are few, but the isogenies: surjective homomorphisms with finite kernel. Two abelian varieties that are isogenous share almost everything of arithmetic interest (their endomorphism algebras, their ℓ -adic representations, their L -functions), differing only by a finite amount of torsion data. This is why isogeny, not isomorphism, is the working equivalence relation of the subject, and it is the analytic shadow of the isogeny theory that Part II will rebuild scheme-theoretically over an arbitrary field.

Two constructions organize this section. The first is the torsion subgroup $X[n]$, the kernel of multiplication by n , which turns out to be a free $\mathbb{Z}/n$ -module of rank $2g$; running over all n (or all powers of a fixed prime) it is the analytic precursor of the Tate module of Part II. The second is the Weil pairing, an alternating pairing on torsion that a polarization manufactures out of the commutator of the theta group of section 4.1. Over $\mathbb{C}$ both constructions are transparent: torsion is $\frac{1}{n}\Lambda/\Lambda$, and the pairing is $e^{2\pi i E}$ read off from the alternating form $E = \text{Im}H$.

Definition 5.2.1: Isogeny

An *isogeny* $f: X \rightarrow Y$ between complex tori is a surjective homomorphism with finite kernel. Equivalently, it is a homomorphism $f: X \rightarrow Y$ with $\dim X = \dim Y$ and $\ker f$ finite. The complex tori X and Y are *isogenous*, written $X \sim Y$, if there exists an isogeny $X \rightarrow Y$.

The equivalence of the two formulations is worth spelling out, since it is the reason isogenies can be detected dimensionally. Recall from proposition 1.1.7 that a homomorphism $f: X \rightarrow Y$ of complex tori $X = V/\Lambda$ and $Y = V'/\Lambda'$ is induced by a $\mathbb{C}$ -linear map $F: V \rightarrow V'$ with $F(\Lambda) \subseteq \Lambda'$. If f is surjective with finite kernel, then F is surjective (its image is a complex subtorus of full measure, hence all of Y) and injective (its kernel is a subspace of V contained in the preimage of the finite set $\ker f$, hence zero), so F is an isomorphism and $\dim X = \dim V = \dim V' = \dim Y$. Conversely, if $\dim X = \dim Y$ and $\ker f$ is finite, then $F: V \rightarrow V'$ is a $\mathbb{C}$ -linear map between spaces of equal dimension whose kernel is contained in the finite set $F^{-1}(\Lambda') \cap (\text{preimage of } \ker f)$, forcing F injective and hence bijective; surjectivity of f follows because $F(V) = V'$ meets every coset of Λ' .

The finiteness of the kernel is what makes an isogeny quasi-invertible, and the order of that kernel is its degree, the numerical invariant that governs it.

Proposition 5.2.2: The Degree of an Isogeny

Let $f: X \rightarrow Y$ be an isogeny of complex tori, induced by the $\mathbb{C}$ -linear isomorphism $F: V \xrightarrow{\sim} V'$ with $F(\Lambda) \subseteq \Lambda'$. Then $F(\Lambda)$ is a full sublattice of Λ' of finite index, and

$$\ker f \cong F^{-1}(\Lambda')/\Lambda \cong \Lambda'/F(\Lambda)$$

is a finite abelian group. Its order

$$\deg f := |\ker f| = [\Lambda' : F(\Lambda)]$$

is the *degree* of f . It is multiplicative: if $g: Y \rightarrow Z$ is a further isogeny, then $\deg(g \circ f) = (\deg g)(\deg f)$.

Proof:

Since F is a $\mathbb{C}$ -linear isomorphism, it carries the full lattice Λ (of rank $2g$) to the full lattice $F(\Lambda) \subseteq \Lambda'$, also of rank $2g$; two full lattices in the same space are commensurable, so $[\Lambda' : F(\Lambda)]$ is finite.

Step 1: Identifying the kernel. A point $v + \Lambda \in X = V/\Lambda$ lies in $\ker f$ precisely when $F(v) \in \Lambda'$, that is, when $v \in F^{-1}(\Lambda')$. Thus

$$\ker f = F^{-1}(\Lambda')/\Lambda$$

as subgroups of V/Λ . Because F is injective, it maps the sublattice $\Lambda \subseteq F^{-1}(\Lambda')$ isomorphically onto $F(\Lambda) \subseteq \Lambda'$ and maps $F^{-1}(\Lambda')$ isomorphically onto Λ' . Passing to quotients,

$$F^{-1}(\Lambda')/\Lambda \xrightarrow{\sim} \Lambda'/F(\Lambda)$$

is an isomorphism of abelian groups, so $\ker f \cong \Lambda'/F(\Lambda)$, a finite group of order $[\Lambda' : F(\Lambda)]$.

Step 2: Multiplicativity. Let $g: Y \rightarrow Z = V''/\Lambda''$ be induced by $G: V' \xrightarrow{\sim} V''$ with $G(\Lambda') \subseteq \Lambda''$. Then $g \circ f$ is induced by $G \circ F$, and the index of $(G \circ F)(\Lambda)$ in Λ'' factors through the intermediate lattice $G(\Lambda')$:

$$[\Lambda'' : (G \circ F)(\Lambda)] = [\Lambda'' : G(\Lambda')] \cdot [G(\Lambda') : G(F(\Lambda))]$$

Since G is injective, $[G(\Lambda') : G(F(\Lambda))] = [\Lambda' : F(\Lambda)]$, and the right-hand side is $(\deg g)(\deg f)$, completing the proof.

The identification $\ker f \cong \Lambda'/F(\Lambda)$ makes the degree computable directly from lattices: an isogeny is, on lattices, an inclusion of one full lattice into another, and its degree is the index of that inclusion. This is the analytic content of the statement that an isogeny is a finite-to-one covering of complex tori of the same dimension, the covering being $\deg f$ sheeted.

Isogeny is genuinely an equivalence relation, and the nontrivial point is symmetry: an isogeny in one direction produces one in the reverse direction. The mechanism is that any isogeny can be inverted up to multiplication by an integer.

Proposition 5.2.3: Isogeny is an Equivalence Relation

The relation $X \sim Y$ of definition 5.2.1 is reflexive, symmetric, and transitive. Concretely, if $f: X \rightarrow Y$ is an isogeny of degree d , then there is an isogeny $g: Y \rightarrow X$ with

$$g \circ f = [d]_X \quad \text{and} \quad f \circ g = [d]_Y$$

where $[d]$ denotes multiplication by d ; in particular $\deg g = d^{2g-1}$.

Proof:

Reflexivity is the identity, and transitivity follows from the multiplicativity of degree in proposition 5.2.2, since a composite of surjections with finite kernels is again one. For symmetry, let f be induced by $F: V \xrightarrow{\sim} V'$ with $F(\Lambda) \subseteq \Lambda'$ of index $d = \deg f$.

The kernel $\ker f = F^{-1}(\Lambda')/\Lambda$ is a finite abelian group of order d , so it is annihilated by d : every element has order dividing $|\ker f| = d$ by Lagrange. This means $d \cdot F^{-1}(\Lambda') \subseteq \Lambda$, that is, $F^{-1}(d\Lambda') \subseteq \Lambda$, hence $d\Lambda' \subseteq F(\Lambda)$. Consider the $\mathbb{C}$ -linear isomorphism $G := dF^{-1}: V' \xrightarrow{\sim} V$. Then

$$G(\Lambda') = dF^{-1}(\Lambda') \subseteq \Lambda$$

so G induces a homomorphism $g: Y \rightarrow X$. By construction $g \circ f$ is induced by $G \circ F = dF^{-1}F = d \cdot \text{id}_V$, which induces $[d]_X$; and $f \circ g$ is induced by $F \circ G = F(dF^{-1}) = d \cdot \text{id}_{V'}$, which induces $[d]_Y$. Both composites are isogenies (degree d^{2g} by the torsion count below), so g is surjective with finite kernel, hence an isogeny. Finally $\deg(g \circ f) = \deg[d]_X = d^{2g}$ by proposition 5.2.4 below, and $\deg(g \circ f) = (\deg g)(\deg f) = (\deg g)d$, so $\deg g = d^{2g}/d = d^{2g-1}$, as we sought to show.

The isogeny g of proposition 5.2.3 is a *quasi-inverse*: it inverts f after clearing the denominator d . This is exactly the sense in which isogenous abelian varieties are “the same up to a finite amount of data”, and it is why one inverts isogenies only after tensoring the endomorphism ring with $\mathbb{Q}$, a passage taken up in section 6.1. The proof leaned on the degree of multiplication by n , which is the next order of business.

Among isogenies the multiplication maps $[n]$ recur most often. Understanding the kernel of $[n]$ amounts to understanding the torsion of X , and this is a computation the lattice description settles at once.

Proposition 5.2.4: The Structure of Torsion

Let $X = V/\Lambda$ be a complex torus of dimension g , and let $n \geq 1$ be an integer. The multiplication map $[n]: X \rightarrow X$ is an isogeny, and its kernel, the *n -torsion subgroup*, is

$$X[n] := \ker[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$$

a free $\mathbb{Z}/n\mathbb{Z}$ -module of rank $2g$, of order n^{2g} . Consequently

$$\deg[n] = |X[n]| = n^{2g}$$

Proof:

Multiplication by n on $X = V/\Lambda$ is induced by the $\mathbb{C}$ -linear isomorphism $n \cdot \text{id}_V: V \rightarrow V$, which certainly carries Λ into itself, so $[n]$ is a homomorphism induced by an invertible linear map, hence dimensions agree.

A point $v + \Lambda$ lies in $\ker[n]$ precisely when $nv \in \Lambda$, that is, when $v \in \frac{1}{n}\Lambda$. Therefore

$$X[n] = \frac{1}{n}\Lambda/\Lambda$$

Multiplication by n carries $\frac{1}{n}\Lambda$ isomorphically onto Λ and carries Λ onto $n\Lambda$, so it induces an isomorphism

$$\frac{1}{n}\Lambda/\Lambda \xrightarrow{\sim} \Lambda/n\Lambda$$

Now $\Lambda \cong \mathbb{Z}^{2g}$ is a free abelian group of rank $2g$ (it is a full lattice in $V = \mathbb{C}^g \cong \mathbb{R}^{2g}$), so

$$\Lambda/n\Lambda \cong \mathbb{Z}^{2g}/n\mathbb{Z}^{2g} \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$$

a free $\mathbb{Z}/n\mathbb{Z}$ -module of rank $2g$. Its order is n^{2g} , which is finite, so $[n]$ has finite kernel and is an isogeny with $\deg[n] = |X[n]| = n^{2g}$, completing the proof.

The formula $\deg[n] = n^{2g}$ is the analytic prototype of a statement that in Part II requires the theorem of the cube for its proof: over an arbitrary field there is no lattice to count, and the degree of $[n]$ must be extracted from the behavior of line bundles under $[n]$ (in the algebraic development of Part II). Here the lattice makes it immediate. The identification $X[n] = \frac{1}{n}\Lambda/\Lambda$ also exhibits the torsion as the reduction of the lattice: choosing a $\mathbb{Z}$ -basis $\lambda_1, \dots, \lambda_{2g}$ of Λ , the n^{2g} points $\frac{1}{n}(a_1\lambda_1 + \dots + a_{2g}\lambda_{2g})$ with $a_i \in \{0, 1, \dots, n-1\}$ are exactly the n -torsion points. Assembling the groups $X[n]$ over the powers of a fixed prime ℓ , along the maps $[\ell]: X[\ell^{m+1}] \rightarrow X[\ell^m]$, produces the analytic Tate module $\varprojlim_m X[\ell^m] \cong (\mathbb{Z}_\ell)^{2g}$, whose algebraic incarnation as a Galois representation is one of the central objects of Part II.

Example 5.2.5: Torsion on an Elliptic Curve

For $g = 1$, take $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$. The proposition gives

$$E[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^2$$

generated by the images of $\frac{1}{n}$ and $\frac{\tau}{n}$, and $\deg[n] = n^2$. This matches the classical count for elliptic curves: the multiplication-by- n map on an elliptic curve has degree n^2 , and its kernel $E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$ when n is invertible in the ground field (see *Elliptic Curves* [3] for the algebraic proof via the division polynomials, which recovers the same group without reference to a lattice). The two generators $\frac{1}{n}, \frac{\tau}{n}$ are the analytic avatars of the two “directions” in the n -torsion that the Weil pairing will detect below.

Every isogeny has a dual, and the duality is contravariant and degree-preserving. The construction is pullback of line bundles: an isogeny $f: X \rightarrow Y$ pulls back a degree-zero bundle on Y to one on X , and under the identification $\hat{X} \cong \text{Pic}^0(X)$ of proposition 5.1.3 this is a homomorphism $\hat{f}: \hat{Y} \rightarrow \hat{X}$. Analytically it is transparent.

Proposition 5.2.6: The Dual Isogeny

Let $f: X \rightarrow Y$ be an isogeny of complex tori. Then the dual homomorphism $\hat{f}: \hat{Y} \rightarrow \hat{X}$ of proposition 5.1.4 is again an isogeny, and

$$\deg \hat{f} = \deg f$$

Moreover there is an isogeny $g: Y \rightarrow X$ with $g \circ f = [\deg f]_X$ and $f \circ g = [\deg f]_Y$, so that $\hat{f}$ and f fit into the duality of theorem 5.1.5: under $\hat{\hat{X}} = X$, the double dual $\hat{\hat{f}}$ is f itself.

Proof:

Write f as induced by $F: V \xrightarrow{\sim} V'$ with $F(\Lambda) \subseteq \Lambda'$ of index $d = \deg f$. Recall from definition 5.1.1 that $\hat{X} = \hat{V}/\hat{\Lambda}$ with $\hat{V} = \overline{V}^*$ the space of $\mathbb{C}$ -antilinear functionals and $\hat{\Lambda} = \{\ell \in \hat{V} : \text{Im} \ell(\lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}$, and from proposition 5.1.4 that $\hat{f}$ is induced by the transpose $\hat{F}: \hat{V}' \rightarrow \hat{V}$, $\hat{F}(\ell) = \ell \circ F$.

Step 1: $\hat{F}$ is an isomorphism. Since $F: V \rightarrow V'$ is a $\mathbb{C}$ -linear isomorphism, precomposition $\ell \mapsto \ell \circ F$ is a $\mathbb{C}$ -linear isomorphism $\overline{V'}^* \rightarrow \overline{V}^*$, that is, $\hat{F}: \hat{V}' \xrightarrow{\sim} \hat{V}$. It carries $\hat{\Lambda}'$ into $\hat{\Lambda}$: if $\ell \in \hat{\Lambda}'$, then for every $\lambda \in \Lambda$ one has $\text{Im}(\ell \circ F)(\lambda) = \text{Im} \ell(F\lambda) \in \mathbb{Z}$ because $F\lambda \in F(\Lambda) \subseteq \Lambda'$. Thus $\hat{F}(\hat{\Lambda}') \subseteq \hat{\Lambda}$ and $\hat{f}$ is a homomorphism induced by a linear isomorphism, hence has $\dim \hat{Y} = \dim \hat{X}$ and finite kernel, that is, an isogeny.

Step 2: The degree is preserved. By proposition 5.2.2, $\deg \hat{f} = [\hat{\Lambda} : \hat{F}(\hat{\Lambda}')] = [\hat{\Lambda} : \hat{F}(\hat{\Lambda}')] = [\Lambda^\vee : \hat{F}((\Lambda')^\vee)] = [\Lambda' : F(\Lambda)] = \deg f$. The pairing $\langle \ell, v \rangle = \text{Im} \ell(v)$ identifies $\hat{\Lambda}$ with the dual lattice $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ and $\hat{\Lambda}'$ with $(\Lambda')^\vee$, under which $\hat{F}$ becomes the transpose of the inclusion $F(\Lambda) \hookrightarrow \Lambda'$. For a finite-index inclusion $M \hookrightarrow N$ of free $\mathbb{Z}$ -modules of the same rank, the transpose $N^\vee \hookrightarrow M^\vee$ has the same index, since the elementary divisors $d_1, \dots, d_{2g}$ of the inclusion (with $\prod d_i = [N : M]$) are unchanged under transposition. Hence

$$\deg \hat{f} = [\Lambda^\vee : \hat{F}((\Lambda')^\vee)] = [\Lambda' : F(\Lambda)] = \deg f$$

Step 3: The quasi-inverse and biduality. The isogeny g with $g \circ f = [d]_X$ and $f \circ g = [d]_Y$ is the one produced in proposition 5.2.3. Dualizing $g \circ f = [d]_X$ and using contravariant functoriality (proposition 5.1.4) together with $\widehat{[d]} = [d]$, one gets $\hat{f} \circ \hat{g} = [d]_{\hat{X}}$, confirming $\hat{f}$ is an isogeny with quasi-inverse $\hat{g}$. Under the canonical isomorphism $\hat{\hat{X}} = X$ of theorem 5.1.5, the double transpose $\hat{\hat{f}} = F$, so $\hat{\hat{f}} = f$, completing the proof.

The dual isogeny is the analytic model for the operation that in Part II is defined by the universal property of the Picard scheme, and the equality $\deg \hat{f} = \deg f$ is the first hint of a duality that a polarization will make into an honest self-duality: the polarization isogeny φ_L of section 5.3 satisfies $\hat{\varphi}_L = \varphi_L$, so an abelian variety with a polarization is nearly self-dual, and exactly self-dual when the polarization is principal.

The dual isogeny puts $\hat{Y}[n]$ in duality with $X[n]$ once a polarization is present, and this duality is the

Weil pairing. Its cleanest source is the theta group of section 4.1: the commutator of the Heisenberg extension is an alternating pairing on $K(L)$, and specializing L to L^n realizes the n -torsion inside $K(L^n)$ and reads off a pairing valued in the n -th roots of unity. The following definition records the pairing between $X[n]$ and the n -torsion of the dual; the polarization then folds it into a pairing on $X[n]$ alone.

Definition 5.2.7: The Weil Pairing

Let $X = V/\Lambda$ be a complex torus of dimension g and $n \geq 1$. Write $\mu_n = \{z \in \mathbb{C}^* : z^n = 1\}$ for the group of n -th roots of unity. The *Weil pairing* is the map

$$e_n : X[n] \times \hat{X}[n] \longrightarrow \mu_n$$

defined as follows. Identify $\hat{X} = \hat{V}/\hat{\Lambda}$ with the antilinear-functional torus of definition 5.1.1. A point of $X[n]$ is $\bar{v} = v + \Lambda$ with $nv \in \Lambda$, and a point of $\hat{X}[n]$ is $\bar{\ell} = \ell + \hat{\Lambda}$ with $n\ell \in \hat{\Lambda}$. Set

$$e_n(\bar{v}, \bar{\ell}) := \exp(2\pi i n \operatorname{Im} \ell(v))$$

This is well defined and μ_n -valued. Replacing v by $v + \lambda$ with $\lambda \in \Lambda$ changes the argument by $2\pi i n \operatorname{Im} \ell(\lambda) = 2\pi i \operatorname{Im}(n\ell)(\lambda) \in 2\pi i \mathbb{Z}$, since $n\ell \in \hat{\Lambda}$; replacing ℓ by $\ell + \mu$ with $\mu \in \hat{\Lambda}$ changes it by $2\pi i \operatorname{Im} \mu(nv) \in 2\pi i \mathbb{Z}$, since $nv \in \Lambda$. And $(e_n(\bar{v}, \bar{\ell}))^n = \exp(2\pi i \operatorname{Im}((n\ell)(nv))) = 1$, because $n\ell \in \hat{\Lambda}$ and $nv \in \Lambda$ give $\operatorname{Im}((n\ell)(nv)) \in \mathbb{Z}$.

The definition matches the theta-group construction of section 4.1. Recall from proposition 4.1.7 that a positive line bundle $L = L(H, \chi)$ has theta group $\mathcal{G}(L)$, a central extension of $K(L)$ by $\mathbb{C}^*$, whose commutator is the nondegenerate alternating pairing $e^L(\bar{v}, \bar{w}) = e^{2\pi i E(v, w)}$ on $K(L)$. Applying this to L^n and using $\ker \varphi_L = K(L)$ (which section 5.3 will establish) identifies $\hat{X}[n]$ with a subgroup pairing against $X[n]$, and the pairing e_n above is the resulting duality pairing $X[n] \times \hat{X}[n] \rightarrow \mu_n$. The two routes, the antilinear-functional formula and the theta-group commutator, give the same e_n ; the formula above is the more explicit, and the theta-group description is the one that survives into the algebraic theory of Part II.

Interpreting e_n requires knowing it is a perfect duality, which is the content of the next result.

Proposition 5.2.8: Nondegeneracy of the Weil Pairing

The Weil pairing $e_n : X[n] \times \hat{X}[n] \rightarrow \mu_n$ of definition 5.2.7 is bilinear and nondegenerate: it is a homomorphism in each variable, and the induced maps

$$X[n] \longrightarrow \operatorname{Hom}(\hat{X}[n], \mu_n), \quad \hat{X}[n] \longrightarrow \operatorname{Hom}(X[n], \mu_n)$$

are isomorphisms. It is also compatible with dual isogenies: for an isogeny $f : X \rightarrow Y$ and n -torsion points $\bar{v} \in X[n]$, $\bar{\ell} \in \hat{Y}[n]$,

$$e_n^Y(f(\bar{v}), \bar{\ell}) = e_n^X(\bar{v}, \hat{f}(\bar{\ell}))$$

Proof:

Step 1: Bilinearity. The formula $e_n(\bar{v}, \bar{\ell}) = \exp(2\pi i n \operatorname{Im} \ell(v))$ is additive in v because ℓ is $\mathbb{R}$ -linear, so $\operatorname{Im} \ell(v_1 + v_2) = \operatorname{Im} \ell(v_1) + \operatorname{Im} \ell(v_2)$; and additive in ℓ because $\operatorname{Im}(\ell_1 + \ell_2)(v) = \operatorname{Im} \ell_1(v) + \operatorname{Im} \ell_2(v)$. The exponential converts these sums to products, so e_n is a homomorphism in each variable.

Step 2: Nondegeneracy. Both $X[n]$ and $\hat{X}[n]$ have order n^{2g} by proposition 5.2.4 (applied to X and to $\hat{X}$, both of dimension g), and $\operatorname{Hom}(\hat{X}[n], \mu_n)$ has order n^{2g} as well since $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$. So it suffices to show the left kernel is trivial: if $\bar{v} \in X[n]$ pairs trivially with every $\bar{\ell} \in \hat{X}[n]$, then $\bar{v} = 0$. Suppose $nv \in \Lambda$ and $n \operatorname{Im} \ell(v) \in \mathbb{Z}$ for all ℓ with $n\ell \in \hat{\Lambda}$. Writing $w := nv \in \Lambda$, the condition $n \operatorname{Im} \ell(v) \in \mathbb{Z}$ becomes $\operatorname{Im} \ell(w) \in \mathbb{Z}$ for all $\ell \in \frac{1}{n}\hat{\Lambda}$, since $n \operatorname{Im} \ell(v) = \operatorname{Im} \ell(nv) = \operatorname{Im} \ell(w)$. Recall $\hat{\Lambda} = \{\ell : \operatorname{Im} \ell(\Lambda) \subseteq \mathbb{Z}\}$ is the dual lattice of Λ under the pairing $(\ell, u) \mapsto \operatorname{Im} \ell(u)$, which is a perfect $\mathbb{R}$ -bilinear pairing $\hat{V} \times V \rightarrow \mathbb{R}$ restricting to a perfect $\mathbb{Z}$ -pairing $\hat{\Lambda} \times \Lambda \rightarrow \mathbb{Z}$. Under this perfect pairing, $\operatorname{Im} \ell(w) \in \mathbb{Z}$ for all $\ell \in \frac{1}{n}\hat{\Lambda}$ forces $w \in n\Lambda$. Hence $v \in \Lambda$ and $\bar{v} = 0$ in $X[n]$. The left-hand map is thus injective between groups of the same order, an isomorphism; by symmetry so is the right-hand map.

Step 3: Compatibility with dual isogenies. Let f be induced by $F: V \rightarrow V'$, so $\hat{f}$ is induced by $\hat{F}(\ell) = \ell \circ F$ (proposition 5.1.4). For $\bar{v} \in X[n]$ and $\bar{\ell} \in \hat{Y}[n]$,

$$e_n^Y(f\bar{v}, \bar{\ell}) = \exp(2\pi i n \operatorname{Im} \ell(Fv)) = \exp(2\pi i n \operatorname{Im}(\ell \circ F)(v)) = e_n^X(\bar{v}, \hat{f}\bar{\ell})$$

which is the stated adjunction, completing the proof.

Nondegeneracy is what makes the Weil pairing an instrument rather than a curiosity: it identifies the dual torsion $\hat{X}[n]$ with the Pontryagin dual $\operatorname{Hom}(X[n], \mu_n)$, so that any structure carried by $X[n]$ is reflected in $\hat{X}[n]$ and conversely. The adjunction $e_n^Y(f\bar{v}, \bar{\ell}) = e_n^X(\bar{v}, \hat{f}\bar{\ell})$ says the Weil pairing turns an isogeny into its dual, which is the analytic origin of the compatibility that governs the algebraic Weil pairing of Part II. When a polarization $\varphi_L: X \rightarrow \hat{X}$ is present, composing e_n with φ_L in the second slot produces an alternating pairing $X[n] \times X[n] \rightarrow \mu_n$, $(\bar{v}, \bar{w}) \mapsto e_n(\bar{v}, \varphi_L(\bar{w}))$; by definition 5.2.7 this equals $\exp(2\pi i n E(w, v))$ with $E = \operatorname{Im} H$, recovering the alternating incarnation of the polarization on torsion. The pairing a polarization thereby places on torsion is the bilinear structure that all subsequent classification (the Rosati involution, Albert's types) will exploit. For $g = 1$ this is the classical Weil pairing $e_n: E[n] \times E[n] \rightarrow \mu_n$ of *Elliptic Curves* [3], alternating and nondegenerate, whose values encode the Galois action on torsion.

The example that ties the section together is multiplication by n itself, the isogeny whose degree, kernel, and self-duality have been computed piecemeal above.

Example 5.2.9: Multiplication by n as an Isogeny

Fix $X = V/\Lambda$ of dimension g and $n \geq 1$. Multiplication by n ,

$$[n]: X \longrightarrow X$$

is an isogeny with the following features, drawn together from the section.

1. *Degree.* By proposition 5.2.4, $\deg[n] = |X[n]| = n^{2g}$. For $g = 1$ this is n^2 , matching example 5.2.5 and the classical elliptic-curve degree.
2. *Kernel.* $\ker[n] = X[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$, the full n -torsion.

3. *Self-duality.* The dual of $[n]: X \rightarrow X$ is $\widehat{[n]} = [n]: \hat{X} \rightarrow \hat{X}$: indeed $[n]$ is induced by $n \cdot \text{id}_V$, whose transpose on $\hat{V} = \overline{V}^*$ is again $n \cdot \text{id}_{\hat{V}}$, so $\widehat{[n]} = [n]$ and $\deg \widehat{[n]} = n^{2g} = \deg [n]$, consistent with proposition 5.2.6.
4. *Quasi-inverse.* The quasi-inverse of $[n]$ from proposition 5.2.3 is $[n]$ itself: $[n] \circ [n] = [n^2] = [\deg [n]/n^{2g-1}]$ is not the identity, but $[n]$ becomes invertible after inverting n , that is, in $\text{End}(X) \otimes \mathbb{Q}$, where $[n]^{-1} = [1/n]$. This is the concrete reason isogenies are inverted only rationally.

The multiplication maps $[n]$ are the source of all the torsion $X[n]$, hence of the Tate module and of the Weil pairing: every n -torsion point is a point of $\ker [n]$, and the pairing e_n of definition 5.2.7 lives on this kernel and its dual. The endomorphism $[n]$ commutes with every homomorphism of tori, so its kernels $X[n]$ carry a canonical action of $\text{End}(X)$, and this is the module structure through which the endomorphism algebra of section 6.1 will act on torsion.

Isogeny is the working equivalence of abelian varieties: it collapses the isomorphism classes into a coarser, more tractable classification in which every abelian variety is a product of simple ones (section 5.4), and in which the endomorphism ring becomes a semisimple algebra over $\mathbb{Q}$. The Weil pairing supplies the other half of the structure. Torsion alone is a bare $(\mathbb{Z}/n)^{2g}$; a polarization, through φ_L and the pairing e_n , endows it with a nondegenerate alternating form, and it is this pairing, not the torsion group itself, that a polarization contributes. The dual torus $\hat{X}$ of definition 5.1.1 is the object that makes the pairing a duality: e_n pairs $X[n]$ against $\hat{X}[n]$ perfectly, and a polarization is what turns that duality into a self-duality of X .

5.3 The Polarization Isogeny and Its Dual

A polarization has so far worn two faces. In definition 3.1.3 it was a positive Hermitian form $H > 0$ on V whose imaginary part is integral on Λ ; equivalently, by the correspondence between Hermitian and alternating forms, a positive class in the Néron-Severi group. Both descriptions live on the linear-algebra side of the torus. The dual torus $\hat{X}$ of definition 5.1.1 now supplies a third: a polarization is a homomorphism from X to $\hat{X}$, built from the line bundle it comes from. This section constructs that homomorphism, computes its kernel and degree, and identifies the one property that separates the maps to $\hat{X}$ arising from polarizations from arbitrary homomorphisms $X \rightarrow \hat{X}$, namely that they are symmetric under biduality.

The construction is the geometric shadow of the map $x \mapsto t_x^* L \otimes L^{-1}$ that already appeared implicitly when the group $\Lambda(L)$ was defined. Translating a line bundle L by a point x and comparing the result to L produces a line bundle whose first Chern class vanishes, hence a point of $\text{Pic}^0(X) = \hat{X}$. Letting x vary gives a homomorphism $X \rightarrow \hat{X}$, and over $\mathbb{C}$ this homomorphism is an isogeny precisely when L is nondegenerate, an isomorphism precisely when the polarization is principal.

Definition 5.3.1: The polarization isogeny

Let $L = L(H, \chi)$ be a line bundle on $X = V/\Lambda$. For $x \in X$ let $t_x: X \rightarrow X$ denote translation by x . The *polarization homomorphism* associated to L is

$$\varphi_L: X \rightarrow \hat{X} = \text{Pic}^0(X), \quad x \mapsto t_x^* L \otimes L^{-1}$$

When L is a polarization (that is, $H > 0$ with $E = \text{Im}H$ integral on Λ), φ_L is called the *polarization isogeny*.

The map is well defined: for every x the class $c_1(t_x^* L \otimes L^{-1})$ vanishes, because translation acts trivially on cohomology (a translation is homotopic to the identity on the torus), so $t_x^* L$ and L have the same first Chern class and their difference lands in $\text{Pic}^0(X) = \hat{X}$. That φ_L is a homomorphism, rather than merely a map of sets, is the theorem of the square in its analytic form; it will be transparent from the linear-algebra description in proposition 5.3.3, where φ_L is induced by an honest $\mathbb{R}$ -linear map of vector spaces.

5.3.2 Convention: two maps named φ The subscripted symbol φ_L of definition 5.3.1 is a homomorphism from X to the *dual torus* $\hat{X}$. It should not be confused with the map $\varphi_{|L^{\otimes n}|}: X \rightarrow \mathbb{P}^N$ of Chapter 4, the morphism to *projective space* defined by the linear system of a suitable power of L (the Lefschetz embedding, theorem 4.4.2). Both are written with the letter φ by long tradition, but their targets differ: φ_L lands in $\hat{X}$, while $\varphi_{|L^{\otimes n}|}$ lands in a projective space. The context, and the presence or absence of a linear system in the subscript, always distinguishes them.

To compute with φ_L one needs its description on the universal cover, where translation and the factor of automorphy become explicit. The result is that φ_L is linear, induced by the map sending v to the antilinear functional $H(v, -)$. This is where the Hermitian form of the polarization enters the picture concretely, and it is the identity from which every subsequent computation of kernel and degree flows.

Proposition 5.3.3: Analytic form of the polarization homomorphism

Let $L = L(H, \chi)$ on $X = V/\Lambda$. The homomorphism $\varphi_L: X \rightarrow \hat{X} = \hat{V}/\hat{\Lambda}$ is induced by the $\mathbb{R}$ -linear map

$$\varphi_H: V \rightarrow \hat{V} = \overline{V}^*, \quad v \mapsto H(v, -)$$

Here $H(v, -)$ is $\mathbb{C}$ -antilinear in its argument, so it lies in $\hat{V} = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$, and φ_H carries Λ into $\hat{\Lambda}$, because $\text{Im}H(v, -) = E(v, -)$ is integral on Λ .

Proof:

Step 1: The target is $\hat{V}$. Fix $v \in V$. The assignment $w \mapsto H(v, w)$ is a function $V \rightarrow \mathbb{C}$. Since H is conjugate-linear in its second variable (definition 3.1.3 and the convention that a Hermitian form is linear in the first and conjugate-linear in the second), $H(v, \alpha w) = \bar{\alpha} H(v, w)$ for $\alpha \in \mathbb{C}$, so $H(v, -)$ is a $\mathbb{C}$ -antilinear functional, an element of $\hat{V} = \overline{V}^*$. The map $\varphi_H: v \mapsto H(v, -)$ is $\mathbb{C}$ -linear in v because H is linear in its first variable, and in particular $\mathbb{R}$ -linear.

Step 2: φ_H carries Λ into $\hat{\Lambda}$. Recall from definition 5.1.1 that

$$\hat{\Lambda} = \{ \ell \in \hat{V} : \text{Im} \ell(\lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda \}$$

Let $\lambda \in \Lambda$. For every $\mu \in \Lambda$,

$$\text{Im}(\varphi_H(\lambda)(\mu)) = \text{Im} H(\lambda, \mu) = E(\lambda, \mu)$$

and $E(\lambda, \mu) \in \mathbb{Z}$ because $E = \text{Im} H$ is integral on Λ (this integrality is exactly the condition that H defines a class in $\text{NS}(X)$, definition 3.1.3). Hence $\varphi_H(\lambda) \in \hat{\Lambda}$, so $\varphi_H(\Lambda) \subseteq \hat{\Lambda}$ and φ_H descends to a homomorphism of tori $V/\Lambda \rightarrow \hat{V}/\hat{\Lambda}$.

Step 3: The descended map is φ_L . It remains to identify the homomorphism induced by φ_H with $x \mapsto t_x^* L \otimes L^{-1}$. Fix $x \in X$ with a lift $u \in V$. The bundle $t_x^* L$ has factor of automorphy $a_{(H, \chi)}(\lambda, v + u)$, so $t_x^* L \otimes L^{-1}$ has factor

$$\frac{a_{(H, \chi)}(\lambda, v + u)}{a_{(H, \chi)}(\lambda, v)} = \exp(\pi H(u, \lambda))$$

independent of v : the $\chi(\lambda)$ and $\frac{\pi}{2} H(\lambda, \lambda)$ factors cancel, and $H(v + u, \lambda) - H(v, \lambda) = H(u, \lambda)$ by linearity in the first slot. This factor has trivial Hermitian part, so $t_x^* L \otimes L^{-1} \in \text{Pic}^0(X)$, and it is equivalent, modulo the holomorphic coboundary used in the modulus reduction of proposition 3.4.5, to its unitary part, the character

$$\lambda \mapsto \exp(2\pi i \text{Im} H(u, \lambda))$$

on Λ (using $H(u, \lambda) - \overline{H(u, \lambda)} = 2i \text{Im} H(u, \lambda)$), which is $\exp(2\pi i \text{Im} \varphi_H(u)(\lambda))$. Under the identification $\text{Pic}^0(X) \cong \hat{V}/\hat{\Lambda}$ of proposition 5.1.3, sending $\ell \in \hat{V}$ to the character $\lambda \mapsto \exp(2\pi i \text{Im} \ell(\lambda))$, this is precisely the image of $\varphi_H(u) \bmod \hat{\Lambda}$. Thus $\varphi_L(x) = \varphi_H(u) \bmod \hat{\Lambda}$, as we sought to show.

The proposition turns a map defined by translating line bundles into a map of vector spaces given by a single formula, $v \mapsto H(v, -)$. Everything about φ_L that one wants to know, whether it is an isogeny, what its kernel is, how it interacts with the dual, is now a question about the linear map φ_H and the Hermitian form H . In particular φ_L is injective (as a map of tori) exactly when φ_H is injective on V , which holds iff H is nondegenerate as a Hermitian form; this is the analytic root of the equivalence between nondegeneracy of H and φ_L being an isogeny, made precise next.

The kernel of φ_L is now within reach. A point x lies in $\ker \varphi_L$ iff a lift u of x satisfies $\varphi_H(u) \in \hat{\Lambda}$, and unwinding what membership in $\hat{\Lambda}$ means recovers exactly the group $\Lambda(L)$ from Chapter 4. The kernel is therefore the finite group $K(L)$, and its order is the determinant of E , which is the square of the Pfaffian.

Proposition 5.3.4: Kernel and degree of the polarization isogeny

Let $L = L(H, \chi)$ with $H > 0$ of type $(d_1, \dots, d_g)$ (definition 3.2.3). Then

$$\ker \varphi_L = \Lambda(L)/\Lambda = K(L)$$

(with $\Lambda(L)$ and $K(L)$ as in definition 4.1.2). Consequently φ_L is an isogeny iff H is nondegenerate, and in that case

$$\deg \varphi_L = |K(L)| = \det E = (d_1 d_2 \cdots d_g)^2$$

Proof:

By proposition 5.3.3, φ_L is induced by the $\mathbb{R}$ -linear map $\varphi_H: V \rightarrow \hat{V}$, $v \mapsto H(v, -)$. A point $x = u \bmod \Lambda$ lies in $\ker \varphi_L$ iff $\varphi_H(u) \in \hat{\Lambda}$, that is, iff

$$\operatorname{Im} H(u, \lambda) = E(u, \lambda) \in \mathbb{Z} \quad \text{for all } \lambda \in \Lambda$$

This is precisely the defining condition of $\Lambda(L) = \{v \in V : E(v, \lambda) \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}$ from definition 4.1.2. Hence the set of lifts u with $x \in \ker \varphi_L$ is $\Lambda(L)$, and

$$\ker \varphi_L = \Lambda(L)/\Lambda = K(L)$$

Since H is positive definite it is in particular nondegenerate, so $E = \operatorname{Im} H$ is a nondegenerate alternating form; then $\Lambda(L)$ is a full lattice containing Λ with finite index, and $K(L)$ is finite. A homomorphism of tori of equal dimension with finite kernel is an isogeny (definition 5.2.1), and φ_L is such because φ_H is an $\mathbb{R}$ -linear isomorphism $V \rightarrow \hat{V}$ (both have real dimension $2g$ and φ_H is injective by nondegeneracy of H), hence surjective on tori. If instead H were degenerate, φ_H would have nonzero kernel in V and φ_L would have positive-dimensional kernel, hence fail to be an isogeny; this proves the stated equivalence.

For the degree, $\deg \varphi_L = |\ker \varphi_L| = |K(L)|$ by proposition 5.2.2. The order of $K(L)$ was computed in proposition 4.1.3: choosing a symplectic basis in which E has matrix $\begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix}$ with $D = \operatorname{diag}(d_1, \dots, d_g)$ (definition 3.2.3), one finds $K(L) \cong (\bigoplus_i \mathbb{Z}/d_i)^2$, so

$$|K(L)| = (d_1 d_2 \cdots d_g)^2 = \det \begin{pmatrix} 0 & D \\ -D & 0 \end{pmatrix} = \det E$$

completing the proof.

The kernel of the polarization isogeny is the same finite group $K(L)$ that carried the commutator pairing of the theta group in Chapter 4. The two appearances are not a coincidence: the pairing e^L on $K(L)$ is, up to sign, the Weil pairing pulled through φ_L , and the nondegeneracy of that pairing is the nondegeneracy of H seen inside the finite group. The degree formula $\deg \varphi_L = (d_1 \cdots d_g)^2 = (\operatorname{Pf} E)^2$ makes the point that the polarization isogeny is never injective unless the polarization is principal: the failure of injectivity is measured by the type, and it disappears exactly when the type is $(1, \dots, 1)$.

That last observation is important enough to record as the isogeny-theoretic definition of a polarization. A polarization, presented so far as a Hermitian form or a Néron-Severi class, can equally be given as the homomorphism φ_L to the dual, and the principal polarizations are exactly those for which φ_L is

an isomorphism.

Definition 5.3.5: Polarization as an isogeny to the dual

A *polarization* on X may be presented as the isogeny $\varphi_L: X \rightarrow \hat{X}$ attached (via definition 5.3.1) to an ample line bundle L ; its *degree* is $\deg \varphi_L = (d_1 \cdots d_g)^2$, where $(d_1, \dots, d_g)$ is the type of the polarization (definition 3.2.3). The polarization is *principal* precisely when $\varphi_L: X \rightarrow \hat{X}$ is an isomorphism, equivalently when its type is $(1, \dots, 1)$.

The equivalence of "principal" with " φ_L an isomorphism" is immediate from proposition 5.3.4: φ_L is an isomorphism iff $\deg \varphi_L = 1$ iff $d_1 \cdots d_g = 1$ iff every $d_i = 1$, since the d_i are positive integers. A principal polarization thus identifies X with its own dual, canonically once the polarization is fixed. This is the analytic prototype of the autoduality that Jacobians will exhibit in Chapter 12: the theta divisor of a curve furnishes a principal polarization, hence an isomorphism $\text{Jac}(C) \cong \widehat{\text{Jac}(C)}$.

What distinguishes the homomorphisms $X \rightarrow \hat{X}$ coming from line bundles among all homomorphisms to the dual is a symmetry condition. Biduality (theorem 5.1.5) identifies $\hat{\hat{X}}$ canonically with X ; under this identification the dual of a homomorphism $f: X \rightarrow \hat{X}$ is again a homomorphism $\hat{f}: X \rightarrow \hat{X}$, and one may ask whether $\hat{f} = f$. For $f = \varphi_L$ the answer is yes: the polarization isogeny is symmetric. This is the property that makes φ_L a polarization rather than a general isogeny to the dual.

Proposition 5.3.6: The polarization isogeny is symmetric

Under the canonical identification $\hat{\hat{X}} = X$ of theorem 5.1.5, the dual homomorphism $\hat{\varphi}_L: \hat{X} \rightarrow X$ equals φ_L . That is, φ_L is symmetric.

Proof:

Work with the analytic descriptions. By proposition 5.3.3, φ_L is induced by $\varphi_H: V \rightarrow \hat{V}$, $v \mapsto H(v, -)$. The dual homomorphism $\hat{\varphi}_L$ is, on universal covers, the transpose $\varphi_H^*: \hat{V} \rightarrow \hat{V}$, followed by the biduality identification $\hat{\hat{V}} \cong V$; recall from proposition 5.1.4 that dualizing a homomorphism of tori corresponds analytically to taking the $\mathbb{C}$ -antilinear transpose of the inducing map.

Unwind the transpose explicitly. The bidual space $\hat{\hat{V}} = \overline{(\hat{V}^*)}^*$ is canonically V via the evaluation pairing $v \leftrightarrow (\ell \mapsto \overline{\ell(v)})$, which is the analytic content of theorem 5.1.5. For $v, w \in V$, the transpose φ_H^* acts by

$$(\varphi_H^*(w))(v) = \overline{\varphi_H(v)(w)} = \overline{H(v, w)} = H(w, v)$$

where the last equality is the Hermitian symmetry $H(w, v) = \overline{H(v, w)}$. Thus $\varphi_H^*(w) = H(w, -) = \varphi_H(w)$ for every $w \in V$, so $\varphi_H^* = \varphi_H$ under the biduality identification. Passing to the induced maps of tori, $\hat{\varphi}_L = \varphi_L$, as we sought to show.

The symmetry $\hat{\varphi}_L = \varphi_L$ is the exact analytic image of the symmetry of the Hermitian form: $H(w, v) = \overline{H(v, w)}$ says that H is Hermitian, and it is this identity, and no more, that forces $\varphi_H^* = \varphi_H$. A general homomorphism $X \rightarrow \hat{X}$ corresponds to an $\mathbb{R}$ -bilinear form on V with no symmetry constraint and need not be symmetric; the symmetric ones are exactly those coming

from Hermitian forms, that is, from classes in $\text{NS}(X)$, and the positive symmetric ones are the polarizations. Symmetry is therefore the algebraic feature that will single out polarizations in Part II, where φ_L is redefined without reference to $\mathbb{C}$ and the Hermitian form is no longer available; the condition $\hat{\varphi}_L = \varphi_L$ survives verbatim and becomes the definition of a symmetric homomorphism to the dual.

The elliptic-curve case closes the circle with the earlier chapters, where a polarization first appeared as a positive real number times the standard form and then as a positive integer generating $\text{NS}(E)$. On an elliptic curve every principal polarization gives a degree-one map, hence an isomorphism, so the polarization isogeny recovers the classical fact that an elliptic curve is its own dual.

Example 5.3.7: The polarization isogeny of an elliptic curve

Let $E = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$, $\tau \in \mathbb{H}$, and take the principal polarization: the Hermitian form

$$H(v, w) = \frac{v \bar{w}}{\text{Im}\tau}$$

with $E_0 := \text{Im}H$ the alternating form having $E_0(1, \tau) = 1$ on the basis $\{1, \tau\}$ of Λ (the generator of $\text{NS}(E) \cong \mathbb{Z}$, of type (1)). Since $g = 1$, the type is the single integer $d_1 = 1$, so by proposition 5.3.4,

$$\deg \varphi_L = (d_1)^2 = 1$$

and $\varphi_L: E \rightarrow \hat{E}$ is an isomorphism. Concretely, $\varphi_H: \mathbb{C} \rightarrow \hat{\mathbb{C}} = \overline{\mathbb{C}}^*$ sends v to the antilinear functional $w \mapsto v\bar{w}/\text{Im}\tau$, which is a bijection because H is nondegenerate; and $\ker \varphi_L = \Lambda(L)/\Lambda$ is trivial because E_0 already pairs Λ to $\mathbb{Z}$ nondegenerately (so $\Lambda(L) = \Lambda$). This reproduces the identification $\hat{E} \cong E$ of example 5.1.2 and matches the $g = 1$ statement of example 5.2.9 that E carries a principal polarization. The degree is the square of the Pfaffian $\text{Pf}(E_0) = d_1 = 1$, consistent with the general formula $\deg \varphi_L = (\text{Pf } E)^2$.

Across the three appearances the same object has been redescribed without change of content. A polarization is a positive Hermitian form H on V with $\text{Im}H$ integral on Λ (definition 3.1.3); equivalently, a positive class in $\text{NS}(X)$; and now, equivalently again, a symmetric isogeny $\varphi_L: X \rightarrow \hat{X}$ to the dual torus, self-dual under biduality, whose degree is the square of the Pfaffian $\text{Pf}(E) = d_1 \cdots d_g$ (definition 3.2.3). The isogeny description is the one that transports to arbitrary fields: it uses only the dual torus, translation of line bundles, and the notion of a symmetric homomorphism, none of which require the complex numbers, and it is the form in which polarizations reappear in Part II when the Hermitian form is no longer available.

5.4 Poincaré Complete Reducibility

The polarization isogeny φ_L of definition 5.3.1 is more than a bridge from an abelian variety to its dual: it is a rigidifying structure that lets one split an abelian variety into indivisible pieces. The guiding analogy is linear algebra. A finite-dimensional vector space carrying a positive-definite inner product decomposes as an orthogonal direct sum of any subspace and its orthogonal complement; there is nothing to obstruct the splitting because the inner product is nondegenerate everywhere it is restricted. An abelian variety with a polarization behaves the same way, with one adjustment forced by the geometry: the pieces meet not in the zero subspace but in a finite group, so the decomposition is a direct sum only up to isogeny. This is Poincaré's complete reducibility theorem, and it reduces

the classification of abelian varieties up to isogeny to the classification of the indivisible ones, the simple abelian varieties. The endomorphism algebra of a simple abelian variety turns out to be a division algebra, and Chapter 6 will read off from Albert's classification exactly which division algebras can occur.

The objects to be split off are the abelian subvarieties. A subtorus of a complex torus is a connected complex subgroup; when the ambient torus is an abelian variety, the interesting subtori are those that are themselves abelian varieties.

Definition 5.4.1: Abelian Subvariety

Let $X = V/\Lambda$ be an abelian variety. An *abelian subvariety* of X is a complex subtorus $Y \subseteq X$ that is itself an abelian variety. Concretely, $Y = W/(\Lambda \cap W)$ for a complex subspace $W \subseteq V$ with $\Lambda \cap W$ a full lattice in W , and Y is required to be polarizable.

Two facts make abelian subvarieties easy to produce and to polarize. First, an abelian subvariety is exactly what arises as the image or the connected component of the kernel of a homomorphism: if $f: X \rightarrow X'$ is a homomorphism of abelian varieties, then $f(X)$ is an abelian subvariety of X' , and the identity component $(\ker f)^0$ is an abelian subvariety of X . This is immediate on the level of the linear map inducing f (proposition 1.1.7): the image and kernel of a $\mathbb{C}$ -linear $F: V \rightarrow V'$ carrying Λ into Λ' are complex subspaces meeting the lattices in full lattices. Second, a subtorus of an abelian variety inherits a polarization by restriction. If H is the positive Hermitian form of a polarization on X and $W \subseteq V$ is the subspace defining Y , then $H|_{W \times W}$ is again positive definite, and $\text{Im}(H|_{W \times W}) = (\text{Im } H)|_{W \times W}$ is integral on the full lattice $\Lambda \cap W$. Thus $H|_W$ is a polarization on Y , and every abelian subvariety of an abelian variety is automatically polarizable: the requirement of polarizability in definition 5.4.1 is a consequence of being a subtorus of a polarized torus, not an extra hypothesis to be checked.

Example 5.4.2: Abelian Subvarieties of a Product

Let A and B be abelian varieties of dimensions a and b , and consider $X = A \times B$, an abelian variety of dimension $a + b$ (the product of two polarizations is a polarization, coordinatewise). The two factors $A \times \{0\}$ and $\{0\} \times B$ are abelian subvarieties: they are the images of the inclusions and the kernels of the two projections. So is the graph of any homomorphism $\psi: A \rightarrow B$, namely $\Gamma_\psi = \{(a, \psi(a)) : a \in A\}$, which is the image of the homomorphism $a \mapsto (a, \psi(a))$ and is isomorphic to A . When $A = B = E$ is an elliptic curve, the diagonal $\Delta = \{(x, x)\} \subset E \times E$ and the antidiagonal $\{(x, -x)\}$ are abelian subvarieties, each a copy of E , and they meet in the finite group $\{(x, x) : x \in E[2]\}$ of order 4. This finiteness of the intersection is the phenomenon the complete reducibility theorem will exploit.

The construction of the complementary subvariety is the mechanism of the theorem; isolating it before the statement makes the proof transparent. Given an abelian subvariety $Y \subseteq X$ and a polarization L on X , the inclusion $\iota: Y \hookrightarrow X$ has a dual $\hat{\iota}: \hat{X} \rightarrow \hat{Y}$ (proposition 5.1.4), which is surjective because ι is injective on a subtorus. Composing the polarization isogeny with this restriction gives a homomorphism

$$\pi = \hat{\iota} \circ \varphi_L: X \xrightarrow{\varphi_L} \hat{X} \xrightarrow{\hat{\iota}} \hat{Y}$$

whose analytic description makes the orthogonality transparent. Recall from proposition 5.3.3 that φ_L is induced by the $\mathbb{R}$ -linear map $v \mapsto H(v, -)$ from V to $\hat{V} = \bar{V}^*$. Under the surjection $\hat{V} \rightarrow \hat{W}$

dual to $W \hookrightarrow V$, the functional $H(v, -)$ restricts to $H(v, -)|_W$. Hence π is induced by

$$v \mapsto H(v, -)|_W \in \overline{W}^*$$

and a vector $v \in V$ lies in the analytic kernel of π precisely when $H(v, w) = 0$ for all $w \in W$, that is, when v is H -orthogonal to the subspace W defining Y . The complement will be the subtorus cut out by exactly this orthogonality condition.

Theorem 5.4.3: Poincaré Complete Reducibility

Let X be an abelian variety with a polarization given by the positive Hermitian form H , and let $Y \subseteq X$ be an abelian subvariety. Set

$$Z = (\ker(\hat{\iota} \circ \varphi_L))^0,$$

the identity component of the kernel of $\hat{\iota} \circ \varphi_L: X \rightarrow \hat{Y}$, where $\iota: Y \hookrightarrow X$ is the inclusion. Then Z is an abelian subvariety of X , called the *complementary abelian subvariety* of Y with respect to the polarization, and:

1. $Y + Z = X$;
2. $Y \cap Z$ is finite.

Consequently the addition map $Y \times Z \rightarrow X$, $(y, z) \mapsto y + z$, is an isogeny, and X is isogenous to $Y \times Z$.

Proof:

Write $X = V/\Lambda$ and $Y = W/(\Lambda \cap W)$ for a complex subspace $W \subseteq V$. Let H be the positive Hermitian form of the polarization and $E = \text{Im}H$ its alternating part.

Step 1: Identify Z with the H -orthogonal complement of W . By the analytic computation preceding the theorem, the homomorphism $\pi = \hat{\iota} \circ \varphi_L$ is induced by the $\mathbb{R}$ -linear map $V \rightarrow \overline{W}^*$, $v \mapsto H(v, -)|_W$. Its kernel as a real subspace is

$$W^\perp = \{v \in V : H(v, w) = 0 \text{ for all } w \in W\}$$

the orthogonal complement of W under the positive-definite Hermitian form H . Because H is $\mathbb{C}$ -linear in its first argument, $W^\perp$ is a complex subspace of V ; because H is positive definite, $V = W \oplus W^\perp$ as complex vector spaces, and $W \cap W^\perp = 0$. The identity component of $\ker \pi$ is the subtorus of X with tangent space $W^\perp$, so

$$Z = W^\perp/(\Lambda \cap W^\perp)$$

provided $\Lambda \cap W^\perp$ is a full lattice in $W^\perp$; this is automatic for the identity component of the kernel of a homomorphism, since $\ker \pi$ is a closed subgroup of the compact torus X and its identity component is a subtorus. Being a subtorus of the polarized X , Z is an abelian subvariety, with polarization $H|_{W^\perp}$ (positive definite, integral imaginary part on $\Lambda \cap W^\perp$).

Step 2: $Y \cap Z$ is finite. On the level of tangent spaces, the intersection $Y \cap Z$ has tangent space $W \cap W^\perp = 0$, since $V = W \oplus W^\perp$. A subtorus with zero tangent space is a point, so the identity component of $Y \cap Z$ is trivial. But $Y \cap Z$ is a closed subgroup of the compact torus X with trivial identity component, hence finite, establishing (2).

Step 3: $Y + Z = X$. The addition map $\mu: Y \times Z \rightarrow X$ is a homomorphism of complex tori, induced on tangent spaces by the addition map $W \oplus W^\perp \rightarrow V$, $(w, w') \mapsto w + w'$. This is an isomorphism of complex vector spaces, precisely because $V = W \oplus W^\perp$. A homomorphism of complex tori whose induced map on tangent spaces is an isomorphism is surjective with finite kernel (its kernel is discrete, hence finite by compactness of the source). Surjectivity gives $Y + Z = \mu(Y \times Z) = X$, establishing (1).

Step 4: *The addition map is an isogeny.* By Step 3 the map $\mu: Y \times Z \rightarrow X$ induces an isomorphism on tangent spaces, so it is a surjective homomorphism with finite kernel between abelian varieties of the same dimension $\dim Y + \dim Z = \dim_{\mathbb{C}} W + \dim_{\mathbb{C}} W^\perp = \dim_{\mathbb{C}} V = \dim X$. This is exactly the definition of an isogeny (definition 5.2.1). Its kernel is $\{(t, -t) : t \in Y \cap Z\}$, of order $|Y \cap Z|$, which is finite by Step 2. Therefore X is isogenous to $Y \times Z$, completing the proof.

The theorem says an abelian subvariety is, up to isogeny, a direct factor. This is false for a general commutative group without a polarization: a subgroup need not have a complement (the subgroup $\mathbb{Z} \subset \mathbb{Z}$ of index 2, or a nonsplit extension of one elliptic curve by another, admits no complementary subgroup). What forces the splitting is the positive-definite form H , which supplies an orthogonal complement in the tangent space; the geometry of the torus then converts the tangent-space splitting $V = W \oplus W^\perp$ into a splitting of X up to the finite ambiguity $Y \cap Z$. The passage from an honest direct-sum decomposition of vector spaces to a mere isogeny of tori is the price of the lattice: Λ need not respect the decomposition $V = W \oplus W^\perp$, so $(\Lambda \cap W) \oplus (\Lambda \cap W^\perp)$ is in general a finite-index sublattice of Λ , and the index is $|Y \cap Z|$.

Two features of the complement deserve emphasis. It depends on the polarization: a different polarization on X can produce a different complement Z for the same Y , since the orthogonality is measured against H . And the construction is symmetric, since $(W^\perp)^\perp = W$ for a nondegenerate Hermitian form; the complement of Z is again Y , so the roles of Y and Z are interchangeable.

Example 5.4.4: Complement of the Diagonal in $E \times E$

Let $E = \mathbb{C}/\Lambda$ be an elliptic curve with its principal polarization H_E (the positive Hermitian form with $\text{Im}H_E$ generating the alternating forms on Λ), and give $X = E \times E$ the product polarization $H = H_E \oplus H_E$ on $V = \mathbb{C}^2$. Take $Y = \Delta$, the diagonal, with tangent space $W = \{(z, z) : z \in \mathbb{C}\}$. The Hermitian form is $H((v_1, v_2), (w_1, w_2)) = H_E(v_1, w_1) + H_E(v_2, w_2)$. A vector (u_1, u_2) is H -orthogonal to all (z, z) precisely when $H_E(u_1, z) + H_E(u_2, z) = 0$ for all z , that is, $H_E(u_1 + u_2, z) = 0$ for all z ; since H_E is nondegenerate, this forces $u_1 + u_2 = 0$. Hence

$$W^\perp = \{(u, -u) : u \in \mathbb{C}\}$$

is the antidiagonal, and the complement of the diagonal is the antidiagonal subvariety $Z = \{(x, -x)\} \cong E$. Their intersection $\Delta \cap Z$ consists of the points (x, x) with $x = -x$, that is, $x \in E[2]$, a group of order 4; and indeed the addition map $\Delta \times Z \rightarrow E \times E$ is an isogeny of degree 4. This recovers the linear-algebra picture in which the diagonal and antidiagonal are orthogonal lines whose lattice sublattice has index 4 in $\Lambda \oplus \Lambda$.

Iterating the theorem splits an abelian variety into indivisible pieces. The pieces that admit no further splitting are the simple abelian varieties.

Definition 5.4.5: Simple Abelian Variety

An abelian variety X of positive dimension is *simple* if it has no abelian subvariety other than 0 and X itself.

The word “simple” is used exactly as in the theory of modules or groups: an object with no nontrivial subobjects of the same kind. An elliptic curve is simple, since a nonzero proper abelian subvariety would have dimension strictly between 0 and 1, which is impossible. In higher dimension simplicity is a genuine restriction: a product $E_1 \times E_2$ of elliptic curves is never simple, and more subtly, an abelian variety can fail to be simple even when it is not visibly a product, if it is merely isogenous to one. Simple abelian varieties are the atoms out of which every abelian variety is assembled up to isogeny.

Corollary 5.4.6: Isotypic Decomposition up to Isogeny

Every abelian variety X of positive dimension is isogenous to a product

$$X \sim X_1^{n_1} \times X_2^{n_2} \times \cdots \times X_r^{n_r}$$

of powers of pairwise non-isogenous simple abelian varieties $X_1, \dots, X_r$. The simple factors X_i (up to isogeny) and their multiplicities n_i are uniquely determined by X , up to permutation of the factors.

Proof:

Step 1: Existence. Argue by induction on $\dim X$. If X is simple, there is nothing to prove ($r = 1$, $n_1 = 1$). Otherwise X has an abelian subvariety Y with $0 \neq Y \neq X$; choosing Y of minimal positive dimension makes Y simple, since any nonzero proper abelian subvariety of Y would be a smaller abelian subvariety of X . Fix a polarization on X and let Z be the complementary abelian subvariety of theorem 5.4.3, so that $X \sim Y \times Z$ with $\dim Z = \dim X - \dim Y < \dim X$. By induction Z is isogenous to a product of powers of simple abelian varieties; multiplying by the simple factor Y and collecting isogenous factors into common powers gives the asserted decomposition of X .

Step 2: Uniqueness. Suppose $X \sim \prod_i X_i^{n_i} \sim \prod_j Y_j^{m_j}$ are two such decompositions into powers of pairwise non-isogenous simple abelian varieties. For a simple abelian variety S and any abelian variety A with simple isogeny factors A_k , one has $\mathrm{Hom}(S, A) \otimes \mathbb{Q} = \bigoplus_k (\mathrm{Hom}(S, A_k) \otimes \mathbb{Q})$; since a homomorphism from the simple S to a simple T is 0 unless $S \sim T$ (proposition 5.4.7 below), only the factors isogenous to S contribute, and viewed as a left vector space over the division algebra $\mathrm{End}^0(S)$,

$$\dim_{\mathrm{End}^0(S)} (\mathrm{Hom}(S, A) \otimes \mathbb{Q})$$

counts precisely the multiplicity of the isogeny class of S in A . Isogenous abelian varieties have isomorphic such Hom-spaces, because an isogeny $A \rightarrow A'$ induces an isomorphism $\mathrm{Hom}(S, A) \otimes \mathbb{Q} \rightarrow \mathrm{Hom}(S, A') \otimes \mathbb{Q}$ (compose with the isogeny and with a quasi-inverse in $\mathrm{Hom}(A', A) \otimes \mathbb{Q}$). Applying this with $S = X_i$ to both decompositions shows that each X_i occurs among the Y_j with the same multiplicity $n_i = m_j$, and conversely; hence the two decompositions agree up to permutation, as we sought to show.

The corollary reduces the classification of abelian varieties up to isogeny to two separate problems:

the classification of the simple abelian varieties themselves, and the bookkeeping of multiplicities. Up to isogeny an abelian variety is thus a formal “monomial” $\prod X_i^{n_i}$ in simple factors, and the entire isogeny category of abelian varieties is a semisimple category whose simple objects are the simple abelian varieties. This is the exact analogue, for abelian varieties, of the decomposition of a semisimple module into isotypic components, and it is the reason the endomorphism algebra will assemble into a product of matrix algebras over division rings.

That last point rests on a single fact about simple abelian varieties: a nonzero endomorphism of a simple abelian variety is invertible, once one is allowed to invert isogenies. This is the input Chapter 6 needs.

Proposition 5.4.7: Endomorphism Algebra of a Simple Abelian Variety

Let X be a simple abelian variety. Then every nonzero homomorphism $f: X \rightarrow Y$ from X to a simple abelian variety Y is an isogeny; in particular every nonzero endomorphism of X is an isogeny. Consequently the endomorphism algebra

$$\text{End}^0(X) = \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$$

is a division algebra over $\mathbb{Q}$.

Proof:

Step 1: A nonzero endomorphism of a simple abelian variety is an isogeny. Let $f \in \text{End}(X)$ be nonzero. The image $f(X)$ is an abelian subvariety of X , and it is nonzero because $f \neq 0$; since X is simple, $f(X) = X$, so f is surjective. The identity component $(\ker f)^0$ is an abelian subvariety of X ; it is not all of X , again because $f \neq 0$, so by simplicity $(\ker f)^0 = 0$. A homomorphism with finite kernel (the kernel has trivial identity component, hence is finite by compactness) that is surjective between varieties of equal dimension is an isogeny (definition 5.2.1). Thus every nonzero $f \in \text{End}(X)$ is an isogeny.

Step 2: Isogenies are invertible in $\text{End}^0(X)$. An isogeny $f: X \rightarrow X$ of degree $d = |\ker f|$ has a quasi-inverse: since $\ker f \subseteq X[d]$ (every element of the finite group $\ker f$ has order dividing d), the multiplication map $[d]$ factors as $[d] = g \circ f$ for a unique homomorphism $g \in \text{End}(X)$. In $\text{End}^0(X)$ this reads $g \circ f = [d] = d \cdot \text{id}$, an invertible scalar, so

$$f^{-1} = \frac{1}{d} g \in \text{End}^0(X)$$

is a two-sided inverse (the same argument on the other side, or the observation that $\text{End}^0(X)$ is finite-dimensional over $\mathbb{Q}$ so a left inverse is a two-sided inverse). Hence every nonzero element of $\text{End}^0(X)$ is invertible.

Step 3: $\text{End}^0(X)$ is a division algebra. $\text{End}^0(X)$ is an associative $\mathbb{Q}$ -algebra with unit id , and it is finite-dimensional over $\mathbb{Q}$: the rational representation ρ_r embeds it into $\text{End}(\Lambda \otimes \mathbb{Q}) \cong M_{2g}(\mathbb{Q})$ (proposition 1.2.6), so $\dim_{\mathbb{Q}} \text{End}^0(X) \leq (2g)^2$. By Steps 1 and 2 every nonzero element is invertible. An associative unital algebra in which every nonzero element is invertible is a division algebra, as we sought to show. The statement for a homomorphism $f: X \rightarrow Y$ between simple abelian varieties follows verbatim: $f(X) = Y$ by simplicity of Y and $(\ker f)^0 = 0$ by simplicity of X , so f is an isogeny.

This is the fact that turns the isotypic decomposition of corollary 5.4.6 into a structure theorem

for the endomorphism algebra. For pairwise non-isogenous simple abelian varieties X_i there are no nonzero homomorphisms between distinct factors, by proposition 5.4.7, so $\text{End}^0(\prod_i X_i^{n_i})$ decomposes as a product $\prod_i \text{End}^0(X_i^{n_i})$, and endomorphisms of the power $X_i^{n_i}$ are $n_i \times n_i$ matrices with entries in $D_i = \text{End}^0(X_i)$. Since each D_i is a division algebra, this yields

$$\text{End}^0(X) \cong \prod_i M_{n_i}(D_i),$$

a product of matrix algebras over division algebras, which is the Wedderburn form of a semisimple algebra. The one input not supplied by pure algebra is that each D_i is a division algebra, and that is exactly proposition 5.4.7. Chapter 6 takes this as its starting point: it introduces the Rosati involution on $\text{End}^0(X)$, shows it is positive, and invokes Albert's classification to determine precisely which division algebras with a positive involution can arise, and hence which endomorphism algebras a simple abelian variety can possess.

Example 5.4.8: Endomorphism Algebra of $E \times E$

Let $E = \mathbb{C}/\Lambda$ be an elliptic curve without complex multiplication, so that $\text{End}^0(E) = \mathbb{Q}$. The product $X = E \times E$ is isogenous to E^2 , a power of the single simple abelian variety E , so it is not simple; its isotypic decomposition (corollary 5.4.6) is $X \sim E^2$ with one simple factor of multiplicity 2. An endomorphism of $E \times E$ is a homomorphism $E^2 \rightarrow E^2$, and by proposition 5.4.7 such a homomorphism is recorded by a matrix of homomorphisms $E \rightarrow E$:

$$\text{End}^0(E \times E) \cong M_2(\text{End}^0(E)) = M_2(\mathbb{Q})$$

Concretely, an endomorphism sends $(x, y) \mapsto (ax + by, cx + dy)$ for $a, b, c, d \in \text{End}^0(E) = \mathbb{Q}$, and composition of endomorphisms corresponds to matrix multiplication. This is a semisimple $\mathbb{Q}$ -algebra of dimension $4 = n^2[D : \mathbb{Q}] = 2^2 \cdot 1$, consistent with the bound $\dim_{\mathbb{Q}} \text{End}^0(X) \leq (2g)^2 = 16$ for $g = 2$; the algebra is not a division algebra (the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is a nonzero non-invertible element), which is the algebraic shadow of the fact that $E \times E$ is not simple. Had E instead had complex multiplication by an imaginary quadratic field K , so $\text{End}^0(E) = K$, the same computation would give $\text{End}^0(E \times E) \cong M_2(K)$, a matrix algebra over the division algebra K .

The two computations in example 5.4.8 display the general pattern in miniature. Splitting an abelian variety by a polarization, via the complementary subvariety of theorem 5.4.3, reduces every abelian variety up to isogeny to a product of powers of simple ones; and the endomorphism algebra of each simple factor is a division algebra by proposition 5.4.7. The classification problem for abelian varieties up to isogeny thereby becomes the classification of simple abelian varieties, and with it the classification of the division algebras that can serve as their endomorphism algebras, the question that Chapter 6 answers through the Rosati involution and Albert's theorem.

Endomorphism Algebras, the Rosati Involution, and Complex Multiplication

The endomorphisms of an abelian variety carry its arithmetic. Chapter 1 already saw that they form a ring of finite rank, constrained by the integrality of the rational representation; Poincaré reducibility then showed that the endomorphism algebra of a simple abelian variety is a division algebra. This chapter determines which division algebras can occur. The instrument is the Rosati involution, an anti-involution of $\text{End}^0(X)$ manufactured from a polarization, whose defining property is positivity: the trace form $(f, g) \mapsto \text{Tr}(fg')$ is positive definite. Positivity is a severe constraint, and Albert's classification extracts its consequences completely, sorting the possible endomorphism algebras into four types of division algebra with involution.

The chapter closes with the fourth and most structured type, complex multiplication. An abelian variety of dimension g has complex multiplication when its endomorphism algebra contains a CM field of degree $2g$, the largest commutative symmetry it can support; such varieties are built directly from the arithmetic of the field through a CM type, and they are the points at which the transcendental theory of Part I meets the arithmetic of Part II most sharply. Complex multiplication is also the gateway to Shimura varieties: the moduli space of Chapter 3 acquires distinguished CM points defined over number fields, and the reflex field that governs them is where the theory of this book hands off to *Shimura Varieties* [6].

6.1 The Endomorphism Algebra

The endomorphism ring $\text{End}(X)$ records every self-map of an abelian variety that respects the group law. It carries the arithmetic of X : for the elliptic curve it distinguishes the generic case from complex multiplication, and in general it governs which isogeny decompositions X admits. As a ring, though, $\text{End}(X)$ is awkward. An isogeny $f: X \rightarrow X$ has finite kernel but no inverse in $\text{End}(X)$, so the ring has no division and its structure resists description. The remedy is to tensor with $\mathbb{Q}$. Poincaré reducibility (theorem 5.4.3) has already resolved X up to isogeny into a product of simple pieces; passing to $\mathbb{Q}$ makes that decomposition visible at the level of the endomorphism ring, converting it into a semisimple algebra whose Wedderburn factors read off the simple constituents of X .

Definition 6.1.1: The Endomorphism Algebra

Let X be an abelian variety. Its *endomorphism algebra* is the $\mathbb{Q}$ -algebra

$$\text{End}^0(X) := \text{End}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$$

obtained by extending scalars from $\mathbb{Z}$ to $\mathbb{Q}$. Its elements are formal fractions f/n with $f \in \text{End}(X)$ and n a positive integer, and multiplication is composition. An element $f \in \text{End}(X)$ is identified with $f \otimes 1$, so $\text{End}(X)$ embeds in $\text{End}^0(X)$.

The passage to $\mathbb{Q}$ does one thing: it inverts isogenies. An isogeny $f: X \rightarrow X$ has, by the isogeny relation of proposition 5.2.6, a complementary isogeny g with $g \circ f = [n]$ and $f \circ g = [n]$ for some $n \geq 1$; in $\text{End}^0(X)$ the element $[n]$ is the invertible scalar n , so $f^{-1} = g/n$. An element of $\text{End}^0(X)$ is thus a unit precisely when it is (a rational multiple of) an isogeny. Endomorphisms that were only isogenies onto their image in $\text{End}(X)$ become genuine units in $\text{End}^0(X)$, and the ring acquires the division needed for the structure theory to apply. The trade is that $\text{End}^0(X)$ forgets the lattice $\text{End}(X)$ inside it, retaining only the isogeny class of X ; this is precisely the information the endomorphism algebra is meant to carry.

For the elliptic curve $E = \mathbb{C}/\Lambda$ the two levels are already familiar from *Elliptic Curves* [3]. Generically $\text{End}(E) = \mathbb{Z}$ and $\text{End}^0(E) = \mathbb{Q}$; when E has complex multiplication by an order in an imaginary quadratic field K , $\text{End}(E)$ is that order and $\text{End}^0(E) = K$. Tensoring with $\mathbb{Q}$ discards the choice of order and keeps the field K , which is what determines the isogeny class. The worked cases appear in example 6.1.5 below.

Theorem 6.1.2: Semisimplicity of the Endomorphism Algebra

Let X be an abelian variety isogenous to $\prod_i X_i^{n_i}$, where the X_i are pairwise non-isogenous simple abelian varieties (corollary 5.4.6). Then

$$\text{End}^0(X) \cong \prod_i M_{n_i}(D_i), \quad D_i := \text{End}^0(X_i)$$

where each D_i is a division algebra over $\mathbb{Q}$ (proposition 5.4.7). In particular $\text{End}^0(X)$ is a semisimple $\mathbb{Q}$ -algebra, and the pair of data (D_i, n_i) is an isogeny invariant of X .

Proof:

Isogenous abelian varieties have isomorphic endomorphism algebras: an isogeny $\alpha: X \rightarrow X'$ induces $f \mapsto \alpha f \alpha^{-1}$, an isomorphism $\text{End}^0(X) \rightarrow \text{End}^0(X')$ of $\mathbb{Q}$ -algebras, since α^{-1} exists in End^0 by proposition 5.2.6. Thus $\text{End}^0(X)$ depends only on the isogeny class, and it suffices to compute it for $X = \prod_i X_i^{n_i}$.

Step 1: No maps between distinct isotypic blocks. Let S and T be simple abelian varieties with S not isogenous to T . A homomorphism $h: S \rightarrow T$ has image an abelian subvariety of T ; simplicity of T forces the image to be 0 or all of T . If the image is all of T then $\ker h$ is a proper abelian subvariety of the simple S , hence finite, so h is an isogeny, contradicting $S \not\sim T$. Therefore $h = 0$, and $\text{Hom}(S, T) = 0$; tensoring with $\mathbb{Q}$, $\text{Hom}^0(S, T) = 0$ as well. Consequently a homomorphism between the blocks $X_i^{n_i}$ and $X_j^{n_j}$ vanishes for $i \neq j$, and $\text{End}^0(X)$ splits as the product of the endomorphism algebras of the individual blocks:

$$\text{End}^0\left(\prod_i X_i^{n_i}\right) \cong \prod_i \text{End}^0(X_i^{n_i})$$

Step 2: A single block is a matrix algebra. Fix one simple factor $S = X_i$ and write $D = \text{End}^0(S)$, a division algebra by proposition 5.4.7. An endomorphism of S^n is, by distributing over the n summands, an $n \times n$ array of homomorphisms $S \rightarrow S$: the component $h_{k\ell} \in \text{Hom}^0(S, S) = D$ is the composite $S \xrightarrow{\iota_\ell} S^n \xrightarrow{f} S^n \xrightarrow{\pi_k} S$, where ι_ℓ and π_k are the inclusion of and projection onto the ℓ -th and k -th factors. The assignment $f \mapsto (h_{k\ell})$ is additive, and composition of endomorphisms corresponds to matrix multiplication because $\pi_k \iota_\ell = \delta_{k\ell} \text{id}_S$. It is bijective, with inverse sending a matrix $(h_{k\ell})$ to the endomorphism $f = \sum_{k,\ell} \iota_k h_{k\ell} \pi_\ell$. Hence

$$\text{End}^0(S^n) \cong M_n(D)$$

Step 3: Assembly and semisimplicity. Combining the two steps,

$$\text{End}^0(X) \cong \prod_i \text{End}^0(X_i^{n_i}) \cong \prod_i M_{n_i}(D_i)$$

Each $M_{n_i}(D_i)$ is a simple algebra with the division algebra D_i as its underlying skew field, so the product is semisimple; this is exactly the form delivered by the Wedderburn-Artin theorem of *Core Representation Theory* [2], run here in reverse from the geometric decomposition. The simple factors of a semisimple algebra, together with their multiplicities and their skew fields, are determined up to isomorphism by the algebra; since the isotypic decomposition $\prod_i X_i^{n_i}$ is unique up to isogeny and permutation (corollary 5.4.6), the data (D_i, n_i) is an isogeny invariant of X , completing the proof.

Ordinarily one is handed an abstract semisimple algebra and decomposes it into matrix algebras over division rings; here the geometry supplies the decomposition, and the endomorphism algebra inherits it. The division algebras D_i are the arithmetic content: the integers n_i merely record how many copies of each simple factor occur, while the D_i encode the internal symmetry of the simple pieces themselves. Classifying endomorphism algebras of abelian varieties therefore reduces to two tasks: describing which division algebras arise as End^0 of a simple abelian variety, and bounding

their size. The Rosati involution (section 6.2) and Albert's theorem (section 6.3) carry out the first; the dimension bound below begins the second.

The reduction also explains why simplicity is the atomic notion. A simple abelian variety is one that cannot be broken up further by isogeny, and its endomorphism algebra is a bare division algebra with no matrix structure at all. Everything else is built from these atoms by taking powers (which introduces the matrix algebras M_{n_i}) and products (which introduces the direct product). The next proposition constrains how large the whole assembly can be.

Before stating it, recall the source of the constraint. Every endomorphism acts on the rational homology $\Lambda \otimes \mathbb{Q}$, and this action is faithful and $\mathbb{Q}$ -linear on a space of dimension $2g$. The endomorphism algebra therefore sits inside a $(2g)^2$ -dimensional matrix algebra, which caps its own dimension.

Proposition 6.1.3: Finite Dimensionality of $\text{End}^0(X)$

Let X be an abelian variety of dimension g . Then $\text{End}^0(X)$ is a finite-dimensional $\mathbb{Q}$ -algebra, and

$$\dim_{\mathbb{Q}} \text{End}^0(X) \leq (2g)^2$$

Proof:

The rational representation of definition 1.2.3 attaches to each endomorphism its induced action on rational homology,

$$\rho_r: \text{End}^0(X) \longrightarrow \text{End}(\Lambda \otimes_{\mathbb{Z}} \mathbb{Q})$$

a homomorphism of $\mathbb{Q}$ -algebras. Since Λ has rank $2g$, the target is $\text{End}(\Lambda \otimes \mathbb{Q}) \cong M_{2g}(\mathbb{Q})$, of $\mathbb{Q}$ -dimension $(2g)^2$.

The representation ρ_r is faithful. An endomorphism $f \in \text{End}(X)$ is determined by its analytic action $\rho_a(f)$ on V , which is in turn determined by the restriction of $\rho_r(f)$ to Λ under $V = \Lambda \otimes \mathbb{R}$; concretely, if $\rho_r(f) = 0$ then f kills the lattice, so $f = 0$ as a map of tori. The same holds after tensoring with $\mathbb{Q}$: if $\rho_r(\xi) = 0$ for $\xi \in \text{End}^0(X)$, write $\xi = f/n$ with $f \in \text{End}(X)$, whence $\rho_r(f) = n \rho_r(\xi) = 0$ and $f = 0$, so $\xi = 0$. That ρ_r is well defined and integral valued is the content of proposition 1.2.6, whose characteristic polynomial computation shows $\rho_r(f)$ has integer characteristic polynomial of degree $2g$.

A faithful homomorphism embeds $\text{End}^0(X)$ as a $\mathbb{Q}$ -subalgebra of $M_{2g}(\mathbb{Q})$, so

$$\dim_{\mathbb{Q}} \text{End}^0(X) \leq \dim_{\mathbb{Q}} M_{2g}(\mathbb{Q}) = (2g)^2$$

which is finite, as we sought to show.

Finite dimensionality is what makes the Wedderburn structure of theorem 6.1.2 meaningful: a semisimple algebra of infinite dimension would have infinitely many or infinite-dimensional factors, and no classification could follow. The bound $(2g)^2$ is the size of the ambient matrix algebra $M_{2g}(\mathbb{Q})$, and it is rarely attained; for a simple abelian variety with $\text{End}^0(X) = \mathbb{Q}$ the dimension is 1, at the opposite extreme. The bound does, however, force $[D_i : \mathbb{Q}]$ and the matrix sizes n_i to satisfy $\sum_i n_i^2 [D_i : \mathbb{Q}] \leq (2g)^2$, a numerical constraint that Albert's classification sharpens into an exact table.

With semisimplicity and the dimension bound in hand, the structural vocabulary can be fixed. The organizing dichotomy is whether the algebra is a division algebra, equivalently whether X is simple.

Definition 6.1.4: Simplicity and the Endomorphism Structure

An abelian variety X is simple (definition 5.4.5) if and only if $\text{End}^0(X)$ is a division algebra. In general $\text{End}^0(X)$ is a semisimple $\mathbb{Q}$ -algebra, and its *endomorphism structure* is the collection of Wedderburn data (D_i, n_i) of theorem 6.1.2: the division algebras $D_i = \text{End}^0(X_i)$ over $\mathbb{Q}$ attached to the simple isogeny factors X_i , together with their multiplicities n_i . This structure is an isogeny invariant of X .

The equivalence packaged into the definition unpacks theorem 6.1.2: X is simple exactly when the isotypic decomposition has a single factor $X = X_1$ with $n_1 = 1$, so $\text{End}^0(X) = D_1$ is a bare division algebra with no matrix part. A nonsimple X shows its reducibility in the endomorphism algebra as either a nontrivial matrix factor $M_{n_i}(D_i)$ with $n_i \geq 2$ (a repeated simple factor) or a nontrivial product (distinct simple factors), or both. The endomorphism structure is thus a faithful algebraic shadow of the geometric decomposition, and because it is an isogeny invariant it can be computed in whichever model of the isogeny class is most convenient.

Example 6.1.5: Endomorphism Algebras of Elliptic Curves

For an elliptic curve E ($g = 1$) the dimension bound of proposition 6.1.3 reads $\dim_{\mathbb{Q}} \text{End}^0(E) \leq 4$, and every elliptic curve is simple, so $\text{End}^0(E)$ is a division algebra of $\mathbb{Q}$ -dimension 1, 2, or 4. The three cases are exactly the possibilities catalogued in *Elliptic Curves* [3].

1. *The generic case.* $\text{End}^0(E) = \mathbb{Q}$, of dimension 1. Here $\text{End}(E) = \mathbb{Z}$: the only endomorphisms are the multiplication-by- n maps. Over $\mathbb{C}$ this is the situation for a lattice $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$ with τ not lying in an imaginary quadratic field, which is the case for all but a countable set of τ .
2. *Complex multiplication.* $\text{End}^0(E) = K$, an imaginary quadratic field, of dimension 2. This occurs when τ generates an imaginary quadratic field $K = \mathbb{Q}(\tau)$; then $\text{End}(E)$ is an order in K , and $\text{End}^0(E) = K$. The curve E has *complex multiplication* by K , the $g = 1$ instance of the CM abelian varieties of section 6.4.
3. *The quaternionic case.* $\text{End}^0(E)$ a quaternion algebra over $\mathbb{Q}$, of dimension 4. This does not arise over $\mathbb{C}$: a 1-dimensional complex torus has commutative endomorphism algebra. It occurs only for supersingular elliptic curves over a field of characteristic p , where $\text{End}^0(E)$ is the quaternion algebra over $\mathbb{Q}$ ramified exactly at p and ∞ . The phenomenon belongs to the algebraic theory of Part II and is taken up there.

The reducible case $E \times E$ illustrates the matrix structure. Its isotypic decomposition is a single simple factor E with multiplicity 2 (example 5.4.8), so by theorem 6.1.2

$$\text{End}^0(E \times E) \cong M_2(\text{End}^0(E))$$

In the generic case this is $M_2(\mathbb{Q})$, of dimension 4, well below the bound of proposition 6.1.3 for $g = \dim(E \times E) = 2$, which is $(2 \cdot 2)^2 = 16$; when E has CM by K it is $M_2(K)$, of dimension 8. More generally, for a product $\prod_i E_i^{n_i}$ of elliptic curves with the E_i pairwise non-isogenous, the endomorphism algebra is $\prod_i M_{n_i}(\text{End}^0(E_i))$, a product of matrix algebras over fields (in the CM and generic cases) that makes the isogeny decomposition legible at a glance.

The three elliptic cases are the smallest instance of Albert's four-type classification, previewed here so the general statement has a concrete anchor. In the language of section 6.3, the generic $\text{End}^0(E) = \mathbb{Q}$ is a totally real field of Type I, the CM field is a Type IV center, and the supersingular quaternion algebra is of Type III (totally definite). The general lesson is already visible: $\text{End}^0(X)$ is a semisimple $\mathbb{Q}$ -algebra, and Wedderburn reduces its classification to that of the division algebras D_i . Which division algebras can occur is not arbitrary; the Rosati involution equips each with a positive anti-involution, and Albert's theorem (theorem 6.3.4) determines the resulting list.

6.2 The Rosati Involution and Its Positivity

The endomorphism algebra $\text{End}^0(X)$ of the previous section is a semisimple $\mathbb{Q}$ -algebra, but so far it carries no extra structure tied to the geometry of X . A polarization supplies that structure. The polarization isogeny $\varphi_L: X \rightarrow \hat{X}$ of definition 5.3.1 lets one transport an endomorphism f of X across to the dual and back, producing a second endomorphism f' . The resulting operation $f \mapsto f'$ is an anti-involution of $\text{End}^0(X)$, and its associated bilinear form $(f, g) \mapsto \text{Tr}(fg')$ is positive definite. This single positivity property is the one strong constraint a polarization places on $\text{End}^0(X)$, and Albert's classification (section 6.3) is exactly the extraction of its consequences.

Throughout, L is a fixed polarization on the abelian variety X , so that $\varphi_L: X \rightarrow \hat{X}$ is an isogeny (definition 5.3.5). In the isogeny category φ_L is invertible: it has finite kernel and equal source and target dimension, so it becomes an isomorphism after tensoring $\text{Hom}(X, \hat{X})$ with $\mathbb{Q}$, and $\varphi_L^{-1} \in \text{Hom}^0(\hat{X}, X) := \text{Hom}(\hat{X}, X) \otimes_{\mathbb{Z}} \mathbb{Q}$ is well defined. The dual construction $f \mapsto \hat{f}$ of proposition 5.1.4 extends $\mathbb{Q}$ -linearly to isogeny classes as well.

Definition 6.2.1: Rosati Involution

Let L be a polarization on X with polarization isogeny $\varphi_L: X \rightarrow \hat{X}$ (definition 5.3.1). The *Rosati involution* associated to L is the map

$$': \text{End}^0(X) \rightarrow \text{End}^0(X), \quad f \mapsto f' := \varphi_L^{-1} \circ \hat{f} \circ \varphi_L$$

where $\hat{f}: \hat{X} \rightarrow \hat{X}$ is the dual of f (proposition 5.1.4) and $\varphi_L^{-1} \in \text{Hom}^0(\hat{X}, X)$ is the inverse of φ_L in the isogeny category.

The composite $\varphi_L^{-1} \circ \hat{f} \circ \varphi_L$ reads off the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & X \\ \downarrow \varphi_L & & \downarrow \varphi_L \\ \hat{X} & \xrightarrow{\hat{f}} & \hat{X} \end{array}$$

run backwards along the right edge: apply φ_L to land in $\hat{X}$, apply $\hat{f}$, then return along φ_L^{-1} . When φ_L is an honest isomorphism (the principal case) f' is again an honest endomorphism; in general f' lives in $\text{End}^0(X)$ and need not preserve the integral lattice $\text{End}(X)$. The construction depends on L only through φ_L , and different polarizations generally give different involutions. What follows shows that $f \mapsto f'$ reverses products and squares to the identity, so that $\text{End}^0(X)$ is an algebra with involution, and then that the involution is positive.

Proposition 6.2.2: The Rosati Involution Is an Anti-Involution

The Rosati involution $f \mapsto f'$ of definition 6.2.1 is a $\mathbb{Q}$ -linear anti-involution of $\text{End}^0(X)$:

1. $(f + g)' = f' + g'$ and $(cf)' = cf'$ for $c \in \mathbb{Q}$;
2. $(fg)' = g'f'$;
3. $(f')' = f$;
4. $[n]' = [n]$ for every $n \in \mathbb{Z}$, and more generally $c' = c$ for every $c \in \mathbb{Q} \subseteq \text{End}^0(X)$.

The motivating point is that all four properties are formal consequences of one fact about dualization: it is a contravariant $\mathbb{Q}$ -linear functor (proposition 5.1.4), so it fixes scalars, reverses composites, and squares to the identity under biduality. The Rosati involution conjugates dualization by φ_L , and conjugation preserves each of these features. The self-duality $\hat{\varphi}_L = \varphi_L$ of proposition 5.3.6 is what makes the squaring property $(f')' = f$ come out clean.

Proof:

We verify the four properties in turn, using that $g \mapsto \hat{g}$ is $\mathbb{Q}$ -linear, contravariant, and compatible with biduality (proposition 5.1.4), and that under the canonical identification $\hat{\hat{X}} = X$ (theorem 5.1.5) one has $\hat{\hat{f}} = f$.

(1) *Linearity.* Dualization is additive and $\mathbb{Q}$ -linear, so $\widehat{f + g} = \hat{f} + \hat{g}$ and $\widehat{cf} = c\hat{f}$. Conjugating by φ_L preserves sums and $\mathbb{Q}$ -scalars because composition in $\text{End}^0(X)$ is $\mathbb{Q}$ -bilinear:

$$(f + g)' = \varphi_L^{-1}(\hat{f} + \hat{g})\varphi_L = \varphi_L^{-1}\hat{f}\varphi_L + \varphi_L^{-1}\hat{g}\varphi_L = f' + g'$$

and likewise $(cf)' = cf'$.

(2) *Anti-multiplicativity.* Contravariance of dualization gives $\widehat{fg} = \hat{g}\hat{f}$. Hence

$$(fg)' = \varphi_L^{-1}\widehat{fg}\varphi_L = \varphi_L^{-1}\hat{g}\hat{f}\varphi_L = (\varphi_L^{-1}\hat{g}\varphi_L)(\varphi_L^{-1}\hat{f}\varphi_L) = g'f'$$

where the inserted $\varphi_L\varphi_L^{-1} = \text{id}$ between $\hat{g}$ and $\hat{f}$ splits the conjugate of the product into the product of the conjugates in reversed order.

(3) *Involutivity.* By definition $f' = \varphi_L^{-1}\hat{f}\varphi_L$, so applying the involution again,

$$(f')' = \varphi_L^{-1}(\widehat{f'})\varphi_L = \varphi_L^{-1}\widehat{\varphi_L^{-1}\hat{f}\varphi_L}\varphi_L$$

Dualizing the inner composite reverses it and applies hats: $\widehat{\varphi_L^{-1}\hat{f}\varphi_L} = \hat{\varphi}_L\hat{\hat{f}}\widehat{\varphi_L^{-1}} = \hat{\varphi}_L f (\hat{\varphi}_L)^{-1}$, using $\hat{\hat{f}} = f$ under biduality and $\widehat{\varphi_L^{-1}} = (\hat{\varphi}_L)^{-1}$. Now $\hat{\varphi}_L = \varphi_L$ by proposition 5.3.6 (here one identifies $\hat{\hat{X}} = X$, so that $\hat{\varphi}_L: X \rightarrow \hat{X}$ is compared with $\varphi_L: X \rightarrow \hat{X}$). Substituting,

$$(f')' = \varphi_L^{-1}(\varphi_L f \varphi_L^{-1})\varphi_L = f$$

the two conjugating pairs cancelling.

(4) *Scalars are fixed.* For $c \in \mathbb{Q}$ the endomorphism c commutes with every isogeny and satisfies $\hat{c} = c$ (dualization is $\mathbb{Q}$ -linear and fixes the identity, so it fixes each rational multiple of it). Hence

$$c' = \varphi_L^{-1}\hat{c}\varphi_L = \varphi_L^{-1}c\varphi_L = c$$

since c commutes past φ_L . In particular $[n]' = [n]$, completing the proof.

An algebra equipped with such a map is an algebra with anti-involution, and $\text{End}^0(X)$ is one for every choice of polarization. Property (2) is the substantive one: the Rosati involution is genuinely an *anti*-automorphism, not an automorphism, so it interacts with the multiplicative structure the way transposition interacts with matrix multiplication. That analogy is exact: for $X = E \times E$ a product of elliptic curves without complex multiplication, carrying the product polarization, $\text{End}^0(X) \cong M_2(\mathbb{Q})$ and the Rosati involution is the transpose, since the H -adjoint of Step 1 of theorem 6.2.3 computes to the conjugate transpose and H is real here. Property (4) records that the central copy of $\mathbb{Q}$ is fixed pointwise, which will let one restrict the involution to the center of each simple factor in Albert's classification.

The reason the Rosati involution deserves a name of its own, rather than being one anti-involution among many, is the positivity of its associated trace form. Recall from proposition 1.2.6 that every $f \in \text{End}^0(X)$ has a rational representation $\rho_r(f)$ acting on $\Lambda \otimes \mathbb{Q}$, whose characteristic polynomial has degree $2g$ and rational (in fact, for $f \in \text{End}(X)$, integer) coefficients; write $\text{Tr}(f)$ for the trace of $\rho_r(f)$, a $\mathbb{Q}$ -valued linear functional on $\text{End}^0(X)$. Pairing f against g' through this trace produces a symmetric bilinear form, and the content of the next theorem is that this form is definite.

Theorem 6.2.3: Positivity of the Rosati Involution

Let $X = V/\Lambda$ be an abelian variety with polarization L and associated Rosati involution $f \mapsto f'$. The bilinear form

$$\langle f, g \rangle := \text{Tr}(fg'), \quad f, g \in \text{End}^0(X)$$

is a symmetric, positive-definite bilinear form on the $\mathbb{Q}$ -vector space $\text{End}^0(X)$. Equivalently,

$$\text{Tr}(ff') > 0 \quad \text{for every } f \in \text{End}^0(X), f \neq 0$$

Positivity is proved analytically. The rational trace Tr is computed most transparently from the analytic representation ρ_a of definition 1.2.2, and the Rosati involution has an equally concrete analytic description as the adjoint with respect to the positive Hermitian form H underlying the polarization. Once both are in analytic terms, $\text{Tr}(ff')$ becomes twice the real part of the trace of $\rho_a(f)$ times its H -adjoint, which is a weighted sum of squared absolute values of matrix entries, hence positive.

Proof:

Write $F := \rho_a(f) \in \text{End}_{\mathbb{C}}(V)$ for the analytic representation of f (definition 1.2.2), the $\mathbb{C}$ -linear map on $V = \mathbb{C}^g$ induced by f . Let H be the positive-definite Hermitian form on V defining the polarization L (definition 3.1.3), linear in the first variable and conjugate-linear in the second.

Step 1: The Rosati involution is the H -adjoint. Denote by F^* the adjoint of F with respect to H , that is, the unique $\mathbb{C}$ -linear endomorphism of V with

$$H(Fv, w) = H(v, F^*w) \quad \text{for all } v, w \in V$$

which exists and is unique because H is nondegenerate. The claim is

$$\rho_a(f') = F^*$$

so that the Rosati involution is analytically the H -adjoint. To see this, recall from proposi-

Step 2: The rational trace as an analytic trace. By proposition 1.2.5, the rational representation and the analytic representation are related by $\rho_r \otimes_{\mathbb{Q}} \mathbb{R} \cong \rho_a \oplus \overline{\rho_a}$. Taking traces of $\rho_r(h)$ for any $h \in \text{End}^0(X)$,

where $\text{tr}_{\mathbb{C}}$ is the ordinary trace of a $\mathbb{C}$ -linear endomorphism of V . Applying this to $h = ff'$ and using Step 1,

Step 3: Positivity via a Hermitian trace. Choose an H -orthonormal $\mathbb{C}$ -basis $e_1, \dots, e_g$ of V , so that $H(e_j, e_k) = \delta_{jk}$; such a basis exists because $H > 0$. Write $F = (F_{jk})$ for the matrix of F in this basis, $Fe_k = \sum_j F_{jk} e_j$. With respect to an H -orthonormal basis the H -adjoint F^* is the conjugate transpose: $(F^*)_{jk} = \overline{F_{kj}}$. Therefore FF^* has diagonal entries

which are real and nonnegative, so

is a real number. Consequently

with equality if and only if every entry F_{jk} vanishes, i.e. $F = \rho_a(f) = 0$. Since ρ_a is injective on $\text{End}^0(X)$ (proposition 1.1.7: an endomorphism is determined by its analytic lift), $F = 0$ forces $f = 0$. Hence $\text{Tr}(ff') > 0$ whenever $f \neq 0$.

Step 4: Symmetry. First, $\mathrm{Tr}(h) = \mathrm{Tr}(h')$ for every $h \in \mathrm{End}^0(X)$: by Step 2, $\mathrm{Tr}(h) = 2\mathrm{Re} \mathrm{tr}_{\mathbb{C}} \rho_a(h)$ and $\mathrm{Tr}(h') = 2\mathrm{Re} \mathrm{tr}_{\mathbb{C}} (\rho_a(h))^*$ (using Step 1), and the trace of the H -adjoint is the complex conjugate of the trace, so the two real parts agree. Second, $(gf')' = f''g' = fg'$ by proposition 6.2.2. Combining the two,

so the form is symmetric. Combined with Step 3, it is symmetric and positive definite, as we sought to show.

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for a simple abelian variety, all but a short list of division algebras with involution, and it fixes the type of involution each can carry. Extracting that list is precisely Albert's classification (section 6.3), which reads off from positivity alone the four possible types of $\text{End}^0(X)$.

The positive-definite form of theorem 6.2.3 organizes $\text{End}^0(X)$ into a Euclidean space, and it is convenient to name it.

Definition 6.2.4: Rosati Form

The *Rosati form* associated to a polarization L on X is the positive-definite symmetric bilinear form

$$\langle f, g \rangle := \text{Tr}(fg')$$

on the $\mathbb{Q}$ -vector space $\text{End}^0(X)$, where $f \mapsto f'$ is the Rosati involution of L (definition 6.2.1) and Tr is the trace of the rational representation ρ_r (proposition 1.2.6). It makes $\text{End}^0(X)$ a finite-dimensional $\mathbb{Q}$ -vector space with a positive-definite inner product, with respect to which the Rosati involution is self-adjoint: $\langle fh, g \rangle = \langle h, f'g \rangle$.

The self-adjointness $\langle fh, g \rangle = \langle h, f'g \rangle$ is immediate from the trace identity $\text{Tr}(fhg') = \text{Tr}(h(f'g'))$, using $(f'g)' = g'f'' = g'f$ and the cyclic invariance of the trace; it says the Rosati involution is the adjoint operation for the Rosati inner product, matching Step 1 of the proof at the level of $\text{End}^0(X)$ rather than of V . This inner-product structure is what makes positivity usable: a positive-definite form has no isotropic vectors, so any element f with $f' = f$ and $\text{Tr}(f^2) = 0$ must vanish, and the self-adjoint elements form a formally real subspace. These are the levers Albert's classification pulls.

The elliptic-curve case grounds the whole discussion, and there the Rosati involution is a familiar operation in disguise.

Example 6.2.5: The Rosati Involution on an Elliptic Curve

Let $E = \mathbb{C}/\Lambda$ be an elliptic curve with its principal polarization, so $g = 1$ and $\varphi_L: E \rightarrow \hat{E}$ is an isomorphism (example 5.3.7). By example 6.1.5 the endomorphism algebra $\text{End}^0(E)$ is one of two kinds: $\mathbb{Q}$, or an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$ (the CM case). The analytic representation identifies $\text{End}^0(E)$ with a subalgebra of $\text{End}_{\mathbb{C}}(V) = \mathbb{C}$ (definition 1.2.2): an endomorphism induced by multiplication by $\alpha \in \mathbb{C}$ has $\rho_a = \alpha$.

By Step 1 of the proof of theorem 6.2.3, the Rosati involution is the H -adjoint on $V = \mathbb{C}$. Here H is a positive real multiple of the standard Hermitian form $H(v, w) = cv\bar{w}$ with $c > 0$, and the adjoint of multiplication by α with respect to $H(v, w) = cv\bar{w}$ is multiplication by $\bar{\alpha}$, since

$$H(\alpha v, w) = c\alpha v\bar{w} = c v\overline{\alpha w} = H(v, \bar{\alpha}w)$$

Thus $\rho_a(f') = \overline{\rho_a(f)}$, and the two cases read off at once.

1. If $\text{End}^0(E) = \mathbb{Q}$, every $f = \rho_a^{-1}(q)$ with $q \in \mathbb{Q} \subseteq \mathbb{R}$ satisfies $\bar{q} = q$, so $f' = f$: the Rosati involution is the *identity*. Positivity says $\text{Tr}(f^2) > 0$ for $f \neq 0$; since $\text{Tr}(f) = 2\text{Re}(q) = 2q$ (Step 2 of the proof, with $g = 1$) and $\text{Tr}(f^2) = 2q^2$, this is the elementary inequality $2q^2 > 0$ for $q \neq 0$.
2. If $\text{End}^0(E) = K = \mathbb{Q}(\sqrt{-d})$ is imaginary quadratic, an element f with $\rho_a(f) = \alpha \in K \subset \mathbb{C}$ has $f' = \rho_a^{-1}(\bar{\alpha})$, and $\bar{\alpha}$ is the image of α under the nontrivial automorphism of K : the Rosati involution is *complex conjugation* on K . The Rosati form is then

$$\text{Tr}(ff') = 2\text{Re}(\alpha\bar{\alpha}) = 2|\alpha|^2 = 2N_{K/\mathbb{Q}}(\alpha)$$

twice the field norm $N_{K/\mathbb{Q}}(\alpha) = \alpha\bar{\alpha}$ of K . Positivity $\text{Tr}(ff') > 0$ for $\alpha \neq 0$ is exactly the positivity of the norm form of an imaginary quadratic field, which is positive definite because K has no real embedding.

In both cases the Rosati involution is the unique positive anti-involution the field carries: the identity on the totally real $\mathbb{Q}$, complex conjugation on the CM field K . This is the $g = 1$ shadow of Albert's dichotomy, with $\mathbb{Q}$ a Type I example and K a Type IV example (theorem 6.1.2 organizes the general simple case into which these fit).

The elliptic-curve computation exhibits in miniature the mechanism the general theory rests on. The Rosati involution restricts to complex conjugation on the CM field, and its positivity is the positivity of the norm form; for a totally real endomorphism field the involution is trivial and positivity is the positivity of the sum of squares of the real embeddings. In the general case a simple abelian variety has $\text{End}^0(X)$ a division algebra (proposition 5.4.7), and the Rosati involution restricts to a positive involution on it whose fixed field is totally real. Positivity forces the fixed field to be totally real and forces the ambient division algebra into one of four shapes; Albert's classification (section 6.3) records which four.

6.3 Albert's Classification

Among all finite-dimensional division algebras over $\mathbb{Q}$, only a handful admit a positive involution at all, and Albert enumerated them completely. The result is that the endomorphism algebra of a simple abelian variety, which by proposition 5.4.7 is a division algebra and by theorem 6.2.3 carries a positive involution, cannot be an arbitrary division algebra: it belongs to one of exactly four families, sorted by its center and by the totally real field the involution fixes. This section states and proves the classification, deferring only the exhaustive numerical constraint table to the standard references.

The right object to classify is not the division algebra alone but the pair consisting of the algebra and its involution.

Definition 6.3.1: Division Algebra with Positive Involution

A *division algebra with positive involution* over $\mathbb{Q}$ is a pair $(D, ')$ where D is a finite-dimensional division algebra over $\mathbb{Q}$ whose center K is a field, and $': D \rightarrow D$ is a $\mathbb{Q}$ -linear anti-involution (so $(a + b)' = a' + b'$, $(ab)' = b'a'$, and $a'' = a$) that is *positive*: the symmetric bilinear form $(a, b) \mapsto \text{Tr}_{D/\mathbb{Q}}(ab')$ on D is positive definite, where $\text{Tr}_{D/\mathbb{Q}}$ is the trace of left multiplication on the $\mathbb{Q}$ -vector space D (equivalently $\text{Tr}_{K/\mathbb{Q}}$ composed with d times the reduced trace $\text{Trd}_{D/K}$, since $\text{Tr}_{D/K} = d \text{Trd}_{D/K}$ for $d = [D: K]^{1/2}$). Write $K_0 = \{a \in K \mid a' = a\}$ for the fixed field of the involution inside the center.

The definition abstracts exactly the structure the Rosati involution supplies. For a simple abelian variety X , the algebra $D = \text{End}^0(X)$ is a division algebra by proposition 5.4.7, and the Rosati involution of definition 6.2.1 is a positive $\mathbb{Q}$ -linear anti-involution by theorem 6.2.3, with the positive form there being $(f, g) \mapsto \text{Tr}(fg')$. So $(\text{End}^0(X), ')$ is a division algebra with positive involution in the sense above, and classifying these pairs classifies the algebras that can occur.

The first structural fact is that the fixed field K_0 is forced to be totally real. Positivity is the whole of the input.

Proposition 6.3.2: The Fixed Field Is Totally Real

Let $(D, ')$ be a division algebra with positive involution over $\mathbb{Q}$, with center K and fixed field $K_0 = K'$. Then K_0 is a totally real number field: every embedding $K_0 \hookrightarrow \mathbb{C}$ has image in $\mathbb{R}$. Moreover the center K is either equal to K_0 (the involution fixes the center) or is a totally imaginary quadratic extension of K_0 (a CM field, on which $'$ restricts to complex conjugation).

Proof:

Step 1: Positivity on the center. The restriction of $'$ to K is a $\mathbb{Q}$ -linear involution of the field K , so its fixed field K_0 is a subfield with $[K : K_0] \in \{1, 2\}$. Because D is a division algebra over $\mathbb{Q}$, the trace form $(a, b) \mapsto \text{Tr}_{D/\mathbb{Q}}(ab')$ restricts to a positive definite form on K_0 : for $a \in K_0$ nonzero, $a' = a$, so $\text{Tr}_{D/\mathbb{Q}}(aa') = \text{Tr}_{D/\mathbb{Q}}(a^2) > 0$ by positivity.

Step 2: Every embedding of K_0 is real. Fix an embedding $\sigma : K_0 \hookrightarrow \mathbb{C}$ and suppose it were not real, i.e. $\sigma(K_0) \not\subseteq \mathbb{R}$. The trace $\text{Tr}_{K_0/\mathbb{Q}}(a^2)$ is a positive multiple of $\sum_{\sigma} \sigma(a)^2$, the sum running over the $[K_0 : \mathbb{Q}]$ embeddings of K_0 into $\mathbb{C}$, each counted once (the reduced trace on K_0 is the field trace, and the $D/\mathbb{Q}$ trace is a positive integer multiple of it). If some σ were complex, its conjugate $\bar{\sigma}$ is a distinct embedding, and one can choose $a \in K_0$ with $\sigma(a)$ a nonzero purely imaginary number to make the pair's contribution $\sigma(a)^2 + \bar{\sigma}(a)^2 = 2\text{Re}(\sigma(a)^2)$ negative while the real embeddings contribute arbitrarily little. Concretely, approximating the idempotent-like element that isolates the σ -component (which exists since $K_0 \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^r \times \mathbb{C}^s$ as an $\mathbb{R}$ -algebra, with s the number of complex places) produces $a \in K_0$ with $\text{Tr}_{K_0/\mathbb{Q}}(a^2) < 0$, contradicting Step 1. Hence $s = 0$: every embedding is real and K_0 is totally real.

Step 3: The center over K_0 . If $[K : K_0] = 1$ then $K = K_0$ is totally real and $'$ fixes the center. If $[K : K_0] = 2$, then $K = K_0(\alpha)$ with $\alpha' = -\alpha$ (choose α in the $'$ -anti-invariant line), and $\beta := \alpha^2 \in K_0$ is fixed. The same positivity argument, applied place by place at each real embedding σ of K_0 (as in Step 2, isolating the σ -component), forces $\sigma(\beta) = \sigma(\alpha)^2 < 0$ at every real place: were $\sigma(\beta) > 0$ for some σ , one could produce an anti-invariant $a \in K$ with $\text{Tr}_{K/\mathbb{Q}}(aa') < 0$, contradicting positivity. Thus $\beta = \alpha^2 \in K_0$ is totally negative, so $K = K_0(\sqrt{\beta})$ is totally imaginary over the totally real K_0 . Thus K is a CM field and $'$, sending $\alpha \mapsto -\alpha$, is complex conjugation at every archimedean place, completing the proof.

The proposition already carves the possibilities into two worlds. Either the center is the totally real field K_0 , and the involution is of the *first kind* (trivial on the center), or the center is a CM field with the involution of the *second kind* (complex conjugation on the center). Albert's theorem refines the first kind into three sub-cases according to whether D is commutative or a quaternion algebra and, in the quaternion case, whether the involution is orthogonal or symplectic. The result is four types.

Definition 6.3.3: The Four Albert Types

Let $(D, ')$ be a division algebra with positive involution over $\mathbb{Q}$, with center K and totally real fixed field K_0 . The pair is of one of the following four types.

1. *Type I.* $D = K = K_0$ is a totally real field and $'$ is the identity.
2. *Type II.* $K = K_0$ is totally real, D is a quaternion division algebra over K_0 that is *totally indefinite* (split at every real place, $D \otimes_{K_0, \sigma} \mathbb{R} \cong M_2(\mathbb{R})$ for every embedding σ), and $'$ is of *orthogonal type*.
3. *Type III.* $K = K_0$ is totally real, D is a quaternion division algebra over K_0 that is *totally definite* (ramified at every real place, $D \otimes_{K_0, \sigma} \mathbb{R}$ is the Hamilton quaternions for every embedding σ), and $'$ is of *symplectic type*.
4. *Type IV.* K is a CM field (a totally imaginary quadratic extension of the totally real K_0), D is a division algebra with center K , and $'$ is of the *second kind*, restricting to complex conjugation on K .

Here the Hamilton quaternions are the unique division algebra of dimension 4 over $\mathbb{R}$, and the involution kind (orthogonal or symplectic) refers to the type of the symmetric bilinear form fixed by $'$ on the reduced 2×2 representation of the split quaternion algebra.

The distinction between orthogonal and symplectic involutions in Types II and III is the involution's behavior on a quaternion algebra D over K_0 . Every K_0 -linear involution of a quaternion algebra is either the standard (conjugation) involution or a twist of it; the standard involution is symplectic, and a nonstandard one is orthogonal. Positivity selects: a totally indefinite quaternion algebra admits a positive involution only of orthogonal type, and a totally definite one only of symplectic type, which is why the two quaternionic cases split cleanly into Types II and III.

Theorem 6.3.4: Albert's Classification

Every division algebra with positive involution $(D, ')$ over $\mathbb{Q}$ is of exactly one of the four types of definition 6.3.3. The type is determined by the pair (K, K_0) and the kind of the involution:

- if D is commutative, then $D = K$; it is Type I when $K = K_0$ and the maximal Type IV case (a field) when K is CM;
- if D is noncommutative with $K = K_0$, then D is a quaternion algebra over K_0 and the type is II or III according as the positive involution is orthogonal or symplectic, equivalently according as D is totally indefinite or totally definite;
- if D is noncommutative with K a CM field, the type is IV.

Consequently the endomorphism algebra $\text{End}^0(X)$ of a simple abelian variety X , together with its Rosati involution, is of one of these four types.

Proof:

Let $(D, ')$ be a division algebra with positive involution, center K , fixed field K_0 . By proposition 6.3.2, K_0 is totally real, and either $K = K_0$ or K is a CM field with $'$ complex conjugation on it. These two cases are the involutions of the first and second kind. The proof is a case analysis on $[K : K_0]$ and on the dimension $[D : K]$, which for a division algebra with center a field is a perfect square d^2 (Wedderburn theory of central simple algebras, *Core Representation Theory* [2]).

Step 1: Second kind gives Type IV. If K is a CM field, then by definition the involution restricts to complex conjugation on the center, and $(D, ')$ is of Type IV whatever the value of d . When $d = 1$ the algebra $D = K$ is the field itself (the maximal commutative Type IV case, realized by CM abelian varieties); when $d > 1$ it is a genuine division algebra over the CM field K . No further constraint on the kind is needed, since complex conjugation on a CM field is the unique nontrivial involution fixing K_0 .

Step 2: First kind, commutative, is Type I. Suppose $K = K_0$ and D is commutative, so $D = K = K_0$ is a totally real field. The involution is a $\mathbb{Q}$ -linear involution of a field fixing $K_0 = K$ pointwise, hence trivial. Positivity is automatic and the trace form on a totally real field is positive definite. This is Type I.

Step 3: First kind, noncommutative, is quaternionic. Suppose $K = K_0$ and D is noncommutative of degree $d = [D : K]^{1/2} > 1$. The existence of an involution of the first kind already forces D to have order dividing 2 in the Brauer group of K : the involution gives an isomorphism $D \cong D^{\text{op}}$, and D^{op} is the Brauer inverse of D , so $[D] = [D]^{-1}$ and $2[D] = 0$ in $\text{Br}(K)$ (*Core Representation Theory* [2]). This alone does not bound the degree: exponent 2 only forces d to be a power of 2, and there exist central division algebras of exponent 2 and degree 4 over a number field (Albert's biquaternion algebras). What rules those out here is not the involution alone but its *positivity*. This is a theorem of Albert: a division algebra over a totally real number field that carries a positive involution of the first kind has degree $d \leq 2$. The obstruction is global. At each real place σ the completion $D \otimes_{K_0, \sigma} \mathbb{R}$ is a real central simple algebra (a matrix algebra over $\mathbb{R}$ or over the Hamilton quaternions), and a first-kind involution positive at σ exists on such a completion in every degree; positivity at a single place does not cap the degree, exactly as the transpose is a positive first-kind involution on $M_n(\mathbb{R})$ for every n . The constraint is that a *single* $\mathbb{Q}$ -algebra involution must be positive at *all* real places of K_0 simultaneously while D remains a division algebra globally, and Albert's analysis shows this is impossible once $d > 2$; the argument is a real-place analysis of positive first-kind involutions on the division algebra, carried out in Albert's *Structure of Algebras* and reproduced in Mumford's *Abelian Varieties*. Combined with $d > 1$ this gives $d = 2$, so D is a quaternion algebra over K_0 .

Step 4: Orthogonal versus symplectic. An involution of the first kind on a quaternion algebra D is either the standard involution $x \mapsto \bar{x}$ (whose fixed space is the center K_0 , of dimension 1 over K_0) or a nonstandard one (whose fixed space of $'$ -symmetric elements has dimension 3). The standard involution is *symplectic*; a nonstandard involution is *orthogonal*. At each real place σ of K_0 , positivity of the involution forces a compatibility with the local invariant of D : the split quaternion algebra $M_2(\mathbb{R})$ carries a positive involution only of orthogonal type, while the ramified quaternion algebra (the Hamilton quaternions) carries a positive involution only of

symplectic type (the standard conjugation $x \mapsto \bar{x}$ is positive because the reduced trace satisfies $\text{Trd}(x\bar{x}) = 2 \text{Nrd}(x) > 0$). Since a positive involution over $\mathbb{Q}$ is positive at every real place simultaneously, either D is split at every real place and $'$ is everywhere orthogonal (Type II, totally indefinite), or D is ramified at every real place and $'$ is everywhere symplectic (Type III, totally definite). The two cannot mix, because the involution's kind is a global datum: it is orthogonal or symplectic as an involution of the $\mathbb{Q}$ -algebra D , and this single choice must be positive at every real place at once.

The four cases exhaust the possibilities of proposition 6.3.2 and are mutually exclusive by their invariants (K, K_0, kind) , so every $(D, ')$ is of exactly one type. Applying this to $D = \text{End}^0(X)$ with its Rosati involution, which is a division algebra with positive involution when X is simple, gives the final assertion, completing the proof.

Albert's theorem is the exact answer to the question that opened this chapter: which division algebras arise as End^0 of a simple abelian variety? The space of finite-dimensional division algebras over $\mathbb{Q}$ is a continuum, indexed by centers and by local invariants at every place; positivity cuts it down to four families, each governed by a totally real field K_0 and a small amount of quaternionic or CM data. The refinement of this coarse classification into precise numerical bounds relates three integers: the degree $[K_0 : \mathbb{Q}]$, the degree $d = [D : K]^{1/2}$, and the dimension g of the abelian variety. The next proposition records these constraints, citing the exhaustive table to the standard sources.

Proposition 6.3.5: Structural Constraints on the Type

Let X be a simple abelian variety of dimension g over an algebraically closed field, $D = \text{End}^0(X)$ its endomorphism algebra, K the center of D , K_0 the totally real fixed field of the Rosati involution, $e_0 = [K_0 : \mathbb{Q}]$, and $d = [D : K]^{1/2}$ the degree of D over its center. Then K_0 is totally real, $(D, ')$ is of one of the four types of theorem 6.3.4, and the following divisibility constraints relate the invariants to g :

- Type I: $D = K_0$, $e_0 | g$;
- Type II: $d = 2$, $K = K_0$, $2e_0 | g$;
- Type III: $d = 2$, $K = K_0$, $2e_0 | g$;
- Type IV: $[K : K_0] = 2$, and $e_0 d^2 | g$.

In particular $[D : \mathbb{Q}]$ divides $2g$ in Types I, II, III (where $[D : \mathbb{Q}] = e_0 d^2$ with $d \in \{1, 2\}$) and $\text{End}^0(X)$ has $\mathbb{Q}$ -dimension at most $(2g)^2$ in all cases (proposition 6.1.3). The precise upper bounds in each type, and the finer constraints in positive characteristic, are as in Mumford's *Abelian Varieties* and Birkenhake-Lange's *Complex Abelian Varieties*.

Proof:

That K_0 is totally real and that $(D, ')$ is of one of the four types is proposition 6.3.2 and theorem 6.3.4. The divisibility constraints come from the action of D on the rational homology $H_1(X, \mathbb{Q})$, a $\mathbb{Q}$ -vector space of dimension $2g$ which by proposition 1.2.6 carries the faithful rational representation ρ_r of $\text{End}^0(X)$. Because X is simple, $H_1(X, \mathbb{Q})$ is a module over the division algebra D , hence a free D -module, so $[D : \mathbb{Q}] = e_0 d^2 [K : K_0]$ divides $\dim_{\mathbb{Q}} H_1(X, \mathbb{Q}) = 2g$. Reading off the value $[K : K_0] \in \{1, 2\}$ and $d \in \{1, 2, \dots\}$ in each type yields the stated

divisibilities. In Types II and III, $[D: \mathbb{Q}] = e_0 d^2 [K: K_0] = e_0 \cdot 4 \cdot 1 = 4e_0$, so $4e_0 \mid 2g$, which is exactly $2e_0 \mid g$; no analytic refinement is needed. In Type IV, $[D: \mathbb{Q}] = e_0 d^2 \cdot 2$, so $2e_0 d^2 \mid 2g$, i.e. $e_0 d^2 \mid g$ directly. Only Type I needs the refinement: there $[D: \mathbb{Q}] = e_0$ gives $e_0 \mid 2g$, and $e_0 \mid g$ follows from the fact that the analytic representation is a form of the rational one of half the dimension.

The Type I refinement from $e_0 \mid 2g$ to $e_0 \mid g$ requires tracking how the analytic representation ρ_a of definition 1.2.2 sits inside $\rho_r \cong \rho_a \oplus \bar{\rho}_a$ (proposition 1.2.5): the tangent space $V = H^{1,0}$ is a D -module of dimension g over $\mathbb{C}$, and the compatibility of a positive involution with a CM or totally real structure constrains the multiplicities of the simple $D \otimes \mathbb{R}$ -summands in V . The full bookkeeping producing the exhaustive table of admissible (e_0, d, g) in each type, together with the reduction data $[K: K_0]$ and the multiplicity restrictions on V , is a finite but intricate case analysis; it is carried out in full in Mumford's *Abelian Varieties* and Birkenhake-Lange's *Complex Abelian Varieties*, and is cited here rather than reproduced, per the batch contract's sanctioned deferral. What is proved in full above is the qualitative content: the four types, the totally real fixed field, and the divisibility of $[D: \mathbb{Q}]$ into $2g$, which is what the applications in this book require, completing the proof.

The constraints already rule out many possibilities. For an elliptic curve, $g = 1$, so $[D: \mathbb{Q}] \mid 2$: the endomorphism algebra is $\mathbb{Q}$ (Type I with $e_0 = 1$) or an imaginary quadratic field (Type IV with $e_0 = 1, d = 1$), and no quaternion algebra can occur over $\mathbb{C}$, matching example 6.1.5. Every type is realized somewhere in the range $g \geq 1$, and the following example exhibits a witness for each.

Example 6.3.6: Realizations of the Four Types

Each Albert type occurs as the endomorphism algebra of a simple abelian variety.

1. *Type I.* The generic elliptic curve $E = \mathbb{C}/\Lambda$ with $\text{End}^0(E) = \mathbb{Q}$ is Type I with $K_0 = \mathbb{Q}$, $e_0 = 1$, and $g = 1$. A higher-dimensional Type I example is a *real multiplication* abelian surface: a simple abelian surface X with $g = 2$ whose endomorphism algebra is a real quadratic field $K_0 = \mathbb{Q}(\sqrt{m})$, $m > 0$, so $e_0 = 2 = g$, saturating the constraint $e_0 \mid g$. Such surfaces populate the Hilbert modular surfaces and are the first place beyond elliptic curves where a nontrivial totally real endomorphism field appears.
2. *Type IV.* A CM elliptic curve, for instance $E: y^2 = x^3 - x$ with $\text{End}^0(E) = \mathbb{Q}(i)$, is Type IV with the CM field $K = \mathbb{Q}(i)$, $K_0 = \mathbb{Q}$, $d = 1$, and $g = 1$; the Rosati involution is complex conjugation on $\mathbb{Q}(i)$. The general Type IV case with $D = K$ a field is a CM abelian variety $X_\Phi = \mathbb{C}^\Phi / \Phi(\mathcal{O}_K)$ for a CM field K of degree $2g$ and a CM type Φ , the construction taken up in section 6.4; there K is a maximal commutative semisimple subalgebra of $\text{End}^0(X)$ of dimension $2g$, the maximal Type IV situation. Noncommutative Type IV, with D a genuine division algebra over a CM field, first occurs at $g \geq 4$, since the constraint $e_0 d^2 \mid g$ of proposition 6.3.5 already forces $4 \mid g$ in the minimal case $e_0 = 1, d = 2$.
3. *Type II.* A simple abelian surface with indefinite quaternionic multiplication realizes Type II: $g = 2$, $K_0 = \mathbb{Q}$, $e_0 = 1$, and D an indefinite ($D \otimes \mathbb{R} \cong M_2(\mathbb{R})$) quaternion division algebra over $\mathbb{Q}$ with $2e_0 = 2 = g$. These are the fake elliptic curves, abelian surfaces with multiplication by an indefinite quaternion order, parametrized by Shimura curves; they are the smallest-dimensional quaternionic case.
4. *Type III.* The divisibility $2e_0 \mid g$ permits $g = 2$ for a definite quaternion algebra ($D \otimes \mathbb{R}$ the

Hamilton quaternions), but this necessary constraint is not sufficient: over $\mathbb{C}$ a definite quaternion algebra over $\mathbb{Q}$ does not occur as End^0 of a simple abelian *surface*. Indeed the simple abelian surfaces over $\mathbb{C}$ have endomorphism algebra $\mathbb{Q}$, a real quadratic field (Type I), an indefinite quaternion algebra (Type II), or a quartic CM field (Type IV); the definite quaternion case is excluded by the multiplicity restrictions on the tangent space (Shimura; Birkenhake-Lange, *Complex Abelian Varieties*, Ch. 9). A genuine Type III witness first appears in higher dimension: with $K_0 = \mathbb{Q}$ and $d = 2$ the smallest simple abelian variety with End^0 a definite quaternion algebra over $\mathbb{Q}$ has $g = 4$, the finer bound being part of the exhaustive table deferred to those references.

6.4 Complex Multiplication

Albert’s classification (theorem 6.3.4) fixes the shape of the algebras that can occur as $\text{End}^0(X)$, but it says nothing about which abelian variety realizes a given algebra, nor how large the algebra can grow before the abelian variety it acts on is pinned down. The extremal case is the one where the commutative part of $\text{End}^0(X)$ is as large as the rational representation permits: a field of degree $2g$ acting on a g -dimensional abelian variety. This is complex multiplication. Its construction is purely a matter of embedding a number field into $\mathbb{C}^g$, yet the resulting tori turn out to be defined over number fields and to generate abelian extensions of those fields, exactly as the j -invariant of a CM elliptic curve generates the Hilbert class field of an imaginary quadratic field.

This section builds the CM abelian varieties from a field and a combinatorial datum, the CM type, and identifies them inside Albert’s Type IV. The arithmetic main theorem, which governs their fields of definition and the reciprocity law relating the two, is the subject of the downstream volume and is only stated here.

The fields that can act are constrained by the positivity of the Rosati involution: an involution of the second kind must fix a totally real field and conjugate on the field itself. The class of fields with this property has a name.

Definition 6.4.1: CM Field

A *CM field* is a totally imaginary quadratic extension K of a totally real number field K_0 . Thus K_0 is a number field all of whose embeddings $K_0 \hookrightarrow \mathbb{C}$ land in $\mathbb{R}$, and $K = K_0(\sqrt{-\delta})$ for some totally positive $\delta \in K_0$, so that $[K : \mathbb{Q}] = 2[K_0 : \mathbb{Q}]$ and no embedding of K into $\mathbb{C}$ has image in $\mathbb{R}$. Complex conjugation on $\mathbb{C}$ induces, via any embedding $K \hookrightarrow \mathbb{C}$, a single automorphism of K independent of the embedding, the *complex conjugation* of K , written $a \mapsto \bar{a}$; its fixed field is K_0 .

The independence of the conjugation from the chosen embedding is what makes a CM field behave like $\mathbb{C}$ over $\mathbb{R}$: there is an intrinsic “real part” K_0 and an intrinsic conjugation, and every place at infinity is complex. For $K_0 = \mathbb{Q}$ the definition recovers the imaginary quadratic fields, the fields that govern the $g = 1$ theory (*Elliptic Curves* [3]). See *Algebraic Number Theory* [1] for totally real and CM fields, their rings of integers, and the fact that the conjugation is well defined.

Example 6.4.2: The Cyclotomic Field $\mathbb{Q}(\zeta_n)$

For $n \geq 3$ the cyclotomic field $K = \mathbb{Q}(\zeta_n)$ is a CM field. Its maximal real subfield is $K_0 = \mathbb{Q}(\zeta_n + \zeta_n^{-1})$, which is totally real of degree $\varphi(n)/2$, and $K = K_0(\zeta_n)$ is a totally imaginary quadratic extension, since ζ_n satisfies $x^2 - (\zeta_n + \zeta_n^{-1})x + 1 = 0$ with discriminant $(\zeta_n + \zeta_n^{-1})^2 - 4 = (\zeta_n - \zeta_n^{-1})^2 < 0$ under every real embedding of K_0 . Complex conjugation is $\zeta_n \mapsto \zeta_n^{-1} = \bar{\zeta}_n$. The smallest case $n = 3$ gives $K = \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$, imaginary quadratic, with $K_0 = \mathbb{Q}$: this is the field of complex multiplication of the curve $y^2 = x^3 + 1$.

To turn a CM field into an abelian variety we must embed it into a complex vector space, and a $\mathbb{C}$ -algebra homomorphism $K \rightarrow \mathbb{C}^g$ is a choice of g of the $2g$ embeddings of K . The embeddings come in conjugate pairs, and choosing one from each pair is precisely the datum that makes the embedded lattice compatible with a complex structure.

Definition 6.4.3: CM Type

Let K be a CM field with $[K : \mathbb{Q}] = 2g$, so that $|\text{Hom}(K, \mathbb{C})| = 2g$. A *CM type* for K is a subset $\Phi \subset \text{Hom}(K, \mathbb{C})$ such that

$$\Phi \sqcup \bar{\Phi} = \text{Hom}(K, \mathbb{C})$$

where $\bar{\Phi} = \{ \bar{\varphi} : \varphi \in \Phi \}$ and $\bar{\varphi}(a) := \overline{\varphi(a)}$. Equivalently, Φ contains exactly one embedding from each conjugate pair $\{ \varphi, \bar{\varphi} \}$. In particular $|\Phi| = g = [K_0 : \mathbb{Q}]$. The pair (K, Φ) is a *CM pair* of degree $2g$.

A CM type is a system of representatives for the complex conjugation acting on the $2g$ embeddings. Since conjugation is fixed-point-free on $\text{Hom}(K, \mathbb{C})$ (no embedding is real, by total imaginarity), the $2g$ embeddings split into g conjugate pairs, and a CM type picks one from each: there are exactly 2^g of them. The choice matters. Different CM types on the same field produce non-isomorphic, in general non-isogenous, abelian varieties, and the type is the combinatorial shadow of the analytic structure.

Example 6.4.4: CM Types on an Imaginary Quadratic Field

Let K be imaginary quadratic, so $[K : \mathbb{Q}] = 2$ and $g = 1$. There are two embeddings $\varphi, \bar{\varphi} : K \rightarrow \mathbb{C}$, forming a single conjugate pair, so there are exactly $2^1 = 2$ CM types, $\Phi = \{ \varphi \}$ and $\Phi' = \{ \bar{\varphi} \}$. The two are interchanged by conjugation and give complex-conjugate, hence isomorphic, elliptic curves. For a genuinely higher-dimensional example take $K = \mathbb{Q}(\zeta_5)$, a CM field of degree 4 with $g = 2$ (example 6.4.2). The four embeddings send ζ_5 to the four primitive fifth roots of unity $\zeta, \zeta^2, \zeta^3, \zeta^4$, paired by conjugation as $\{ \zeta, \zeta^4 \}$ and $\{ \zeta^2, \zeta^3 \}$ since $\zeta^4 = \bar{\zeta}$ and $\zeta^3 = \bar{\zeta}^2$. Choosing $\Phi = \{ \zeta \mapsto \zeta, \zeta \mapsto \zeta^2 \}$ gives one of the $2^2 = 4$ CM types.

The action of K on the associated abelian variety is what deserves the name “complex multiplication”: the field multiplies the tangent space in g independent ways at once. Before constructing the variety, the defining property is worth isolating, so that it can be checked against Albert’s list.

Definition 6.4.5: Abelian Variety with Complex Multiplication

An abelian variety X of dimension g has *complex multiplication* if its endomorphism algebra $\text{End}^0(X)$ contains a commutative semisimple $\mathbb{Q}$ -subalgebra of dimension $2g$ over $\mathbb{Q}$; when that subalgebra is a field K , one says X has *complex multiplication by K* , and $2g = [K : \mathbb{Q}]$ is forced. Since the rational representation ρ_r is faithful of degree $2g$ (proposition 1.2.6), a commutative subalgebra of $\text{End}^0(X)$ has $\mathbb{Q}$ -dimension at most $2g$, so complex multiplication is the maximal commutative case.

The bound $\dim_{\mathbb{Q}} \leq 2g$ comes directly from the faithfulness of ρ_r : a commutative semisimple algebra acting faithfully on the $2g$ -dimensional $\mathbb{Q}$ -vector space $\Lambda \otimes \mathbb{Q}$ has dimension at most $2g$, because a maximal commutative subalgebra of $\text{End}_{\mathbb{Q}}(\Lambda \otimes \mathbb{Q}) \cong M_{2g}(\mathbb{Q})$ that is a product of fields is a maximal étale subalgebra, of dimension equal to the size $2g$ of a simultaneous eigenbasis after extending scalars (*Core Algebra* [7]). Complex multiplication saturates this bound. When X is simple the relevant subalgebra is a field, and the definition asks for a maximal subfield of degree $2g$; the general definition allows a product of CM fields, matching the reducible case. As will be seen in proposition 6.4.10, positivity of the Rosati form forces such a maximal field to be a CM field, which is why the name attaches to that class.

We now construct, from a CM pair, an abelian variety carrying the action. The construction is the direct generalization of $E = \mathbb{C}/\mathcal{O}_K$ from the imaginary quadratic case.

Theorem 6.4.6: Construction of CM Abelian Varieties

Let (K, Φ) be a CM pair with $[K : \mathbb{Q}] = 2g$ and $|\Phi| = g$, and let $\mathcal{O}_K$ be the ring of integers of K . Embed K into $\mathbb{C}^{\Phi} \cong \mathbb{C}^g$ by

$$\Phi: K \longrightarrow \mathbb{C}^{\Phi}, \quad a \longmapsto (\varphi(a))_{\varphi \in \Phi}$$

Then:

1. $\Phi(\mathcal{O}_K)$ is a full lattice in $\mathbb{C}^{\Phi}$, so $X_{\Phi} := \mathbb{C}^{\Phi}/\Phi(\mathcal{O}_K)$ is a complex torus of dimension g .
2. K acts on X_{Φ} by $\mathbb{Q}$ -algebra endomorphisms: for $k \in K$, multiplication by the diagonal matrix $\text{diag}(\varphi(k))_{\varphi \in \Phi}$ on $\mathbb{C}^{\Phi}$ preserves $\Phi(K)$ and induces $K \hookrightarrow \text{End}^0(X_{\Phi})$. In particular X_{Φ} has complex multiplication by K .
3. X_{Φ} is an abelian variety.

Proof:

Step 1: The lattice is full. The map $\Phi: K \rightarrow \mathbb{C}^{\Phi}$ is $\mathbb{Q}$ -linear, so it suffices to show that $\Phi(\mathcal{O}_K)$ spans $\mathbb{C}^{\Phi}$ over $\mathbb{R}$, equivalently that Φ extends to an $\mathbb{R}$ -linear isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} \mathbb{C}^{\Phi}$. Now $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$ decomposes according to the archimedean places of K ; since K is totally imaginary of degree $2g$, it has $r_1 = 0$ real places and $r_2 = g$ complex places, so $K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{C}^g$ as an $\mathbb{R}$ -algebra, of real dimension $2g$. The g complex places correspond bijectively to the conjugate pairs $\{\varphi, \bar{\varphi}\}$, and a CM type Φ selects one embedding from each pair; the map $a \otimes 1 \mapsto (\varphi(a))_{\varphi \in \Phi}$ is exactly the resulting isomorphism $K \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} \mathbb{C}^{\Phi}$, since each complex place contributes the pair $(\varphi, \bar{\varphi})$ and retaining φ records the full complex coordinate. Hence $\Phi(K \otimes \mathbb{R}) = \mathbb{C}^{\Phi}$, and $\Phi(\mathcal{O}_K)$, the image of a full $\mathbb{Z}$ -lattice $\mathcal{O}_K \subset K$ of rank $2g$, is a full lattice of rank $2g$ in $\mathbb{C}^{\Phi}$.

Step 2: K acts by endomorphisms. Fix $k \in K$ and let $M_k := \text{diag}(\varphi(k))_{\varphi \in \Phi} \in \text{End}_{\mathbb{C}}(\mathbb{C}^{\Phi})$. For $a \in \mathcal{O}_K$,

$$M_k \cdot \Phi(a) = (\varphi(k)\varphi(a))_{\varphi \in \Phi} = (\varphi(ka))_{\varphi \in \Phi} = \Phi(ka)$$

so M_k carries $\Phi(\mathcal{O}_K)$ into $\Phi(k\mathcal{O}_K) \subseteq \Phi(K)$. If $k \in \mathcal{O}_K$ this lands in $\Phi(\mathcal{O}_K)$, so M_k preserves the lattice and, being $\mathbb{C}$ -linear, descends to a holomorphic endomorphism of X_{Φ} (proposition 1.1.7). This gives a ring homomorphism $\mathcal{O}_K \rightarrow \text{End}(X_{\Phi})$, $k \mapsto M_k$, injective because $M_k = 0$ forces $\varphi(k) = 0$, hence $k = 0$. Tensoring with $\mathbb{Q}$ yields an injection of $\mathbb{Q}$ -algebras $K = \mathcal{O}_K \otimes \mathbb{Q} \hookrightarrow \text{End}^0(X_{\Phi})$: for general $k \in K$ some integer multiple nk lies in $\mathcal{O}_K$, and $M_k = \frac{1}{n}M_{nk}$ is a genuine element of $\text{End}^0(X_{\Phi})$. Since $\dim_{\mathbb{Q}} K = 2g = 2 \dim X_{\Phi}$ and K is a field, X_{Φ} has complex multiplication by K in the sense of definition 6.4.5.

Step 3: X_{Φ} is polarizable. It remains to exhibit a positive Hermitian form H on $\mathbb{C}^{\Phi}$ whose imaginary part is integral on $\Phi(\mathcal{O}_K)$, i.e. a polarization, so that X_{Φ} is an abelian variety (definition 3.2.1). Choose $\xi \in K$ purely imaginary for the conjugation, $\bar{\xi} = -\xi$, with $\text{Im}\varphi(\xi) < 0$ for every $\varphi \in \Phi$; such a ξ exists because $K = K_0(\xi_0)$ with $\xi_0 = -\bar{\xi}_0$, and multiplying ξ_0 by a suitable totally positive element of K_0 arranges all g imaginary parts to have the same sign, which we take negative by replacing ξ with $-\xi$ if needed (an approximation available since K_0 is dense in $K_0 \otimes \mathbb{R} = \mathbb{R}^g$ and the totally positive elements form an open cone). Define, for $a, b \in K$,

$$E(\Phi(a), \Phi(b)) := \text{Tr}_{K/\mathbb{Q}}(\xi a \bar{b})$$

extended $\mathbb{R}$ -bilinearly to $\mathbb{C}^{\Phi} = \Phi(K) \otimes \mathbb{R}$. Then E takes integer values on $\Phi(\mathcal{O}_K) \times \Phi(\mathfrak{d}^{-1}\mathcal{O}_K)$ for the inverse different $\mathfrak{d}^{-1}$, and after rescaling ξ by an integer $E(\Phi(\mathcal{O}_K), \Phi(\mathcal{O}_K)) \subseteq \mathbb{Z}$. It is alternating: $E(\Phi(b), \Phi(a)) = \text{Tr}(\xi b \bar{a}) = \text{Tr}(\bar{\xi} b \bar{a}) = \text{Tr}(\bar{\xi} b a) = -\text{Tr}(\xi a \bar{b}) = -E(\Phi(a), \Phi(b))$, using $\text{Tr}(\bar{z}) = \text{Tr}(z)$ and $\bar{\xi} = -\xi$. In the coordinates of $\mathbb{C}^{\Phi}$ the trace runs over the pairs $\{\varphi, \bar{\varphi}\}$, contributing $\varphi(\xi a \bar{b})$ and its conjugate, so $E(\Phi(a), \Phi(b)) = \sum_{\varphi \in \Phi} (\varphi(\xi)\varphi(a)\bar{\varphi}(b) + \overline{\varphi(\xi)\varphi(a)\bar{\varphi}(b)}) = \sum_{\varphi} 2\text{Re}(\varphi(\xi)\varphi(a)\bar{\varphi}(b))$. Writing $\varphi(\xi) = it_{\varphi}$ with $t_{\varphi} = \text{Im}\varphi(\xi) < 0$, each summand is $2\text{Re}(it_{\varphi}\varphi(a)\bar{\varphi}(b)) = -2t_{\varphi}\text{Im}(\varphi(a)\bar{\varphi}(b))$, so $E(\Phi(a), \Phi(b)) = \sum_{\varphi} (-2t_{\varphi})\text{Im}(\varphi(a)\bar{\varphi}(b))$, which is the imaginary part of the Hermitian form

$$H(u, w) = \sum_{\varphi \in \Phi} (-2t_{\varphi}) u_{\varphi} \overline{w_{\varphi}}$$

on $\mathbb{C}^{\Phi}$; here H is diagonal with entries $-2t_{\varphi} = -2\text{Im}\varphi(\xi) > 0$, so $H > 0$. Thus $E = \text{Im}H$ is a polarization and X_{Φ} is an abelian variety, completing the proof.

The construction is entirely explicit: a CM field, a choice of half of its embeddings, and the lattice of algebraic integers embedded through them. Every step is linear algebra over K , yet the output is an abelian variety with an action of K so large that K is a maximal commutative subalgebra of its endomorphism algebra. The positivity in Step 3 is where the CM hypothesis earns its keep: a totally imaginary ξ with all g embeddings on the same side of the real axis exists precisely because K is totally imaginary over the totally real K_0 , and this is the analytic incarnation of the Rosati positivity that will reappear in proposition 6.4.10. For $g = 1$ this is nothing but $E = \mathbb{C}/\mathcal{O}_K$ with K imaginary quadratic, the CM elliptic curve of *Elliptic Curves* [3], recorded below in example 6.4.11.

Example 6.4.7: The CM Abelian Surface from $\mathbb{Q}(\zeta_5)$

Take $K = \mathbb{Q}(\zeta_5)$, $g = 2$, with the CM type $\Phi = \{\varphi_1, \varphi_2\}$, $\varphi_1(\zeta_5) = \zeta$, $\varphi_2(\zeta_5) = \zeta^2$ (where $\zeta = e^{2\pi i/5}$), from example 6.4.4. The ring of integers is $\mathcal{O}_K = \mathbb{Z}[\zeta_5]$, with $\mathbb{Z}$ -basis $1, \zeta_5, \zeta_5^2, \zeta_5^3$. Embedding through Φ sends this basis to the columns

$$\begin{pmatrix} 1 & \zeta & \zeta^2 & \zeta^3 \\ 1 & \zeta^2 & \zeta^4 & \zeta^6 \end{pmatrix} = \begin{pmatrix} 1 & \zeta & \zeta^2 & \zeta^3 \\ 1 & \zeta^2 & \zeta^4 & \zeta \end{pmatrix}$$

(using $\zeta^6 = \zeta$), a 2×4 period matrix whose four columns span a full lattice $\Lambda = \Phi(\mathbb{Z}[\zeta_5])$ in $\mathbb{C}^2$. The surface $X_\Phi = \mathbb{C}^2/\Lambda$ carries an action of $\mathbb{Z}[\zeta_5]$: multiplication by ζ_5 acts as $\text{diag}(\zeta, \zeta^2)$ on $\mathbb{C}^2$, cyclically permuting the basis columns since $\zeta_5 \cdot \zeta_5^j = \zeta_5^{j+1}$. Thus $\mathbb{Q}(\zeta_5) \hookrightarrow \text{End}^0(X_\Phi)$, a field of degree $4 = 2 \dim X_\Phi$, so X_Φ is a simple abelian surface with complex multiplication by $\mathbb{Q}(\zeta_5)$. It is isogenous to the Jacobian of the genus-two quotient curve $y^2 = x^5 - 1$, on which ζ_5 acts by $x \mapsto \zeta_5 x$, one of the oldest examples of a higher-dimensional CM abelian variety.

The construction depends on a second field, hidden in the CM type, that controls the arithmetic: the field generated by the traces of K under Φ . This is the reflex field, and it is the field over which the CM abelian variety and its moduli point are naturally defined.

Definition 6.4.8: Reflex Field and Reflex Type

Let (K, Φ) be a CM pair. Extend each $\varphi \in \Phi$ to an embedding of the Galois closure $\tilde{K}$ of K into $\mathbb{C}$, and let $G = \text{Gal}(\tilde{K}/\mathbb{Q})$ act on $\text{Hom}(K, \mathbb{C})$. The *reflex field* K^* of (K, Φ) is the subfield of $\mathbb{C}$ fixed by

$$\{\sigma \in G : \sigma\Phi = \Phi\}$$

equivalently, $K^* = \mathbb{Q}(\{\text{Tr}_\Phi(a) := \sum_{\varphi \in \Phi} \varphi(a) : a \in K\})$, the field generated over $\mathbb{Q}$ by the “ Φ -traces”. The reflex field is itself a CM field. Regarding a CM type as the subset $\tilde{\Phi} = \{\sigma \in G : \sigma|_K \in \Phi\}$ of G that induces it, the associated *reflex type* Φ^* on K^* is the set of restrictions to K^* of the inverses,

$$\Phi^* = \{\sigma^{-1}|_{K^*} : \sigma \in G, \sigma|_K \in \Phi\}$$

a set of $[K_0^* : \mathbb{Q}]$ embeddings $K^* \hookrightarrow \mathbb{C}$ (the restriction $\sigma^{-1}|_{K^*}$ depending only on the coset, so the count is correct), making (K^*, Φ^*) a CM pair, the *reflex* of (K, Φ) .

When K is abelian over $\mathbb{Q}$, as for the cyclotomic examples, the Galois action is transparent and K^* is a subfield of K itself; in general K^* may differ from K , and the passage $(K, \Phi) \mapsto (K^*, \Phi^*)$ is an involution up to isogeny. The reflex field is where the arithmetic lives: the CM abelian variety X_Φ has a model over a number field, and the reflex norm $N_{\Phi^*} : K^* \rightarrow K$ transports idele classes of K^* to the Galois action on the torsion of X_Φ . This is the content of the CM main theorem, stated below but not proved here.

6.4.9 Key Ideas: The CM Main Theorem A CM abelian variety X_Φ with CM by (K, Φ) is defined over a number field, and its field of moduli is a specific abelian extension of the reflex field K^* . The Galois action on the torsion points $X_\Phi[\mathfrak{m}]$ is described by the *reflex norm* through the reciprocity map of class field theory: an idele class of K^* acts on X_Φ and its torsion via its reflex norm to K , generalizing the statement that the j -invariant of a CM elliptic curve generates the Hilbert class field of the imaginary quadratic field. The proof requires the full apparatus of class field theory and the theory of canonical models, developed in *Class Field Theory* [9] and *Shimura Varieties* [6]; it lies beyond the scope of this book, whose contribution is the construction X_Φ and its endomorphisms, established in full in theorem 6.4.6.

It remains to place CM abelian varieties inside Albert's classification and confirm that a maximal commutative subfield must be a CM field, which is what makes definition 6.4.1 the right notion.

Proposition 6.4.10: CM Abelian Varieties Have Endomorphism Algebra of Type IV

Let X be a simple abelian variety of dimension g with complex multiplication by a field K of degree $2g$, so $K = \text{End}^0(X)$ is its full endomorphism algebra. Then K is a CM field, and $\text{End}^0(X)$ is of Albert Type IV with K a maximal subfield equal to the center. Conversely, every CM field of degree $2g$ arises as $\text{End}^0(X)$ for a simple CM abelian variety of dimension g .

Proof:

Since X is simple, $\text{End}^0(X)$ is a division algebra (proposition 5.4.7); as it contains the field K of maximal degree $2g$ and $\dim_{\mathbb{Q}} \text{End}^0(X) \leq 2g$ (proposition 1.2.6), the containment is an equality $\text{End}^0(X) = K$. Thus $\text{End}^0(X)$ is a field, so its center is K itself and it falls under Albert Type IV, the only type with commutative endomorphism algebra of degree $2g$ (theorem 6.3.4).

It remains to show K is a CM field. The Rosati involution $f \mapsto f'$ attached to any polarization of X (theorem 6.2.3) restricts to an involution of the field $K = \text{End}^0(X)$, and its fixed field $K_0 := \{k \in K : k' = k\}$ is a subfield. Two cases arise. If the Rosati involution is trivial on K , then $K = K_0$ and the positivity $\text{Tr}(kk') = \text{Tr}(k^2) > 0$ for all $k \neq 0$ forces every $\mathbb{Q}$ -embedding of K to send k^2 to a positive real for every k , so K is totally real; but then K would be a totally real field of degree $2g$ acting on the simple X as an Albert Type I algebra, whose numerical constraint requires $[K : \mathbb{Q}] = [K_0 : \mathbb{Q}]$ to divide g (proposition 6.3.5), impossible for $2g \nmid g$ with $g \geq 1$. Hence the involution is nontrivial, $[K : K_0] = 2$, and K_0 is its fixed field. Positivity gives, for $k \in K_0 \setminus \{0\}$, $\text{Tr}(kk') = \text{Tr}(k^2) > 0$, so K_0 is totally real; and for $k \in K$ with $k' = \bar{k}$ the second-kind involution restricts to complex conjugation on K under every embedding, so K is totally imaginary over K_0 . Therefore K is a CM field, its conjugation the Rosati involution.

The converse is theorem 6.4.6: given a CM field K of degree $2g$ and any CM type Φ , the abelian variety X_Φ has $K \hookrightarrow \text{End}^0(X_\Phi)$, and choosing Φ so that X_Φ is simple (which holds when Φ is a *primitive* type, one not induced from a proper CM subfield) yields $\text{End}^0(X_\Phi) = K$ exactly. This completes the proof.

The proposition closes the classification loop: the maximal commutative case of Type IV is exactly complex multiplication by a CM field, and definition 6.4.1 is not an arbitrary choice but the class of

fields the Rosati involution allows to act maximally. The nontriviality of the Rosati involution on K is the algebraic counterpart of the totally imaginary ξ in Step 3 of theorem 6.4.6: both express that a CM field carries an intrinsic conjugation fixing a totally real subfield.

Example 6.4.11: CM Elliptic Curves

The case $g = 1$ is the origin of the whole story. Let K be an imaginary quadratic field, $K = \mathbb{Q}(\sqrt{-d})$ with $d > 0$ squarefree, and $\mathcal{O}_K$ its ring of integers. There is a unique conjugate pair of embeddings $\{\varphi, \bar{\varphi}\} = \text{Hom}(K, \mathbb{C})$, so a CM type is a single embedding $\Phi = \{\varphi\}$, and the construction of theorem 6.4.6 produces

$$E = X_\Phi = \mathbb{C}/\varphi(\mathcal{O}_K) = \mathbb{C}/\mathcal{O}_K$$

an elliptic curve. Its endomorphism algebra is $\text{End}^0(E) = K$, of degree $2 = 2 \dim E$, so E has complex multiplication by K and, by proposition 6.4.10, $\text{End}^0(E)$ is of Type IV with K its own center: the Rosati involution is complex conjugation on K , and $K_0 = \mathbb{Q}$. This is exactly the CM theory of *Elliptic Curves* [3], where the deeper arithmetic is proved: the j -invariant $j(E)$ is an algebraic integer, generating the Hilbert class field of K over K , and the torsion of E generates the ray class fields, the imaginary quadratic instance of the CM main theorem. For $d = 1$, $K = \mathbb{Q}(i)$ and $E = \mathbb{C}/\mathbb{Z}[i]$ is the curve $y^2 = x^3 - x$; for $d = 3$, $K = \mathbb{Q}(\zeta_3)$ and $E = \mathbb{C}/\mathbb{Z}[\zeta_3]$ is $y^2 = x^3 + 1$.

Part II

Abelian Varieties over Arbitrary Fields

Part I built the theory of abelian varieties on the complex numbers, where a lattice $\Lambda \subset V = \mathbb{C}^g$ makes every construction explicit: an abelian variety was a polarizable complex torus, and its line bundles, its dual, and its endomorphisms were computations with the pair (V, Λ) . None of that scaffolding survives over a general field. There is no universal cover, no lattice to count, and no Hermitian form; an abelian variety must be built and studied as an algebraic object in its own right.

This Part rebuilds the theory over an arbitrary field k , defining an abelian variety to be a complete connected group variety and extracting its structure from the geometry of that group alone. The device that replaces the lattice is the theorem of the cube, from which the rigidity of the group law, the behavior of line bundles under translation, and the degree n^{2g} of multiplication by n all follow. ?? lays the groundwork: it fixes the notion of an abelian variety as a group scheme, proves the rigidity lemma that forces the group law to be commutative and every morphism to be a translate of a homomorphism, and sets up the line bundles and the Picard functor on which the theorem of the cube will act. Over $\mathbb{C}$ the algebraic and analytic theories describe the same objects, a comparison this Part makes precise once the dual abelian variety and its polarizations are available algebraically.

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